

Seed-fate prediction: full results

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1 Experimental setup

We group our experiments into three categories.

1.1 Category 1: predicting the fate of a seed

Given a trained diffusion model and a seed x_T , we ask whether we can predict, without running the sampler, if x_T ends in a hallucination and, if not, in which mode. Our approach uses the true score. We take points of known fate in the data space, map them back to the seed space with the true score, and fit a predictor on the result. We then test this predictor against the fates that the learned sampler actually produces.

Data. We use a mixture of K Gaussians in $\mathbb{R}^d$, whose true score is known in closed form:

$$p(x) = \sum_{i=1}^K \pi_i \mathcal{N}(\mu_i, \sigma^2 I), \quad (1)$$

with $\sigma = 0.1$. Let R_{99} be the radius of the ball containing 99% of the mass of one component. We place the means μ_i on a sphere of radius $2\sqrt{d}/2$, at least $3R_{99}$ apart, so that these balls never overlap.

We consider two versions of the mixture. In the *unweighted* version, all modes are equally likely, $\pi_i = 1/K$. In the *weighted* version, we draw K weights from $U[0.3, 0.8]$ once and normalise them to sum to one, and each run assigns them to the modes in a random order. The most likely mode is then at most $8/3$ times as likely as the least likely one. The two versions are separate experiments, each with its own trained models. The weighted version tests whether the predictors still work when some modes attract more seeds than others.

The fate of a seed is defined by where the sampler takes it. If x_0 lands within R_{99} of a mean μ_k , the fate of x_T is mode k ; if it lands in none of the balls, it is a hallucination, with label -1 .

Model. We train a residual MLP with four blocks to predict the noise (DDIM) or the velocity (flow matching), and sample with $T_{\text{train}} = 500$ steps (Figure 1). The width is 256 for $d \leq 64$ and 1024 for $d \geq 128$, since the smaller network cannot resolve the modes in high dimension. A model that hallucinates on more than 1.5% of seeds is considered under-trained and is retrained once with $1.7\times$ the steps. Table 1 lists the remaining training details.

Samplers. We study two generative processes, and for one of them four numerical solvers. All samplers are deterministic, so each seed has exactly one fate.

DDIM. A variance-preserving diffusion with a linear noise schedule, β from 10^{-4} to 0.02. The network predicts the noise added to a data point, and we sample with the deterministic DDIM update. This update is Euler’s method applied to the probability-flow ODE of the diffusion.

Flow matching. The optimal-transport path of Lipman et al. (2023), which moves a noise sample z to a data point x along a straight line, $x_t = (1 - t)z + tx$ for $t \in [0, 1]$, up to a small $\sigma_{\min} = 10^{-4}$. The network predicts the velocity $x - z$, and we sample by integrating it from $t = 0$ to $t = 1$ with Euler steps.

Heun, *RK45* and *DPM-Solver++(2M)*. These are not new models. They solve the same probability-flow ODE with the trained DDIM network, but with a more accurate solver than DDIM’s Euler step. Heun’s method is second order and uses two network evaluations per step. RK45 is a fixed-step Dormand–Prince method, fifth order, with six evaluations per step. DPM-Solver++(2M) (Lu et al., 2022) is a second-order multistep solver that works on the network’s estimate of x_0 and uses one evaluation per step. All three use the same 500-step grid as DDIM. Since they share DDIM’s network, they also share its anchors and differ only in the ground truth. Comparing them with DDIM therefore shows how the solver alone changes the fates.

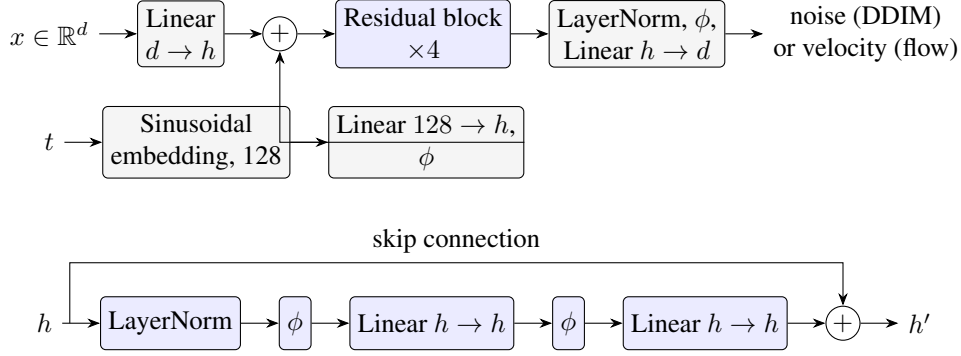


Figure 1: **Score network.** Top: full network, with hidden width h . Bottom: one residual block. ϕ is LeakyReLU(0.2).

Table 1: **Score network: architecture and training.**

Hidden width	256 for $d \leq 64$; 1024 for $d \geq 128$
Residual blocks	4
Time embedding	sinusoidal, dimension 128
Activation	LeakyReLU(0.2)
Output	noise (DDIM) or velocity (flow matching)
Training set	10^5 samples from p
Optimiser	Adam, learning rate 10^{-3} , cosine decay to 10^{-5}
Batch size	512
Gradient clipping	norm 1
Weight EMA	decay 0.999
Training steps	$30,000 (1 + d/16)(1 + K/16)$
Retraining	once, with $1.7 \times$ the steps, if hallucination rate $> 1.5\%$
Sampling steps	500

Anchors. Anchors are the training data of the predictors: seeds with a known fate. For a budget of b anchors per mode, we sample b points uniformly inside each R_{99} ball, labelled with that mode, and $b/2$ points in the band between R_{99} and $R_{99} + 2\sigma$, labelled -1 . We then map all of them from x_0 to x_T by running the sampler’s ODE backwards with the true score (for flow matching, the true velocity), using Heun’s method with T steps (Figure 2, top).

Ground truth. We draw $50,000 \cdot K$ test seeds from $\mathcal{N}(0, I)$ and label each by running the learned sampler (Figure 2, bottom). Scaling with K keeps the number of test seeds per mode fixed. The anchors come from

the true score while the ground truth comes from the learned model, so a high score means that the true score predicts the learned model’s behaviour. Since T enters only the anchors, the ground truth is the same for every T .

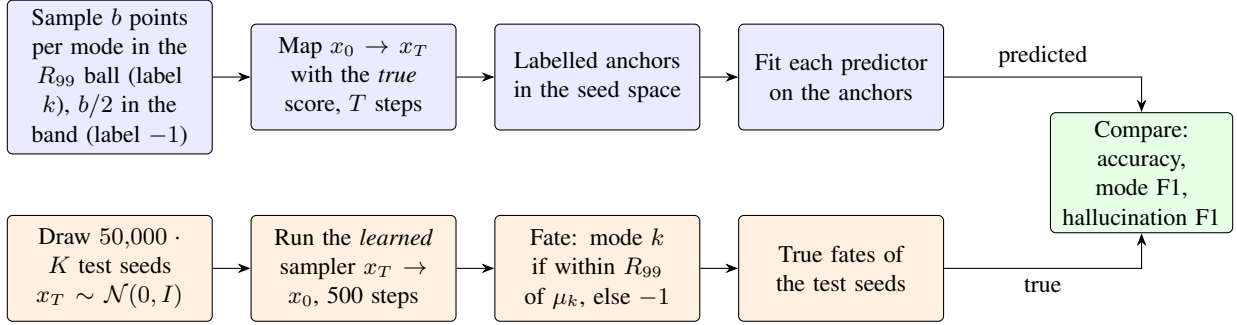


Figure 2: **Evaluation protocol.** Top: anchors are placed around each mode and mapped to the seed space with the true score; the predictors are fit on them. Bottom: test seeds are labelled by the learned sampler. The predictors are evaluated on the same test seeds.

Metrics. We report overall accuracy over the $K + 1$ classes, mode F1 averaged over the K modes, and hallucination F1. Hallucinations make up only a few percent of seeds, so a predictor that never predicts them still reaches high accuracy; hallucination F1 measures whether it finds them. All results are averaged over three runs, each with new means, a new network initialisation and new test seeds.

Predictors. We compare four predictors. Each makes a different assumption about how fates are arranged in the seed space. Table 2 lists their settings.

kNN. Assumes that nearby seeds share a fate. A seed receives the most common label among its 10 nearest anchors.

Altered kNN. Assumes that a hallucination is a seed whose nearest anchors come from different modes. It therefore uses mode anchors only, and no hallucination anchors. Around each mean, we place $b/5$ points on each of five spheres, with radii from $0.3R_{99}$ to $1.5R_{99}$ in steps of $0.3R_{99}$. All points carry that mode’s label, with weights decreasing linearly from 0.84 to 0.2, so that anchors closer to the mean count more. As before, we map them to the seed space with the true score.

For a seed x , let w_j and y_j be the weight and label of its j -th nearest anchor, $j = 1, \dots, 10$. The score of mode c is

$$s_c = \frac{\sum_j w_j [y_j = c]}{\sum_j w_j}, \quad (2)$$

and $p = \text{softmax}(s/0.1)$. When the neighbours agree, p is concentrated on one mode; when they disagree, it is spread out. We measure this with the confidence $1 - H(p)/\log K$, where H is the entropy. A seed whose confidence falls below a threshold θ is a hallucination; otherwise it is assigned to the mode with the largest p_c .

Since the anchors contain no hallucinations, θ cannot be learned from them. Instead, we set it on 5,000 separate seeds labelled with the true score: we try 100 values between the 0.5% and 50% quantiles of their confidence and keep the one with the highest hallucination F1.

Quadratic network. Assumes that the boundaries between fates are quadratic surfaces. It is a linear classifier with $K + 1$ outputs on x and all products $x_i x_j$ ($i \leq j$).

Polar network. Assumes that the fate depends mainly on the direction of the seed, and that hallucinations lie on the boundaries between modes. The first holds because seeds concentrate near $\|x_T\| = \sqrt{d}$. We therefore write a seed as a direction $u = x/\|x\|$ and a radial offset $r = \|x\| - \sqrt{d}$. A linear layer maps (u, r) to a vector z of size 256, and the K mode logits are a linear function of z , z^2 and z^3 (element-wise). The hallucination logit grows as the two best modes get closer, that is, near a boundary:

$$\ell_{\text{hall}} = c + a r - b(\ell_1 - \ell_2), \quad (3)$$

where ℓ_1 and ℓ_2 are the two largest mode logits and a, c and $b > 0$ are learned.

Both networks are trained with cross-entropy on all anchors, each as an ensemble of three models.

Table 2: **Predictor settings.**

Predictor	Setting	Value
kNN	neighbours	10
Altered kNN	neighbours	10
	spheres per mode	5, radius $0.3R_{99}$ to $1.5R_{99}$
	anchor weights	0.84 to 0.2, linear in the radius
	softmax temperature	0.1
	threshold θ	best hallucination F1 on 5,000 seeds
Quadratic	features	x_i and $x_i x_j, i \leq j$
Polar	hidden size, degree	256, 3
Quadratic and polar	optimiser	AdamW, learning rate 3×10^{-3} , one-cycle
	weight decay, batch size	10^{-4} , 2,048
	epochs	30 (at least 3,000 steps)
	ensemble	3 models

Experiment grid. We run this protocol for every combination of sampler, mixture weights, K , d , T and anchor budget in Table 3; each combination is one entry of the result tables.

Table 3: **Experiment grid.**

Sampler	DDIM, flow matching; Heun, RK45, DPM-Solver++(2M) (DDIM network)
Mixing weights	uniform; random (drawn from $U[0.3, 0.8]$, normalised)
Number of modes K	2, 4, 8, 16
Dimension d	2, 4, 8, 16, 32, 64, 128, 256 ($K = 16, d = 2$ not run)
True-score steps T	250, 500, 750
Anchors per mode b	2,000, 5,000, 10,000, 15,000, 20,000
Test seeds	$50,000 \cdot K$ (100,000 to 800,000)
Calibration seeds	5,000 (altered kNN only)
Runs	3

1.2 Category 2

Claude in this category we presnet the mnist experimets thank you

You have to first describe the autoencoder training the same thing latent space and that we should how well it works

The question that you must have now is that you have three output splits we do algorithm 1 on latent space and that we actually do all of this on three datasets right

(See experiment 2 somewhere back in desktop to get a sense of what we are trying to do) Next question that you must do is that how we are generating the visualization and the grid to show how it plays in the r99 or everything cutoffs ok and everything as category 2 maybe these sections as experiment 1 and experiment -2 and put in j thank you

1.3 Category 3: comparison with prior work

We compare our method with baselines from prior work on mode accuracy and hallucination F1.

Table 4: **Unweighted DDIM, 2,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.0	98.5	98.5	96.0	96.2	97.2	98.1	98.8	98.9	98.9	99.5	99.5
2	4	91.7	91.6	91.6	88.9	89.5	90.2	97.9	98.8	98.8	98.6	99.5	99.4
2	8	82.6	82.0	81.8	83.9	84.9	87.0	97.1	98.6	98.6	97.5	99.3	99.2
2	16	80.3	78.9	78.8	83.6	74.4	66.2	95.1	97.8	97.8	95.5	97.9	97.8
2	32	80.3	74.9	74.3	76.1	73.9	72.6	84.3	95.5	95.7	92.3	96.5	96.7
2	64	80.8	71.5	70.2	62.9	80.0	77.6	36.2	60.7	64.9	85.2	94.4	94.7
2	128	82.9	69.3	66.3	53.6	70.9	73.2	29.6	39.5	41.0	72.4	93.4	94.2
2	256	83.9	67.4	62.5	55.2	64.7	68.1	24.1	37.3	38.9	56.0	91.8	93.1
4	2	91.6	97.5	96.3	90.6	96.1	94.7	90.7	93.3	92.7	92.1	98.1	96.7
4	4	83.8	84.0	83.7	80.9	87.4	86.8	91.4	93.9	93.1	92.4	97.7	96.6
4	8	70.3	69.4	69.2	82.1	82.7	81.4	93.5	95.2	94.8	94.3	97.6	97.1
4	16	64.4	62.1	62.0	83.2	81.5	82.2	94.1	95.5	95.4	94.5	95.5	95.3
4	32	63.1	56.8	56.4	80.3	71.7	67.0	86.8	92.5	92.7	91.6	93.6	93.3
4	64	64.5	51.7	50.5	68.9	75.8	73.9	40.8	71.0	73.4	85.7	91.7	91.5
4	128	67.9	47.6	44.9	51.3	67.5	61.5	20.3	29.4	31.1	69.8	91.2	91.4
4	256	70.5	45.0	40.1	51.3	65.8	58.4	14.2	21.4	21.3	49.6	88.0	90.1
8	2	95.7	96.7	96.5	94.6	95.2	93.6	82.6	83.5	83.6	96.2	96.9	96.6
8	4	79.2	79.3	78.7	84.8	85.1	84.5	88.2	90.2	89.4	90.3	96.0	94.9
8	8	60.8	60.2	60.0	77.6	78.9	76.1	91.5	92.6	92.2	93.8	96.4	95.7
8	16	51.2	49.4	49.3	78.5	76.0	76.7	91.2	92.4	92.0	93.2	94.5	93.9
8	32	48.8	43.0	42.8	78.9	73.8	73.7	85.8	89.5	89.5	91.3	92.8	92.0
8	64	49.8	37.0	36.2	78.8	68.3	66.6	47.0	71.0	72.3	86.7	90.2	89.7
8	128	53.8	32.4	30.6	59.6	54.2	58.6	15.1	29.8	32.7	70.2	88.5	88.8
8	256	57.3	28.9	25.6	51.7	53.6	43.2	9.0	16.2	16.2	46.5	86.6	88.1
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	77.1	77.4	77.2	82.8	84.8	79.1	87.4	88.8	88.8	92.1	94.6	94.1
16	8	52.9	52.4	52.1	73.2	76.4	75.2	88.0	89.6	89.3	91.4	94.2	93.4
16	16	40.8	39.6	39.6	74.6	72.5	72.2	89.2	90.7	90.5	92.7	93.4	93.0
16	32	36.8	32.5	32.4	77.0	70.3	70.0	83.4	87.1	87.2	90.9	91.8	91.3
16	64	36.9	26.6	26.2	75.7	63.5	61.8	50.5	69.5	70.5	86.4	89.1	88.7
16	128	40.9	21.7	20.7	53.3	49.1	45.2	12.0	34.4	37.9	71.4	86.7	86.5
16	256	45.1	18.2	16.3	51.1	43.5	35.8	5.0	15.0	15.7	44.4	81.9	84.4

2 Tables: overall accuracy

Each table reports overall seed-fate accuracy (% , mean over 3 seeds) on the (K, d) grid, $d \leq 256$, for one anchor budget, with $T = 250, 500, 750$ as sub-columns of each predictor; the best predictor at each T is in bold. Heatmaps show overall accuracy, mode F1 and hallucination F1 for each predictor, one per T and anchor budget. $K = 16, d = 2$ was not run (—).

2.1 Unweighted DDIM

Table 5: **Unweighted DDIM, 5,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.6	99.1	99.1	97.0	97.6	97.7	98.1	98.8	98.8	99.0	99.5	99.5
2	4	93.2	93.2	93.2	84.1	92.6	92.5	98.0	98.9	98.9	98.5	99.5	99.4
2	8	84.2	83.5	83.5	83.8	79.4	85.1	97.5	98.8	98.8	97.6	99.4	99.3
2	16	80.6	79.3	79.4	76.2	86.1	85.8	95.8	98.3	98.3	96.6	99.0	98.9
2	32	80.7	75.9	75.4	77.8	75.9	71.6	87.3	97.1	97.2	93.5	98.5	98.5
2	64	81.8	72.6	71.3	65.1	77.1	78.4	46.2	92.0	93.0	86.2	97.0	97.2
2	128	83.8	70.2	67.4	57.5	66.0	61.1	30.0	43.7	46.9	71.5	95.3	95.8
2	256	84.9	68.6	63.5	58.8	70.5	50.7	25.1	37.1	39.5	51.6	94.4	95.5
4	2	92.0	98.4	96.9	91.4	97.3	95.9	90.8	93.2	92.6	92.1	98.1	96.7
4	4	86.5	87.0	86.8	84.3	87.7	88.5	91.3	93.9	93.1	92.5	97.9	96.7
4	8	72.6	71.6	71.3	82.0	81.2	85.0	93.8	95.3	94.9	94.6	98.1	97.4
4	16	65.5	63.3	63.2	79.9	83.5	83.0	95.0	96.3	96.2	96.3	97.5	97.2
4	32	64.1	58.3	58.0	81.9	82.0	81.2	90.2	95.0	95.1	94.2	96.8	96.5
4	64	65.8	53.1	52.0	68.5	77.7	75.5	63.2	89.0	89.6	88.4	95.4	95.2
4	128	69.2	49.1	46.3	51.7	71.3	68.5	23.9	43.7	49.7	69.5	94.0	94.3
4	256	72.3	46.3	41.2	54.4	62.8	53.7	16.4	25.5	27.5	47.4	91.6	93.1
8	2	96.3	97.6	97.3	95.5	95.9	96.1	82.3	83.2	83.3	96.2	96.9	96.6
8	4	82.8	83.4	82.9	85.7	87.2	86.0	88.1	90.2	89.5	90.3	96.2	95.0
8	8	63.6	63.0	62.9	79.5	78.5	76.6	91.7	92.9	92.5	94.2	96.9	96.3
8	16	52.8	51.0	50.9	78.1	76.1	76.6	92.1	93.3	92.9	94.6	96.2	95.4
8	32	50.1	44.6	44.3	82.0	74.6	71.9	89.7	92.7	92.5	93.7	95.9	95.1
8	64	51.3	38.5	37.7	74.8	69.7	68.6	70.8	85.5	85.8	89.6	94.3	93.8
8	128	55.5	33.7	31.8	51.8	62.5	60.2	21.2	48.2	53.8	72.2	93.1	93.2
8	256	59.4	30.1	26.6	53.4	42.9	44.2	10.9	22.4	25.8	47.4	91.1	92.0
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	81.4	81.8	81.6	85.1	87.1	85.5	87.4	88.9	88.8	92.1	94.7	94.2
16	8	56.3	55.7	55.4	75.0	75.2	75.9	88.3	89.8	89.5	91.9	94.8	94.0
16	16	42.9	41.8	41.7	75.2	73.5	71.6	90.1	91.5	91.2	93.8	94.8	94.4
16	32	38.2	34.0	33.9	78.2	72.2	71.3	87.5	90.5	90.5	93.0	94.2	93.8
16	64	38.5	28.0	27.6	77.2	66.5	65.1	72.4	83.0	83.4	90.0	93.0	92.6
16	128	42.8	22.9	21.8	65.1	58.6	55.7	20.0	50.4	54.6	74.1	91.3	91.1
16	256	47.4	19.1	17.1	51.6	36.6	32.4	6.9	23.7	29.0	49.2	88.0	89.5

Table 6: **Unweighted DDIM, 10,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.8	99.4	99.4	97.9	97.5	98.3	98.0	98.7	98.8	99.0	99.5	99.5
2	4	94.2	94.3	94.3	85.8	91.6	92.7	98.2	99.0	99.0	98.5	99.5	99.4
2	8	85.6	84.6	84.5	83.1	90.3	77.8	97.5	98.9	98.9	97.6	99.4	99.4
2	16	81.1	79.7	79.6	81.0	83.0	82.7	96.0	98.6	98.5	96.9	99.2	99.2
2	32	81.1	76.5	76.0	75.3	79.9	85.7	88.7	97.8	97.9	94.2	99.1	99.1
2	64	82.1	73.2	72.0	64.0	78.8	61.5	50.4	95.1	95.6	86.7	98.3	98.4
2	128	84.6	71.0	68.1	59.8	66.2	60.0	21.7	69.1	75.5	70.9	96.7	97.0
2	256	85.6	69.3	64.2	60.8	61.0	71.8	25.1	38.2	41.6	48.8	95.6	96.5
4	2	92.2	98.8	97.1	91.9	97.7	96.3	90.8	93.3	92.8	92.1	98.2	96.7
4	4	88.3	89.2	88.8	88.1	89.0	85.0	91.5	93.9	93.1	92.5	97.9	96.7
4	8	74.5	73.4	73.0	82.2	84.5	86.0	94.0	95.4	95.0	94.7	98.2	97.5
4	16	66.5	64.4	64.3	80.5	81.5	83.5	95.4	96.7	96.5	96.6	97.9	97.6
4	32	64.9	59.2	58.9	81.8	80.3	80.1	91.6	96.2	96.2	95.1	98.0	97.5
4	64	66.6	54.1	53.0	64.2	77.1	67.6	67.7	92.6	93.0	89.7	97.0	96.7
4	128	70.3	50.1	47.3	54.1	65.7	57.8	24.7	74.2	77.8	70.4	96.0	96.2
4	256	73.3	47.1	41.9	56.9	65.9	61.4	18.0	29.6	34.4	46.7	93.8	95.2
8	2	96.6	97.9	97.7	95.9	97.0	97.0	82.5	83.3	83.4	96.2	96.9	96.6
8	4	85.0	86.0	85.4	86.4	88.7	84.8	88.2	90.2	89.5	90.3	96.2	95.0
8	8	65.9	65.3	65.1	80.8	80.3	74.0	91.7	92.9	92.6	94.3	97.1	96.4
8	16	54.1	52.3	52.1	76.2	76.9	78.0	92.4	93.6	93.1	95.0	96.8	95.9
8	32	51.2	45.8	45.6	81.0	77.1	76.9	91.1	93.9	93.7	94.2	96.8	96.0
8	64	52.3	39.6	38.8	79.5	69.5	70.7	76.4	89.8	89.9	91.0	95.6	95.1
8	128	56.9	34.7	32.8	54.5	66.2	62.1	29.4	72.5	74.5	74.5	95.2	95.1
8	256	60.7	31.0	27.4	56.0	54.0	47.5	12.4	29.3	37.3	48.7	94.2	94.6
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	84.2	84.9	84.7	87.1	87.7	85.2	87.9	89.3	89.2	92.2	94.8	94.2
16	8	59.2	58.6	58.4	76.6	78.1	78.1	88.7	90.2	89.7	91.9	95.1	94.3
16	16	44.5	43.4	43.3	71.5	69.8	73.7	90.9	92.1	91.8	94.2	95.5	95.0
16	32	39.3	35.2	35.1	78.5	72.8	72.2	89.7	92.2	92.1	93.4	95.1	94.7
16	64	39.5	28.9	28.5	76.7	68.4	67.2	79.8	88.2	88.4	90.8	94.3	93.8
16	128	44.1	23.8	22.7	63.7	60.6	57.1	33.3	70.3	72.0	75.9	92.8	92.3
16	256	48.9	19.8	17.6	54.2	44.7	38.9	9.3	31.8	40.5	51.7	91.0	91.3

Table 7: **Unweighted DDIM, 15,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.9	99.5	99.4	98.0	98.0	98.4	98.1	98.8	98.8	99.0	99.5	99.5
2	4	94.8	94.9	94.9	91.5	91.4	89.0	98.1	99.0	99.0	98.5	99.5	99.4
2	8	86.1	85.2	85.3	82.6	90.1	87.5	97.6	98.9	98.9	97.6	99.4	99.3
2	16	81.5	80.2	80.1	69.9	79.7	86.4	96.1	98.7	98.6	96.9	99.2	99.2
2	32	81.3	76.8	76.4	78.6	84.7	85.4	88.9	98.1	98.1	94.4	99.2	99.2
2	64	82.4	73.7	72.5	61.1	74.8	71.3	51.2	96.0	96.4	87.0	98.8	98.8
2	128	84.7	71.3	68.4	61.0	68.7	64.0	10.7	83.1	87.4	70.3	97.0	97.3
2	256	85.9	69.7	64.6	61.9	72.9	59.9	24.0	39.7	43.9	47.6	96.2	97.0
4	2	92.3	98.9	97.2	92.1	98.0	96.5	90.8	93.3	92.8	92.1	98.2	96.7
4	4	89.1	90.2	89.8	88.3	89.7	87.1	91.5	93.9	93.1	92.5	97.9	96.7
4	8	75.2	74.2	73.9	85.1	80.5	86.5	94.0	95.4	94.9	94.8	98.3	97.6
4	16	67.0	64.9	64.8	82.7	81.9	82.2	95.5	96.8	96.6	96.7	98.1	97.7
4	32	65.3	59.8	59.5	85.4	80.4	80.8	92.0	96.6	96.6	95.3	98.2	97.8
4	64	67.1	54.8	53.7	76.7	78.2	73.9	68.5	93.9	94.2	90.2	97.3	97.0
4	128	70.6	50.5	47.8	55.5	74.4	68.0	18.5	82.6	85.0	69.7	96.7	96.9
4	256	73.8	47.6	42.3	58.2	63.4	47.6	17.8	34.5	41.9	46.3	95.2	96.3
8	2	96.7	98.0	97.8	96.3	97.6	97.2	82.5	83.4	83.5	96.2	96.9	96.5
8	4	86.1	87.4	86.8	87.1	88.5	87.7	88.3	90.2	89.4	90.4	96.2	95.0
8	8	67.2	66.5	66.3	81.3	81.0	82.3	91.8	93.0	92.6	94.4	97.2	96.5
8	16	54.8	53.0	52.9	78.0	77.5	77.2	92.5	93.7	93.2	95.1	97.0	96.0
8	32	51.7	46.4	46.1	82.1	76.3	76.5	91.6	94.4	94.1	94.3	97.0	96.1
8	64	52.9	40.2	39.4	76.3	72.9	70.6	78.1	91.4	91.5	91.2	95.9	95.4
8	128	57.4	35.1	33.3	55.9	63.0	57.9	29.1	80.2	81.4	74.1	95.9	95.7
8	256	61.5	31.3	27.7	54.9	57.6	49.2	13.7	35.0	44.7	49.0	95.2	95.6
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	85.6	86.4	86.2	87.7	87.3	87.4	88.6	89.8	89.5	92.2	95.0	94.4
16	8	60.7	60.1	59.8	78.6	79.6	79.2	89.9	91.0	90.2	91.9	95.3	94.5
16	16	45.4	44.3	44.3	71.6	71.7	73.3	92.1	93.2	92.8	94.4	95.8	95.4
16	32	39.9	35.8	35.6	78.6	71.6	72.2	91.5	93.5	93.4	93.8	95.3	95.0
16	64	40.1	29.5	29.1	81.4	69.8	68.7	84.3	91.0	91.1	90.3	94.9	94.5
16	128	44.7	24.1	23.1	54.0	57.5	57.6	41.1	80.3	80.9	77.7	94.0	93.5
16	256	49.8	20.1	17.9	52.6	49.8	37.2	11.0	36.9	46.2	52.7	91.8	92.0

Table 8: **Unweighted DDIM, 20,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.0	99.5	99.5	98.6	98.5	98.4	98.1	98.8	98.9	99.0	99.5	99.5
2	4	95.2	95.2	95.2	90.9	91.1	91.4	98.2	98.9	99.0	98.5	99.5	99.4
2	8	86.5	85.6	85.5	83.5	90.4	84.6	97.6	98.9	98.8	97.6	99.4	99.4
2	16	81.5	80.5	80.4	80.1	83.5	85.6	96.3	98.7	98.6	96.9	99.3	99.2
2	32	81.5	77.0	76.7	76.7	81.8	80.1	89.0	98.3	98.3	94.5	99.2	99.2
2	64	82.5	73.9	72.8	61.4	84.5	80.3	51.0	96.5	97.0	87.2	98.9	99.0
2	128	84.9	71.7	68.7	61.9	73.7	72.8	9.3	88.3	91.3	70.6	97.6	97.8
2	256	86.0	70.0	64.8	62.7	60.8	54.6	21.7	41.9	47.2	46.7	96.7	97.5
4	2	92.4	98.9	97.3	91.9	98.1	96.7	90.8	93.2	92.7	92.1	98.2	96.6
4	4	89.7	90.9	90.6	87.4	89.1	90.0	91.6	94.0	93.1	92.5	97.9	96.7
4	8	75.9	74.8	74.4	85.1	86.5	86.6	94.0	95.4	94.9	94.7	98.3	97.6
4	16	67.2	65.2	65.2	83.2	82.6	80.7	95.5	96.8	96.6	96.6	98.1	97.7
4	32	65.6	60.2	59.9	83.0	80.8	81.9	92.2	96.8	96.8	95.3	98.3	97.8
4	64	67.3	55.1	54.0	77.7	76.8	71.9	68.9	94.5	94.8	90.3	97.6	97.3
4	128	70.9	50.8	48.1	56.3	71.9	67.3	17.3	86.2	88.1	69.4	97.0	97.1
4	256	74.1	47.9	42.7	58.8	66.6	63.1	17.9	39.6	48.3	46.6	95.6	96.5
8	2	96.7	98.1	97.9	96.5	97.5	97.3	83.2	84.2	84.3	96.3	97.0	96.7
8	4	86.8	88.4	87.7	86.4	89.3	87.4	88.8	90.5	89.6	90.4	96.3	95.1
8	8	68.1	67.4	67.2	81.1	82.8	82.6	92.2	93.2	92.8	94.4	97.2	96.5
8	16	55.4	53.6	53.5	76.3	76.8	77.5	93.0	94.0	93.5	95.2	97.2	96.2
8	32	51.9	46.6	46.4	81.2	76.8	75.7	92.3	94.8	94.5	94.4	97.0	96.2
8	64	53.2	40.6	39.9	82.8	75.1	72.1	80.1	92.6	92.6	91.5	96.2	95.6
8	128	57.9	35.5	33.6	56.8	62.7	65.3	29.1	84.5	85.3	75.3	95.9	95.8
8	256	61.8	31.6	28.0	53.2	46.0	48.8	14.6	40.4	49.9	50.3	95.2	95.5
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	86.5	87.5	87.2	87.8	88.7	88.9	88.4	89.5	89.2	92.2	95.1	94.5
16	8	61.7	61.1	60.9	78.7	77.3	79.6	90.4	91.3	90.5	91.9	95.4	94.6
16	16	45.9	44.9	44.9	73.6	72.1	72.0	92.7	93.8	93.3	94.4	96.0	95.5
16	32	40.2	36.1	36.1	78.1	72.3	71.4	92.4	94.3	94.1	93.9	95.6	95.2
16	64	40.6	29.9	29.5	80.1	69.9	69.5	86.4	92.5	92.5	90.1	95.2	94.8
16	128	45.2	24.4	23.3	59.0	57.7	53.1	45.1	84.9	85.2	79.0	94.5	94.0
16	256	50.3	20.4	18.2	50.9	44.0	42.8	12.3	41.9	50.5	53.6	92.8	92.6

Table 9: **Weighted DDIM, 2,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.3	98.5	98.3	95.5	96.9	96.7	95.0	97.8	98.5	96.7	99.4	98.6
2	4	90.7	91.4	91.5	87.2	90.1	86.4	94.7	97.7	98.4	96.1	99.4	98.8
2	8	81.7	81.7	81.9	79.3	78.6	82.5	93.1	96.4	97.0	95.1	99.2	98.7
2	16	79.0	78.4	78.6	79.7	85.0	85.7	91.5	95.7	96.4	94.0	98.0	97.5
2	32	79.0	74.4	74.2	78.3	83.3	73.0	82.4	94.4	95.2	91.1	96.6	96.5
2	64	79.5	70.7	69.7	62.7	79.2	72.6	37.2	62.0	66.1	83.9	94.2	94.9
2	128	81.7	68.9	66.2	53.6	78.7	67.6	29.4	39.5	41.1	71.4	92.9	94.0
2	256	83.0	67.0	62.3	55.2	64.5	54.8	24.0	37.2	38.8	55.6	91.0	92.3
4	2	91.5	97.6	96.4	90.8	95.8	95.6	89.2	93.0	93.0	91.8	98.0	96.6
4	4	82.5	84.0	83.8	82.8	84.9	87.0	88.0	92.9	92.5	90.2	97.5	96.2
4	8	68.9	68.8	68.8	75.8	81.2	81.7	89.5	93.9	94.1	92.4	97.5	96.6
4	16	62.8	61.7	62.0	74.6	81.0	81.1	88.5	93.1	94.0	90.5	95.1	95.1
4	32	61.8	56.2	56.1	79.1	76.4	78.0	82.7	90.2	91.2	89.7	93.4	93.5
4	64	63.5	51.6	50.7	65.8	74.7	72.6	40.8	71.0	73.7	82.6	91.8	92.0
4	128	67.1	47.6	45.2	50.7	66.9	57.0	20.2	29.4	31.2	68.2	90.2	91.0
4	256	69.8	45.0	40.2	51.0	67.5	55.3	14.2	21.4	21.5	48.9	86.5	88.8
8	2	93.9	96.7	96.1	93.2	95.7	94.9	80.1	81.6	81.8	94.1	96.6	96.1
8	4	77.7	78.8	78.5	79.6	86.2	85.5	85.3	89.4	89.2	89.2	96.0	94.8
8	8	59.4	59.2	59.2	75.6	77.8	77.7	88.6	91.4	91.4	92.4	96.2	95.6
8	16	49.9	48.7	48.7	74.2	75.0	74.0	86.3	90.4	90.7	90.7	94.4	93.7
8	32	48.0	42.9	42.9	73.9	72.5	72.7	81.6	87.8	88.5	88.5	92.8	92.0
8	64	49.0	36.8	36.1	74.3	62.5	63.5	46.1	70.4	72.2	84.7	90.2	89.9
8	128	53.0	32.3	30.7	56.3	57.7	53.0	15.0	29.6	32.6	66.6	87.2	88.4
8	256	56.5	28.8	25.7	50.9	50.7	43.3	8.9	16.2	16.2	45.4	84.8	87.6
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	76.4	76.8	76.7	83.5	82.9	83.1	86.3	87.5	87.5	92.6	94.7	94.2
16	8	51.7	51.6	51.5	69.7	73.1	73.1	86.0	88.7	88.6	91.1	94.2	93.5
16	16	39.8	39.0	39.1	70.5	70.2	70.2	86.1	89.6	89.8	90.9	93.5	93.2
16	32	36.1	32.2	32.1	71.0	68.6	67.4	80.2	85.9	86.4	88.6	91.7	91.3
16	64	36.3	26.4	26.0	71.7	62.4	61.6	49.6	69.0	70.3	85.0	89.1	88.9
16	128	40.1	21.5	20.6	51.8	48.3	50.8	11.8	34.1	37.7	68.7	86.9	86.6
16	256	44.5	18.0	16.2	50.1	40.3	33.9	5.0	14.9	15.7	43.5	82.3	84.5

2.2 Weighted DDIM

Table 10: **Weighted DDIM, 5,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.6	99.1	98.6	96.0	97.3	97.6	95.1	97.9	98.7	96.7	99.4	98.6
2	4	92.1	93.0	93.2	89.9	90.8	88.5	94.6	97.7	98.4	96.1	99.4	98.8
2	8	83.5	83.5	83.5	79.8	83.9	82.6	93.3	96.5	97.1	95.2	99.3	98.8
2	16	79.4	78.9	79.2	76.3	82.3	82.8	91.6	95.9	96.7	93.8	98.8	98.4
2	32	79.5	75.4	75.3	76.4	77.6	81.4	84.5	95.6	96.6	91.9	98.4	98.0
2	64	80.2	72.0	71.0	64.9	78.0	79.5	46.7	91.6	93.0	85.3	97.0	97.0
2	128	82.6	69.9	67.3	57.5	61.3	67.3	29.9	43.7	47.0	70.5	94.9	95.8
2	256	84.0	68.2	63.4	58.9	74.9	51.1	25.1	37.1	39.6	51.4	93.4	94.4
4	2	91.9	98.5	96.9	91.2	97.2	96.3	89.3	92.9	92.8	91.8	98.1	96.6
4	4	85.3	87.1	87.0	85.2	83.0	89.3	88.0	93.0	92.5	90.3	97.7	96.3
4	8	71.2	71.1	71.0	76.7	82.1	81.9	89.6	94.1	94.3	92.8	98.0	96.8
4	16	64.2	63.0	63.2	73.8	80.7	81.7	88.7	93.5	94.6	91.8	97.4	96.8
4	32	62.8	57.6	57.6	77.3	79.9	80.3	84.9	92.0	93.2	90.4	96.0	96.2
4	64	64.6	53.1	52.3	68.0	76.7	75.0	61.1	87.5	89.0	85.2	94.1	94.3
4	128	68.3	49.0	46.6	51.2	73.8	61.3	23.7	43.5	49.7	68.0	93.8	93.3
4	256	71.2	46.2	41.4	54.1	59.9	53.2	16.3	25.5	27.6	46.9	90.6	92.7
8	2	94.5	97.6	96.8	93.4	95.9	95.8	79.8	81.4	81.6	94.1	96.6	96.1
8	4	81.3	83.2	82.8	77.1	83.2	82.8	85.2	89.3	89.2	89.2	96.1	95.0
8	8	62.4	62.3	62.2	76.0	77.1	76.7	88.8	91.7	91.7	92.7	96.9	96.2
8	16	51.5	50.4	50.4	73.0	73.9	73.0	86.8	91.1	91.3	91.8	96.1	95.2
8	32	49.2	44.4	44.3	76.9	73.5	73.8	84.2	90.3	91.1	89.5	95.7	94.9
8	64	50.4	38.3	37.7	70.3	68.9	67.8	67.9	84.2	85.3	87.2	93.9	93.7
8	128	54.5	33.6	31.9	50.7	57.1	59.9	21.0	47.9	53.8	68.3	92.5	92.7
8	256	58.6	30.0	26.7	52.6	50.6	44.2	10.8	22.4	25.8	46.6	90.8	91.4
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	81.0	81.5	81.3	85.5	86.2	84.3	86.5	87.7	87.7	92.6	94.8	94.3
16	8	55.2	55.0	54.9	71.0	72.5	75.0	86.2	88.9	88.8	91.4	94.8	94.0
16	16	41.9	41.1	41.2	71.6	68.4	71.1	86.9	90.4	90.6	91.8	94.9	94.5
16	32	37.4	33.6	33.6	74.1	69.9	69.4	83.3	88.8	89.3	89.6	94.0	93.7
16	64	37.7	27.7	27.4	70.6	61.8	60.1	69.6	82.0	82.9	87.5	92.2	92.3
16	128	42.1	22.7	21.7	55.2	57.5	54.8	19.8	50.1	54.6	71.3	90.8	90.8
16	256	46.8	19.0	17.1	50.7	36.4	33.5	6.9	23.6	28.9	48.2	88.4	89.0

Table 11: **Weighted DDIM, 10,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.7	99.4	98.7	96.4	97.9	97.8	95.2	98.0	98.7	96.7	99.4	98.6
2	4	93.2	94.2	94.3	88.9	85.4	90.6	94.6	97.7	98.4	96.1	99.4	98.8
2	8	84.5	84.4	84.6	83.6	81.2	83.3	93.3	96.5	97.1	95.2	99.3	98.8
2	16	80.2	79.3	79.5	77.8	79.0	83.3	91.9	96.0	96.9	93.8	99.1	98.5
2	32	79.9	75.9	75.8	74.3	84.9	67.8	85.4	95.9	97.0	91.6	99.0	98.2
2	64	80.7	72.7	71.9	67.8	79.0	72.3	49.5	94.4	95.6	85.2	98.1	98.0
2	128	83.6	70.5	68.0	59.8	67.7	72.1	21.6	68.9	75.3	70.0	96.6	97.0
2	256	84.7	69.0	64.1	60.8	63.5	53.4	25.2	38.3	41.8	48.4	94.4	95.4
4	2	92.1	98.8	97.1	91.7	98.0	96.7	89.2	92.9	92.9	91.8	98.1	96.6
4	4	86.9	89.2	88.9	84.0	88.0	88.0	88.2	93.0	92.5	90.3	97.7	96.3
4	8	72.9	72.8	72.7	77.8	84.2	80.3	89.7	94.1	94.3	92.8	98.1	96.9
4	16	65.0	64.2	64.4	72.8	80.9	83.0	88.8	93.7	94.8	91.9	97.8	97.2
4	32	63.5	58.7	58.7	77.2	78.6	80.7	85.5	92.7	93.9	90.4	97.0	97.3
4	64	65.2	54.0	53.4	66.7	77.6	77.1	64.2	90.6	92.3	86.5	95.7	95.7
4	128	69.0	50.0	47.6	53.6	62.5	66.0	24.4	73.6	77.7	68.2	95.8	95.0
4	256	72.4	47.1	42.2	56.5	63.6	62.0	17.9	29.5	34.5	46.2	93.8	93.3
8	2	94.8	97.9	97.0	94.0	97.2	96.5	79.9	81.5	81.7	94.2	96.6	96.1
8	4	83.5	85.9	85.5	80.9	84.9	86.9	85.2	89.3	89.1	89.2	96.1	95.0
8	8	64.8	64.7	64.5	76.1	75.7	80.1	88.8	91.7	91.7	92.8	97.1	96.3
8	16	52.8	51.7	51.7	72.8	74.2	75.6	87.2	91.3	91.6	92.0	96.6	95.6
8	32	50.3	45.6	45.6	76.0	75.5	75.4	84.9	91.1	92.0	89.7	96.4	95.8
8	64	51.5	39.4	38.8	76.5	70.9	70.2	72.5	88.0	89.2	87.6	94.4	94.6
8	128	55.9	34.6	32.9	53.2	60.5	61.5	28.8	71.9	74.5	71.0	94.5	94.2
8	256	59.8	31.0	27.5	55.1	58.6	44.6	12.3	29.2	37.2	47.8	93.9	93.4
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	84.0	84.6	84.4	86.2	87.0	86.4	86.9	88.1	88.1	92.7	94.9	94.4
16	8	58.1	57.9	57.7	73.7	76.7	74.4	86.8	89.3	89.2	91.5	95.1	94.3
16	16	43.5	42.8	42.9	69.2	71.6	70.0	87.7	91.2	91.4	92.1	95.5	95.1
16	32	38.4	34.8	34.8	73.4	70.3	70.6	85.0	90.3	90.9	89.8	94.9	94.5
16	64	38.8	28.6	28.3	74.0	66.8	66.4	76.0	86.8	87.6	87.3	93.0	93.2
16	128	43.3	23.5	22.5	61.3	59.1	56.7	32.7	69.9	71.9	73.3	92.0	91.8
16	256	48.2	19.6	17.6	53.1	44.9	43.0	9.3	31.6	40.4	50.4	91.1	90.8

Table 12: **Weighted DDIM, 15,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.8	99.5	98.7	96.5	98.2	97.9	95.1	97.9	98.6	96.7	99.4	98.6
2	4	93.6	94.7	95.0	90.2	90.9	92.5	94.6	97.7	98.4	96.1	99.4	98.8
2	8	85.0	85.1	85.2	81.6	84.3	81.4	93.3	96.5	97.1	95.2	99.3	98.8
2	16	80.7	79.8	80.0	69.2	79.0	75.8	91.9	96.0	96.9	93.8	99.1	98.5
2	32	80.0	76.3	76.3	75.7	85.2	84.2	85.3	96.0	97.1	91.5	99.0	98.2
2	64	81.2	73.0	72.2	61.5	79.4	82.4	50.1	95.1	96.2	85.3	98.3	98.3
2	128	83.7	71.0	68.3	61.0	68.6	62.4	10.6	82.8	87.5	69.5	97.1	97.3
2	256	85.2	69.4	64.5	61.9	69.8	72.6	24.1	39.7	44.0	47.0	95.0	95.9
4	2	92.2	98.9	97.2	92.0	98.0	96.7	89.2	92.9	92.9	91.8	98.1	96.7
4	4	87.7	90.3	90.0	83.2	90.1	90.4	88.3	93.1	92.6	90.4	97.7	96.3
4	8	74.0	73.9	73.6	81.2	82.1	82.8	89.7	94.1	94.3	92.8	98.2	97.0
4	16	65.5	64.8	65.1	75.7	76.0	76.9	88.8	93.8	94.9	92.0	97.9	97.2
4	32	64.1	59.3	59.3	75.0	80.5	74.9	85.8	92.9	94.1	90.5	97.1	97.5
4	64	65.6	54.5	53.8	69.6	78.1	78.4	64.7	91.4	93.2	86.5	96.2	96.1
4	128	69.3	50.4	48.0	54.9	73.8	66.0	18.3	81.9	85.0	67.4	96.5	95.2
4	256	72.8	47.6	42.6	57.8	68.0	53.1	17.8	34.4	41.9	46.2	94.8	93.3
8	2	94.9	98.0	97.2	94.6	97.5	96.6	80.0	81.6	81.8	94.2	96.6	96.1
8	4	84.6	87.4	86.9	84.6	87.0	88.4	85.2	89.3	89.2	89.3	96.1	95.0
8	8	66.1	66.0	65.8	77.7	79.3	80.7	88.8	91.7	91.7	92.9	97.1	96.4
8	16	53.5	52.4	52.5	73.5	75.7	75.8	87.3	91.5	91.7	92.1	96.8	95.8
8	32	50.7	46.0	46.1	75.2	74.8	74.2	85.2	91.4	92.3	89.5	96.4	95.8
8	64	51.9	39.9	39.3	69.2	70.2	69.4	73.7	89.2	90.4	87.7	94.4	94.7
8	128	56.5	35.1	33.3	54.6	59.3	53.5	28.4	79.3	81.3	69.9	94.9	94.6
8	256	60.7	31.3	27.8	54.1	60.5	49.9	13.6	34.8	44.7	47.9	94.8	94.2
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	85.5	86.2	86.0	87.2	87.8	86.8	87.6	88.7	88.6	92.7	95.0	94.6
16	8	59.5	59.4	59.2	76.1	76.4	76.6	88.2	90.4	89.9	91.5	95.3	94.5
16	16	44.3	43.7	43.8	68.3	71.7	69.8	89.0	92.5	92.6	92.3	95.9	95.4
16	32	39.0	35.3	35.3	73.9	69.3	70.3	86.7	91.7	92.3	90.5	95.1	94.7
16	64	39.4	29.1	28.8	76.5	68.0	65.7	79.8	89.3	90.2	86.6	94.0	94.2
16	128	43.9	23.9	22.9	52.5	49.6	53.0	40.2	79.6	80.7	75.3	92.4	92.7
16	256	49.1	20.0	17.9	51.7	44.2	37.0	10.9	36.7	46.2	51.6	92.0	91.5

Table 13: **Weighted DDIM, 20,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.8	99.5	98.7	96.6	98.5	98.1	95.1	97.9	98.6	96.7	99.4	98.6
2	4	94.0	95.1	95.3	90.7	90.4	90.6	94.7	97.8	98.5	96.1	99.4	98.8
2	8	85.2	85.5	85.5	83.9	87.0	90.5	93.4	96.5	97.2	95.2	99.3	98.8
2	16	80.8	80.1	80.3	75.3	78.5	76.2	91.9	96.0	96.9	93.8	99.1	98.5
2	32	80.2	76.6	76.6	75.4	81.2	80.7	85.3	96.0	97.2	91.3	99.0	98.3
2	64	81.0	73.1	72.2	66.2	71.3	63.6	50.4	95.5	96.7	85.4	98.4	98.4
2	128	83.8	71.2	68.6	61.8	73.4	73.3	9.3	88.0	91.3	69.4	97.5	97.7
2	256	85.2	69.7	64.7	62.7	63.3	55.1	21.8	42.0	47.3	46.3	95.6	96.6
4	2	92.2	99.0	97.2	92.0	98.2	96.7	89.3	92.9	92.9	91.8	98.1	96.6
4	4	88.1	91.0	90.6	86.5	90.8	90.7	88.2	93.1	92.5	90.3	97.7	96.3
4	8	74.6	74.5	74.3	81.0	83.8	84.3	89.7	94.0	94.2	92.9	98.2	97.0
4	16	65.8	65.2	65.5	78.1	80.9	76.6	88.8	93.8	94.9	92.0	98.0	97.2
4	32	64.3	59.6	59.7	76.1	79.7	79.3	85.9	93.0	94.3	90.7	97.3	97.5
4	64	66.0	55.1	54.4	67.8	78.9	78.9	65.0	91.9	93.8	87.2	96.1	95.9
4	128	69.7	50.7	48.3	55.7	70.5	68.6	17.0	85.3	88.0	67.2	96.6	95.3
4	256	73.0	47.9	43.0	58.5	62.9	56.3	17.8	39.4	48.2	46.6	94.9	93.1
8	2	95.0	98.0	97.2	94.6	97.5	96.8	80.8	82.5	82.7	94.2	96.8	96.2
8	4	85.3	88.4	87.9	80.8	89.4	88.2	85.9	89.8	89.5	89.3	96.1	95.0
8	8	67.0	66.8	66.7	78.9	79.5	79.8	89.3	92.1	92.1	92.9	97.2	96.4
8	16	54.1	53.0	53.1	73.2	75.7	75.8	87.7	91.9	92.1	92.2	97.0	95.9
8	32	50.9	46.3	46.3	76.0	74.5	75.2	85.8	91.9	92.8	89.5	96.5	95.9
8	64	52.3	40.4	39.8	70.5	72.2	70.5	75.1	90.1	91.4	87.3	94.3	94.6
8	128	56.9	35.4	33.6	55.5	64.3	61.4	28.4	83.3	85.1	71.7	94.9	94.7
8	256	61.0	31.6	28.1	52.5	49.1	48.8	14.5	40.2	49.9	49.4	94.8	94.1
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	86.4	87.3	87.0	86.8	88.1	86.2	87.6	88.7	88.5	92.7	95.1	94.7
16	8	60.5	60.4	60.2	76.0	76.4	76.4	88.9	90.9	90.3	91.6	95.4	94.6
16	16	44.9	44.3	44.4	71.2	70.2	71.5	89.9	93.4	93.3	92.3	96.0	95.5
16	32	39.4	35.7	35.7	73.0	69.8	69.3	87.5	92.6	93.2	90.8	95.3	94.7
16	64	39.8	29.6	29.2	74.8	68.5	67.7	81.4	90.7	91.6	85.7	94.7	94.7
16	128	44.4	24.2	23.2	57.1	57.8	53.4	43.9	83.9	84.9	77.3	92.6	92.9
16	256	49.5	20.3	18.2	50.1	44.9	35.1	12.2	41.8	50.5	52.2	92.9	92.1

Table 14: **Unweighted Flow matching, 2,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.5	98.5	98.5	96.6	96.6	96.6	99.0	99.0	99.0	99.7	99.7	99.7
2	4	91.5	91.5	91.5	90.2	90.2	90.2	98.7	98.7	98.7	99.5	99.5	99.5
2	8	81.8	81.8	81.8	89.0	89.0	89.0	98.7	98.7	98.7	99.3	99.3	99.3
2	16	78.8	78.8	78.8	79.4	69.7	69.7	97.9	97.9	97.9	97.9	97.9	97.9
2	32	74.3	74.3	74.3	72.4	63.2	63.2	95.9	95.9	95.9	96.8	96.8	96.8
2	64	69.9	70.0	70.0	76.8	76.8	76.8	67.4	65.7	65.7	95.2	94.9	94.8
2	128	66.0	65.9	65.9	72.6	72.6	72.6	40.8	41.2	41.2	94.5	94.5	94.5
2	256	62.2	62.2	62.2	66.8	66.8	66.7	38.9	38.9	38.9	93.5	93.4	93.4
4	2	97.5	97.5	97.5	96.4	96.4	96.4	93.4	93.4	93.4	98.1	98.1	98.1
4	4	83.9	83.9	83.9	89.4	89.4	89.4	94.1	94.1	94.1	97.9	97.9	97.9
4	8	69.2	69.2	69.2	83.2	83.2	83.2	95.2	95.2	95.2	97.6	97.6	97.6
4	16	62.0	62.0	62.0	82.9	82.9	82.9	95.7	95.7	95.7	95.3	95.3	95.3
4	32	56.3	56.2	56.2	67.5	65.7	65.7	92.7	92.7	92.7	93.5	93.5	93.5
4	64	50.2	50.3	50.3	74.0	74.0	74.0	74.6	74.0	74.0	92.4	91.6	91.6
4	128	44.5	44.4	44.4	63.3	63.3	63.3	30.9	31.0	31.0	92.0	91.9	91.9
4	256	39.5	39.5	39.5	52.8	52.7	52.7	21.2	21.2	21.2	90.4	90.4	90.4
8	2	96.9	96.9	96.9	96.0	96.0	96.0	83.4	83.4	83.4	97.1	97.1	97.1
8	4	79.3	79.3	79.3	86.6	86.6	86.5	90.2	90.2	90.2	96.4	96.4	96.4
8	8	60.0	59.9	59.9	78.3	78.3	78.3	92.6	92.6	92.6	96.3	96.3	96.3
8	16	49.2	49.2	49.2	76.4	76.4	76.4	92.6	92.6	92.6	94.3	94.3	94.3
8	32	42.7	42.7	42.7	73.3	73.3	73.3	89.8	89.8	89.8	92.7	92.7	92.7
8	64	36.1	36.2	36.2	66.2	66.2	66.2	72.8	72.9	72.9	90.4	90.2	90.2
8	128	30.2	30.2	30.2	57.6	57.6	57.6	32.8	33.1	33.1	89.5	89.5	89.5
8	256	25.3	25.3	25.3	43.6	43.6	43.6	16.0	16.0	16.0	88.3	88.2	88.3
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	77.3	77.3	77.3	84.9	84.7	84.7	89.0	89.0	89.0	95.0	95.0	95.0
16	8	52.2	52.2	52.2	74.8	74.8	74.8	89.6	89.6	89.6	94.3	94.3	94.3
16	16	39.4	39.4	39.4	72.1	72.1	72.1	90.7	90.7	90.7	93.3	93.3	93.3
16	32	32.2	32.2	32.2	69.6	69.7	69.7	87.3	87.3	87.3	91.7	91.7	91.7
16	64	26.1	26.1	26.1	62.0	62.0	62.0	70.9	71.0	71.0	89.2	89.2	89.2
16	128	20.5	20.5	20.5	46.7	46.7	46.7	38.0	38.2	38.2	86.6	86.7	86.6
16	256	16.2	16.1	16.1	36.3	36.2	36.2	15.6	15.6	15.6	84.4	84.5	84.4

2.3 Unweighted Flow matching

Table 15: **Unweighted Flow matching, 5,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.1	99.1	99.1	97.7	97.7	97.7	98.9	98.9	98.9	99.7	99.7	99.7
2	4	93.1	93.1	93.1	90.2	90.2	90.2	98.9	98.9	98.9	99.5	99.5	99.5
2	8	83.4	83.4	83.4	82.4	82.2	82.2	98.9	98.9	98.9	99.5	99.5	99.5
2	16	79.4	79.4	79.4	85.9	85.9	85.9	98.4	98.4	98.4	99.0	99.0	99.0
2	32	75.4	75.4	75.4	84.9	84.9	84.9	97.4	97.4	97.4	98.6	98.6	98.6
2	64	71.2	71.2	71.2	78.2	78.2	78.2	93.4	93.3	93.3	97.3	97.4	97.4
2	128	66.9	67.0	67.0	63.3	63.3	63.3	47.8	47.6	47.6	96.2	96.2	96.2
2	256	63.1	63.1	63.1	55.4	55.4	55.4	39.8	39.8	39.8	95.9	95.9	95.9
4	2	98.4	98.4	98.4	96.8	96.8	96.8	93.3	93.3	93.3	98.2	98.2	98.2
4	4	87.0	87.0	87.0	88.6	87.6	87.6	94.1	94.1	94.1	98.0	98.0	98.0
4	8	71.4	71.4	71.4	85.5	85.5	85.5	95.4	95.4	95.4	98.2	98.2	98.2
4	16	63.1	63.1	63.1	82.3	82.2	82.2	96.5	96.5	96.5	97.5	97.5	97.5
4	32	57.9	57.8	57.8	80.8	80.8	80.8	95.1	95.1	95.1	96.9	96.9	96.9
4	64	51.8	51.8	51.8	75.3	75.3	75.3	89.8	89.9	89.9	95.1	95.3	95.3
4	128	45.8	46.0	46.0	67.8	67.8	67.8	50.7	51.0	51.0	94.7	94.8	94.8
4	256	40.7	40.7	40.7	51.3	51.2	51.2	27.5	27.4	27.4	93.5	93.5	93.5
8	2	97.8	97.8	97.8	96.2	96.5	96.4	83.1	83.1	83.1	97.1	97.1	97.1
8	4	83.5	83.5	83.4	87.8	87.8	87.9	90.3	90.3	90.3	96.6	96.6	96.6
8	8	62.8	62.8	62.8	80.7	80.7	80.7	92.9	92.9	92.9	96.9	96.9	96.9
8	16	50.8	50.8	50.8	76.1	76.1	76.1	93.5	93.5	93.5	96.1	96.1	96.1
8	32	44.2	44.3	44.3	75.8	75.8	75.8	92.9	93.0	93.0	95.9	95.9	95.9
8	64	37.7	37.6	37.6	66.8	66.8	66.8	86.3	86.3	86.3	94.8	94.8	94.8
8	128	31.5	31.5	31.5	54.4	54.4	54.4	54.4	54.5	54.5	93.6	93.8	93.8
8	256	26.2	26.2	26.2	47.9	47.9	47.9	26.0	26.0	26.0	92.5	92.5	92.5
16	2	–	–	–	–	–	–	–	–	–	–	–	–
16	4	81.8	81.8	81.8	84.8	84.8	84.8	89.0	89.0	89.0	95.1	95.1	95.1
16	8	55.5	55.5	55.5	75.8	75.8	75.8	89.8	89.8	89.8	94.9	94.9	94.9
16	16	41.6	41.5	41.5	71.6	71.6	71.6	91.5	91.5	91.5	94.7	94.7	94.7
16	32	33.8	33.8	33.8	71.7	71.7	71.7	90.5	90.5	90.5	94.2	94.2	94.2
16	64	27.5	27.5	27.5	65.6	65.6	65.6	83.7	83.8	83.8	93.2	93.3	93.3
16	128	21.6	21.5	21.5	52.2	52.2	52.2	55.3	55.4	55.4	91.7	91.6	91.6
16	256	17.0	17.0	17.0	38.9	38.9	38.8	29.3	29.3	29.3	89.3	89.3	89.3

Table 16: **Unweighted Flow matching, 10,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.5	99.5	99.5	97.8	97.8	97.8	98.9	98.9	98.9	99.7	99.7	99.7
2	4	94.3	94.3	94.3	87.1	87.1	87.1	99.0	99.0	99.0	99.5	99.5	99.5
2	8	84.5	84.6	84.5	83.2	83.2	83.4	98.9	98.9	98.9	99.5	99.5	99.5
2	16	80.0	80.0	80.0	83.0	83.0	83.0	98.6	98.6	98.6	99.2	99.2	99.2
2	32	75.9	76.0	76.0	72.6	72.6	72.6	98.0	98.0	98.0	99.2	99.2	99.2
2	64	72.0	71.9	71.9	71.4	76.1	76.1	95.9	95.8	95.8	98.5	98.5	98.5
2	128	67.7	67.8	67.8	70.8	76.3	76.3	76.4	76.2	76.2	97.1	97.2	97.2
2	256	63.8	63.8	63.8	70.5	70.5	70.5	42.2	42.2	42.2	97.0	97.0	97.0
4	2	98.8	98.8	98.8	97.8	97.7	97.7	93.4	93.4	93.4	98.2	98.1	98.1
4	4	89.1	89.1	89.1	86.5	86.5	86.5	94.1	94.2	94.2	98.1	98.1	98.1
4	8	73.0	73.1	73.0	86.2	86.2	86.2	95.5	95.5	95.5	98.3	98.3	98.3
4	16	64.3	64.3	64.3	84.1	84.1	84.1	96.8	96.8	96.8	97.9	97.9	97.9
4	32	58.8	58.8	58.8	76.2	76.2	76.2	96.2	96.2	96.2	98.1	97.9	97.9
4	64	52.8	52.8	52.8	68.0	66.5	66.5	93.2	93.2	93.2	97.2	97.1	97.1
4	128	46.9	47.0	47.0	65.1	70.7	70.7	78.4	78.3	78.3	96.5	96.6	96.5
4	256	41.5	41.5	41.5	56.6	56.9	56.9	35.2	35.2	35.2	95.5	95.5	95.5
8	2	98.1	98.1	98.1	97.0	97.1	97.2	83.3	83.3	83.3	97.0	97.0	97.0
8	4	86.1	86.1	86.1	87.5	87.5	87.5	90.3	90.3	90.3	96.6	96.6	96.6
8	8	65.1	65.0	65.1	79.4	79.4	79.4	92.9	92.9	92.9	97.1	97.1	97.1
8	16	52.2	52.2	52.2	77.3	77.3	77.3	93.8	93.8	93.8	96.7	96.7	96.7
8	32	45.5	45.3	45.3	76.8	76.9	76.9	94.1	94.1	94.1	96.9	96.9	96.9
8	64	38.8	38.7	38.7	71.3	68.8	68.8	90.4	90.4	90.4	96.3	96.3	96.3
8	128	32.5	32.6	32.6	58.0	56.3	56.3	75.2	75.1	75.1	95.7	95.7	95.7
8	256	27.0	27.0	27.0	43.8	43.8	43.8	38.3	38.3	38.3	94.6	94.6	94.6
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	84.9	84.9	84.9	85.5	85.2	85.2	89.5	89.5	89.5	95.2	95.2	95.2
16	8	58.4	58.4	58.4	77.6	77.6	77.6	90.2	90.2	90.2	95.2	95.2	95.2
16	16	43.2	43.2	43.2	69.5	69.5	69.5	92.2	92.2	92.2	95.3	95.3	95.3
16	32	35.0	34.8	34.8	71.5	71.5	71.5	92.2	92.2	92.2	95.1	95.2	95.2
16	64	28.5	28.4	28.4	67.5	67.5	67.5	88.7	88.7	88.7	94.4	94.7	94.6
16	128	22.4	22.4	22.4	55.8	54.4	54.4	72.6	72.5	72.5	93.4	93.3	93.2
16	256	17.5	17.5	17.5	40.5	40.4	40.4	41.3	41.3	41.3	91.8	91.8	91.8

Table 17: **Unweighted Flow matching, 15,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.6	99.6	99.6	98.4	98.5	98.5	98.9	98.9	98.9	99.7	99.7	99.7
2	4	94.9	94.9	94.9	91.4	91.4	91.4	99.0	99.0	99.0	99.5	99.5	99.5
2	8	85.3	85.2	85.3	89.9	89.9	89.9	98.9	99.0	98.9	99.5	99.5	99.5
2	16	80.1	80.1	80.1	78.3	74.1	74.3	98.7	98.7	98.7	99.3	99.3	99.3
2	32	76.4	76.3	76.3	85.3	80.8	80.8	98.3	98.3	98.3	99.3	99.4	99.4
2	64	72.2	72.4	72.4	81.6	66.6	66.6	96.5	96.6	96.6	98.9	98.9	98.9
2	128	68.2	68.3	68.3	71.6	62.9	62.9	88.3	88.3	88.3	97.7	97.7	97.7
2	256	64.0	64.0	64.0	51.6	51.6	51.6	44.8	44.8	44.8	97.4	97.4	97.4
4	2	98.9	98.9	98.9	98.1	98.1	98.1	93.4	93.4	93.4	98.2	98.2	98.2
4	4	90.1	90.1	90.1	87.9	87.9	87.9	94.2	94.2	94.2	98.1	98.1	98.1
4	8	73.9	73.9	73.9	86.5	86.5	86.5	95.5	95.5	95.5	98.4	98.4	98.4
4	16	64.7	64.7	64.7	83.9	83.9	83.9	96.9	96.9	96.9	98.0	98.0	98.0
4	32	59.4	59.3	59.3	82.0	80.6	80.6	96.5	96.5	96.5	98.4	98.3	98.3
4	64	53.3	53.5	53.4	77.6	75.4	75.4	94.3	94.3	94.3	97.8	97.5	97.5
4	128	47.5	47.5	47.5	71.4	68.0	67.9	85.6	85.7	85.7	97.3	97.2	97.2
4	256	41.9	41.9	41.9	55.6	55.6	55.6	43.0	43.0	43.0	96.5	96.5	96.5
8	2	98.3	98.3	98.3	97.6	97.6	97.6	83.4	83.4	83.4	97.1	97.0	97.1
8	4	87.5	87.5	87.5	89.8	89.8	89.8	90.3	90.3	90.3	96.6	96.6	96.6
8	8	66.3	66.3	66.3	82.0	82.0	82.0	93.0	92.9	93.0	97.2	97.2	97.2
8	16	52.8	52.8	52.8	78.4	78.4	78.4	93.9	93.9	93.9	96.9	96.9	96.9
8	32	46.1	46.0	46.0	76.6	76.1	76.1	94.5	94.5	94.5	97.0	97.1	97.1
8	64	39.3	39.3	39.3	72.9	71.1	71.1	91.9	91.9	91.9	96.8	96.8	96.7
8	128	33.0	33.1	33.1	51.0	54.5	54.5	81.8	81.8	81.8	96.4	96.3	96.3
8	256	27.3	27.3	27.3	42.7	42.7	42.7	45.7	45.8	45.8	95.7	95.7	95.7
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	86.5	86.5	86.5	89.0	89.0	89.0	89.9	89.9	89.9	95.3	95.3	95.3
16	8	59.9	59.9	59.9	79.4	79.4	79.4	90.9	90.9	90.9	95.5	95.5	95.5
16	16	44.2	44.1	44.1	73.6	73.4	73.4	93.2	93.2	93.2	95.7	95.7	95.7
16	32	35.5	35.5	35.5	71.4	70.9	70.9	93.5	93.5	93.5	95.3	95.4	95.4
16	64	28.9	29.0	29.0	66.7	68.3	68.3	91.4	91.5	91.5	95.3	95.3	95.3
16	128	22.9	22.9	22.9	56.3	57.3	57.3	81.3	81.4	81.4	93.9	94.0	94.0
16	256	17.8	17.8	17.8	41.4	41.4	41.4	47.0	47.1	47.1	92.4	92.4	92.4

Table 18: **Unweighted Flow matching, 20,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.6	99.6	99.6	98.6	98.6	98.6	98.9	98.9	98.9	99.7	99.7	99.7
2	4	95.2	95.2	95.2	93.7	93.8	93.8	99.0	99.0	99.0	99.5	99.5	99.5
2	8	85.5	85.5	85.5	86.5	86.5	86.5	98.9	99.0	98.9	99.5	99.5	99.5
2	16	80.2	80.2	80.2	81.5	82.5	82.5	98.8	98.8	98.8	99.4	99.4	99.4
2	32	76.7	76.6	76.6	76.8	78.0	78.0	98.5	98.4	98.4	99.4	99.4	99.4
2	64	72.5	72.7	72.7	81.7	80.8	80.8	97.1	97.1	97.1	99.1	99.2	99.2
2	128	68.6	68.5	68.5	72.1	63.5	63.5	91.8	91.7	91.7	97.9	98.0	98.0
2	256	64.4	64.4	64.4	58.8	58.8	58.8	48.4	48.4	48.4	97.6	97.7	97.6
4	2	98.9	98.9	98.9	98.2	98.3	98.3	93.4	93.4	93.4	98.2	98.2	98.2
4	4	90.8	90.8	90.8	89.6	89.6	89.6	94.2	94.2	94.2	98.1	98.1	98.1
4	8	74.5	74.5	74.5	87.4	87.4	87.4	95.5	95.5	95.5	98.4	98.5	98.4
4	16	65.1	65.1	65.1	82.3	82.3	82.3	97.0	97.0	97.0	98.1	98.1	98.1
4	32	59.8	59.7	59.7	80.1	81.7	81.7	96.7	96.7	96.7	98.4	98.5	98.5
4	64	53.9	53.8	53.8	75.9	78.6	78.6	95.0	95.0	95.0	97.8	97.9	97.9
4	128	47.9	47.9	47.9	65.6	61.7	61.7	88.4	88.4	88.4	97.7	97.5	97.5
4	256	42.2	42.2	42.2	60.4	60.4	60.4	49.7	49.7	49.7	97.0	97.0	97.0
8	2	98.3	98.3	98.3	97.5	97.7	97.7	84.1	84.1	84.1	97.2	97.2	97.1
8	4	88.4	88.4	88.4	89.4	89.4	89.4	90.6	90.6	90.6	96.7	96.7	96.7
8	8	67.2	67.2	67.2	82.5	82.5	82.5	93.2	93.2	93.2	97.3	97.3	97.3
8	16	53.3	53.3	53.3	77.3	77.2	77.2	94.2	94.2	94.2	97.1	97.1	97.1
8	32	46.3	46.4	46.4	76.8	76.6	76.6	95.0	95.0	95.0	97.1	97.2	97.2
8	64	39.8	39.8	39.8	72.4	72.8	72.8	93.1	93.1	93.1	96.9	97.0	97.0
8	128	33.3	33.3	33.3	52.0	61.0	61.0	85.5	85.5	85.5	96.5	96.7	96.7
8	256	27.6	27.6	27.6	47.5	47.5	47.5	51.2	51.2	51.2	95.9	95.9	95.9
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	87.5	87.5	87.5	89.4	89.4	89.4	89.7	89.7	89.7	95.5	95.5	95.5
16	8	60.9	60.9	60.9	78.8	79.2	79.2	91.2	91.2	91.2	95.6	95.6	95.6
16	16	44.8	44.7	44.7	72.0	72.0	72.0	93.8	93.8	93.8	95.8	95.8	95.8
16	32	35.9	35.9	35.9	72.0	71.3	71.3	94.2	94.2	94.2	95.5	95.6	95.6
16	64	29.4	29.4	29.4	69.4	67.7	67.7	92.9	92.9	92.9	95.8	95.7	95.7
16	128	23.2	23.2	23.2	58.5	55.5	55.5	85.4	85.4	85.4	94.6	94.7	94.7
16	256	18.1	18.1	18.1	45.5	44.0	44.0	51.4	51.4	51.4	92.9	92.8	92.8

Table 19: **Weighted Flow matching, 2,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.5	98.5	98.5	97.8	97.8	97.8	97.6	97.6	97.6	99.5	99.5	99.5
2	4	91.2	91.2	91.2	89.1	89.1	86.2	97.6	97.6	97.6	99.5	99.5	99.5
2	8	81.8	81.8	81.8	88.6	89.4	89.4	96.2	96.2	96.2	99.3	99.3	99.3
2	16	78.4	78.4	78.4	85.1	85.1	85.1	95.1	95.1	95.1	98.0	98.0	98.0
2	32	74.0	74.0	74.0	81.5	81.5	81.5	94.1	94.1	94.1	96.8	96.8	96.8
2	64	69.4	69.4	69.4	77.0	77.0	77.0	64.9	64.9	66.6	93.0	93.1	94.0
2	128	65.4	65.4	65.4	72.9	72.9	72.9	41.4	41.4	41.4	91.9	91.9	91.9
2	256	61.7	61.7	61.6	46.1	46.1	60.1	39.0	39.0	39.1	89.7	89.6	89.5
4	2	97.6	97.6	97.6	95.6	95.6	95.6	93.0	93.0	93.0	98.2	98.2	98.2
4	4	84.0	84.0	84.0	84.6	87.0	87.0	92.8	92.8	92.8	97.7	97.7	97.7
4	8	68.8	68.7	68.7	79.3	79.3	79.3	93.8	93.8	93.8	97.5	97.5	97.5
4	16	61.8	61.7	61.7	80.3	80.3	80.3	92.6	92.6	92.6	95.2	95.2	95.2
4	32	55.8	55.8	55.8	76.9	76.9	76.9	89.9	89.9	89.9	93.4	93.4	93.4
4	64	50.6	50.5	50.4	71.3	71.3	71.3	73.2	73.3	73.7	91.6	91.6	91.4
4	128	44.8	44.8	44.8	57.5	57.5	57.5	31.1	31.1	31.1	90.0	89.9	90.0
4	256	39.9	39.9	39.9	46.9	46.9	51.7	21.4	21.4	21.4	85.6	85.5	84.9
8	2	97.0	97.0	97.0	95.0	95.1	95.1	81.4	81.4	81.4	96.9	97.0	97.0
8	4	78.9	78.9	78.9	86.2	86.2	86.2	89.4	89.4	89.4	96.3	96.3	96.3
8	8	59.1	59.1	59.1	78.8	78.8	78.8	91.4	91.4	91.4	96.3	96.3	96.3
8	16	48.5	48.5	48.5	72.8	74.2	74.2	90.2	90.2	90.2	94.2	94.2	94.2
8	32	42.6	42.6	42.6	71.6	71.6	71.6	87.7	87.7	87.7	92.8	92.8	92.8
8	64	36.0	36.0	35.9	62.6	62.6	62.6	72.3	72.3	72.3	90.2	90.2	90.3
8	128	30.2	30.2	30.2	56.3	56.3	56.3	32.7	32.7	32.7	88.4	88.4	88.4
8	256	25.3	25.3	25.3	42.5	42.5	41.3	16.0	15.9	15.9	85.9	85.9	85.7
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	76.8	76.8	76.8	83.6	83.6	83.6	87.6	87.6	87.6	95.1	95.1	95.1
16	8	51.4	51.4	51.4	72.2	72.2	72.2	88.7	88.7	88.7	94.3	94.3	94.3
16	16	38.8	38.8	38.8	69.3	69.3	69.3	89.6	89.6	89.6	93.4	93.4	93.4
16	32	31.9	31.9	31.9	66.9	66.9	66.9	86.0	86.0	86.0	91.8	91.8	91.8
16	64	25.8	25.8	25.8	61.1	61.1	61.1	70.5	70.5	70.5	89.1	89.1	89.2
16	128	20.2	20.2	20.2	49.3	49.3	49.3	37.8	37.8	37.8	86.1	86.1	86.1
16	256	16.0	15.9	16.0	38.0	36.6	36.6	15.4	15.4	15.4	83.7	83.7	83.9

2.4 Weighted Flow matching

Table 20: **Weighted Flow matching, 5,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.1	99.1	99.1	97.8	97.8	97.8	97.7	97.7	97.7	99.5	99.5	99.5
2	4	93.0	93.0	93.0	91.5	91.5	91.5	97.6	97.6	97.6	99.5	99.5	99.5
2	8	83.4	83.4	83.4	88.3	88.3	88.3	96.3	96.3	96.3	99.5	99.5	99.5
2	16	79.1	79.1	79.1	85.6	85.5	85.6	95.3	95.3	95.3	98.9	98.9	98.9
2	32	75.1	75.1	75.1	79.1	79.1	79.1	95.0	95.0	95.0	98.5	98.5	98.5
2	64	71.0	70.9	70.8	78.2	78.2	78.2	92.3	92.2	92.5	96.9	96.8	97.0
2	128	66.5	66.5	66.5	58.2	58.2	58.2	47.7	47.7	47.7	94.3	94.3	94.3
2	256	62.7	62.7	62.6	49.3	49.2	49.2	40.0	40.0	40.0	90.6	90.6	90.5
4	2	98.5	98.5	98.5	97.6	97.6	97.6	93.0	93.0	93.0	98.2	98.2	98.2
4	4	87.2	87.2	87.2	89.6	89.6	89.6	92.9	92.9	92.9	97.9	97.9	97.9
4	8	71.0	71.0	71.0	81.6	81.6	81.6	94.0	94.0	94.0	98.0	98.0	98.0
4	16	63.0	63.0	63.0	77.3	77.3	77.3	93.0	93.0	93.0	97.5	97.5	97.5
4	32	57.3	57.3	57.3	78.9	78.9	79.5	91.5	91.5	91.5	96.1	96.2	96.1
4	64	51.9	51.8	51.9	74.6	74.6	74.6	87.7	87.6	87.6	94.6	94.4	93.7
4	128	46.2	46.2	46.2	63.7	63.7	63.7	50.7	50.7	50.7	93.9	93.9	93.9
4	256	41.0	41.0	41.0	60.7	60.6	58.4	27.6	27.6	27.7	90.3	90.3	90.7
8	2	98.0	98.0	98.0	96.5	96.5	96.5	81.2	81.2	81.2	96.9	96.9	96.9
8	4	83.2	83.2	83.2	82.3	82.2	82.2	89.4	89.4	89.4	96.4	96.4	96.4
8	8	62.1	62.1	62.1	76.5	76.5	76.5	91.7	91.7	91.7	96.9	96.9	96.9
8	16	50.2	50.2	50.2	73.0	73.0	73.0	90.9	90.9	90.9	96.0	96.0	96.0
8	32	44.0	44.0	44.0	72.4	72.5	72.5	89.9	89.9	89.9	95.7	95.7	95.7
8	64	37.4	37.5	37.5	67.0	66.9	66.9	84.8	84.7	84.8	93.8	93.9	94.1
8	128	31.5	31.5	31.5	55.7	55.7	55.7	54.2	54.2	54.2	92.5	92.5	92.5
8	256	26.3	26.2	26.3	45.9	47.4	44.4	25.8	25.8	25.8	91.3	91.3	91.1
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	81.5	81.5	81.5	84.8	84.8	84.8	87.8	87.8	87.8	95.2	95.2	95.2
16	8	54.8	54.8	54.8	74.4	74.6	74.6	88.9	88.9	88.9	94.9	94.9	94.9
16	16	41.0	41.0	41.0	71.2	71.2	71.2	90.4	90.4	90.4	94.8	94.8	94.8
16	32	33.3	33.3	33.3	69.6	69.5	69.5	88.7	88.7	88.7	94.1	94.1	94.1
16	64	27.1	27.1	27.1	59.4	59.1	59.1	82.6	82.6	82.6	92.5	92.5	92.3
16	128	21.3	21.3	21.3	50.9	50.8	50.8	55.2	55.2	55.2	90.2	90.2	90.2
16	256	16.8	16.8	16.8	36.9	36.9	38.3	28.9	28.9	29.0	89.3	89.2	89.0

Table 21: **Weighted Flow matching, 10,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.4	99.4	99.4	98.3	98.4	98.4	97.8	97.8	97.8	99.5	99.5	99.5
2	4	94.2	94.2	94.2	90.9	90.9	88.0	97.6	97.6	97.6	99.6	99.6	99.6
2	8	84.4	84.4	84.4	83.7	83.7	83.7	96.3	96.3	96.3	99.5	99.5	99.5
2	16	79.7	79.3	79.3	86.0	86.0	86.0	95.4	95.4	95.4	99.1	99.1	99.1
2	32	75.5	75.5	75.5	72.4	67.9	67.9	95.2	95.2	95.2	99.1	99.1	99.1
2	64	71.6	71.7	71.7	62.2	62.2	81.6	94.4	94.4	94.5	97.7	97.7	97.9
2	128	67.4	67.4	67.4	76.6	76.6	76.6	75.6	75.6	75.6	95.8	95.8	95.8
2	256	63.4	63.4	63.4	57.9	57.9	57.9	42.4	42.4	42.4	91.6	91.6	91.2
4	2	98.8	98.8	98.8	97.8	97.8	97.8	92.9	92.9	92.9	98.2	98.2	98.2
4	4	89.2	89.2	89.2	85.1	85.1	85.1	93.0	93.0	93.0	97.9	98.0	97.9
4	8	72.7	72.7	72.7	81.7	81.7	81.7	94.0	94.0	94.0	98.2	98.2	98.2
4	16	64.3	64.2	64.2	82.4	82.4	82.4	93.2	93.2	93.2	97.9	97.9	97.9
4	32	58.3	58.3	58.3	78.4	78.3	78.3	92.1	92.1	92.1	97.1	97.1	97.1
4	64	53.0	53.0	53.0	72.1	74.1	76.2	90.1	90.1	90.1	95.7	95.5	95.3
4	128	47.2	47.2	47.2	65.8	65.7	65.7	77.3	77.3	77.3	95.8	95.8	95.8
4	256	41.9	41.8	41.7	51.8	51.8	60.4	35.2	35.2	35.0	94.0	94.0	94.2
8	2	98.4	98.4	98.4	97.5	97.5	97.5	81.3	81.3	81.3	96.9	96.9	96.9
8	4	86.0	86.0	86.0	85.6	85.6	85.6	89.4	89.4	89.4	96.4	96.4	96.4
8	8	64.5	64.5	64.5	78.0	78.0	78.0	91.7	91.7	91.7	97.2	97.2	97.1
8	16	51.6	51.6	51.6	75.3	76.1	76.1	91.1	91.1	91.1	96.5	96.5	96.5
8	32	45.4	45.3	45.3	74.2	74.2	74.2	90.7	90.7	90.7	96.4	96.4	96.4
8	64	38.5	38.5	38.6	66.6	66.6	69.3	88.2	88.2	88.3	94.6	94.5	94.7
8	128	32.6	32.6	32.6	57.0	57.0	57.0	74.5	74.5	74.5	93.8	93.7	93.7
8	256	27.0	27.0	27.0	45.1	45.1	47.4	38.0	38.0	37.9	93.1	93.1	93.2
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	84.7	84.7	84.7	86.9	86.9	86.9	88.3	88.3	88.3	95.3	95.3	95.3
16	8	57.7	57.7	57.7	74.9	74.9	74.9	89.4	89.4	89.4	95.2	95.2	95.2
16	16	42.6	42.6	42.6	70.7	70.7	70.7	91.1	91.2	91.2	95.4	95.5	95.5
16	32	34.5	34.5	34.5	70.1	70.1	70.1	90.2	90.2	90.2	95.0	95.0	95.0
16	64	28.0	28.0	28.1	65.2	65.1	65.6	87.0	87.0	87.0	93.5	93.5	93.1
16	128	22.1	22.1	22.1	52.8	52.7	52.7	72.1	72.1	72.1	91.3	91.3	91.4
16	256	17.3	17.3	17.3	37.0	35.0	44.6	41.1	41.1	41.1	90.7	90.7	90.9

Table 22: **Weighted Flow matching, 15,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.6	99.6	99.6	98.5	98.5	98.5	97.7	97.7	97.7	99.5	99.5	99.5
2	4	94.8	94.8	94.8	92.6	92.6	92.6	97.6	97.6	97.6	99.6	99.6	99.6
2	8	84.9	84.9	84.9	83.1	83.1	83.1	96.3	96.3	96.3	99.5	99.5	99.5
2	16	79.9	79.8	79.9	75.2	75.2	75.2	95.4	95.4	95.4	99.2	99.1	99.1
2	32	76.0	76.0	76.0	85.6	85.6	85.6	95.3	95.2	95.2	99.1	99.1	99.1
2	64	72.0	72.0	71.9	71.8	71.8	82.3	94.9	94.9	94.9	98.0	97.8	98.0
2	128	67.9	67.9	67.9	68.8	68.8	68.8	87.8	87.8	87.8	96.5	96.5	96.5
2	256	63.6	63.6	63.7	52.1	52.1	65.0	45.2	45.2	44.9	91.8	91.7	91.6
4	2	98.9	98.9	98.9	98.2	98.2	98.2	93.0	93.0	93.0	98.2	98.2	98.2
4	4	90.3	90.3	90.3	90.2	91.0	91.0	93.0	93.0	93.0	97.9	97.9	97.9
4	8	73.5	73.5	73.5	82.7	82.7	82.7	93.9	93.9	93.9	98.3	98.3	98.3
4	16	64.8	64.9	64.9	74.9	74.9	74.9	93.2	93.2	93.2	98.0	98.0	98.0
4	32	59.0	58.9	58.9	71.4	71.5	71.4	92.3	92.3	92.3	97.2	97.2	97.2
4	64	53.6	53.5	53.5	74.5	74.5	75.6	90.8	90.7	90.8	96.0	95.5	95.8
4	128	47.7	47.7	47.7	64.4	65.1	64.4	84.4	84.4	84.4	95.9	95.9	95.9
4	256	42.2	42.2	42.1	55.7	55.7	52.1	42.8	42.8	42.8	95.5	95.5	95.4
8	2	98.5	98.5	98.5	97.8	97.6	97.6	81.4	81.4	81.4	96.9	96.9	96.9
8	4	87.4	87.4	87.4	89.1	89.1	89.1	89.4	89.4	89.4	96.4	96.4	96.4
8	8	65.8	65.8	65.8	81.4	81.4	81.4	91.8	91.8	91.8	97.2	97.2	97.2
8	16	52.3	52.3	52.3	75.4	75.4	75.4	91.3	91.2	91.3	96.8	96.7	96.7
8	32	45.8	45.8	45.8	73.9	73.9	73.9	91.0	91.0	91.0	96.5	96.5	96.5
8	64	39.1	39.1	39.1	70.6	70.6	68.5	89.4	89.4	89.3	94.8	94.6	94.6
8	128	33.1	33.0	33.0	58.7	58.7	58.7	80.8	80.9	80.9	93.8	93.8	93.8
8	256	27.4	27.4	27.3	41.8	41.8	47.9	45.5	45.5	45.6	93.9	93.9	93.8
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	86.3	86.3	86.3	87.8	87.8	87.8	88.9	88.9	88.9	95.5	95.5	95.5
16	8	59.1	59.1	59.1	76.7	76.7	76.7	90.3	90.3	90.3	95.4	95.4	95.4
16	16	43.5	43.6	43.5	69.5	69.3	69.3	92.5	92.5	92.5	95.7	95.8	95.8
16	32	35.1	35.0	35.0	70.2	70.2	70.2	91.5	91.5	91.5	95.3	95.3	95.3
16	64	28.5	28.5	28.5	66.2	66.2	66.7	89.4	89.4	89.3	94.5	94.3	94.4
16	128	22.6	22.6	22.6	53.1	53.1	53.1	80.5	80.5	80.5	91.5	91.5	91.5
16	256	17.6	17.6	17.6	40.6	39.8	35.4	46.9	46.9	47.0	90.7	90.9	91.1

Table 23: **Weighted Flow matching, 20,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.6	99.6	99.6	98.6	98.6	98.6	97.7	97.7	97.7	99.5	99.5	99.5
2	4	95.1	95.1	95.1	90.3	90.2	90.2	97.6	97.6	97.6	99.6	99.6	99.6
2	8	85.3	85.3	85.3	89.8	89.7	89.6	96.3	96.3	96.3	99.5	99.5	99.5
2	16	80.0	80.2	80.1	79.1	79.2	79.2	95.4	95.4	95.4	99.2	99.1	99.1
2	32	76.4	76.4	76.4	81.4	81.4	81.4	95.3	95.3	95.3	99.1	99.1	99.1
2	64	72.2	72.3	71.9	82.4	68.1	77.4	95.2	95.2	95.2	97.9	98.0	98.0
2	128	68.1	68.1	68.1	69.3	69.3	69.3	91.2	91.2	91.2	96.6	96.6	96.6
2	256	64.0	64.0	63.9	65.6	65.6	59.5	48.6	48.6	48.4	92.4	92.4	92.7
4	2	99.0	99.0	99.0	98.2	98.1	98.4	93.0	93.0	93.0	98.2	98.2	98.2
4	4	91.0	90.9	90.9	91.0	91.0	91.0	93.0	93.0	93.0	97.9	97.9	97.9
4	8	74.2	74.2	74.2	82.8	82.8	82.8	93.9	93.9	93.9	98.3	98.3	98.3
4	16	65.2	65.3	65.2	75.5	75.5	75.5	93.3	93.2	93.2	98.1	98.1	98.1
4	32	59.2	59.2	59.2	78.0	77.9	77.9	92.3	92.3	92.3	97.3	97.3	97.3
4	64	54.0	53.9	54.1	75.7	74.9	76.2	91.1	91.1	91.1	95.9	95.4	95.7
4	128	48.0	48.0	48.0	57.3	57.3	57.3	86.7	86.7	86.7	96.2	96.2	96.2
4	256	42.4	42.4	42.5	55.3	55.3	57.7	49.4	49.4	49.3	95.5	95.5	95.8
8	2	98.6	98.6	98.6	98.0	98.0	98.0	82.3	82.3	82.3	97.1	97.0	97.1
8	4	88.4	88.4	88.4	88.5	88.6	88.6	89.9	89.9	89.9	96.5	96.5	96.5
8	8	66.6	66.6	66.6	81.6	81.6	81.6	92.2	92.2	92.2	97.3	97.3	97.3
8	16	52.8	52.8	52.9	74.7	74.7	74.7	91.8	91.7	91.7	96.9	96.9	96.9
8	32	46.1	46.1	46.1	74.6	74.6	74.6	91.4	91.4	91.4	96.6	96.6	96.6
8	64	39.5	39.5	39.6	69.4	69.9	69.5	90.2	90.2	90.2	94.9	94.5	94.6
8	128	33.3	33.3	33.3	59.0	59.0	59.0	84.2	84.2	84.2	93.7	93.7	93.7
8	256	27.6	27.6	27.7	48.2	50.1	45.9	51.0	51.0	51.0	93.9	93.9	93.7
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	87.4	87.3	87.3	88.8	88.8	88.8	88.9	88.9	88.9	95.6	95.6	95.6
16	8	60.1	60.1	60.1	76.5	76.5	76.5	90.8	90.8	90.8	95.6	95.6	95.6
16	16	44.1	44.1	44.1	69.9	69.9	69.9	93.3	93.3	93.3	95.9	95.9	95.9
16	32	35.4	35.4	35.4	70.1	70.1	70.1	92.4	92.4	92.4	95.3	95.3	95.3
16	64	29.0	29.0	29.0	64.0	64.3	67.2	90.6	90.6	90.6	95.0	94.9	95.1
16	128	22.9	22.9	22.9	48.9	48.9	48.9	84.2	84.2	84.2	92.0	92.0	92.0
16	256	17.8	17.8	17.9	42.8	42.8	42.1	51.3	51.3	51.4	91.1	91.1	91.3

Table 24: **Unweighted Heun, 2,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.0	98.4	98.4	95.9	96.1	97.2	98.2	98.8	98.8	99.0	99.5	99.6
2	4	91.6	91.5	91.4	88.8	89.3	90.1	97.9	98.7	98.7	98.6	99.5	99.4
2	8	82.5	81.8	81.7	83.8	84.8	86.9	97.1	98.5	98.4	97.7	99.2	99.2
2	16	80.1	78.7	78.6	83.4	74.3	66.2	95.2	97.6	97.6	95.7	97.7	97.7
2	32	79.9	74.6	74.0	76.0	73.8	72.5	84.5	95.2	95.5	92.6	96.3	96.4
2	64	80.5	71.1	69.6	62.9	79.7	77.3	37.8	63.0	67.2	85.8	94.3	94.6
2	128	82.4	68.7	65.6	53.7	70.5	72.8	30.6	39.6	40.6	73.8	93.2	93.8
2	256	82.7	66.5	61.8	55.2	64.4	60.6	24.5	37.2	38.4	57.3	91.2	91.9
4	2	91.7	97.5	96.2	90.6	96.0	94.6	90.8	93.3	92.7	92.1	98.1	96.6
4	4	83.8	83.9	83.6	80.9	87.3	86.7	91.5	93.9	93.1	92.5	97.7	96.6
4	8	70.2	69.3	69.0	82.0	82.6	81.3	93.5	95.1	94.7	94.4	97.6	97.0
4	16	64.2	62.0	61.9	83.1	81.4	82.1	94.1	95.4	95.3	94.7	95.4	95.1
4	32	62.9	56.5	56.2	80.1	71.6	66.9	87.0	92.3	92.5	91.9	93.4	93.1
4	64	64.2	51.5	50.2	68.8	75.5	73.5	41.8	71.7	74.1	86.6	92.1	91.9
4	128	67.3	47.1	44.4	51.4	67.2	61.2	20.6	29.4	30.9	70.3	91.0	91.4
4	256	69.7	44.5	39.6	51.5	65.3	54.2	14.8	21.7	21.3	50.7	87.6	89.3
8	2	95.7	96.7	96.5	94.5	95.2	93.5	82.7	83.5	83.6	96.2	96.9	96.6
8	4	79.2	79.2	78.7	84.8	85.0	84.4	88.3	90.2	89.4	90.3	96.0	94.8
8	8	60.8	60.1	59.9	77.5	78.8	76.1	91.5	92.6	92.2	93.8	96.3	95.7
8	16	51.1	49.3	49.2	78.4	75.9	76.6	91.2	92.4	91.9	93.3	94.4	93.8
8	32	48.6	42.9	42.6	78.8	73.6	73.5	85.9	89.4	89.3	91.5	92.6	91.9
8	64	49.5	36.9	36.0	78.6	68.0	66.3	47.6	71.2	72.5	87.4	90.2	89.7
8	128	53.4	32.0	30.3	59.6	54.1	58.3	15.5	30.0	32.7	71.2	88.6	88.8
8	256	56.7	28.6	25.4	51.8	53.3	43.8	9.6	16.7	16.6	47.5	86.4	87.4
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	77.1	77.4	77.2	82.7	84.8	79.0	87.5	88.9	88.8	92.1	94.7	94.1
16	8	52.8	52.3	52.1	73.2	76.3	75.1	88.1	89.6	89.3	91.4	94.2	93.4
16	16	40.7	39.6	39.5	74.5	72.4	72.2	89.3	90.6	90.4	92.7	93.3	92.9
16	32	36.6	32.4	32.3	76.9	70.2	69.9	83.5	87.0	87.1	91.1	91.6	91.1
16	64	36.8	26.5	26.1	75.5	63.3	61.6	51.1	69.6	70.6	86.9	88.9	88.5
16	128	40.4	21.6	20.6	53.4	49.0	45.1	12.4	34.4	37.8	71.7	86.2	86.2
16	256	44.8	18.0	16.2	51.3	43.4	36.8	5.7	15.5	16.2	45.5	81.9	83.6

2.5 Unweighted Heun

Table 25: **Unweighted Heun, 5,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.6	99.0	99.0	96.9	97.5	97.6	98.1	98.7	98.8	99.0	99.5	99.6
2	4	93.2	93.1	93.0	84.0	92.5	92.4	98.1	98.8	98.8	98.6	99.5	99.4
2	8	84.0	83.3	83.3	83.7	79.3	85.0	97.5	98.7	98.6	97.8	99.3	99.2
2	16	80.4	79.1	79.0	76.2	85.9	85.6	95.9	98.1	98.0	96.9	98.9	98.8
2	32	80.4	75.6	75.4	77.6	75.8	71.5	87.5	96.8	97.0	94.0	98.3	98.2
2	64	81.3	72.2	70.8	65.1	76.9	78.2	48.0	91.7	92.7	86.7	96.6	96.7
2	128	83.2	69.6	66.6	57.6	71.2	55.3	30.9	44.1	47.2	72.5	95.1	95.3
2	256	83.7	67.7	62.6	58.8	70.0	69.5	25.6	37.1	39.6	52.9	93.8	94.5
4	2	92.1	98.4	96.8	91.4	97.3	95.8	90.9	93.2	92.6	92.1	98.2	96.6
4	4	86.5	87.0	86.7	84.3	87.6	88.4	91.4	93.9	93.1	92.6	97.8	96.7
4	8	72.5	71.5	71.2	81.9	81.2	84.9	93.9	95.3	94.8	94.7	98.0	97.3
4	16	65.3	63.1	63.0	79.8	83.3	82.9	95.1	96.2	96.0	96.5	97.4	97.1
4	32	63.9	58.1	57.8	81.8	81.8	81.0	90.3	94.8	94.9	94.6	96.6	96.5
4	64	65.5	53.0	51.8	68.5	77.4	75.2	63.8	88.8	89.3	88.9	94.8	94.5
4	128	68.7	48.5	45.8	51.8	71.4	66.8	24.3	44.3	49.8	71.2	94.2	94.1
4	256	71.5	45.8	40.7	54.5	62.5	48.9	16.8	25.8	27.7	48.5	91.2	92.6
8	2	96.3	97.5	97.3	95.5	95.8	96.0	82.3	83.2	83.3	96.2	96.9	96.6
8	4	82.8	83.4	82.8	85.7	87.2	85.9	88.2	90.3	89.5	90.4	96.2	95.0
8	8	63.6	63.0	62.8	79.4	78.4	76.6	91.8	92.9	92.5	94.2	96.9	96.2
8	16	52.6	50.9	50.8	78.0	76.0	76.5	92.2	93.2	92.8	94.7	96.1	95.3
8	32	49.9	44.4	44.2	81.8	74.5	71.8	89.8	92.6	92.4	93.9	95.7	94.9
8	64	51.0	38.3	37.5	74.7	69.5	68.3	71.3	85.4	85.7	90.1	94.2	93.6
8	128	55.2	33.4	31.6	51.9	61.1	58.7	21.6	48.2	53.5	73.2	93.3	92.8
8	256	58.8	29.8	26.3	53.5	43.0	48.4	11.4	22.8	26.1	48.5	91.0	91.8
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	81.3	81.8	81.6	85.1	87.0	85.5	87.5	88.9	88.9	92.2	94.7	94.2
16	8	56.2	55.7	55.4	75.0	75.2	75.8	88.3	89.8	89.5	91.9	94.8	93.9
16	16	42.8	41.7	41.7	75.1	73.4	71.5	90.1	91.5	91.2	93.8	94.7	94.3
16	32	38.0	33.9	33.8	78.1	72.0	71.2	87.6	90.4	90.3	93.1	94.1	93.5
16	64	38.2	27.8	27.4	77.0	66.3	64.9	72.7	82.9	83.3	90.4	92.6	92.2
16	128	42.4	22.7	21.6	58.8	55.5	52.7	20.3	50.5	54.6	74.9	91.5	91.2
16	256	47.0	19.0	16.9	51.8	36.8	41.5	7.5	24.1	29.2	50.2	88.2	89.2

Table 26: **Unweighted Heun, 10,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.8	99.3	99.3	97.8	97.4	98.2	98.1	98.7	98.8	99.0	99.6	99.6
2	4	94.2	94.2	94.1	85.7	91.5	92.6	98.2	98.9	98.9	98.6	99.5	99.4
2	8	85.4	84.4	84.4	83.0	90.2	77.7	97.6	98.7	98.7	97.8	99.3	99.2
2	16	81.0	79.5	79.5	80.9	82.8	82.6	96.2	98.3	98.2	97.1	99.1	99.0
2	32	80.7	76.2	75.8	75.2	79.8	85.4	89.0	97.6	97.6	94.6	99.0	98.9
2	64	81.5	72.7	71.6	68.1	78.0	74.6	50.6	94.9	95.4	87.1	98.1	98.0
2	128	83.5	70.3	67.4	59.7	71.9	68.0	22.2	68.6	75.0	71.6	96.4	96.4
2	256	84.4	68.4	63.3	60.7	60.8	52.8	25.6	38.3	41.5	50.1	95.1	95.6
4	2	92.3	98.7	97.0	91.9	97.7	96.2	90.9	93.3	92.8	92.2	98.2	96.6
4	4	88.3	89.1	88.7	88.1	88.9	84.9	91.6	94.0	93.1	92.6	97.9	96.7
4	8	74.4	73.1	72.9	82.1	84.4	85.9	94.0	95.3	94.8	94.8	98.2	97.5
4	16	66.3	64.2	64.1	80.4	81.4	83.4	95.5	96.5	96.2	96.7	97.8	97.4
4	32	64.6	59.0	58.8	81.7	80.1	79.9	91.8	96.0	95.9	95.4	97.8	97.2
4	64	66.1	53.9	52.9	69.9	76.9	71.9	67.3	92.3	92.7	90.2	96.8	96.6
4	128	69.8	49.8	46.9	54.2	63.2	64.8	25.4	73.8	77.5	71.2	96.1	95.9
4	256	72.5	46.6	41.5	56.9	65.4	58.5	18.3	29.8	34.5	47.8	93.8	94.6
8	2	96.6	97.9	97.7	95.9	96.9	96.9	82.5	83.3	83.5	96.3	96.9	96.6
8	4	85.0	86.0	85.4	86.4	88.6	84.7	88.3	90.3	89.5	90.4	96.2	95.0
8	8	65.8	65.3	65.0	80.7	80.2	74.0	91.8	93.0	92.5	94.3	97.0	96.4
8	16	53.8	52.1	52.1	76.1	76.8	77.8	92.5	93.5	93.0	95.1	96.7	95.8
8	32	51.0	45.7	45.4	80.8	76.9	76.7	91.2	93.8	93.5	94.4	96.7	95.6
8	64	52.2	39.5	38.7	78.9	71.0	69.4	76.4	89.6	89.8	91.5	95.4	94.9
8	128	56.6	34.5	32.6	54.6	64.5	56.4	29.8	72.4	74.5	74.6	95.3	95.0
8	256	60.1	30.7	27.1	56.1	53.8	49.5	12.8	29.5	37.4	49.8	94.4	94.2
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	84.2	84.9	84.6	87.1	87.7	85.1	87.9	89.4	89.3	92.2	94.8	94.2
16	8	59.1	58.6	58.3	76.6	78.0	78.1	88.8	90.2	89.7	91.9	95.1	94.2
16	16	44.3	43.3	43.4	71.5	69.8	73.6	90.9	92.1	91.8	94.2	95.4	94.9
16	32	39.2	35.1	34.8	78.4	72.6	72.1	89.8	92.1	92.0	93.5	94.9	94.4
16	64	39.4	28.9	28.4	80.3	68.1	67.3	80.0	88.0	88.2	91.5	93.8	93.3
16	128	43.9	23.6	22.5	60.4	50.9	52.4	33.8	70.3	71.9	76.8	93.1	92.6
16	256	48.5	19.6	17.5	54.3	44.7	42.5	9.8	32.0	40.7	52.7	91.2	91.7

Table 27: **Unweighted Heun, 15,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.9	99.4	99.4	97.9	97.9	98.3	98.1	98.8	98.8	99.0	99.6	99.5
2	4	94.8	94.8	94.8	91.4	91.2	88.9	98.2	98.9	98.9	98.6	99.5	99.4
2	8	85.9	85.1	85.3	82.5	89.9	87.4	97.7	98.7	98.7	97.8	99.3	99.2
2	16	81.5	80.0	79.8	69.9	79.6	86.3	96.4	98.4	98.3	97.1	99.1	99.0
2	32	81.0	76.5	76.0	78.5	84.5	86.0	89.2	97.8	97.8	94.8	99.2	99.1
2	64	82.0	73.1	72.0	65.7	80.4	67.0	51.8	95.8	96.1	87.7	98.6	98.5
2	128	83.9	70.7	67.8	61.0	82.0	77.1	11.5	83.0	87.2	71.5	97.1	97.0
2	256	84.7	68.7	63.6	61.8	72.3	53.8	24.6	39.7	44.0	48.9	95.9	96.1
4	2	92.4	98.9	97.1	92.1	98.0	96.4	90.8	93.3	92.8	92.2	98.2	96.6
4	4	89.1	90.1	89.8	88.3	89.6	87.0	91.6	94.0	93.1	92.6	97.8	96.7
4	8	75.1	74.0	73.8	85.0	80.4	86.4	94.0	95.3	94.9	94.8	98.2	97.5
4	16	66.6	64.7	64.6	82.6	81.8	82.0	95.6	96.6	96.3	96.8	97.9	97.5
4	32	65.1	59.6	59.2	85.2	80.2	80.4	92.2	96.3	96.3	95.6	98.1	97.5
4	64	66.6	54.5	53.4	76.9	79.1	71.6	68.8	93.6	93.9	90.7	97.5	97.1
4	128	70.4	50.2	47.4	55.4	68.8	68.0	19.5	82.7	84.8	71.0	96.9	96.8
4	256	73.0	47.1	41.9	58.2	63.1	54.9	18.3	34.6	41.9	47.5	95.3	95.8
8	2	96.7	98.0	97.8	96.3	97.4	97.1	82.6	83.4	83.6	96.2	96.9	96.6
8	4	86.1	87.4	86.7	87.0	88.4	87.6	88.3	90.3	89.5	90.4	96.2	95.0
8	8	67.2	66.5	66.3	81.2	80.9	82.2	91.9	93.0	92.6	94.4	97.1	96.4
8	16	54.6	52.9	52.8	77.8	77.4	77.1	92.6	93.6	93.1	95.1	96.8	95.9
8	32	51.5	46.2	45.8	82.0	76.2	76.5	91.7	94.2	94.0	94.5	96.8	95.9
8	64	52.7	40.0	39.2	79.8	71.2	69.8	78.4	91.2	91.3	91.7	95.9	95.4
8	128	57.3	35.0	33.0	55.9	62.4	55.5	29.7	80.0	81.1	76.0	95.8	95.8
8	256	60.8	31.0	27.4	55.0	57.3	51.4	14.1	35.1	44.8	50.1	95.5	95.3
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	85.6	86.4	86.2	87.6	87.2	87.4	88.6	89.8	89.5	92.2	95.0	94.4
16	8	60.6	60.0	59.8	78.5	79.5	79.2	89.9	91.0	90.3	91.9	95.3	94.4
16	16	45.2	44.2	44.2	71.6	71.7	73.2	92.1	93.1	92.7	94.3	95.7	95.2
16	32	39.7	35.6	35.4	78.5	71.5	70.9	91.6	93.4	93.3	93.9	95.1	94.9
16	64	40.0	29.4	29.0	81.3	69.5	68.6	84.6	90.8	90.9	90.8	94.6	94.2
16	128	44.6	24.0	22.9	61.8	60.1	52.2	41.8	80.3	80.8	78.2	93.7	93.3
16	256	49.4	20.0	17.8	52.8	49.6	41.7	11.4	37.0	46.2	53.7	92.2	91.6

Table 28: **Unweighted Heun, 20,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.0	99.5	99.5	98.5	98.4	98.3	98.1	98.8	98.8	99.0	99.6	99.5
2	4	95.2	95.1	95.1	90.8	91.0	91.3	98.2	98.9	98.9	98.6	99.5	99.4
2	8	86.3	85.4	85.4	83.4	90.3	84.6	97.7	98.7	98.7	97.8	99.3	99.2
2	16	81.6	80.2	79.7	80.0	83.4	85.5	96.4	98.4	98.4	97.2	99.2	99.1
2	32	81.2	76.7	76.3	76.6	81.6	83.7	89.3	98.0	97.9	94.9	99.2	99.1
2	64	82.1	73.4	72.2	68.8	84.4	82.2	51.9	96.2	96.4	87.7	98.7	98.6
2	128	84.0	71.0	68.2	61.6	73.4	72.5	10.1	87.6	90.7	71.4	97.5	97.3
2	256	84.8	69.0	63.9	62.5	60.6	54.5	22.4	41.9	47.0	48.1	96.5	96.5
4	2	92.5	98.9	97.2	91.9	98.1	96.6	90.9	93.3	92.7	92.2	98.2	96.6
4	4	89.6	90.8	90.5	87.3	89.0	89.9	91.6	94.0	93.1	92.6	97.9	96.7
4	8	75.8	74.6	74.3	85.0	86.4	86.5	94.1	95.3	94.9	94.8	98.2	97.5
4	16	67.1	65.1	65.0	83.1	82.5	80.6	95.7	96.7	96.3	96.8	98.0	97.7
4	32	65.3	59.9	59.7	82.8	80.7	79.4	92.4	96.5	96.5	95.7	98.1	97.6
4	64	67.0	54.8	53.7	70.4	80.3	78.0	69.4	94.2	94.4	90.6	97.5	97.2
4	128	70.6	50.4	47.8	56.3	67.2	61.9	18.1	85.9	87.7	70.3	97.1	97.2
4	256	73.2	47.4	42.2	58.8	66.2	54.7	18.4	39.6	48.1	47.7	95.9	96.1
8	2	96.8	98.1	97.9	96.5	97.4	97.2	83.2	84.2	84.3	96.3	97.0	96.6
8	4	86.8	88.3	87.6	86.3	89.2	87.2	88.8	90.6	89.6	90.5	96.2	95.0
8	8	68.0	67.3	67.1	81.0	82.8	82.5	92.2	93.3	92.8	94.4	97.2	96.4
8	16	55.1	53.5	53.3	76.3	76.6	77.3	93.0	94.0	93.3	95.2	97.0	96.1
8	32	51.7	46.5	46.3	81.0	76.7	76.3	92.4	94.6	94.4	94.7	96.9	96.0
8	64	53.1	40.5	39.7	80.6	74.4	71.8	80.5	92.4	92.4	91.8	96.0	95.5
8	128	57.5	35.3	33.4	56.9	67.0	56.6	29.6	84.2	84.9	75.1	96.3	95.9
8	256	61.2	31.3	27.7	53.3	46.1	51.5	15.1	40.5	49.7	51.4	95.5	95.6
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	86.5	87.4	87.1	87.8	88.6	88.8	88.5	89.5	89.2	92.3	95.1	94.5
16	8	61.6	61.1	60.8	78.6	77.3	79.5	90.4	91.3	90.5	91.9	95.4	94.6
16	16	45.8	44.8	44.9	73.5	72.1	71.9	92.7	93.7	93.3	94.4	95.8	95.4
16	32	40.1	36.0	35.9	78.0	72.2	71.4	92.5	94.1	94.0	94.1	95.4	95.1
16	64	40.4	29.8	29.3	81.6	70.2	69.2	86.8	92.2	92.2	90.7	95.0	94.6
16	128	45.0	24.3	23.2	60.6	59.3	58.2	45.7	84.7	84.9	79.7	94.6	94.0
16	256	49.8	20.3	18.1	51.1	44.1	36.2	12.8	42.0	50.3	54.6	93.2	92.7

Table 29: **Weighted Heun, 2,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.2	98.4	98.2	95.4	96.8	96.7	95.0	97.8	98.5	96.7	99.4	98.6
2	4	90.7	91.3	91.4	87.2	90.0	86.3	94.7	97.7	98.4	96.2	99.4	98.7
2	8	81.5	81.5	81.7	79.3	78.5	82.5	93.1	96.2	96.9	95.2	99.1	98.6
2	16	78.8	78.2	78.3	79.6	84.8	85.5	91.6	95.5	96.2	94.2	97.8	97.3
2	32	78.7	74.1	73.9	78.1	83.0	72.9	82.6	94.2	94.9	91.5	96.4	96.2
2	64	79.0	70.3	69.3	62.7	78.9	72.4	37.6	62.7	66.8	84.5	93.7	94.3
2	128	81.0	68.3	65.7	53.7	78.1	67.3	29.6	39.4	41.1	72.3	92.6	93.6
2	256	82.0	66.0	61.4	55.2	64.1	54.6	25.2	37.1	38.7	56.8	90.4	91.5
4	2	91.5	97.6	96.3	90.8	95.7	95.5	89.3	93.0	93.0	91.9	98.1	96.5
4	4	82.5	83.9	83.7	82.7	84.9	86.9	88.1	93.0	92.5	90.3	97.5	96.2
4	8	68.8	68.7	68.7	75.7	81.1	81.6	89.5	93.8	94.0	92.5	97.5	96.5
4	16	62.6	61.5	61.8	74.6	80.8	80.9	88.5	93.0	93.8	90.7	95.0	94.9
4	32	61.5	56.0	55.9	78.9	76.2	77.8	82.9	90.0	91.0	90.0	93.2	93.3
4	64	63.2	51.3	50.4	65.8	74.4	72.3	41.2	71.2	73.8	83.4	91.6	91.6
4	128	66.6	47.2	44.9	50.9	66.5	56.9	20.4	29.5	31.3	69.0	89.8	90.5
4	256	69.3	44.5	39.8	51.2	66.9	54.9	15.0	21.7	21.8	50.1	86.1	88.1
8	2	94.0	96.7	96.1	93.2	95.7	94.9	80.1	81.6	81.9	94.2	96.7	96.1
8	4	77.7	78.8	78.5	79.6	86.2	85.4	85.4	89.4	89.2	89.2	96.0	94.8
8	8	59.3	59.2	59.1	75.5	77.7	77.6	88.6	91.4	91.4	92.4	96.2	95.6
8	16	49.8	48.6	48.6	74.1	74.9	73.9	86.3	90.4	90.6	90.8	94.2	93.5
8	32	47.8	42.8	42.7	73.8	72.3	72.5	81.7	87.7	88.3	88.7	92.7	91.8
8	64	48.7	36.6	35.9	74.1	62.4	63.3	46.6	70.7	72.4	85.1	90.1	89.7
8	128	52.6	32.1	30.5	56.3	57.5	52.8	15.3	29.8	32.7	67.4	87.0	88.0
8	256	56.3	28.6	25.5	49.8	50.5	43.2	9.7	16.7	16.6	46.5	84.5	87.1
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	76.4	76.8	76.7	83.5	82.8	83.1	86.4	87.5	87.5	92.6	94.7	94.2
16	8	51.7	51.6	51.4	69.7	73.1	73.0	86.1	88.7	88.6	91.1	94.2	93.5
16	16	39.7	39.0	39.0	70.4	70.2	70.1	86.1	89.6	89.8	90.9	93.4	93.1
16	32	36.0	32.1	32.1	71.0	68.4	67.3	80.3	85.8	86.3	88.7	91.5	91.1
16	64	36.1	26.2	25.9	71.5	62.2	61.4	50.1	69.2	70.5	85.6	89.0	88.7
16	128	39.9	21.4	20.5	52.0	48.1	50.6	12.2	34.2	37.8	69.4	86.8	86.5
16	256	44.3	17.9	16.1	50.6	40.2	33.9	5.8	15.5	16.2	44.1	82.2	84.2

2.6 Weighted Heun

Table 30: **Weighted Heun, 5,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.6	99.1	98.5	95.9	97.2	97.5	95.1	97.9	98.7	96.7	99.4	98.6
2	4	92.1	92.9	93.1	89.8	90.7	88.4	94.7	97.7	98.4	96.2	99.4	98.7
2	8	83.3	83.3	83.4	79.8	83.8	82.5	93.3	96.4	97.0	95.3	99.2	98.7
2	16	79.3	78.7	79.0	76.2	82.2	82.7	91.7	95.7	96.5	94.1	98.8	98.2
2	32	79.8	75.3	75.2	76.3	77.4	81.2	85.0	95.5	96.3	92.2	98.1	97.7
2	64	79.9	71.6	70.7	64.9	77.8	79.2	47.5	91.1	92.5	85.6	96.2	96.5
2	128	81.9	69.3	66.8	57.5	61.2	67.0	30.1	43.7	47.0	71.4	94.5	95.2
2	256	82.7	67.3	62.6	58.7	74.2	51.2	25.7	37.2	39.6	52.5	92.8	93.4
4	2	91.9	98.5	96.8	91.2	96.9	96.2	89.4	92.9	92.9	91.9	98.1	96.6
4	4	85.3	87.1	86.9	85.2	82.9	89.2	88.1	93.0	92.5	90.4	97.7	96.3
4	8	71.1	71.0	70.9	76.7	82.0	81.8	89.7	94.0	94.2	92.8	97.9	96.7
4	16	64.0	62.8	63.1	73.7	80.5	81.5	88.8	93.4	94.4	91.9	97.4	96.7
4	32	62.6	57.5	57.5	77.2	79.6	80.0	84.9	91.8	92.9	91.3	96.3	96.4
4	64	64.2	52.9	52.1	67.9	76.3	74.7	61.6	87.4	88.9	86.2	94.2	93.8
4	128	67.8	48.7	46.2	51.4	73.3	61.0	23.9	43.5	49.6	68.8	93.5	92.9
4	256	70.8	45.7	40.9	54.3	59.6	52.9	16.8	25.7	27.8	48.0	90.2	91.9
8	2	94.5	97.5	96.8	93.4	95.9	95.7	79.8	81.4	81.7	94.1	96.6	96.1
8	4	81.3	83.1	82.8	77.1	83.2	82.8	85.3	89.4	89.2	89.3	96.1	95.0
8	8	62.3	62.2	62.1	76.0	77.0	76.7	88.8	91.7	91.7	92.8	96.8	96.1
8	16	51.5	50.3	50.3	72.9	73.8	73.0	86.9	91.0	91.3	91.8	96.0	95.1
8	32	49.1	44.3	44.3	76.8	73.3	73.6	84.3	90.2	91.0	89.9	95.5	94.6
8	64	50.1	38.0	37.4	70.2	68.6	67.5	68.7	84.2	85.3	87.2	93.4	93.4
8	128	54.2	33.4	31.7	50.9	56.9	59.6	21.2	47.9	53.7	69.1	92.4	92.4
8	256	57.9	29.8	26.5	52.9	50.4	44.1	11.3	22.7	26.1	47.6	90.7	91.0
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	81.0	81.5	81.3	85.5	86.1	84.2	86.6	87.7	87.7	92.6	94.8	94.3
16	8	55.1	55.0	54.8	71.0	72.4	75.0	86.3	88.9	88.8	91.4	94.8	94.0
16	16	41.8	41.2	41.2	71.5	68.4	71.0	86.9	90.4	90.6	91.8	94.8	94.4
16	32	37.3	33.6	33.6	74.0	69.8	69.3	83.5	88.7	89.3	89.9	93.8	93.4
16	64	37.6	27.5	27.2	70.5	61.6	60.0	70.3	82.0	82.9	88.2	92.4	92.2
16	128	41.8	22.5	21.6	55.3	57.1	54.5	20.0	50.1	54.5	72.1	90.8	90.8
16	256	46.3	18.9	16.9	51.2	36.6	33.6	7.3	24.0	29.2	49.9	88.5	88.9

Table 31: **Weighted Heun, 10,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.7	99.4	98.6	96.3	97.8	97.7	95.2	98.0	98.7	96.7	99.4	98.6
2	4	93.1	94.1	94.2	88.8	85.4	90.5	94.6	97.6	98.4	96.2	99.4	98.7
2	8	84.4	84.3	84.4	83.5	81.2	83.2	93.4	96.4	97.0	95.3	99.2	98.7
2	16	79.9	79.2	79.3	77.8	78.8	83.2	91.9	95.8	96.6	94.1	99.1	98.3
2	32	79.9	75.8	75.7	74.3	84.7	67.8	85.3	95.6	96.7	91.9	98.8	98.1
2	64	80.3	72.2	71.5	67.8	77.7	76.5	50.0	94.1	95.2	86.3	97.8	97.7
2	128	82.9	70.0	67.4	59.7	67.4	71.7	22.2	68.7	75.0	70.9	96.3	96.5
2	256	83.5	68.1	63.2	60.8	63.2	53.3	25.1	38.3	41.7	49.8	93.9	94.3
4	2	92.1	98.8	97.0	91.7	98.0	96.6	89.3	92.9	92.9	91.9	98.1	96.6
4	4	86.9	89.1	88.9	83.9	87.9	87.9	88.3	93.1	92.6	90.4	97.7	96.3
4	8	72.8	72.7	72.6	77.8	84.1	80.2	89.7	94.0	94.3	92.9	98.0	96.8
4	16	64.9	64.0	64.2	72.8	80.8	82.8	88.9	93.6	94.6	92.2	97.7	97.0
4	32	63.4	58.6	58.6	77.1	78.5	80.4	85.6	92.4	93.6	91.0	96.8	97.0
4	64	64.9	53.9	53.2	66.8	76.9	76.0	64.3	90.3	91.9	86.9	95.3	95.3
4	128	68.5	49.7	47.2	53.7	62.3	65.6	24.9	73.5	77.5	69.0	95.8	94.7
4	256	71.4	46.6	41.7	56.5	63.2	61.4	18.1	29.7	34.6	47.5	93.6	92.7
8	2	94.8	97.9	97.0	94.0	97.2	96.5	80.0	81.5	81.7	94.2	96.6	96.1
8	4	83.5	85.9	85.5	80.9	84.8	86.8	85.2	89.3	89.2	89.3	96.1	94.9
8	8	64.8	64.6	64.5	76.0	75.6	80.0	88.8	91.7	91.7	92.8	97.0	96.3
8	16	52.8	51.7	51.6	72.7	74.1	75.5	87.1	91.2	91.5	92.2	96.5	95.5
8	32	50.1	45.4	45.4	75.9	75.4	75.2	85.1	91.0	91.9	90.0	96.2	95.4
8	64	51.3	39.3	38.6	71.6	71.0	69.9	72.9	87.8	88.9	88.0	94.2	94.2
8	128	55.6	34.4	32.7	53.4	60.3	61.1	29.3	71.9	74.3	71.7	94.5	94.1
8	256	59.3	30.7	27.2	55.2	58.2	44.6	12.7	29.4	37.3	49.5	94.1	93.2
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	84.0	84.6	84.4	86.2	86.9	86.4	87.0	88.2	88.1	92.7	94.9	94.4
16	8	58.0	57.9	57.7	73.6	76.6	74.4	86.8	89.4	89.2	91.5	95.0	94.2
16	16	43.5	42.9	42.8	69.2	71.5	70.0	87.7	91.1	91.3	92.1	95.4	95.0
16	32	38.2	34.5	34.5	73.3	70.2	70.5	85.1	90.2	90.8	90.0	94.6	94.3
16	64	38.7	28.6	28.2	73.4	65.8	65.4	76.4	86.6	87.5	88.1	92.8	92.9
16	128	43.0	23.4	22.4	61.3	58.8	56.4	33.2	69.8	71.8	74.0	92.1	91.8
16	256	47.7	19.5	17.4	53.3	44.8	42.8	9.7	31.8	40.5	51.4	91.3	90.8

Table 32: **Weighted Heun, 15,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.8	99.5	98.7	96.4	98.1	97.8	95.1	97.9	98.6	96.7	99.4	98.6
2	4	93.6	94.6	94.8	90.1	90.8	92.3	94.6	97.6	98.4	96.2	99.4	98.7
2	8	84.8	84.8	85.0	81.5	84.2	81.3	93.4	96.3	97.0	95.3	99.3	98.7
2	16	80.1	79.5	79.8	69.2	78.9	75.8	92.0	95.8	96.6	94.1	99.1	98.3
2	32	79.9	75.9	75.9	71.6	81.8	83.6	85.6	95.7	96.8	91.9	98.9	98.1
2	64	80.7	72.6	71.8	65.5	83.4	71.3	51.1	94.9	95.9	86.0	98.1	98.0
2	128	83.0	70.4	67.8	60.9	68.3	62.2	11.5	82.7	87.1	70.3	96.9	96.8
2	256	83.8	68.4	63.6	61.9	69.2	71.8	24.7	39.8	44.0	48.3	94.7	94.8
4	2	92.2	98.9	97.1	91.8	97.9	96.6	89.3	92.9	92.9	91.9	98.1	96.6
4	4	87.7	90.2	89.9	83.2	90.0	90.3	88.4	93.1	92.6	90.4	97.7	96.2
4	8	73.9	73.6	73.7	81.1	82.1	82.7	89.8	94.0	94.2	92.9	98.1	96.9
4	16	65.4	64.5	64.9	75.6	75.9	76.8	88.9	93.7	94.7	92.1	97.8	97.0
4	32	63.8	59.0	59.1	74.5	80.1	78.6	85.8	92.6	93.8	91.0	97.0	97.2
4	64	65.3	54.3	53.6	72.8	76.7	73.2	65.3	91.1	92.9	87.2	95.8	95.7
4	128	68.8	50.0	47.6	55.0	73.3	65.6	19.0	81.7	84.8	68.2	96.5	94.9
4	256	72.0	47.1	42.1	57.8	67.4	52.9	18.4	34.5	41.9	47.8	94.9	92.8
8	2	94.9	98.0	97.1	94.6	97.5	96.5	80.1	81.6	81.8	94.2	96.6	96.1
8	4	84.6	87.4	86.9	84.6	86.9	88.3	85.3	89.4	89.2	89.3	96.1	94.9
8	8	66.0	65.8	65.8	77.6	79.2	80.6	88.9	91.7	91.7	92.9	97.0	96.3
8	16	53.6	52.5	52.3	73.4	75.6	75.7	87.3	91.4	91.7	92.3	96.7	95.6
8	32	50.4	45.8	45.8	76.2	74.1	73.6	85.4	91.3	92.2	89.9	96.3	95.6
8	64	51.8	39.8	39.2	75.6	69.8	70.5	74.2	89.1	90.2	88.0	94.2	94.5
8	128	56.1	34.8	33.1	54.7	59.1	53.4	29.1	79.2	81.1	70.7	95.0	94.5
8	256	60.0	31.0	27.5	54.2	60.1	49.6	14.2	34.9	44.6	49.6	95.1	94.0
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	85.4	86.2	86.0	87.1	87.8	86.7	87.7	88.8	88.6	92.7	95.0	94.6
16	8	59.4	59.3	59.2	76.0	76.3	76.5	88.2	90.4	89.9	91.5	95.2	94.5
16	16	44.2	43.6	43.7	68.2	71.7	69.8	89.0	92.5	92.5	92.2	95.7	95.3
16	32	38.8	35.1	35.1	72.7	70.6	70.7	86.7	91.6	92.2	90.6	95.0	94.6
16	64	39.2	29.1	28.8	76.0	67.6	66.6	80.1	89.1	90.0	86.8	93.8	93.9
16	128	43.6	23.8	22.8	52.7	49.6	52.8	40.8	79.5	80.5	76.0	92.5	92.6
16	256	48.6	19.8	17.7	50.5	44.2	37.1	11.4	36.8	46.2	52.9	92.2	91.5

Table 33: **Weighted Heun, 20,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.8	99.5	98.7	96.5	98.4	98.0	95.1	97.9	98.6	96.7	99.4	98.6
2	4	94.0	95.0	95.2	90.6	90.3	90.5	94.7	97.7	98.4	96.2	99.4	98.8
2	8	85.1	85.1	85.5	83.8	86.9	90.3	93.4	96.4	97.0	95.3	99.3	98.7
2	16	80.0	79.3	80.0	75.3	78.4	76.1	91.8	95.7	96.7	94.1	99.2	98.3
2	32	79.9	76.1	76.2	69.0	80.0	83.8	85.6	95.7	96.8	91.8	99.0	98.1
2	64	80.9	72.9	72.1	68.5	83.6	74.0	51.1	95.1	96.2	86.1	98.3	98.0
2	128	83.1	70.6	68.1	61.7	73.0	72.9	10.2	87.8	90.9	70.2	97.4	97.2
2	256	83.9	68.7	63.8	62.6	63.1	55.0	22.8	42.0	47.2	48.0	95.3	95.6
4	2	92.2	99.0	97.2	92.0	98.1	96.6	89.4	93.0	92.9	91.9	98.1	96.6
4	4	88.1	90.9	90.5	86.4	90.7	90.6	88.3	93.1	92.5	90.4	97.7	96.3
4	8	74.5	74.3	74.2	80.9	83.7	84.2	89.8	93.9	94.2	93.0	98.1	96.9
4	16	65.9	65.1	65.3	78.0	80.7	76.5	89.0	93.8	94.7	92.3	97.9	97.1
4	32	64.1	59.4	59.5	75.4	79.2	80.0	86.0	92.7	93.9	91.0	97.1	97.2
4	64	65.6	54.7	54.0	67.2	72.0	76.7	65.7	91.6	93.5	87.1	96.0	95.9
4	128	69.1	50.3	48.0	55.8	70.2	68.2	17.8	85.1	87.7	68.0	96.7	95.1
4	256	72.2	47.4	42.5	58.5	62.5	56.0	18.5	39.5	48.2	47.8	95.1	92.7
8	2	95.0	98.0	97.2	94.6	97.4	96.7	80.8	82.5	82.7	94.3	96.8	96.2
8	4	85.3	88.4	87.8	80.8	89.3	88.1	85.9	89.8	89.5	89.4	96.1	95.0
8	8	66.9	66.8	66.6	78.9	79.4	79.7	89.3	92.1	92.0	92.9	97.1	96.4
8	16	54.1	53.0	52.9	73.1	75.6	75.7	87.8	91.9	92.0	92.4	97.0	95.8
8	32	50.8	46.3	46.3	76.6	74.9	74.5	85.9	91.7	92.6	89.8	96.4	95.7
8	64	52.1	40.2	39.6	74.6	72.4	71.5	75.7	90.0	91.2	87.9	94.0	94.4
8	128	56.5	35.1	33.4	55.5	64.0	61.1	29.2	83.2	84.9	72.5	95.0	94.7
8	256	60.4	31.4	27.9	50.4	49.1	48.6	15.0	40.3	49.8	50.0	95.1	94.0
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	86.4	87.2	87.0	86.8	88.1	86.2	87.7	88.7	88.5	92.8	95.1	94.7
16	8	60.5	60.4	60.2	75.9	76.4	76.4	88.9	90.9	90.2	91.6	95.4	94.6
16	16	44.8	44.2	44.3	71.1	70.2	71.4	89.8	93.3	93.2	92.3	95.9	95.4
16	32	39.2	35.5	35.5	73.2	69.7	68.8	87.6	92.5	93.1	91.1	95.1	94.6
16	64	39.6	29.4	29.1	74.4	68.3	67.7	81.8	90.4	91.3	86.5	94.3	94.3
16	128	44.1	24.1	23.1	57.2	57.6	53.2	44.6	83.7	84.6	78.0	92.7	92.9
16	256	49.1	20.2	18.1	50.8	44.8	35.3	12.8	41.8	50.4	53.0	93.2	92.2

Table 34: **Unweighted RK45, 2,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.0	98.4	98.4	95.9	96.1	97.2	98.2	98.8	98.8	99.0	99.5	99.6
2	4	91.6	91.5	91.4	88.8	89.3	90.1	97.9	98.7	98.7	98.6	99.5	99.4
2	8	82.5	81.8	81.7	83.8	84.9	86.9	97.1	98.5	98.4	97.7	99.2	99.2
2	16	80.1	78.7	78.6	83.5	74.3	66.2	95.2	97.6	97.6	95.7	97.7	97.7
2	32	80.0	74.6	74.0	76.0	73.8	72.5	84.5	95.2	95.5	92.6	96.3	96.4
2	64	80.3	71.1	69.7	62.9	79.7	77.3	36.4	60.6	67.2	85.8	94.1	94.7
2	128	82.2	68.7	65.7	53.7	70.5	72.8	29.7	39.5	41.0	73.2	93.1	93.7
2	256	83.0	66.5	61.7	55.2	64.4	54.2	25.4	37.2	38.7	57.1	91.2	92.3
4	2	91.7	97.5	96.2	90.6	96.0	94.6	90.8	93.3	92.7	92.1	98.1	96.6
4	4	83.8	83.9	83.6	80.9	87.3	86.7	91.5	93.9	93.1	92.5	97.7	96.6
4	8	70.2	69.3	69.0	82.0	82.6	81.3	93.5	95.1	94.7	94.4	97.6	97.0
4	16	64.2	62.0	61.9	83.1	81.4	82.1	94.1	95.4	95.3	94.7	95.3	95.1
4	32	62.9	56.5	56.2	80.1	71.6	66.9	87.0	92.3	92.5	91.9	93.4	93.1
4	64	64.1	51.5	50.2	68.8	75.5	73.5	40.9	70.9	74.1	86.2	91.4	92.0
4	128	67.5	47.3	44.6	51.4	67.2	61.2	20.5	29.5	31.1	70.5	90.9	90.9
4	256	70.0	44.5	39.5	51.5	65.3	46.3	15.0	21.7	21.6	50.7	87.6	89.3
8	2	95.7	96.7	96.5	94.5	95.2	93.5	82.7	83.5	83.6	96.2	96.9	96.6
8	4	79.2	79.2	78.7	84.8	85.0	84.4	88.3	90.2	89.4	90.3	96.0	94.8
8	8	60.8	60.1	59.9	77.5	78.8	76.1	91.5	92.6	92.2	93.8	96.3	95.7
8	16	51.1	49.3	49.2	78.4	75.9	76.6	91.2	92.4	91.9	93.3	94.4	93.8
8	32	48.6	42.9	42.6	78.8	73.6	73.5	85.9	89.4	89.3	91.5	92.6	91.9
8	64	49.6	36.8	36.0	78.6	68.0	66.3	47.2	70.9	72.5	87.2	90.0	89.8
8	128	53.5	32.2	30.3	59.6	54.1	58.3	15.4	29.9	32.8	70.9	88.2	88.4
8	256	57.0	28.6	25.4	50.4	53.3	43.8	9.8	16.7	16.6	47.4	86.4	87.5
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	77.1	77.4	77.2	82.7	84.8	79.0	87.5	88.9	88.8	92.1	94.7	94.1
16	8	52.8	52.3	52.1	73.2	76.3	75.1	88.1	89.6	89.3	91.4	94.2	93.4
16	16	40.7	39.6	39.5	74.5	72.4	72.2	89.3	90.6	90.4	92.7	93.3	92.9
16	32	36.6	32.4	32.3	76.9	70.2	69.9	83.5	87.0	87.1	91.1	91.6	91.1
16	64	36.8	26.5	26.1	75.5	63.3	61.6	50.7	69.4	70.6	86.8	88.9	88.5
16	128	40.6	21.6	20.6	53.4	49.0	45.1	12.3	34.5	38.0	72.1	86.7	86.4
16	256	45.0	18.1	16.2	51.6	43.4	32.8	5.8	15.5	16.2	45.0	81.9	84.0

2.7 Unweighted RK45

Table 35: **Unweighted RK45, 5,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.6	99.0	99.0	96.9	97.5	97.6	98.1	98.7	98.8	99.0	99.5	99.6
2	4	93.2	93.1	93.0	84.0	92.5	92.4	98.1	98.8	98.8	98.6	99.5	99.4
2	8	84.0	83.3	83.3	83.7	79.3	85.0	97.5	98.7	98.6	97.8	99.3	99.2
2	16	80.3	79.0	79.1	76.2	85.9	85.6	95.9	98.1	98.0	96.8	98.9	98.8
2	32	80.8	75.8	75.1	77.6	75.8	71.5	87.5	96.9	96.9	93.8	98.2	98.2
2	64	81.3	72.2	70.8	65.1	76.9	78.2	46.7	91.6	92.7	86.8	96.7	96.7
2	128	83.1	69.7	66.8	57.5	65.8	60.9	30.2	43.7	46.8	72.4	94.9	95.2
2	256	83.7	67.7	62.7	58.7	70.0	57.0	25.8	37.1	39.5	52.8	93.8	94.7
4	2	92.1	98.4	96.8	91.4	97.3	95.8	90.9	93.2	92.6	92.2	98.2	96.6
4	4	86.5	87.0	86.7	84.3	87.6	88.4	91.4	93.9	93.1	92.6	97.8	96.7
4	8	72.5	71.5	71.2	81.9	81.2	84.9	93.9	95.3	94.8	94.7	98.0	97.3
4	16	65.3	63.0	63.1	79.8	83.3	82.9	95.1	96.2	96.0	96.5	97.4	97.0
4	32	64.0	58.1	57.7	81.8	81.8	81.0	90.3	94.9	94.9	94.4	96.8	96.3
4	64	65.5	52.9	51.8	68.5	77.4	75.2	63.6	88.8	89.3	89.0	95.1	94.5
4	128	68.7	48.8	46.0	51.8	70.9	68.1	24.1	43.7	49.7	70.2	93.7	93.9
4	256	71.5	45.8	40.7	54.6	62.5	53.1	16.8	25.8	27.6	48.2	91.2	92.6
8	2	96.3	97.5	97.3	95.5	95.8	96.0	82.3	83.2	83.3	96.2	96.9	96.6
8	4	82.8	83.4	82.8	85.7	87.2	85.9	88.2	90.3	89.5	90.4	96.2	95.0
8	8	63.6	63.0	62.8	79.4	78.4	76.6	91.8	92.9	92.5	94.2	96.9	96.2
8	16	52.6	50.9	50.7	78.0	76.0	76.5	92.2	93.3	92.8	94.7	96.2	95.3
8	32	50.0	44.5	44.2	81.8	74.5	71.8	89.8	92.6	92.4	93.9	95.7	94.9
8	64	51.1	38.4	37.5	74.7	69.5	68.3	71.1	85.4	85.8	90.1	94.1	93.6
8	128	55.2	33.4	31.6	52.0	62.2	59.8	21.4	48.2	53.7	73.0	93.0	93.0
8	256	58.7	29.8	26.4	53.6	43.0	48.0	11.4	22.8	26.0	48.7	91.1	92.0
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	81.3	81.8	81.6	85.1	87.0	85.5	87.5	88.9	88.9	92.1	94.7	94.2
16	8	56.2	55.7	55.4	75.0	75.2	75.8	88.3	89.8	89.5	91.9	94.8	93.9
16	16	42.8	41.7	41.6	75.1	73.4	71.5	90.1	91.5	91.2	93.8	94.8	94.3
16	32	38.0	33.9	33.8	78.1	72.0	71.2	87.6	90.4	90.4	93.2	94.0	93.6
16	64	38.3	27.9	27.4	77.0	66.3	64.9	72.7	82.9	83.3	90.4	92.8	92.2
16	128	42.5	22.8	21.7	65.1	58.3	55.4	20.3	50.4	54.6	74.8	91.4	91.1
16	256	47.0	19.0	17.0	52.1	36.8	37.0	7.4	24.1	29.2	50.7	88.2	88.8

Table 36: **Unweighted RK45, 10,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.8	99.3	99.3	97.8	97.4	98.2	98.0	98.7	98.8	99.0	99.6	99.6
2	4	94.2	94.2	94.1	85.7	91.5	92.6	98.2	98.9	98.9	98.6	99.5	99.4
2	8	85.4	84.5	84.4	83.0	90.2	77.7	97.6	98.7	98.7	97.8	99.3	99.2
2	16	80.9	79.5	79.8	80.9	82.8	82.6	96.1	98.2	98.2	97.1	99.1	99.0
2	32	81.0	76.2	75.8	75.2	79.8	85.4	88.5	97.5	97.6	94.6	99.0	98.9
2	64	81.7	72.8	71.6	64.1	78.5	74.6	50.9	94.8	95.4	87.3	98.0	98.0
2	128	83.9	70.4	67.6	59.8	66.0	59.8	22.3	69.0	75.3	71.7	96.4	96.4
2	256	84.5	68.4	63.5	60.8	60.8	58.8	25.1	38.3	41.5	50.1	95.1	95.7
4	2	92.3	98.7	97.0	91.9	97.7	96.2	90.9	93.3	92.8	92.2	98.2	96.6
4	4	88.3	89.1	88.7	88.1	88.9	84.9	91.6	94.0	93.1	92.6	97.9	96.7
4	8	74.4	73.3	72.9	82.1	84.4	85.9	94.0	95.3	94.8	94.8	98.2	97.5
4	16	66.3	64.1	64.2	80.4	81.4	83.4	95.4	96.5	96.3	96.8	97.8	97.4
4	32	64.9	59.1	58.7	81.7	80.1	79.9	91.6	95.9	96.0	95.2	97.6	97.1
4	64	66.3	53.9	52.9	64.2	76.8	71.9	68.2	92.4	92.7	90.3	96.9	96.6
4	128	69.8	49.7	46.9	54.2	65.5	57.6	25.2	74.1	77.6	71.2	96.0	95.9
4	256	72.4	46.6	41.5	56.9	65.4	54.0	18.1	29.8	34.7	47.7	93.8	94.8
8	2	96.6	97.9	97.7	95.9	96.9	96.9	82.5	83.3	83.5	96.2	96.9	96.6
8	4	85.0	86.0	85.4	86.4	88.6	84.7	88.3	90.3	89.5	90.4	96.2	95.0
8	8	65.8	65.2	65.0	80.7	80.2	74.0	91.8	92.9	92.5	94.3	97.1	96.3
8	16	54.0	52.2	52.1	76.1	76.8	77.8	92.5	93.5	93.0	95.1	96.7	95.7
8	32	50.9	45.6	45.2	80.9	76.9	76.7	91.2	93.8	93.5	94.4	96.5	95.8
8	64	52.1	39.4	38.7	79.3	69.4	69.4	76.7	89.6	89.8	91.5	95.3	94.9
8	128	56.5	34.4	32.6	54.6	65.8	61.8	29.8	72.5	74.4	75.2	95.2	95.0
8	256	60.2	30.7	27.1	56.1	53.8	46.3	12.8	29.5	37.5	50.3	94.4	94.2
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	84.2	84.9	84.6	87.1	87.7	85.1	87.9	89.4	89.3	92.2	94.8	94.2
16	8	59.1	58.5	58.3	76.6	78.0	78.1	88.8	90.2	89.7	91.9	95.1	94.2
16	16	44.5	43.4	43.3	71.5	69.8	73.6	90.9	92.1	91.8	94.2	95.4	94.9
16	32	39.1	34.9	34.8	78.4	72.6	72.1	89.8	92.1	92.0	93.7	94.9	94.5
16	64	39.4	28.8	28.4	76.5	68.2	67.3	80.2	88.0	88.2	91.2	93.9	93.3
16	128	43.8	23.6	22.5	63.7	60.2	56.8	33.7	70.3	71.9	76.6	92.9	92.4
16	256	48.5	19.6	17.5	54.4	44.7	37.4	9.7	31.9	40.6	52.8	91.3	91.5

Table 37: **Unweighted RK45, 15,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.9	99.4	99.4	97.9	97.9	98.3	98.1	98.8	98.8	99.0	99.6	99.5
2	4	94.8	94.8	94.8	91.4	91.2	88.9	98.2	98.9	98.9	98.6	99.5	99.4
2	8	85.9	85.0	85.1	82.5	89.9	87.4	97.7	98.8	98.7	97.8	99.3	99.2
2	16	81.2	79.8	79.8	69.9	79.6	86.3	96.2	98.4	98.3	97.1	99.1	99.0
2	32	81.0	76.4	76.0	71.6	86.8	83.8	89.0	97.8	97.8	94.8	99.2	99.1
2	64	82.0	73.3	72.0	61.2	74.6	67.0	51.8	95.8	96.1	87.6	98.6	98.5
2	128	84.0	70.8	67.8	60.9	68.4	63.8	11.6	82.9	87.1	71.2	96.8	96.7
2	256	84.7	68.8	63.6	61.9	72.4	53.5	24.7	39.7	44.1	48.6	95.9	96.2
4	2	92.4	98.9	97.1	92.1	98.0	96.4	90.8	93.3	92.8	92.2	98.2	96.6
4	4	89.1	90.1	89.8	88.3	89.6	87.0	91.6	94.0	93.1	92.6	97.8	96.7
4	8	75.1	74.0	73.7	85.0	80.4	86.4	94.0	95.3	94.8	94.8	98.2	97.5
4	16	66.9	64.7	64.5	82.6	81.8	82.0	95.6	96.5	96.3	96.8	97.9	97.4
4	32	65.3	59.6	59.2	81.6	80.5	80.5	92.1	96.3	96.4	95.6	98.0	97.5
4	64	66.7	54.5	53.4	76.6	77.9	71.7	69.0	93.6	93.9	90.8	97.3	97.1
4	128	70.1	50.1	47.4	55.6	74.0	67.7	19.2	82.5	84.7	70.5	96.8	96.7
4	256	73.0	47.1	41.9	58.1	63.1	50.0	18.5	34.6	41.9	47.9	95.3	95.8
8	2	96.7	98.0	97.8	96.3	97.4	97.1	82.6	83.4	83.5	96.2	96.9	96.5
8	4	86.1	87.4	86.7	87.0	88.4	87.6	88.3	90.3	89.5	90.4	96.2	95.0
8	8	67.2	66.5	66.2	81.2	80.9	82.2	91.9	93.0	92.6	94.4	97.1	96.4
8	16	54.7	52.9	52.7	77.8	77.4	77.1	92.6	93.6	93.1	95.2	96.9	95.9
8	32	51.3	46.1	46.0	82.1	77.0	76.5	91.7	94.2	94.0	94.5	96.8	96.0
8	64	52.6	40.0	39.2	76.2	72.6	69.8	78.5	91.2	91.3	91.8	95.7	95.4
8	128	57.0	34.9	33.0	56.0	62.8	57.7	29.7	80.1	81.2	74.8	95.9	95.5
8	256	60.7	31.0	27.4	55.0	57.3	50.6	14.3	35.1	44.6	50.5	95.5	95.4
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	85.6	86.4	86.2	87.6	87.2	87.4	88.6	89.8	89.5	92.2	95.0	94.4
16	8	60.6	60.1	59.8	78.5	79.5	79.2	89.9	91.0	90.2	91.9	95.3	94.4
16	16	45.3	44.2	44.3	71.6	71.7	73.2	92.1	93.1	92.8	94.3	95.7	95.2
16	32	39.6	35.6	35.5	77.8	71.0	70.7	91.6	93.4	93.2	93.9	95.1	94.9
16	64	40.0	29.4	29.0	81.1	69.5	68.6	84.7	90.8	90.9	90.8	94.6	94.2
16	128	44.5	24.0	22.9	54.1	57.3	57.3	41.7	80.2	80.7	78.4	94.1	93.6
16	256	49.3	20.0	17.8	51.4	49.6	42.9	11.5	37.0	46.1	54.0	92.2	92.4

Table 38: **Unweighted RK45, 20,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.0	99.5	99.5	98.5	98.4	98.3	98.1	98.8	98.8	99.0	99.6	99.5
2	4	95.2	95.1	95.1	90.8	91.0	91.3	98.2	98.9	98.9	98.6	99.5	99.4
2	8	86.3	85.5	85.4	83.4	90.3	84.6	97.7	98.8	98.7	97.8	99.3	99.2
2	16	81.2	79.7	79.9	80.0	83.4	85.5	96.4	98.5	98.4	97.1	99.2	99.1
2	32	81.2	76.6	76.3	73.7	84.1	82.3	89.3	98.0	97.9	94.9	99.2	99.1
2	64	82.0	73.5	72.2	61.5	80.0	82.2	51.5	96.3	96.4	87.8	98.8	98.6
2	128	84.2	71.1	68.2	61.8	73.3	72.4	10.2	88.1	90.9	71.5	97.5	97.3
2	256	84.8	69.1	64.1	62.7	60.7	60.2	22.8	41.9	47.3	48.2	96.5	96.5
4	2	92.5	98.9	97.2	91.9	98.1	96.6	90.9	93.3	92.7	92.2	98.2	96.6
4	4	89.6	90.8	90.5	87.3	89.0	89.9	91.6	94.0	93.1	92.6	97.9	96.7
4	8	75.8	74.7	74.3	85.0	86.4	86.5	94.1	95.3	94.9	94.8	98.3	97.5
4	16	67.0	65.0	65.0	83.1	82.5	80.6	95.6	96.5	96.4	96.7	98.0	97.5
4	32	65.5	60.0	59.6	80.0	78.4	81.5	92.3	96.5	96.6	95.7	98.1	97.6
4	64	66.9	54.9	53.7	77.5	78.5	78.0	69.4	94.3	94.5	90.9	97.5	97.2
4	128	70.4	50.4	47.8	56.4	71.6	67.0	18.0	86.0	87.8	70.2	97.1	96.9
4	256	73.3	47.4	42.2	59.0	66.2	60.5	18.6	39.6	48.2	47.8	95.9	96.5
8	2	96.8	98.1	97.9	96.5	97.4	97.2	83.2	84.2	84.3	96.3	97.0	96.7
8	4	86.8	88.3	87.7	86.3	89.2	87.2	88.9	90.6	89.6	90.4	96.2	95.0
8	8	68.0	67.3	67.1	81.0	82.8	82.5	92.2	93.2	92.7	94.4	97.2	96.4
8	16	55.2	53.4	53.2	76.3	76.6	77.3	93.0	93.9	93.4	95.3	97.1	96.1
8	32	51.7	46.5	46.4	81.7	72.3	76.5	92.4	94.6	94.4	94.6	96.9	96.0
8	64	53.0	40.4	39.7	82.6	71.7	71.8	80.5	92.3	92.4	92.0	95.9	95.5
8	128	57.5	35.2	33.3	56.9	62.5	64.9	29.8	84.3	85.1	76.0	96.0	95.7
8	256	61.3	31.3	27.7	50.9	46.1	53.8	15.1	40.5	49.9	50.8	95.5	95.6
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	86.5	87.5	87.2	87.8	88.6	88.8	88.5	89.5	89.2	92.2	95.1	94.5
16	8	61.6	61.1	60.8	78.6	77.3	79.5	90.4	91.3	90.5	91.9	95.4	94.5
16	16	45.8	44.9	44.9	73.5	72.1	71.9	92.7	93.8	93.3	94.4	95.9	95.4
16	32	40.0	35.9	36.0	77.1	72.3	71.9	92.5	94.2	94.0	94.2	95.5	95.1
16	64	40.4	29.8	29.3	79.9	69.9	69.3	86.7	92.2	92.3	90.6	95.0	94.6
16	128	44.9	24.3	23.2	59.1	57.5	53.0	45.8	84.7	84.9	79.7	94.6	94.0
16	256	49.9	20.3	18.1	51.6	44.1	41.3	12.8	42.0	50.3	54.4	93.2	92.8

Table 39: **Weighted RK45, 2,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.2	98.4	98.2	95.4	96.8	96.7	95.0	97.8	98.5	96.7	99.4	98.6
2	4	90.7	91.3	91.4	87.2	90.0	86.3	94.7	97.7	98.4	96.2	99.4	98.7
2	8	81.5	81.5	81.7	79.3	78.5	82.5	93.1	96.2	96.9	95.2	99.1	98.6
2	16	78.8	78.2	78.3	79.6	84.8	85.5	91.6	95.5	96.2	94.2	97.8	97.3
2	32	78.7	74.1	73.9	78.1	83.0	72.9	82.6	94.2	94.9	91.5	96.4	96.2
2	64	79.0	70.3	69.3	62.7	78.9	72.4	37.6	62.7	66.8	84.5	93.7	94.4
2	128	81.0	68.3	65.7	53.7	78.1	67.3	29.6	39.4	41.1	72.3	92.6	93.6
2	256	81.9	66.1	61.4	55.2	64.1	54.6	24.4	37.1	38.7	57.0	90.4	91.5
4	2	91.5	97.6	96.3	90.8	95.7	95.5	89.3	93.0	93.0	91.9	98.1	96.5
4	4	82.5	83.9	83.7	82.7	84.9	86.9	88.1	93.0	92.5	90.3	97.5	96.2
4	8	68.8	68.7	68.7	75.7	81.1	81.6	89.5	93.8	94.0	92.5	97.5	96.5
4	16	62.6	61.5	61.8	74.6	80.8	80.9	88.5	93.0	93.8	90.7	95.0	94.9
4	32	61.5	56.0	55.9	78.9	76.2	77.8	82.9	90.0	91.0	90.0	93.2	93.3
4	64	63.2	51.3	50.4	65.8	74.4	72.3	41.2	71.2	73.8	83.4	91.6	91.6
4	128	66.6	47.2	44.9	50.9	66.5	56.9	20.4	29.5	31.3	69.0	89.8	90.5
4	256	69.0	44.5	39.8	51.1	66.9	54.9	14.8	21.7	21.8	50.1	86.1	88.1
8	2	94.0	96.7	96.1	93.2	95.7	94.9	80.1	81.6	81.9	94.2	96.7	96.1
8	4	77.7	78.8	78.5	79.6	86.2	85.4	85.4	89.4	89.2	89.2	96.0	94.8
8	8	59.3	59.2	59.1	75.5	77.7	77.6	88.6	91.4	91.4	92.4	96.2	95.6
8	16	49.8	48.6	48.6	74.1	74.9	73.9	86.3	90.4	90.6	90.8	94.2	93.6
8	32	47.8	42.8	42.7	73.8	72.4	72.5	81.7	87.7	88.3	88.7	92.7	91.8
8	64	48.7	36.6	35.9	74.1	62.4	63.3	46.6	70.7	72.4	85.1	90.1	89.7
8	128	52.6	32.1	30.5	56.3	57.5	52.8	15.3	29.8	32.7	67.4	87.0	88.0
8	256	56.0	28.6	25.5	51.1	50.5	43.2	9.5	16.7	16.6	46.4	84.5	87.1
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	76.4	76.8	76.7	83.5	82.8	83.1	86.4	87.5	87.5	92.6	94.7	94.2
16	8	51.7	51.6	51.4	69.7	73.1	73.0	86.1	88.7	88.6	91.1	94.2	93.5
16	16	39.7	39.0	39.0	70.4	70.2	70.1	86.1	89.6	89.8	90.9	93.4	93.1
16	32	36.0	32.1	32.1	71.0	68.4	67.3	80.3	85.8	86.3	88.7	91.5	91.1
16	64	36.1	26.2	25.9	71.5	62.2	61.4	50.1	69.2	70.5	85.6	89.0	88.7
16	128	39.9	21.4	20.5	52.0	48.1	50.6	12.2	34.2	37.8	69.4	86.8	86.5
16	256	44.1	17.9	16.1	50.3	40.2	33.9	5.7	15.4	16.2	44.5	82.2	84.2

2.8 Weighted RK45

Table 40: **Weighted RK45, 5,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.6	99.1	98.5	95.9	97.2	97.5	95.1	97.9	98.7	96.7	99.4	98.6
2	4	92.1	92.9	93.1	89.8	90.7	88.4	94.7	97.7	98.4	96.2	99.4	98.7
2	8	83.3	83.3	83.4	79.8	83.8	82.5	93.3	96.4	97.0	95.3	99.2	98.7
2	16	79.2	78.7	79.0	76.2	82.2	82.7	91.7	95.7	96.5	94.0	98.7	98.2
2	32	79.8	75.1	75.0	76.3	77.4	81.2	85.0	95.4	96.3	92.2	98.2	97.7
2	64	79.9	71.6	70.7	64.9	77.8	79.2	47.5	91.1	92.5	85.6	96.2	96.5
2	128	82.0	69.3	66.8	57.5	61.2	67.0	30.1	43.7	47.0	71.4	94.6	95.3
2	256	82.9	67.3	62.6	58.8	74.2	51.2	25.5	37.1	39.6	52.7	92.9	93.5
4	2	91.9	98.5	96.8	91.2	96.9	96.2	89.4	92.9	92.9	91.9	98.1	96.6
4	4	85.3	87.1	86.9	85.2	82.9	89.2	88.1	93.0	92.5	90.4	97.7	96.3
4	8	71.1	71.0	70.9	76.7	82.0	81.8	89.7	94.0	94.2	92.8	97.9	96.7
4	16	64.0	62.8	63.1	73.7	80.5	81.5	88.8	93.4	94.4	91.9	97.3	96.7
4	32	62.6	57.4	57.4	77.2	79.6	80.0	84.9	91.9	93.0	91.3	95.8	96.1
4	64	64.2	52.9	52.1	67.9	76.3	74.7	61.6	87.4	88.9	86.2	94.2	93.8
4	128	67.8	48.7	46.2	51.4	73.3	61.0	23.9	43.5	49.6	68.8	93.5	92.9
4	256	70.4	45.7	40.9	54.2	59.7	52.9	16.8	25.7	27.8	48.1	90.2	91.9
8	2	94.5	97.5	96.8	93.4	95.9	95.7	79.8	81.4	81.7	94.2	96.6	96.1
8	4	81.3	83.2	82.8	77.1	83.2	82.8	85.3	89.4	89.2	89.3	96.1	95.0
8	8	62.3	62.2	62.1	76.0	77.0	76.7	88.8	91.7	91.7	92.8	96.8	96.1
8	16	51.4	50.2	50.3	72.9	73.8	73.0	86.9	91.0	91.3	91.8	96.0	95.1
8	32	49.1	44.3	44.2	76.8	73.3	73.6	84.3	90.2	90.9	89.9	95.5	94.7
8	64	50.1	38.0	37.4	70.2	68.6	67.5	68.7	84.2	85.3	87.2	93.4	93.4
8	128	54.2	33.4	31.7	50.9	56.9	59.6	21.2	47.9	53.7	69.1	92.4	92.5
8	256	58.1	29.8	26.5	52.8	50.4	44.1	11.3	22.7	26.0	47.6	90.8	91.0
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	81.0	81.5	81.3	85.5	86.1	84.2	86.6	87.7	87.7	92.6	94.8	94.3
16	8	55.1	55.0	54.8	71.0	72.4	75.0	86.3	88.9	88.8	91.4	94.8	94.0
16	16	41.8	41.1	41.2	71.5	68.4	71.1	86.9	90.4	90.6	91.7	94.8	94.4
16	32	37.3	33.5	33.5	74.0	69.8	69.3	83.5	88.7	89.3	89.9	93.8	93.4
16	64	37.6	27.6	27.2	70.5	61.6	60.0	70.2	82.0	82.9	88.2	92.4	92.2
16	128	41.8	22.5	21.6	55.3	57.1	54.5	20.0	50.1	54.5	72.0	90.8	90.8
16	256	46.4	18.9	16.9	50.9	36.6	33.6	7.4	24.0	29.2	49.3	88.5	88.9

Table 41: **Weighted RK45, 10,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.7	99.4	98.6	96.3	97.8	97.7	95.2	98.0	98.7	96.7	99.4	98.6
2	4	93.1	94.1	94.2	88.8	85.4	90.5	94.6	97.6	98.4	96.2	99.4	98.7
2	8	84.5	84.2	84.4	83.5	81.2	83.2	93.4	96.4	97.0	95.3	99.3	98.7
2	16	79.7	79.3	79.6	77.8	78.8	83.2	91.9	95.7	96.6	94.1	99.0	98.3
2	32	79.8	75.7	75.7	74.3	84.7	67.8	85.5	95.6	96.6	92.0	98.8	98.0
2	64	80.3	72.2	71.5	67.8	77.7	76.5	50.0	94.1	95.2	86.3	97.8	97.7
2	128	82.9	70.0	67.4	59.7	67.5	71.7	22.2	68.7	75.0	70.8	96.3	96.5
2	256	83.6	68.1	63.2	60.7	63.2	53.3	25.6	38.3	41.7	49.8	94.0	94.4
4	2	92.1	98.8	97.0	91.7	98.0	96.6	89.3	92.9	92.9	91.9	98.1	96.6
4	4	86.9	89.1	88.9	83.9	87.9	87.9	88.3	93.1	92.6	90.4	97.7	96.2
4	8	73.1	72.7	72.6	77.8	84.1	80.2	89.7	94.0	94.3	93.0	98.0	96.8
4	16	65.0	64.1	64.4	72.8	80.8	82.9	88.9	93.6	94.6	92.2	97.7	96.9
4	32	63.4	58.4	58.4	77.1	78.5	80.4	85.6	92.5	93.7	91.0	96.6	96.8
4	64	64.9	53.9	53.2	66.8	76.9	76.0	64.3	90.3	91.9	86.8	95.3	95.3
4	128	68.5	49.7	47.2	53.7	62.3	65.7	24.9	73.5	77.5	69.0	95.8	94.7
4	256	71.6	46.6	41.7	56.5	63.2	61.4	18.3	29.7	34.6	47.3	93.7	92.7
8	2	94.8	97.9	97.0	94.0	97.2	96.5	80.0	81.5	81.7	94.2	96.6	96.1
8	4	83.5	85.9	85.5	80.9	84.8	86.8	85.3	89.3	89.2	89.3	96.1	95.0
8	8	64.7	64.7	64.5	76.0	75.6	80.0	88.8	91.7	91.7	92.8	97.0	96.3
8	16	52.8	51.6	51.6	72.7	74.1	75.5	87.1	91.3	91.5	92.2	96.5	95.5
8	32	50.0	45.2	45.2	75.9	75.4	75.2	85.1	90.9	91.8	90.1	96.3	95.6
8	64	51.3	39.3	38.6	71.6	71.0	69.9	72.8	87.8	88.9	88.0	94.2	94.3
8	128	55.6	34.4	32.7	53.4	60.3	61.1	29.3	71.9	74.3	71.7	94.5	94.1
8	256	59.3	30.7	27.2	55.2	58.2	44.6	12.7	29.4	37.3	48.9	94.1	93.3
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	84.0	84.6	84.4	86.2	86.9	86.4	87.0	88.2	88.1	92.7	94.9	94.4
16	8	58.0	57.9	57.7	73.6	76.6	74.4	86.8	89.4	89.2	91.5	95.0	94.2
16	16	43.4	42.8	42.8	69.2	71.5	70.0	87.6	91.1	91.3	92.1	95.4	95.0
16	32	38.3	34.5	34.5	73.3	70.2	70.5	85.1	90.2	90.8	90.1	94.7	94.3
16	64	38.7	28.6	28.2	73.4	65.8	65.4	76.3	86.6	87.5	88.1	92.8	92.9
16	128	43.0	23.4	22.4	61.3	58.8	56.4	33.2	69.8	71.8	74.0	92.1	91.8
16	256	47.8	19.5	17.5	53.3	44.8	42.8	9.7	31.8	40.5	51.4	91.4	90.8

Table 42: **Weighted RK45, 15,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.8	99.5	98.7	96.4	98.1	97.8	95.1	97.9	98.6	96.7	99.4	98.6
2	4	93.6	94.6	94.8	90.1	90.8	92.3	94.6	97.6	98.4	96.2	99.4	98.7
2	8	85.2	84.9	85.0	81.5	84.2	81.3	93.5	96.4	97.0	95.3	99.3	98.7
2	16	80.1	79.5	79.8	69.2	78.9	75.8	92.0	95.8	96.6	94.1	99.1	98.2
2	32	79.8	75.8	75.8	71.6	84.5	84.8	85.6	95.6	96.7	91.8	99.0	98.1
2	64	80.7	72.6	71.8	65.5	83.4	71.3	51.1	94.9	95.9	86.0	98.1	98.0
2	128	83.0	70.4	67.8	60.9	68.3	62.2	11.5	82.7	87.1	70.3	97.0	96.8
2	256	84.0	68.4	63.6	61.8	69.2	71.8	24.6	39.8	44.0	48.4	94.7	94.9
4	2	92.2	98.9	97.1	92.0	97.9	96.6	89.3	92.9	92.9	91.9	98.1	96.6
4	4	87.7	90.2	89.9	83.2	90.0	90.3	88.4	93.1	92.6	90.4	97.7	96.2
4	8	73.9	73.8	73.7	81.1	82.1	82.7	89.6	94.0	94.2	93.0	98.1	96.9
4	16	65.4	64.6	64.8	75.6	75.9	76.8	88.9	93.7	94.7	92.2	97.8	97.0
4	32	63.8	59.1	59.1	74.5	80.6	77.8	85.9	92.7	93.9	91.0	96.9	97.2
4	64	65.3	54.4	53.6	72.8	76.7	73.2	65.3	91.1	92.9	87.2	95.8	95.7
4	128	68.8	50.0	47.6	55.0	73.3	65.6	19.0	81.7	84.8	68.2	96.6	95.0
4	256	72.0	47.1	42.1	57.7	67.5	52.9	18.2	34.5	41.9	47.4	95.0	92.8
8	2	94.9	98.0	97.1	94.6	97.5	96.5	80.1	81.6	81.8	94.2	96.6	96.1
8	4	84.6	87.4	86.9	84.6	86.9	88.3	85.3	89.4	89.2	89.3	96.1	94.9
8	8	66.0	65.9	65.8	77.6	79.2	80.6	88.8	91.7	91.7	92.9	97.1	96.3
8	16	53.5	52.3	52.3	73.5	75.6	75.7	87.3	91.4	91.6	92.3	96.8	95.7
8	32	50.4	45.9	45.9	76.2	74.5	74.5	85.4	91.2	92.2	90.0	96.2	95.6
8	64	51.8	39.8	39.2	75.7	69.8	70.5	74.2	89.1	90.2	88.0	94.2	94.5
8	128	56.1	34.8	33.1	54.7	59.1	53.4	29.1	79.2	81.1	70.7	95.0	94.5
8	256	60.1	31.0	27.5	54.2	60.1	49.6	14.0	34.9	44.6	49.0	95.1	94.0
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	85.4	86.2	86.0	87.1	87.8	86.7	87.7	88.8	88.6	92.7	95.0	94.6
16	8	59.4	59.3	59.2	76.0	76.3	76.5	88.2	90.4	89.9	91.5	95.3	94.5
16	16	44.2	43.6	43.7	68.2	71.7	69.8	89.0	92.4	92.5	92.2	95.7	95.3
16	32	38.8	35.2	35.2	72.7	69.1	70.4	86.7	91.6	92.2	90.7	95.0	94.6
16	64	39.2	29.1	28.8	76.0	67.6	66.6	80.1	89.2	90.0	86.8	93.8	93.9
16	128	43.6	23.8	22.8	52.7	49.6	52.8	40.8	79.5	80.5	76.0	92.5	92.7
16	256	48.6	19.8	17.8	51.8	44.2	37.1	11.4	36.8	46.2	52.6	92.2	91.5

Table 43: **Weighted RK45, 20,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.8	99.5	98.7	96.5	98.4	98.0	95.1	97.9	98.6	96.7	99.4	98.6
2	4	93.9	95.0	95.2	90.6	90.3	90.5	94.7	97.7	98.5	96.2	99.4	98.7
2	8	85.3	85.4	85.5	83.8	86.9	90.3	93.5	96.4	97.0	95.3	99.3	98.7
2	16	80.1	79.6	79.9	75.3	78.4	76.1	92.0	95.8	96.6	94.1	99.1	98.2
2	32	79.9	76.2	76.3	69.0	83.4	84.3	85.7	95.7	96.8	91.8	99.0	98.1
2	64	80.9	72.9	72.1	68.4	83.6	74.0	51.1	95.1	96.2	86.1	98.3	98.0
2	128	83.1	70.6	68.1	61.7	73.0	72.9	10.1	87.8	91.0	70.2	97.4	97.2
2	256	84.1	68.7	63.8	62.5	63.1	55.0	22.4	42.0	47.2	47.7	95.3	95.6
4	2	92.2	99.0	97.2	92.0	98.1	96.6	89.4	93.0	92.9	91.9	98.1	96.6
4	4	88.1	90.9	90.5	86.4	90.7	90.6	88.4	93.1	92.6	90.4	97.7	96.3
4	8	74.5	74.3	74.2	80.9	83.7	84.2	89.7	94.0	94.2	93.0	98.1	96.9
4	16	65.6	64.9	65.2	78.0	80.7	76.5	89.0	93.7	94.8	92.2	97.9	97.0
4	32	64.2	59.5	59.5	75.4	80.0	79.3	86.1	92.7	94.0	91.1	97.1	97.3
4	64	65.6	54.7	54.0	67.2	72.0	76.7	65.7	91.6	93.5	87.1	96.0	95.9
4	128	69.2	50.3	48.0	55.8	70.2	68.2	17.8	85.1	87.7	68.0	96.7	95.1
4	256	72.2	47.4	42.5	58.4	62.5	56.0	18.3	39.5	48.2	47.7	95.2	92.7
8	2	95.0	98.0	97.2	94.6	97.4	96.7	80.8	82.5	82.7	94.2	96.8	96.2
8	4	85.3	88.4	87.9	80.8	89.3	88.1	86.0	89.8	89.5	89.4	96.1	95.0
8	8	66.8	66.7	66.6	78.9	79.4	79.7	89.3	92.1	92.1	92.9	97.1	96.4
8	16	54.0	52.8	52.8	73.1	75.6	75.7	87.8	91.9	92.0	92.4	96.9	95.8
8	32	50.8	46.3	46.3	76.6	74.2	75.1	85.9	91.7	92.6	90.0	96.3	95.7
8	64	52.1	40.2	39.6	74.6	72.4	71.5	75.7	90.0	91.2	87.9	94.0	94.4
8	128	56.5	35.1	33.4	55.5	64.1	61.1	29.2	83.2	84.9	72.5	95.0	94.7
8	256	60.4	31.4	27.9	52.7	49.1	48.6	14.9	40.3	49.8	50.5	95.1	94.0
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	86.4	87.2	87.0	86.8	88.1	86.2	87.7	88.7	88.5	92.8	95.2	94.7
16	8	60.5	60.3	60.2	75.9	76.4	76.4	88.9	90.9	90.2	91.5	95.4	94.6
16	16	44.8	44.3	44.3	71.1	70.2	71.4	89.8	93.3	93.2	92.3	95.8	95.4
16	32	39.2	35.6	35.6	73.2	69.9	70.0	87.6	92.5	93.1	91.1	95.1	94.6
16	64	39.6	29.4	29.1	74.4	68.3	67.7	81.8	90.4	91.3	86.5	94.3	94.3
16	128	44.1	24.1	23.1	57.2	57.6	53.3	44.6	83.7	84.6	78.0	92.7	92.9
16	256	49.1	20.2	18.1	50.3	44.8	35.3	12.7	41.8	50.4	53.2	93.2	92.2

Table 44: **Unweighted DPM-Solver++(2M), 2,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.0	98.4	98.4	95.9	96.0	97.2	98.2	98.8	98.8	99.0	99.5	99.6
2	4	91.6	91.5	91.4	88.8	89.3	90.0	97.9	98.7	98.7	98.6	99.5	99.4
2	8	82.4	81.7	81.6	83.8	84.8	86.9	97.1	98.4	98.4	97.7	99.2	99.1
2	16	80.0	78.6	78.5	83.4	74.3	66.2	95.2	97.5	97.5	95.7	97.7	97.6
2	32	79.8	74.5	73.9	75.9	73.7	72.4	84.6	95.2	95.4	92.7	96.2	96.3
2	64	80.3	71.0	69.5	62.9	79.7	77.2	37.8	60.6	67.2	85.9	94.0	94.5
2	128	82.0	68.6	65.5	53.8	70.4	72.7	29.8	39.5	41.0	73.5	92.9	93.6
2	256	82.4	66.3	61.4	55.2	72.2	67.2	26.0	37.3	38.7	57.3	91.2	92.0
4	2	91.7	97.5	96.2	90.6	96.0	94.5	90.8	93.3	92.7	92.1	98.1	96.6
4	4	83.8	83.9	83.6	80.9	87.3	86.7	91.5	93.9	93.1	92.5	97.7	96.6
4	8	70.2	69.2	69.0	82.0	82.6	81.2	93.5	95.1	94.7	94.4	97.5	97.0
4	16	64.1	61.9	61.8	83.0	81.4	82.0	94.1	95.3	95.2	94.7	95.3	95.1
4	32	62.8	56.5	56.1	80.1	71.5	66.9	87.0	92.3	92.4	92.0	93.3	93.0
4	64	64.1	51.4	50.2	68.8	75.4	73.4	41.8	70.9	74.0	86.8	91.3	91.8
4	128	67.3	47.1	44.3	51.5	67.1	61.1	20.6	29.5	31.0	70.8	90.7	91.1
4	256	69.6	44.2	39.5	51.5	66.0	57.7	15.5	21.8	21.7	51.4	87.5	89.0
8	2	95.7	96.7	96.5	94.5	95.2	93.5	82.7	83.5	83.6	96.2	96.9	96.6
8	4	79.2	79.2	78.6	84.7	85.0	84.4	88.3	90.2	89.4	90.3	96.0	94.8
8	8	60.8	60.1	59.9	77.5	78.8	76.1	91.5	92.6	92.2	93.8	96.3	95.6
8	16	51.1	49.3	49.2	78.4	75.9	76.6	91.2	92.3	91.9	93.3	94.3	93.7
8	32	48.6	42.9	42.6	78.7	73.5	73.5	86.0	89.4	89.3	91.6	92.5	91.8
8	64	49.5	36.8	36.0	78.6	67.9	66.2	47.7	70.9	72.5	87.5	89.9	89.6
8	128	53.4	32.1	30.2	59.6	54.1	58.2	15.5	30.0	33.0	71.1	88.0	88.7
8	256	56.7	28.5	25.3	50.5	47.9	43.0	9.9	16.9	16.7	47.7	86.1	87.3
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	77.1	77.4	77.2	82.7	84.8	79.0	87.5	88.9	88.8	92.1	94.7	94.1
16	8	52.8	52.3	52.1	73.1	76.3	75.1	88.1	89.6	89.3	91.4	94.1	93.4
16	16	40.7	39.5	39.5	74.5	72.4	72.1	89.3	90.6	90.4	92.7	93.3	92.9
16	32	36.6	32.4	32.3	76.9	70.1	69.8	83.5	87.0	87.1	91.1	91.5	91.1
16	64	36.8	26.5	26.1	75.4	63.2	61.5	51.1	69.4	70.6	87.0	88.8	88.4
16	128	40.6	21.6	20.5	53.5	49.0	45.1	12.4	34.6	37.9	72.3	86.6	86.2
16	256	44.8	18.0	16.1	51.6	43.7	35.7	6.0	15.7	16.4	45.3	81.5	83.9

2.9 Unweighted DPM-Solver++(2M)

Table 45: **Unweighted DPM-Solver++(2M), 5,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.6	99.0	99.0	96.9	97.5	97.6	98.1	98.7	98.8	99.0	99.5	99.6
2	4	93.1	93.0	93.0	84.0	92.5	92.3	98.1	98.8	98.8	98.6	99.5	99.4
2	8	84.0	83.2	83.2	83.7	79.3	85.0	97.5	98.6	98.6	97.8	99.3	99.2
2	16	80.3	79.0	79.0	76.2	85.8	85.6	95.9	98.0	98.0	97.0	98.8	98.7
2	32	80.4	75.7	75.1	77.6	75.8	71.5	87.7	96.8	96.8	94.0	98.1	98.1
2	64	81.1	72.1	70.7	65.1	76.8	78.1	48.1	91.7	92.5	86.8	96.6	96.6
2	128	82.9	69.5	66.5	57.5	65.7	55.3	30.3	43.7	47.0	72.6	94.8	95.1
2	256	83.4	67.5	62.4	58.7	74.4	50.7	25.7	37.3	39.4	52.8	93.7	94.2
4	2	92.1	98.4	96.8	91.3	97.3	95.8	90.9	93.2	92.6	92.1	98.2	96.6
4	4	86.5	87.0	86.7	84.3	87.6	88.4	91.4	93.9	93.1	92.6	97.8	96.6
4	8	72.5	71.5	71.1	81.8	81.2	84.9	93.9	95.2	94.8	94.7	98.0	97.3
4	16	65.3	63.0	63.0	79.8	83.3	82.8	95.1	96.1	96.0	96.5	97.3	97.0
4	32	64.0	58.1	57.7	81.7	81.7	80.9	90.4	94.8	94.8	94.8	96.7	96.2
4	64	65.4	52.8	51.8	68.4	77.3	75.1	63.9	88.8	89.3	89.1	95.0	94.3
4	128	68.5	48.6	45.8	51.9	70.8	66.7	24.1	43.7	50.1	70.5	93.5	94.0
4	256	71.2	45.7	40.6	54.6	66.8	53.4	16.9	25.8	27.7	48.5	91.0	92.0
8	2	96.3	97.5	97.3	95.5	95.8	96.0	82.3	83.2	83.4	96.2	96.9	96.6
8	4	82.8	83.4	82.8	85.7	87.2	85.9	88.2	90.3	89.5	90.3	96.2	95.0
8	8	63.5	62.9	62.8	79.4	78.4	76.6	91.8	92.9	92.5	94.2	96.8	96.2
8	16	52.6	50.8	50.7	78.0	75.9	76.5	92.2	93.2	92.8	94.7	96.0	95.2
8	32	50.0	44.5	44.1	81.8	74.4	71.8	89.9	92.5	92.3	93.9	95.6	94.8
8	64	50.9	38.3	37.5	74.6	69.4	68.3	71.4	85.3	85.7	90.3	94.0	93.5
8	128	55.1	33.4	31.5	52.0	62.1	58.7	21.5	48.2	53.6	73.2	92.8	92.9
8	256	58.7	29.7	26.3	53.6	52.0	44.1	11.4	22.8	26.2	48.6	91.2	91.3
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	81.3	81.8	81.6	85.1	87.0	85.4	87.5	88.9	88.9	92.1	94.7	94.2
16	8	56.2	55.7	55.4	75.0	75.1	75.8	88.3	89.8	89.5	91.8	94.8	93.9
16	16	42.7	41.6	41.6	75.1	73.4	71.5	90.1	91.4	91.2	93.8	94.7	94.2
16	32	38.0	33.9	33.8	78.1	72.0	71.1	87.6	90.3	90.3	93.2	93.9	93.5
16	64	38.2	27.9	27.4	77.0	66.2	64.8	72.8	82.9	83.3	90.5	92.7	92.0
16	128	42.5	22.7	21.6	65.1	58.2	52.7	20.3	50.4	54.7	74.9	91.3	90.9
16	256	46.9	19.0	16.9	51.9	42.4	32.6	7.5	24.2	29.3	50.8	87.8	89.2

Table 46: **Unweighted DPM-Solver++(2M), 10,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.8	99.3	99.3	97.8	97.4	98.2	98.1	98.7	98.8	99.0	99.6	99.6
2	4	94.2	94.2	94.1	85.7	91.5	92.5	98.2	98.9	98.9	98.6	99.5	99.4
2	8	85.4	84.4	84.4	83.0	90.1	77.7	97.6	98.7	98.6	97.8	99.3	99.2
2	16	80.9	79.7	79.7	80.8	82.8	82.5	96.1	98.2	98.1	97.1	99.1	99.0
2	32	80.7	76.1	75.5	75.2	79.7	85.3	88.8	97.4	97.5	94.7	98.9	98.8
2	64	81.3	72.7	71.5	68.1	78.4	74.6	50.7	94.7	95.3	87.3	97.9	97.9
2	128	83.7	70.3	67.3	59.8	65.9	76.3	22.4	69.0	74.7	72.0	96.2	96.2
2	256	84.1	68.4	63.1	60.7	61.6	70.8	25.2	38.1	41.5	50.5	95.0	95.1
4	2	92.3	98.7	97.0	91.8	97.7	96.1	90.9	93.3	92.8	92.1	98.2	96.6
4	4	88.3	89.1	88.7	88.0	88.9	84.9	91.6	94.0	93.1	92.6	97.8	96.7
4	8	74.3	73.2	72.9	82.0	84.4	85.9	94.0	95.3	94.8	94.8	98.1	97.4
4	16	66.3	64.2	64.1	80.3	81.4	83.3	95.4	96.4	96.2	96.8	97.7	97.3
4	32	64.6	59.0	58.6	81.6	80.1	79.9	91.7	95.9	95.9	95.5	97.7	97.3
4	64	66.0	53.8	52.8	69.9	76.7	71.8	67.5	92.3	92.6	90.4	96.7	96.5
4	128	69.7	49.6	46.9	54.3	65.4	62.8	25.4	74.1	77.3	71.4	95.8	95.8
4	256	72.1	46.6	41.3	56.9	62.1	60.7	18.1	29.8	34.5	47.9	93.6	94.3
8	2	96.6	97.9	97.7	95.9	96.9	96.9	82.5	83.3	83.5	96.2	96.9	96.6
8	4	85.0	86.0	85.4	86.4	88.6	84.7	88.3	90.3	89.5	90.4	96.2	95.0
8	8	65.8	65.2	64.9	80.7	80.2	74.0	91.8	92.9	92.5	94.3	97.0	96.3
8	16	53.9	52.2	52.0	76.1	76.8	77.8	92.5	93.5	93.0	95.1	96.6	95.7
8	32	50.7	45.4	45.5	80.8	76.9	76.6	91.2	93.7	93.5	94.6	96.4	95.7
8	64	52.1	39.3	38.7	78.9	69.3	69.3	76.5	89.6	89.7	91.7	95.3	94.7
8	128	56.4	34.4	32.6	54.6	65.7	56.7	30.0	72.4	74.3	75.4	95.1	94.8
8	256	59.9	30.6	27.1	53.7	55.3	47.3	12.9	29.6	37.5	50.3	93.9	94.1
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	84.2	84.9	84.6	87.0	87.6	85.1	87.9	89.4	89.3	92.2	94.8	94.2
16	8	59.1	58.5	58.3	76.6	78.0	78.0	88.8	90.2	89.7	91.9	95.1	94.2
16	16	44.4	43.4	43.3	71.5	69.8	73.5	90.9	92.1	91.8	94.2	95.3	94.8
16	32	39.0	34.9	35.0	78.4	72.6	72.1	89.8	92.0	92.0	93.7	94.8	94.4
16	64	39.4	28.7	28.4	80.3	68.1	67.2	80.1	88.0	88.2	91.6	93.9	93.2
16	128	43.8	23.6	22.4	63.7	60.1	53.7	33.8	70.3	71.9	76.8	92.8	92.4
16	256	48.4	19.6	17.5	54.4	38.4	38.8	9.8	32.0	40.6	53.0	91.6	91.1

Table 47: **Unweighted DPM-Solver++(2M), 15,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	98.9	99.4	99.4	97.9	97.9	98.3	98.1	98.8	98.8	99.0	99.6	99.5
2	4	94.8	94.8	94.8	91.4	91.2	88.9	98.2	98.9	98.9	98.6	99.5	99.4
2	8	85.9	85.4	85.3	82.5	89.9	87.4	97.7	98.7	98.6	97.8	99.3	99.2
2	16	81.1	79.8	79.8	69.9	79.6	86.2	96.2	98.3	98.3	97.2	99.1	99.0
2	32	80.7	76.3	76.0	72.4	86.7	85.1	89.3	97.7	97.7	94.9	99.1	98.9
2	64	81.8	73.2	71.9	65.7	74.6	66.9	51.9	95.6	96.0	87.8	98.5	98.3
2	128	83.8	70.6	67.8	60.9	68.4	71.9	11.8	82.8	87.0	71.4	96.6	96.7
2	256	84.3	68.6	63.5	61.8	62.4	59.6	25.0	40.0	43.8	49.4	95.8	95.6
4	2	92.4	98.8	97.1	92.0	98.0	96.3	90.8	93.3	92.8	92.1	98.2	96.6
4	4	89.1	90.1	89.8	88.2	89.6	87.0	91.6	94.0	93.1	92.6	97.8	96.7
4	8	75.1	74.1	73.8	85.0	80.4	86.3	94.1	95.3	94.8	94.8	98.2	97.5
4	16	66.7	64.6	64.5	82.5	81.7	82.0	95.6	96.5	96.3	96.9	97.8	97.4
4	32	65.0	59.5	59.2	82.2	80.4	80.5	92.2	96.2	96.3	95.7	97.9	97.5
4	64	66.5	54.4	53.3	76.9	77.8	71.6	68.9	93.5	93.8	90.8	97.0	97.0
4	128	69.9	50.0	47.4	55.6	73.8	63.5	19.5	82.4	84.8	70.7	96.6	96.5
4	256	72.8	47.0	41.7	58.1	60.3	47.6	18.6	34.6	41.8	48.2	94.8	95.5
8	2	96.7	98.0	97.8	96.3	97.4	97.1	82.6	83.4	83.6	96.2	96.9	96.6
8	4	86.1	87.4	86.7	87.0	88.4	87.6	88.4	90.3	89.5	90.4	96.2	95.0
8	8	67.1	66.5	66.3	81.2	80.9	82.2	91.9	93.0	92.6	94.4	97.1	96.4
8	16	54.6	52.8	52.7	77.8	77.3	77.1	92.6	93.6	93.1	95.2	96.8	95.9
8	32	51.4	46.0	45.9	82.3	77.0	76.3	91.7	94.1	93.9	94.6	96.7	95.8
8	64	52.6	40.0	39.2	79.7	72.5	69.8	78.5	91.1	91.2	91.8	95.7	95.2
8	128	56.9	34.8	33.1	56.1	62.7	63.7	29.9	80.0	81.1	75.0	95.7	95.3
8	256	60.7	31.0	27.3	55.0	60.8	48.9	14.4	35.1	44.7	50.4	95.1	95.2
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	85.6	86.4	86.2	87.6	87.2	87.4	88.6	89.8	89.5	92.2	95.0	94.4
16	8	60.6	60.0	59.8	78.5	79.5	79.1	89.9	91.0	90.2	91.9	95.3	94.4
16	16	45.3	44.3	44.2	71.6	71.6	73.2	92.1	93.1	92.7	94.3	95.6	95.2
16	32	39.7	35.6	35.5	77.4	71.0	72.0	91.6	93.3	93.2	93.9	95.0	94.7
16	64	40.0	29.3	29.0	81.3	69.4	68.6	84.7	90.8	90.8	90.9	94.5	94.1
16	128	44.4	24.0	22.9	54.1	57.2	57.7	41.9	80.2	80.8	78.6	94.0	93.0
16	256	49.2	19.9	17.8	51.4	48.2	37.3	11.6	36.9	46.2	54.3	92.4	91.8

Table 48: **Unweighted DPM-Solver++(2M), 20,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	99.0	99.5	99.5	98.5	98.3	98.3	98.1	98.8	98.8	99.0	99.6	99.5
2	4	95.1	95.1	95.0	90.8	91.0	91.3	98.2	98.9	98.9	98.6	99.5	99.4
2	8	86.3	85.4	85.4	83.4	90.3	84.5	97.7	98.7	98.6	97.8	99.3	99.2
2	16	81.2	79.8	79.8	80.0	83.4	85.4	96.4	98.4	98.3	97.2	99.1	99.0
2	32	80.9	76.5	76.3	77.6	84.0	79.9	89.5	97.9	97.8	95.0	99.1	99.0
2	64	82.0	73.3	72.1	68.8	84.1	82.1	52.0	96.1	96.3	87.8	98.7	98.4
2	128	84.0	70.9	68.1	61.8	73.2	72.4	10.4	88.0	90.6	71.7	97.3	97.1
2	256	84.7	68.8	63.7	62.5	63.1	54.5	23.2	42.3	47.1	49.0	96.2	96.2
4	2	92.5	98.9	97.2	91.9	98.1	96.6	90.9	93.3	92.7	92.2	98.2	96.6
4	4	89.7	90.8	90.4	87.3	88.9	89.9	91.7	94.0	93.1	92.6	97.9	96.7
4	8	75.7	74.7	74.3	84.9	86.3	86.4	94.1	95.3	94.8	94.8	98.2	97.5
4	16	67.0	65.0	64.9	83.0	82.4	80.5	95.7	96.5	96.3	96.9	97.9	97.5
4	32	65.3	59.9	59.7	80.8	78.4	81.6	92.5	96.4	96.5	95.8	98.0	97.5
4	64	66.9	54.7	53.6	70.4	76.4	77.9	69.5	94.2	94.3	90.7	97.4	97.1
4	128	70.3	50.3	47.7	56.5	71.4	56.8	18.2	85.9	87.6	70.4	96.9	96.7
4	256	73.1	47.3	42.1	58.9	57.5	62.4	18.8	39.7	48.2	48.2	95.7	95.8
8	2	96.7	98.1	97.8	96.4	97.4	97.2	83.2	84.2	84.3	96.3	97.0	96.7
8	4	86.8	88.4	87.6	86.3	89.2	87.2	88.9	90.6	89.6	90.4	96.2	95.0
8	8	68.0	67.3	67.1	81.0	82.7	82.5	92.2	93.2	92.8	94.4	97.2	96.4
8	16	55.1	53.3	53.2	76.2	76.6	77.3	93.0	93.9	93.3	95.2	97.0	96.0
8	32	51.7	46.5	46.2	81.5	72.3	75.6	92.5	94.6	94.3	94.8	96.8	95.9
8	64	53.0	40.4	39.7	80.6	74.7	71.7	80.6	92.3	92.3	92.0	95.9	95.3
8	128	57.4	35.2	33.4	57.0	62.4	60.0	30.0	84.3	84.9	76.2	95.8	95.8
8	256	61.1	31.3	27.6	50.9	60.5	48.5	15.0	40.4	49.8	52.6	95.5	95.1
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	86.5	87.5	87.1	87.8	88.6	88.8	88.5	89.5	89.2	92.2	95.1	94.5
16	8	61.6	61.0	60.8	78.6	77.3	79.5	90.4	91.3	90.5	91.9	95.4	94.5
16	16	45.8	44.9	44.8	73.5	72.1	71.9	92.7	93.7	93.3	94.4	95.8	95.3
16	32	40.0	36.0	35.9	77.8	72.3	71.3	92.5	94.1	93.9	94.2	95.3	94.9
16	64	40.4	29.7	29.3	81.5	69.6	69.2	86.9	92.2	92.2	90.8	94.8	94.5
16	128	44.9	24.3	23.2	59.1	57.5	55.9	45.9	84.6	84.9	79.9	94.5	94.0
16	256	49.8	20.2	18.1	51.2	45.1	42.6	12.9	41.9	50.4	54.6	93.0	92.5

Table 49: **Weighted DPM-Solver++(2M), 2,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	k NN			altered k NN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.2	98.4	98.2	95.4	96.8	96.6	95.0	97.8	98.5	96.7	99.4	98.6
2	4	90.7	91.2	91.3	87.2	90.0	86.3	94.7	97.7	98.3	96.2	99.4	98.7
2	8	81.5	81.5	81.6	79.3	78.5	82.5	93.2	96.2	96.8	95.2	99.1	98.6
2	16	78.7	78.1	78.3	79.5	84.7	85.5	91.7	95.4	96.1	94.2	97.8	97.2
2	32	78.6	74.0	73.9	78.1	82.9	72.9	82.7	94.1	94.9	91.6	96.3	96.1
2	64	78.9	70.2	69.2	62.7	78.8	72.3	37.7	61.9	66.0	84.6	93.8	94.4
2	128	81.2	68.1	65.3	53.7	78.0	67.2	30.6	39.7	41.1	72.9	92.4	93.1
2	256	81.5	66.1	61.5	55.2	61.0	54.4	24.6	37.1	38.4	57.4	90.0	91.2
4	2	91.5	97.6	96.3	90.8	95.7	95.5	89.3	93.0	93.0	91.9	98.1	96.5
4	4	82.5	83.9	83.7	82.7	84.9	86.9	88.1	93.0	92.5	90.3	97.5	96.1
4	8	68.8	68.7	68.7	75.7	81.1	81.5	89.5	93.8	94.0	92.5	97.4	96.5
4	16	62.6	61.5	61.8	74.6	80.8	80.9	88.5	92.9	93.8	90.7	95.0	94.9
4	32	61.5	56.0	55.9	78.9	76.2	77.7	82.9	89.9	90.9	90.1	93.1	93.2
4	64	63.1	51.2	50.4	65.8	74.3	72.2	41.2	70.9	73.5	83.5	91.4	91.5
4	128	66.2	47.0	44.7	50.9	66.4	56.8	20.5	29.4	31.1	69.2	89.9	90.9
4	256	68.8	44.4	39.8	51.2	56.5	52.9	14.9	21.6	21.5	50.4	85.5	87.8
8	2	93.9	96.7	96.1	93.1	95.7	94.9	80.1	81.6	81.9	94.2	96.6	96.1
8	4	77.7	78.8	78.5	79.6	86.1	85.4	85.4	89.4	89.2	89.2	96.0	94.8
8	8	59.3	59.1	59.1	75.5	77.6	77.5	88.6	91.4	91.4	92.4	96.2	95.5
8	16	49.8	48.5	48.6	74.0	74.9	73.9	86.3	90.3	90.6	90.8	94.2	93.5
8	32	47.8	42.8	42.7	73.7	72.3	72.4	81.7	87.7	88.3	88.7	92.6	91.7
8	64	48.7	36.6	35.9	74.0	62.4	63.3	46.7	70.3	72.0	85.2	89.9	89.5
8	128	52.5	32.0	30.3	56.3	57.4	52.8	15.4	29.8	32.8	68.4	87.8	88.5
8	256	55.9	28.5	25.3	51.1	50.1	40.6	9.7	16.9	16.8	46.7	83.6	86.7
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	76.4	76.8	76.7	83.5	82.8	83.0	86.4	87.5	87.5	92.6	94.7	94.2
16	8	51.7	51.6	51.4	69.7	73.0	73.0	86.1	88.7	88.6	91.0	94.2	93.5
16	16	39.7	38.9	39.0	70.4	70.1	70.1	86.1	89.6	89.8	90.8	93.4	93.0
16	32	36.0	32.1	32.0	70.9	68.4	67.2	80.3	85.8	86.3	88.8	91.4	91.0
16	64	36.1	26.2	25.9	71.4	62.1	61.3	50.1	68.9	70.2	85.7	88.8	88.5
16	128	39.7	21.3	20.4	52.0	48.1	50.5	12.3	34.2	37.7	69.4	85.9	86.0
16	256	44.0	17.9	16.1	50.4	44.0	34.1	5.9	15.6	16.2	44.8	81.3	83.3

2.10 Weighted DPM-Solver++(2M)

Table 50: **Weighted DPM-Solver++(2M), 5,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.6	99.0	98.5	95.9	97.2	97.5	95.1	97.9	98.6	96.7	99.4	98.6
2	4	92.1	92.9	93.0	89.8	90.6	88.4	94.7	97.6	98.4	96.2	99.4	98.7
2	8	83.3	83.2	83.3	79.7	83.8	82.5	93.4	96.3	96.9	95.4	99.2	98.6
2	16	79.2	78.6	78.9	76.2	82.1	82.7	91.8	95.6	96.4	94.2	98.7	98.1
2	32	79.3	75.0	75.2	76.2	77.4	81.1	84.9	95.3	96.2	92.4	98.1	97.6
2	64	79.8	71.5	70.5	64.9	77.7	79.1	47.7	91.3	92.6	85.7	96.6	96.5
2	128	81.9	69.1	66.4	57.6	63.8	64.3	30.9	44.1	47.1	71.3	94.7	95.2
2	256	82.6	66.9	62.3	58.8	61.1	63.4	25.7	37.3	39.7	53.1	92.5	93.0
4	2	91.9	98.5	96.8	91.2	97.1	96.1	89.4	92.9	92.9	91.9	98.1	96.5
4	4	85.3	87.0	86.9	85.2	82.9	89.1	88.2	93.0	92.5	90.4	97.7	96.2
4	8	71.1	71.0	70.9	76.6	82.0	81.8	89.7	94.0	94.2	92.8	97.9	96.7
4	16	63.9	62.8	63.0	73.7	80.5	81.5	88.8	93.4	94.3	91.9	97.3	96.6
4	32	62.5	57.3	57.5	77.1	79.6	80.0	85.2	91.8	92.8	90.8	95.7	96.3
4	64	64.1	52.8	52.0	67.9	76.2	74.6	61.7	87.2	88.7	86.3	93.7	93.8
4	128	67.5	48.4	46.1	51.4	61.3	67.1	24.2	44.2	50.0	69.6	93.5	93.0
4	256	70.2	45.6	40.8	54.2	55.7	50.2	16.9	25.7	27.9	48.4	90.9	91.8
8	2	94.5	97.5	96.8	93.4	95.8	95.7	79.8	81.4	81.7	94.1	96.6	96.1
8	4	81.3	83.1	82.8	77.1	83.2	82.8	85.3	89.4	89.2	89.3	96.1	94.9
8	8	62.3	62.2	62.1	76.0	77.0	76.7	88.8	91.7	91.7	92.7	96.8	96.1
8	16	51.4	50.2	50.2	72.9	73.8	73.0	86.9	91.0	91.2	91.8	95.9	95.0
8	32	49.1	44.3	44.3	76.7	73.3	73.6	84.3	90.1	90.9	89.9	95.4	94.6
8	64	50.0	38.1	37.5	70.2	68.6	67.5	68.7	84.0	85.1	87.3	93.6	93.3
8	128	54.2	33.3	31.6	50.9	61.1	56.5	21.4	48.0	53.6	70.1	92.5	92.4
8	256	57.9	29.6	26.3	52.8	55.1	50.0	11.4	22.7	26.1	47.9	90.5	91.1
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	81.0	81.5	81.3	85.5	86.1	84.2	86.6	87.7	87.7	92.6	94.8	94.3
16	8	55.1	55.0	54.8	71.0	72.4	74.9	86.3	88.9	88.8	91.4	94.7	93.9
16	16	41.8	41.1	41.1	71.5	68.4	71.0	86.9	90.4	90.6	91.8	94.7	94.3
16	32	37.3	33.5	33.5	74.0	69.7	69.3	83.5	88.7	89.3	89.8	93.7	93.3
16	64	37.5	27.5	27.2	70.5	61.6	60.0	70.3	81.8	82.7	88.3	91.9	91.9
16	128	41.5	22.4	21.4	56.9	52.5	53.5	20.1	50.3	54.7	72.3	90.7	90.5
16	256	46.3	18.7	16.8	50.9	45.0	41.3	7.6	24.1	29.2	49.5	88.1	88.8

Table 51: **Weighted DPM-Solver++(2M), 10,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.7	99.4	98.6	96.3	97.8	97.7	95.2	98.0	98.7	96.7	99.4	98.6
2	4	93.1	94.0	94.2	88.8	85.4	90.5	94.6	97.6	98.4	96.2	99.4	98.7
2	8	84.5	84.2	84.4	83.5	81.2	83.2	93.5	96.3	96.9	95.3	99.2	98.7
2	16	79.9	79.0	79.2	77.7	78.8	83.1	91.9	95.7	96.5	94.2	98.9	98.3
2	32	79.4	75.6	75.6	74.3	84.6	67.8	85.4	95.5	96.6	92.1	98.7	98.0
2	64	80.2	72.2	71.4	67.8	78.7	72.1	50.1	94.1	95.1	86.5	97.7	97.5
2	128	82.4	69.8	67.2	59.7	60.4	71.4	22.2	68.3	74.7	70.8	96.2	96.3
2	256	83.2	67.8	62.9	60.7	70.1	59.1	25.7	38.2	41.6	50.2	93.7	94.2
4	2	92.1	98.8	97.0	91.6	97.9	96.6	89.3	92.9	92.9	91.9	98.1	96.6
4	4	86.9	89.1	88.8	83.9	87.9	87.9	88.3	93.1	92.6	90.4	97.7	96.2
4	8	73.0	72.7	72.6	77.7	84.1	80.2	89.7	94.0	94.2	93.0	98.0	96.8
4	16	64.9	63.9	64.2	72.8	80.8	82.8	88.9	93.5	94.6	92.2	97.6	97.0
4	32	63.4	58.4	58.5	77.1	78.4	80.4	85.9	92.4	93.5	91.0	96.5	96.9
4	64	64.8	53.7	53.0	66.8	77.2	76.7	64.4	90.3	91.9	87.0	95.4	95.3
4	128	68.5	49.6	47.1	53.7	63.5	67.7	25.2	73.1	77.1	69.0	95.5	94.2
4	256	71.4	46.5	41.6	56.5	71.5	61.7	18.4	29.7	34.6	47.7	93.5	92.0
8	2	94.8	97.9	97.0	94.0	97.2	96.4	80.0	81.5	81.7	94.2	96.6	96.1
8	4	83.5	85.9	85.5	80.9	84.8	86.8	85.3	89.4	89.2	89.3	96.1	94.9
8	8	64.7	64.6	64.4	76.0	75.6	80.0	88.8	91.7	91.7	92.8	97.0	96.2
8	16	52.8	51.5	51.5	72.7	74.1	75.4	87.1	91.2	91.5	92.2	96.5	95.4
8	32	49.9	45.1	45.3	75.9	75.3	75.2	85.1	90.9	91.8	90.2	96.2	95.3
8	64	51.3	39.2	38.6	71.5	70.5	69.8	72.9	87.8	88.9	88.1	94.2	94.2
8	128	55.6	34.4	32.7	53.3	64.7	55.1	29.4	71.8	74.3	71.2	94.6	93.9
8	256	59.1	30.4	27.0	55.2	57.2	46.3	12.8	29.4	37.3	49.2	93.7	93.0
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	84.0	84.6	84.4	86.2	86.9	86.4	87.0	88.2	88.1	92.7	94.9	94.4
16	8	58.0	57.8	57.7	73.6	76.6	74.4	86.8	89.3	89.2	91.5	95.0	94.2
16	16	43.5	42.7	42.8	69.2	71.5	69.9	87.7	91.2	91.3	92.1	95.4	94.9
16	32	38.2	34.5	34.5	73.2	70.1	70.4	85.1	90.1	90.7	90.2	94.6	94.2
16	64	38.6	28.5	28.2	73.4	66.4	66.0	76.4	86.6	87.4	88.2	92.7	92.8
16	128	43.0	23.3	22.3	55.0	48.3	53.7	33.3	69.8	71.8	74.0	91.8	91.7
16	256	47.7	19.4	17.4	53.3	42.1	39.6	9.8	31.9	40.6	51.7	91.6	90.6

Table 52: **Weighted DPM-Solver++(2M), 15,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.8	99.5	98.7	96.4	98.1	97.8	95.1	97.9	98.6	96.7	99.4	98.6
2	4	93.5	94.6	94.8	90.1	90.8	92.3	94.6	97.6	98.4	96.2	99.4	98.7
2	8	85.1	84.9	84.9	81.5	84.1	81.3	93.5	96.3	96.9	95.3	99.2	98.7
2	16	80.1	79.5	79.7	69.2	78.9	75.8	92.0	95.7	96.5	94.2	98.9	98.3
2	32	79.1	75.7	75.9	76.5	84.5	83.5	85.7	95.5	96.7	91.9	98.9	98.0
2	64	80.6	72.5	71.7	65.5	79.0	82.0	51.2	94.7	95.6	86.2	98.0	97.8
2	128	82.8	70.2	67.8	60.9	81.5	72.2	11.7	82.7	87.1	70.5	96.9	96.8
2	256	83.7	68.2	63.3	61.7	65.0	60.1	24.8	39.8	44.1	48.8	94.6	94.9
4	2	92.2	98.9	97.1	91.8	97.9	96.6	89.3	92.9	92.9	91.9	98.1	96.6
4	4	87.7	90.2	89.9	83.2	90.0	90.3	88.4	93.1	92.6	90.4	97.7	96.2
4	8	73.8	73.8	73.4	81.1	82.0	82.7	89.6	94.0	94.2	93.0	98.1	96.9
4	16	65.3	64.5	64.8	75.6	75.9	76.8	88.9	93.6	94.7	92.2	97.8	97.0
4	32	63.9	59.1	59.0	72.9	80.5	78.5	86.0	92.6	93.8	91.0	96.9	97.1
4	64	65.2	54.2	53.5	72.8	77.7	77.9	65.4	91.1	92.8	87.4	96.0	95.7
4	128	69.0	50.1	47.6	54.9	72.4	70.4	19.4	81.7	84.7	68.6	96.3	94.9
4	256	71.8	47.0	42.1	57.8	66.2	51.5	18.3	34.6	41.9	47.7	94.7	92.6
8	2	94.9	98.0	97.1	94.5	97.5	96.5	80.1	81.6	81.8	94.2	96.7	96.1
8	4	84.6	87.3	86.8	84.5	86.9	88.3	85.3	89.4	89.2	89.3	96.1	94.9
8	8	66.0	65.9	65.7	77.6	79.2	80.5	88.8	91.7	91.7	92.9	97.0	96.2
8	16	53.6	52.3	52.3	73.4	75.5	75.6	87.3	91.4	91.6	92.3	96.6	95.5
8	32	50.4	45.9	45.8	75.5	74.5	73.6	85.3	91.2	92.2	89.8	96.1	95.5
8	64	51.7	39.7	39.1	75.6	69.9	69.1	74.3	88.9	90.1	88.1	94.1	94.4
8	128	56.2	34.9	33.2	54.6	66.2	58.7	29.2	79.2	81.0	71.8	95.0	94.3
8	256	59.9	31.0	27.5	54.2	59.2	51.3	14.1	35.1	44.7	49.3	94.7	93.6
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	85.4	86.2	86.0	87.1	87.8	86.7	87.7	88.8	88.6	92.7	95.0	94.6
16	8	59.4	59.3	59.1	76.0	76.3	76.5	88.2	90.4	89.9	91.5	95.2	94.4
16	16	44.2	43.6	43.7	68.2	71.6	69.8	89.0	92.5	92.5	92.2	95.7	95.3
16	32	38.8	35.1	35.1	72.4	69.1	70.7	86.7	91.5	92.1	90.7	94.9	94.6
16	64	39.2	29.0	28.7	75.9	67.6	65.4	80.2	89.0	89.9	86.9	93.7	93.8
16	128	43.7	23.8	22.7	59.6	58.6	56.1	41.1	79.5	80.6	76.3	92.0	92.0
16	256	48.5	19.8	17.8	51.9	43.1	42.7	11.5	36.8	46.1	52.9	92.0	90.9

Table 53: **Weighted DPM-Solver++(2M), 20,000 anchors per mode.** Overall seed-fate accuracy (% , mean over 3 seeds) at $T = 250, 500, 750$ (sub-columns); best predictor per T in bold.

K	d	kNN			altered kNN			quadratic			polar		
		250	500	750	250	500	750	250	500	750	250	500	750
2	2	96.8	99.5	98.7	96.5	98.4	97.9	95.1	97.9	98.6	96.7	99.4	98.6
2	4	94.0	95.0	95.1	90.6	90.3	90.5	94.7	97.7	98.4	96.2	99.4	98.7
2	8	85.2	85.3	85.2	83.8	86.9	90.3	93.5	96.3	97.0	95.3	99.2	98.7
2	16	79.9	79.8	80.0	75.3	78.4	76.1	91.8	95.7	96.6	94.1	98.9	98.3
2	32	79.4	76.1	76.1	75.9	83.4	83.7	85.7	95.6	96.8	91.9	98.9	98.0
2	64	80.8	72.5	71.7	68.4	71.1	63.5	51.2	95.1	96.1	86.2	98.1	97.9
2	128	83.0	70.6	68.0	61.5	68.5	72.7	10.2	87.3	90.6	70.4	97.2	97.0
2	256	83.8	68.4	63.5	62.4	67.1	54.9	22.6	42.0	47.1	48.1	94.8	95.0
4	2	92.2	99.0	97.1	92.0	98.1	96.6	89.4	93.0	92.9	91.9	98.1	96.6
4	4	88.1	90.9	90.5	86.4	90.7	90.5	88.3	93.1	92.5	90.4	97.7	96.2
4	8	74.4	74.3	74.1	80.9	83.6	84.1	89.7	94.0	94.1	93.0	98.1	96.9
4	16	65.8	65.0	65.3	77.9	80.7	76.5	89.0	93.6	94.7	92.3	97.8	97.0
4	32	64.1	59.4	59.4	75.0	79.9	79.9	86.0	92.7	93.9	91.1	97.0	97.1
4	64	65.6	54.8	54.0	67.2	78.5	78.4	65.8	91.6	93.4	87.2	95.9	95.5
4	128	69.1	50.2	47.9	55.7	64.4	59.1	18.0	84.7	87.4	68.3	96.5	94.9
4	256	72.0	47.3	42.4	58.4	64.1	60.9	18.4	39.5	48.1	48.0	94.9	92.7
8	2	95.0	98.0	97.2	94.6	97.4	96.7	80.8	82.5	82.7	94.2	96.8	96.2
8	4	85.3	88.3	87.8	80.8	89.3	88.1	85.9	89.8	89.5	89.4	96.1	95.0
8	8	66.8	66.7	66.6	78.8	79.4	79.7	89.3	92.1	92.0	92.9	97.1	96.3
8	16	54.0	52.8	52.9	73.0	75.5	75.6	87.8	91.8	92.0	92.4	96.8	95.7
8	32	50.6	46.2	46.2	75.7	74.2	74.5	85.9	91.7	92.5	89.9	96.2	95.7
8	64	52.0	40.1	39.5	74.6	71.8	70.3	75.8	89.9	91.1	88.0	94.0	94.3
8	128	56.4	35.2	33.4	55.4	61.8	57.9	29.1	83.0	84.7	71.1	95.0	94.7
8	256	60.3	31.4	27.8	52.7	56.0	48.7	15.0	40.1	49.7	50.8	95.0	94.0
16	2	—	—	—	—	—	—	—	—	—	—	—	—
16	4	86.4	87.2	87.0	86.8	88.1	86.1	87.7	88.7	88.5	92.8	95.2	94.7
16	8	60.4	60.3	60.2	75.9	76.4	76.4	88.9	90.9	90.2	91.5	95.4	94.5
16	16	44.7	44.2	44.3	71.1	70.1	71.4	89.8	93.3	93.2	92.3	95.8	95.4
16	32	39.1	35.5	35.5	73.0	69.9	68.8	87.7	92.4	93.0	91.2	95.0	94.5
16	64	39.6	29.4	29.1	74.3	68.2	67.4	81.9	90.3	91.2	86.6	94.4	94.4
16	128	44.1	24.0	23.0	58.6	57.7	57.4	44.8	83.7	84.5	78.1	92.5	92.8
16	256	49.0	20.1	18.0	50.4	39.9	36.8	12.8	41.8	50.3	53.5	92.7	92.0

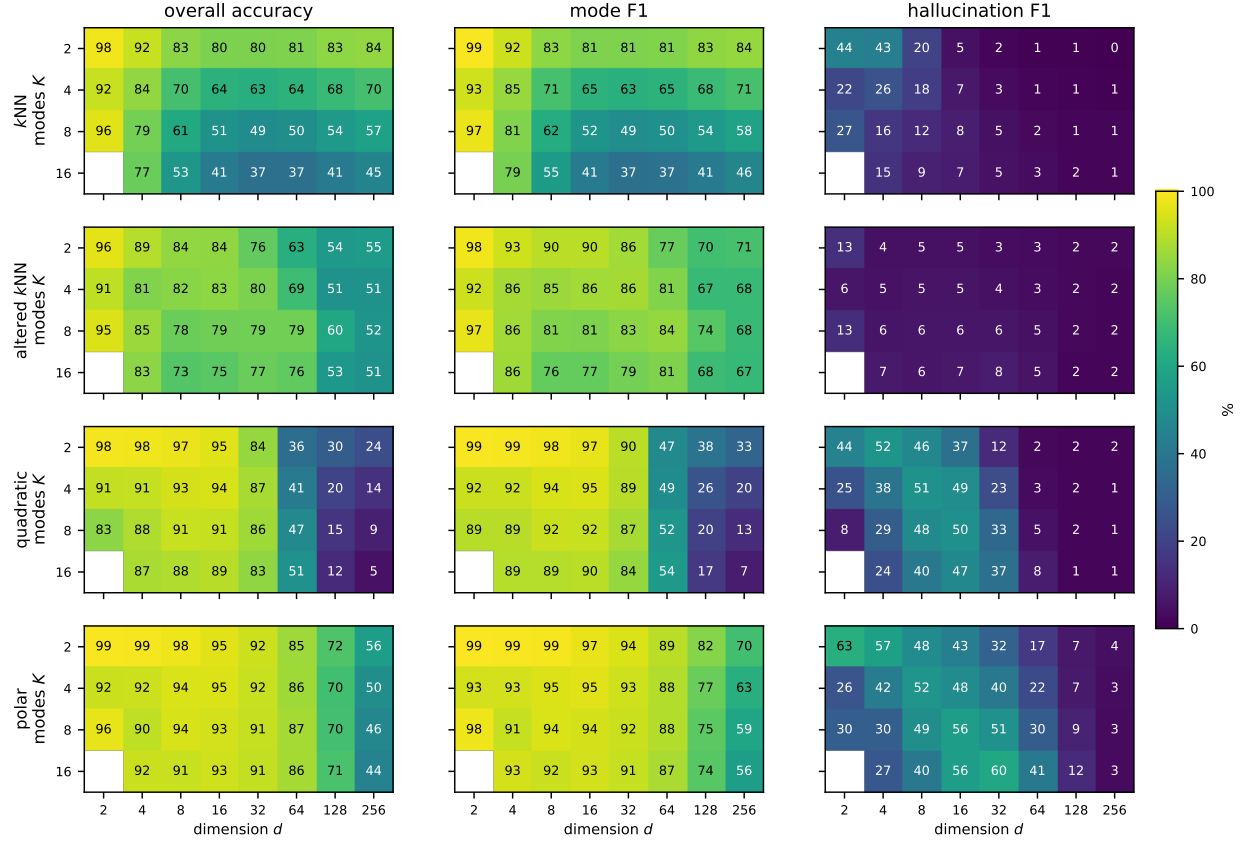


Figure 3: **Unweighted DDIM, $T = 250$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3 Heatmaps

3.1 Unweighted DDIM, $T = 250$

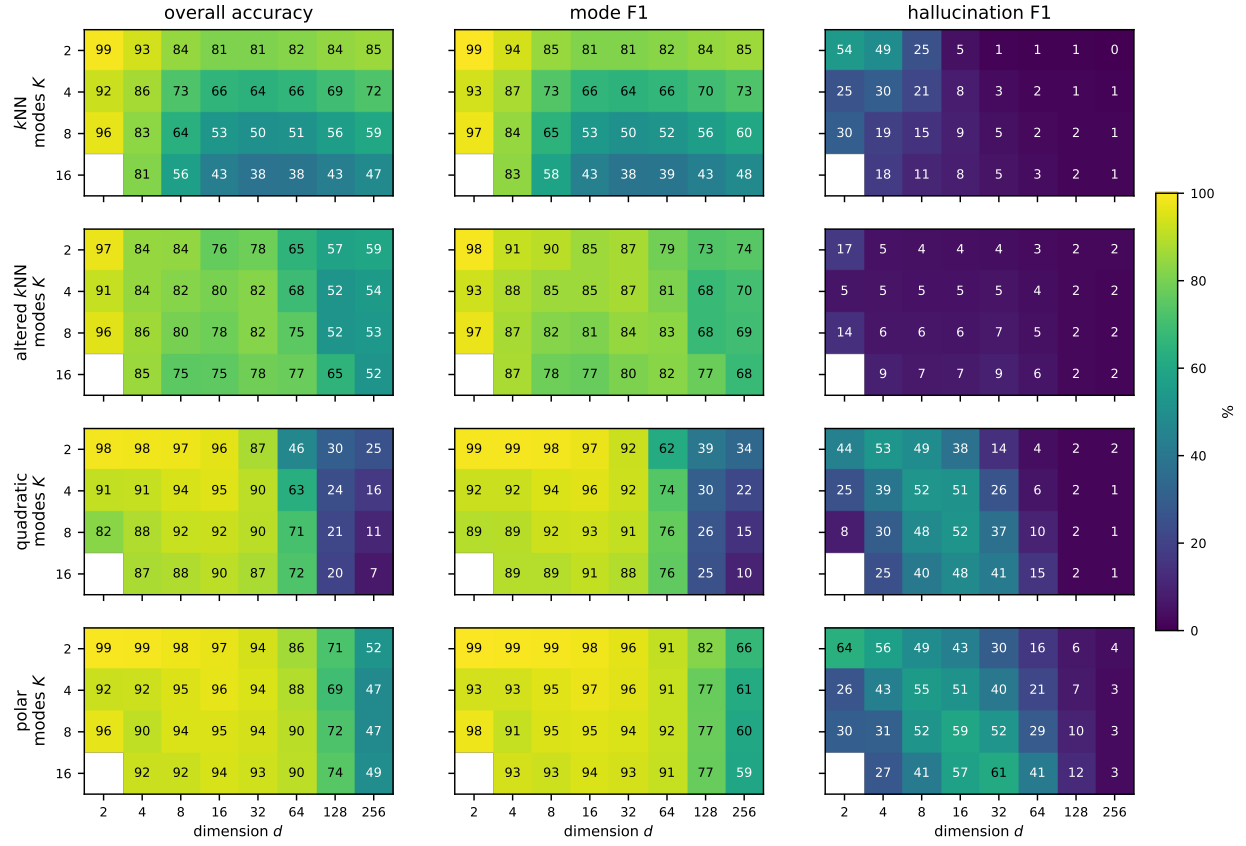


Figure 4: **Unweighted DDIM, $T = 250$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

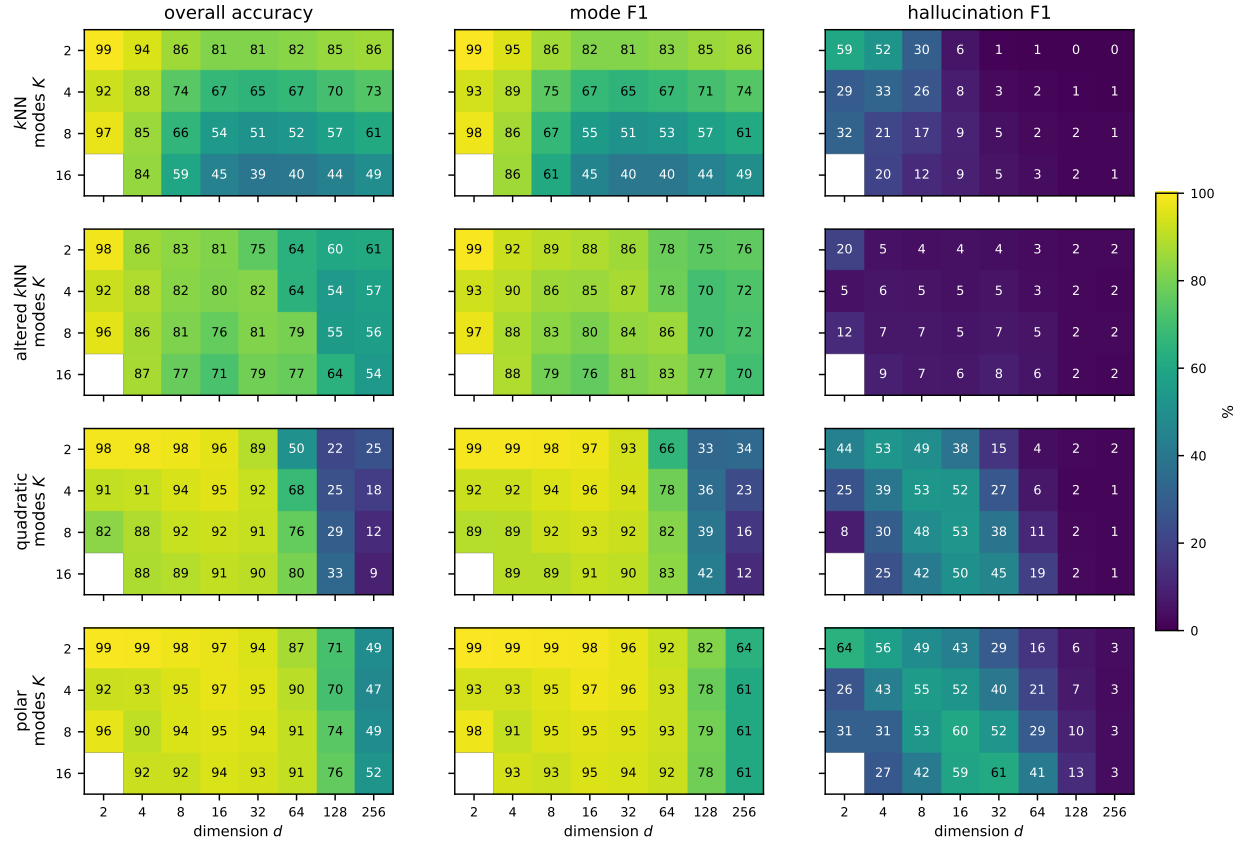


Figure 5: **Unweighted DDIM, $T = 250$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

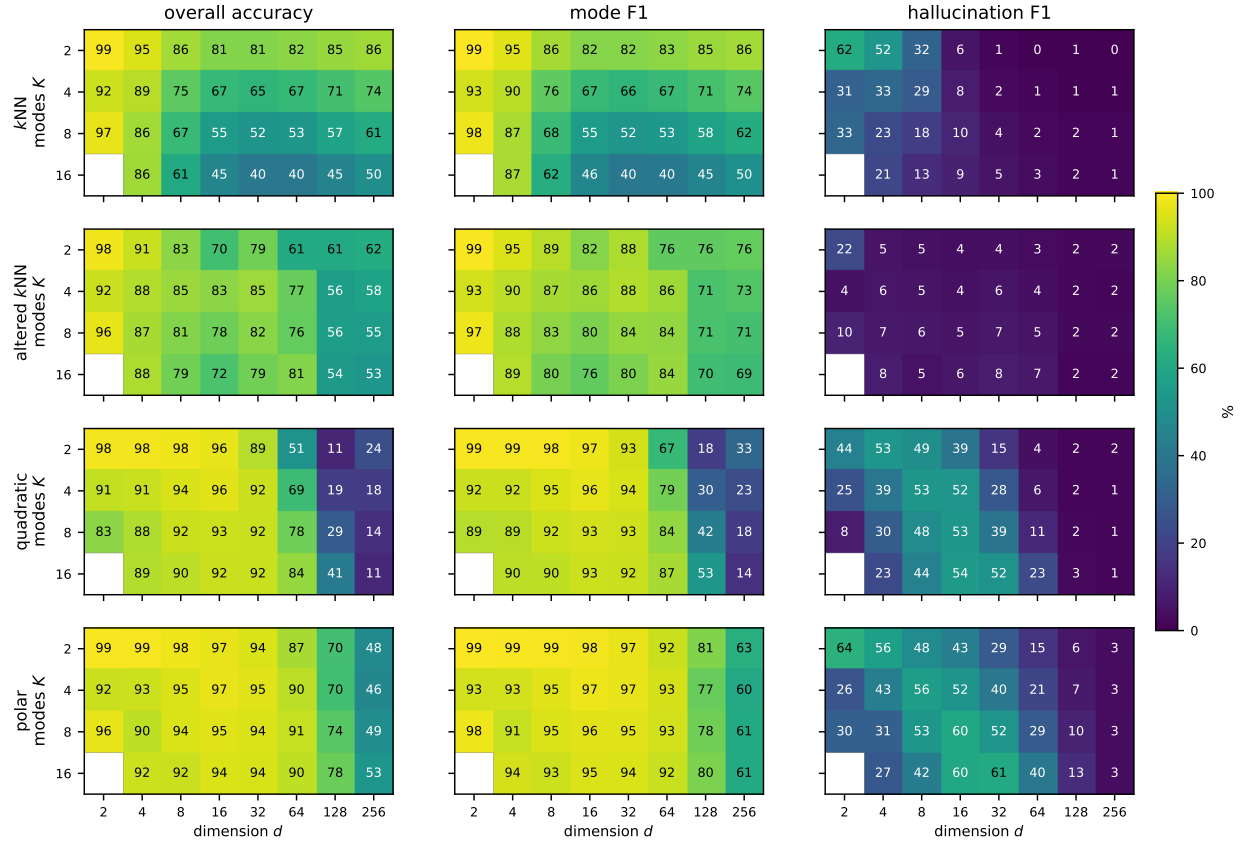


Figure 6: **Unweighted DDIM, $T = 250$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

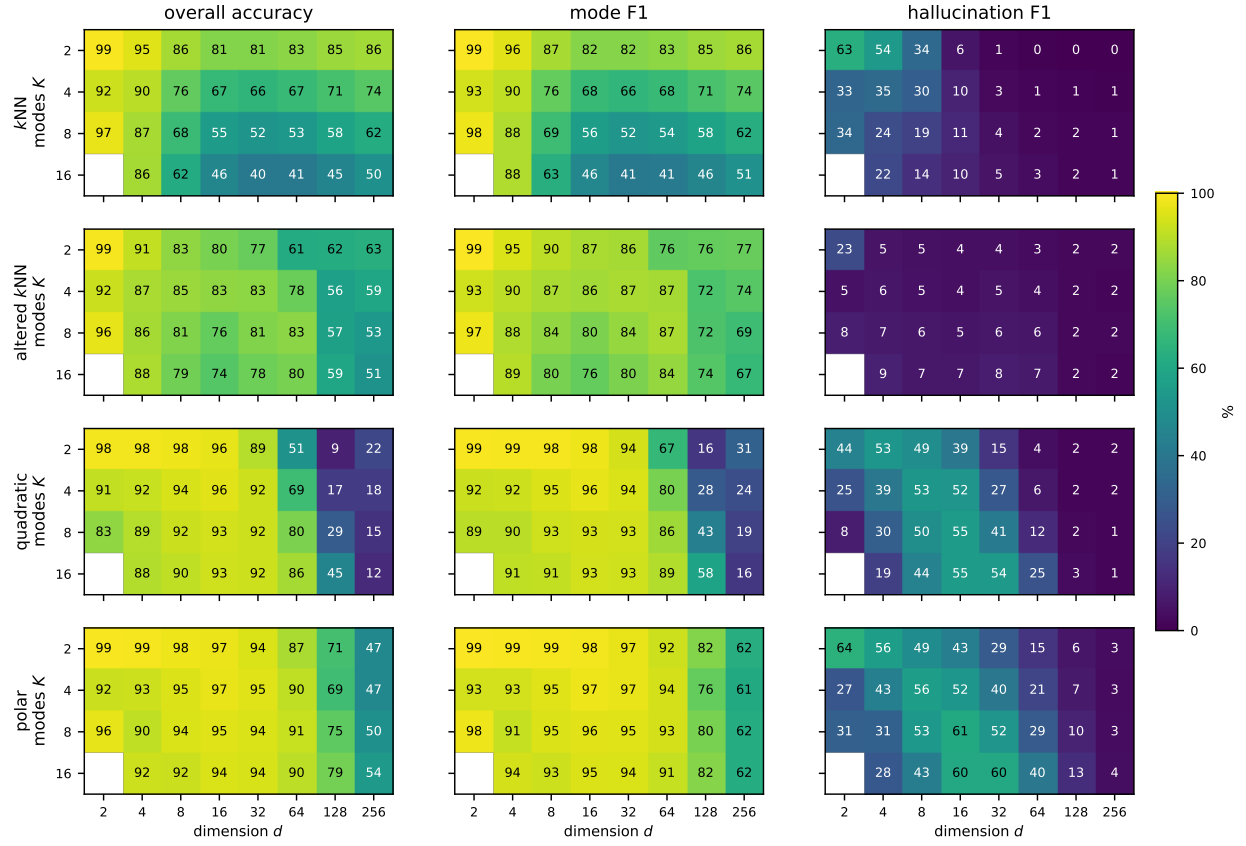


Figure 7: **Unweighted DDIM, $T = 250$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

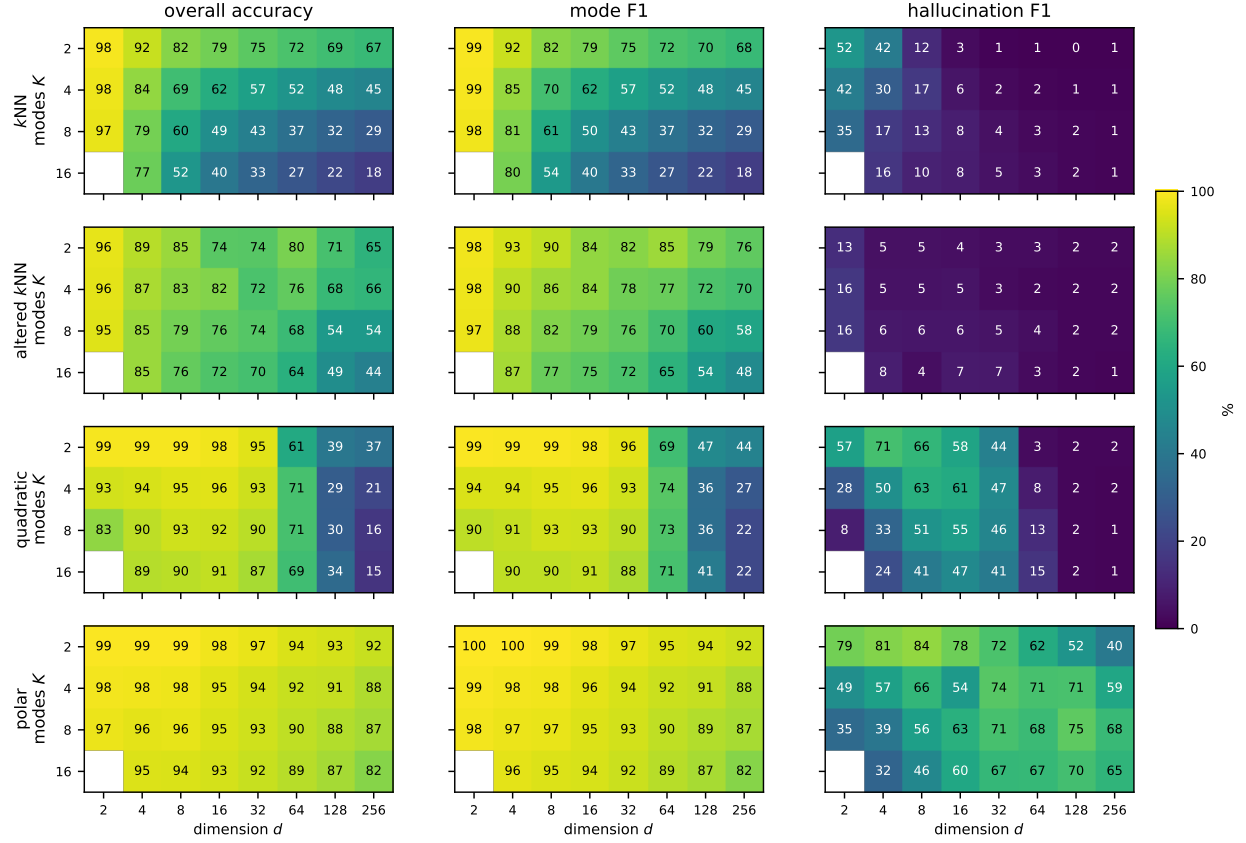


Figure 8: **Unweighted DDIM, $T = 500$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.2 Unweighted DDIM, $T = 500$

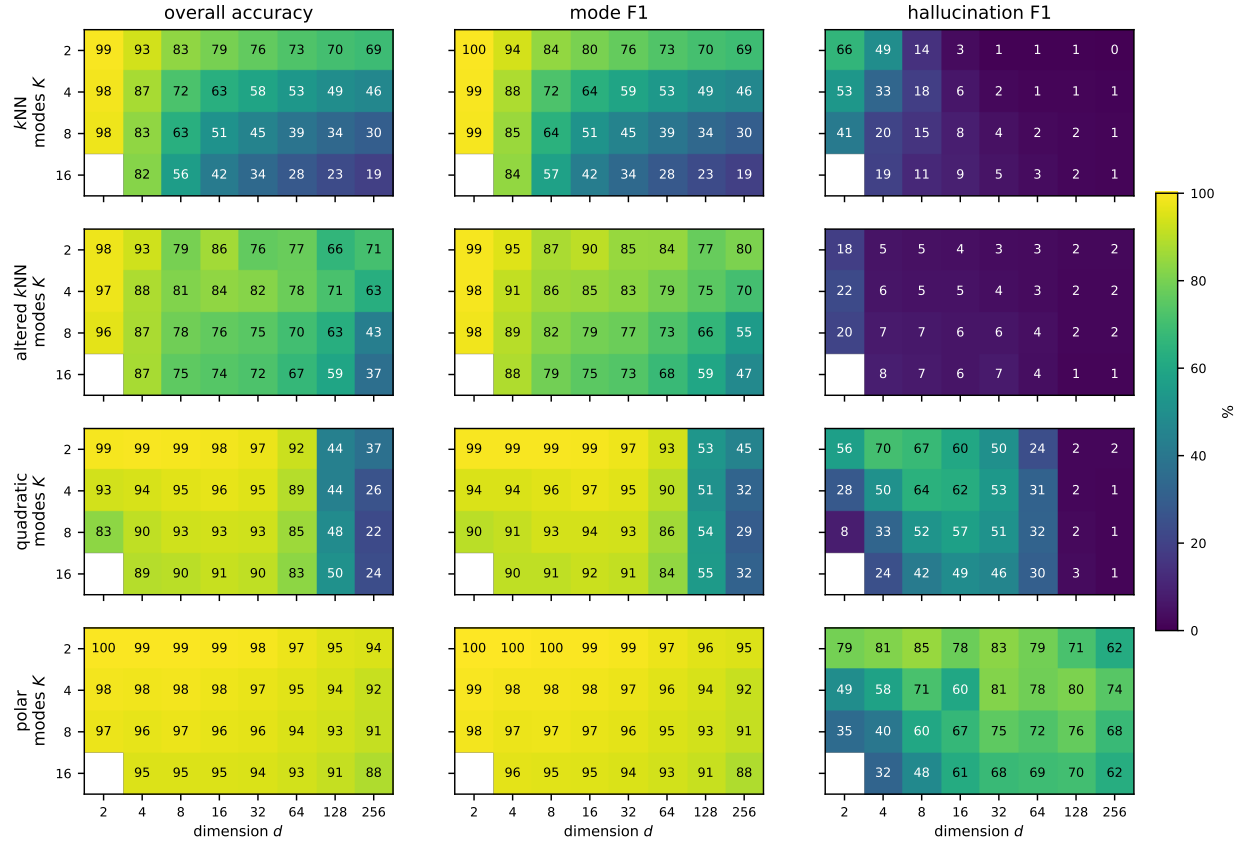


Figure 9: **Unweighted DDIM, $T = 500$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

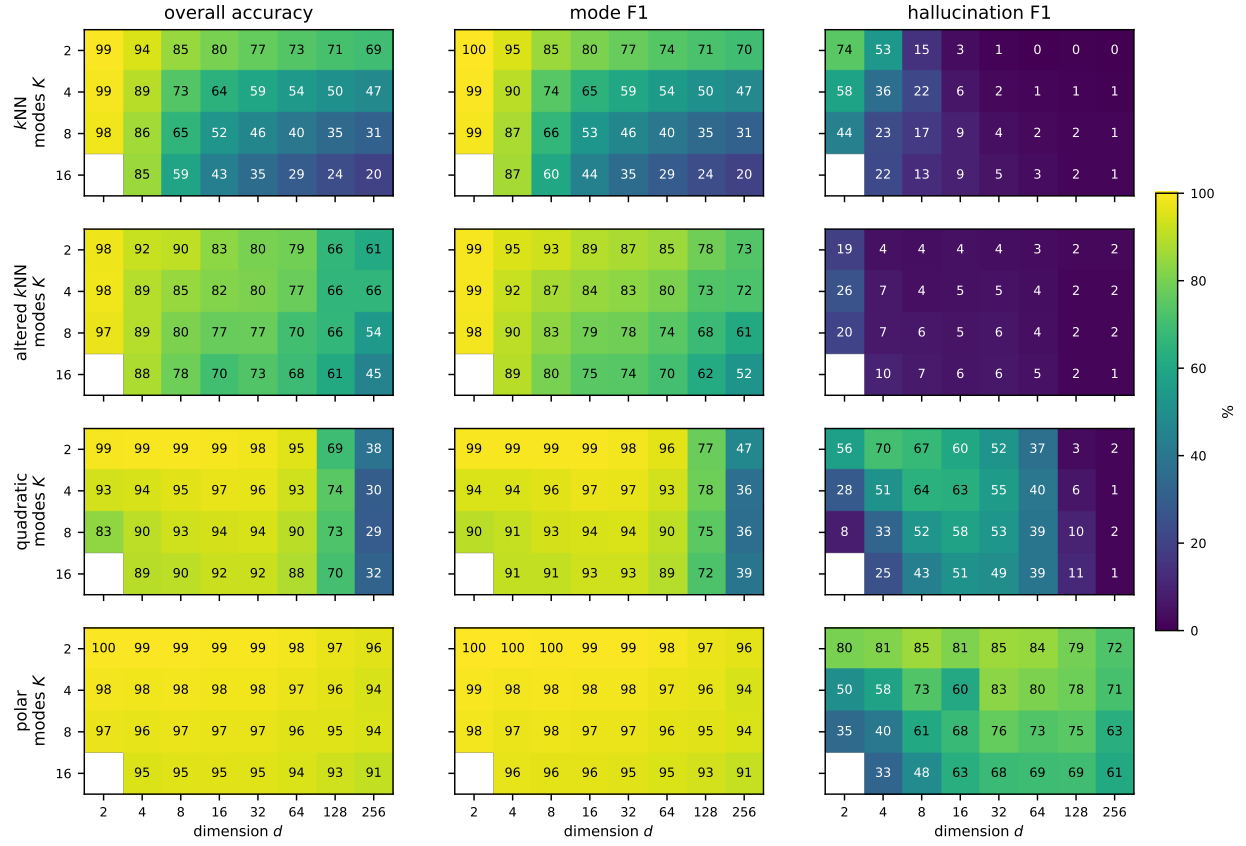


Figure 10: **Unweighted DDIM, $T = 500$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

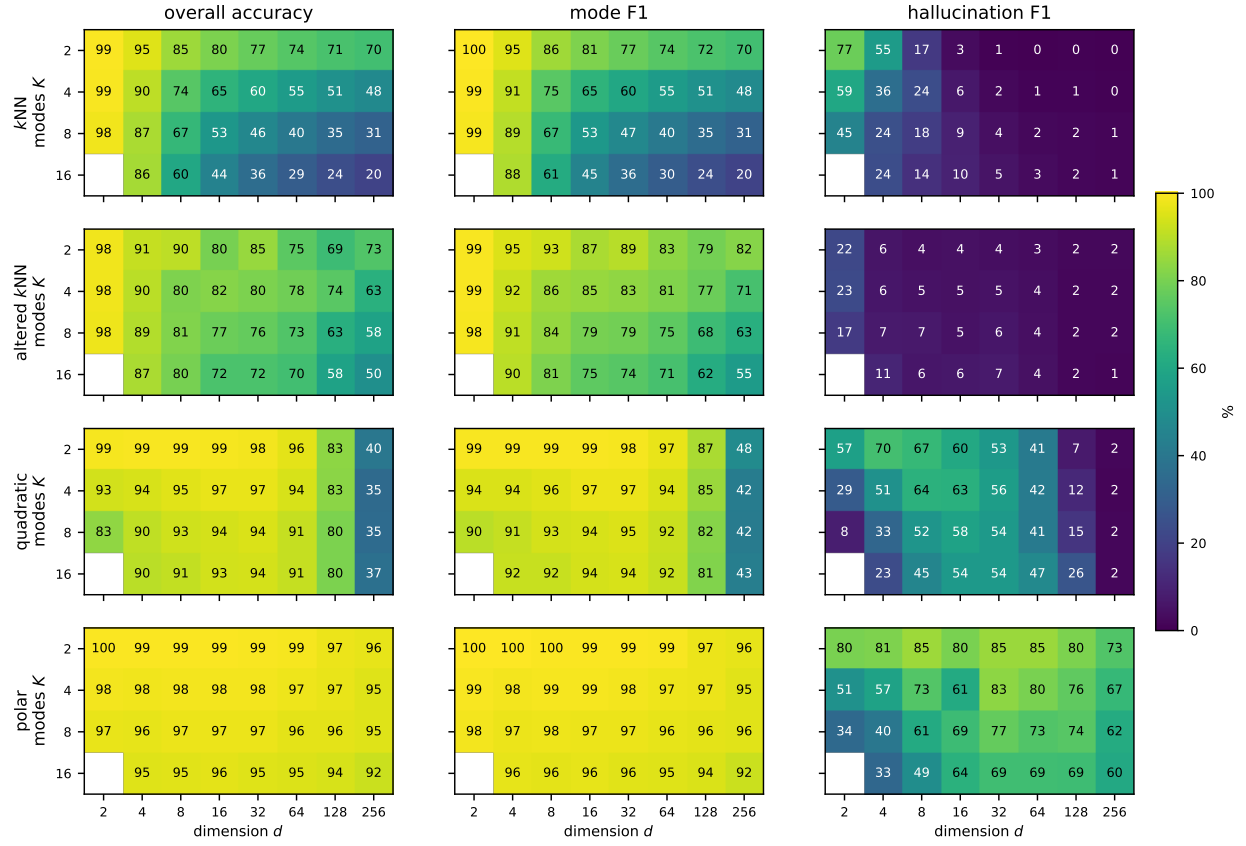


Figure 11: **Unweighted DDIM**, $T = 500$, **15,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

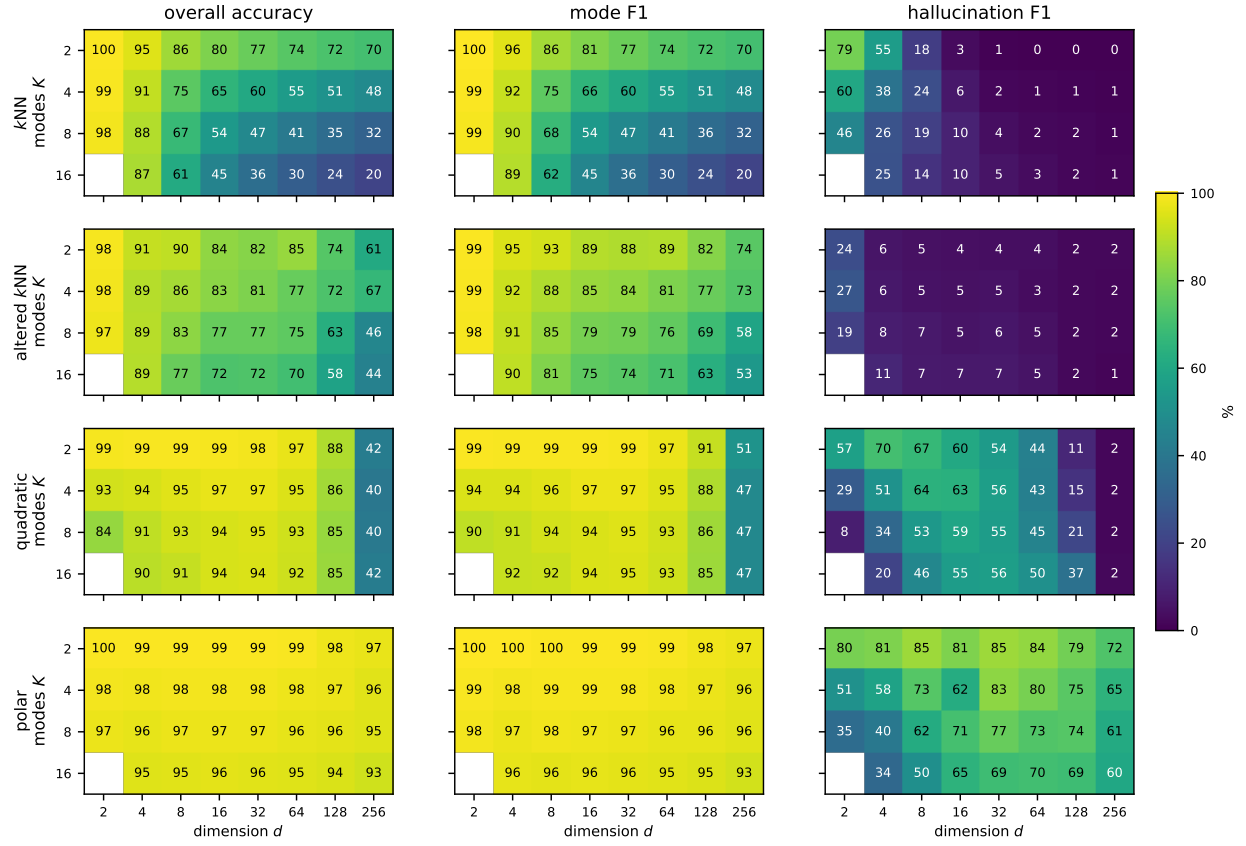


Figure 12: **Unweighted DDIM, $T = 500$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

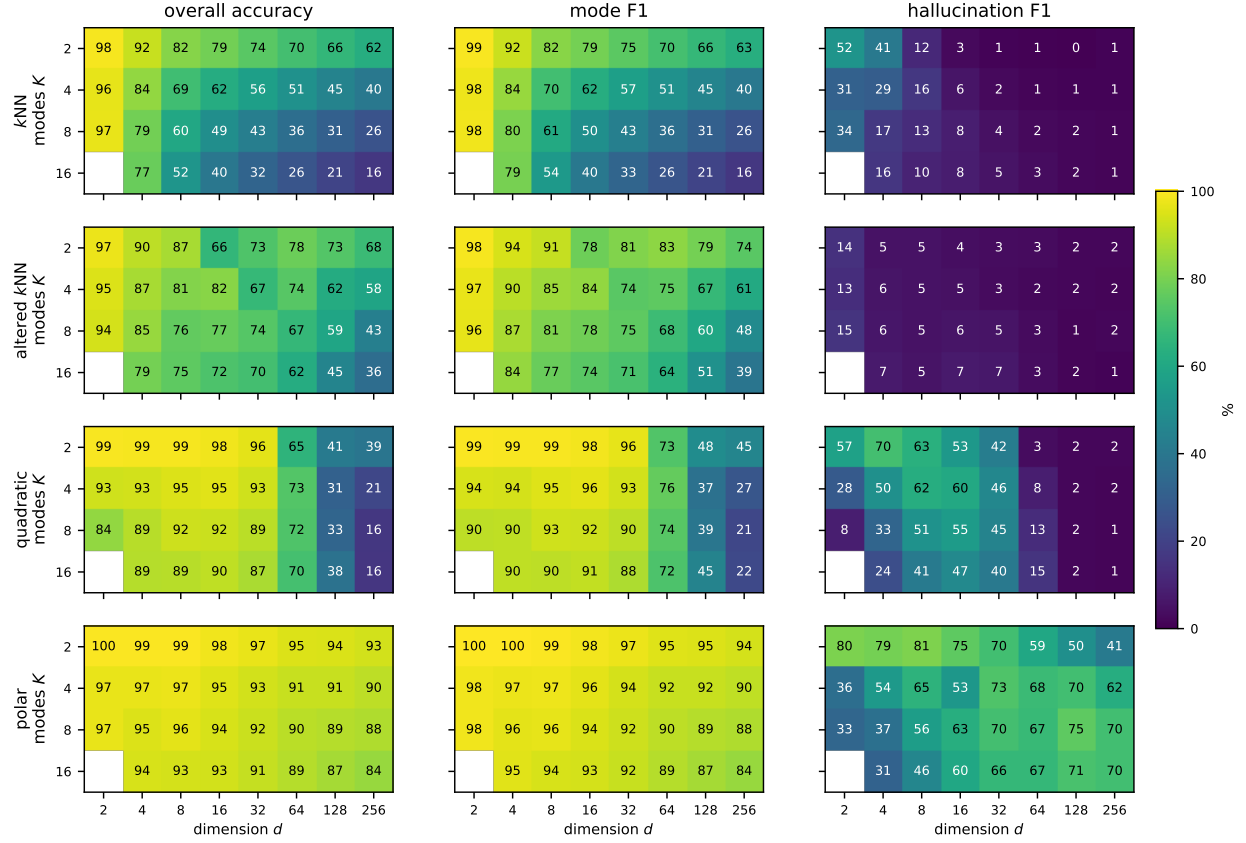


Figure 13: **Unweighted DDIM, $T = 750$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.3 Unweighted DDIM, $T = 750$

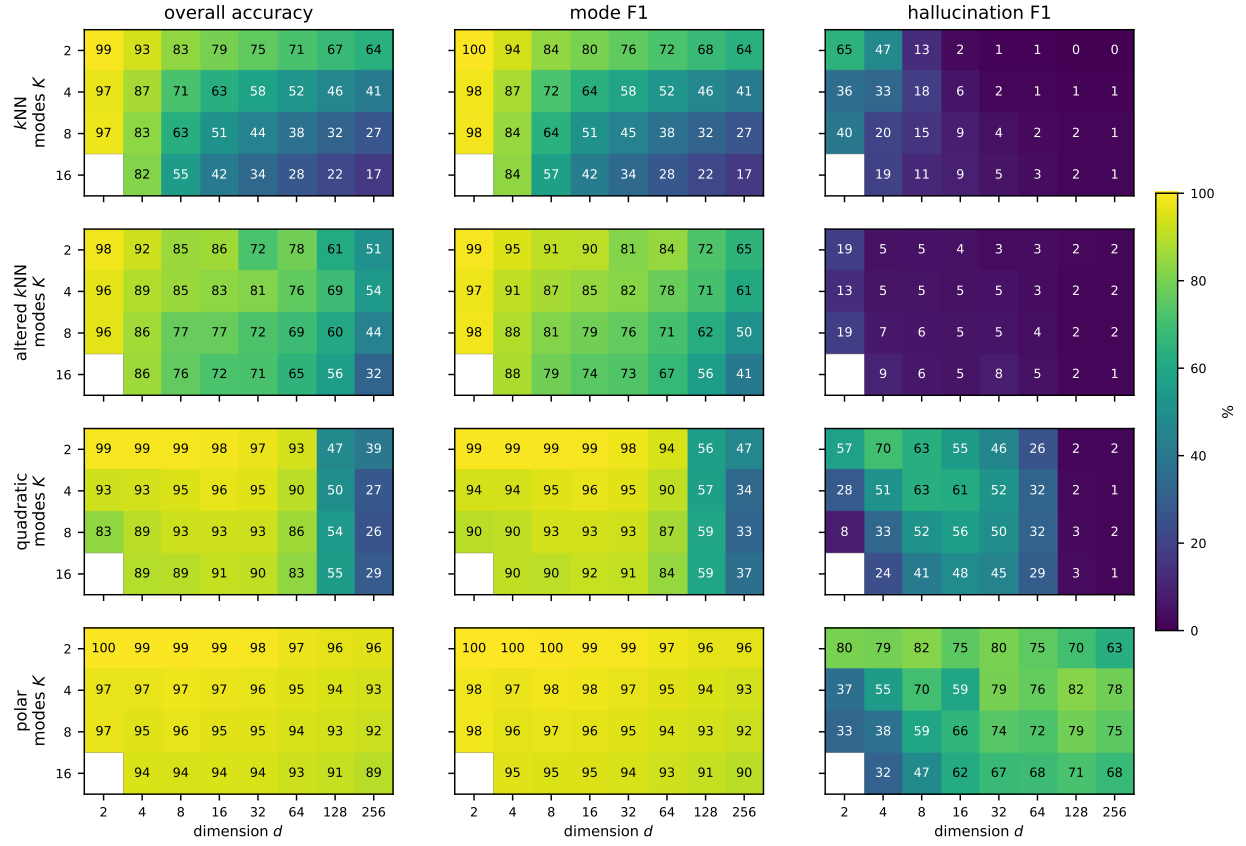


Figure 14: **Unweighted DDIM, $T = 750$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

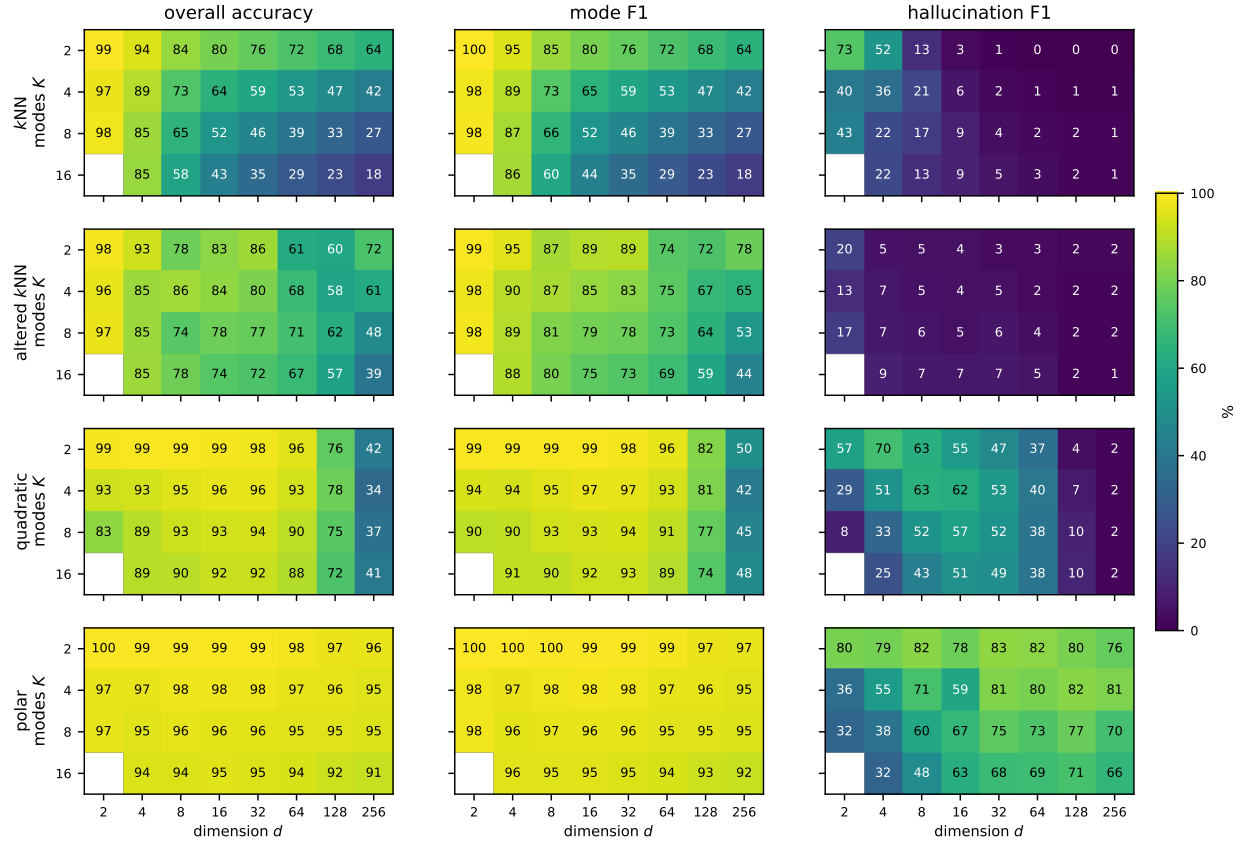


Figure 15: **Unweighted DDIM, $T = 750$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

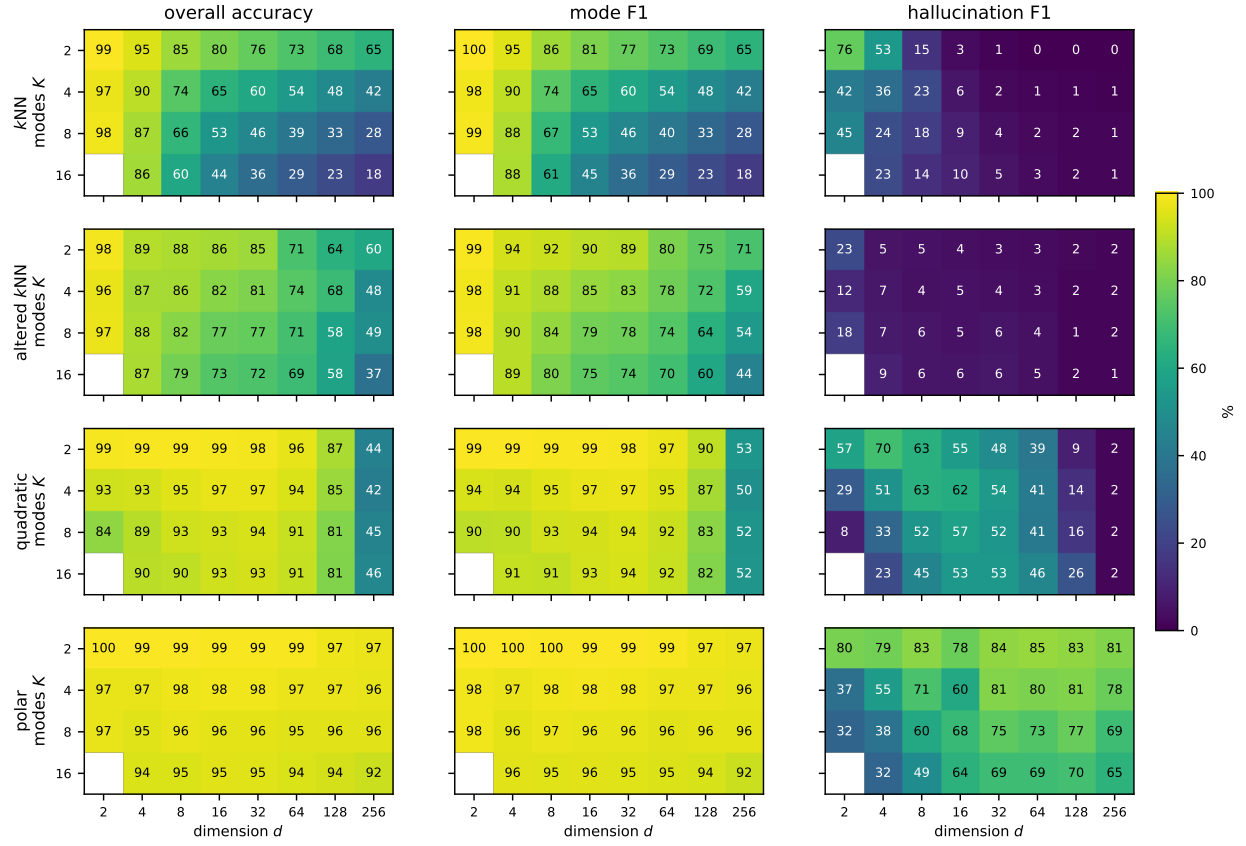


Figure 16: **Unweighted DDIM, $T = 750$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

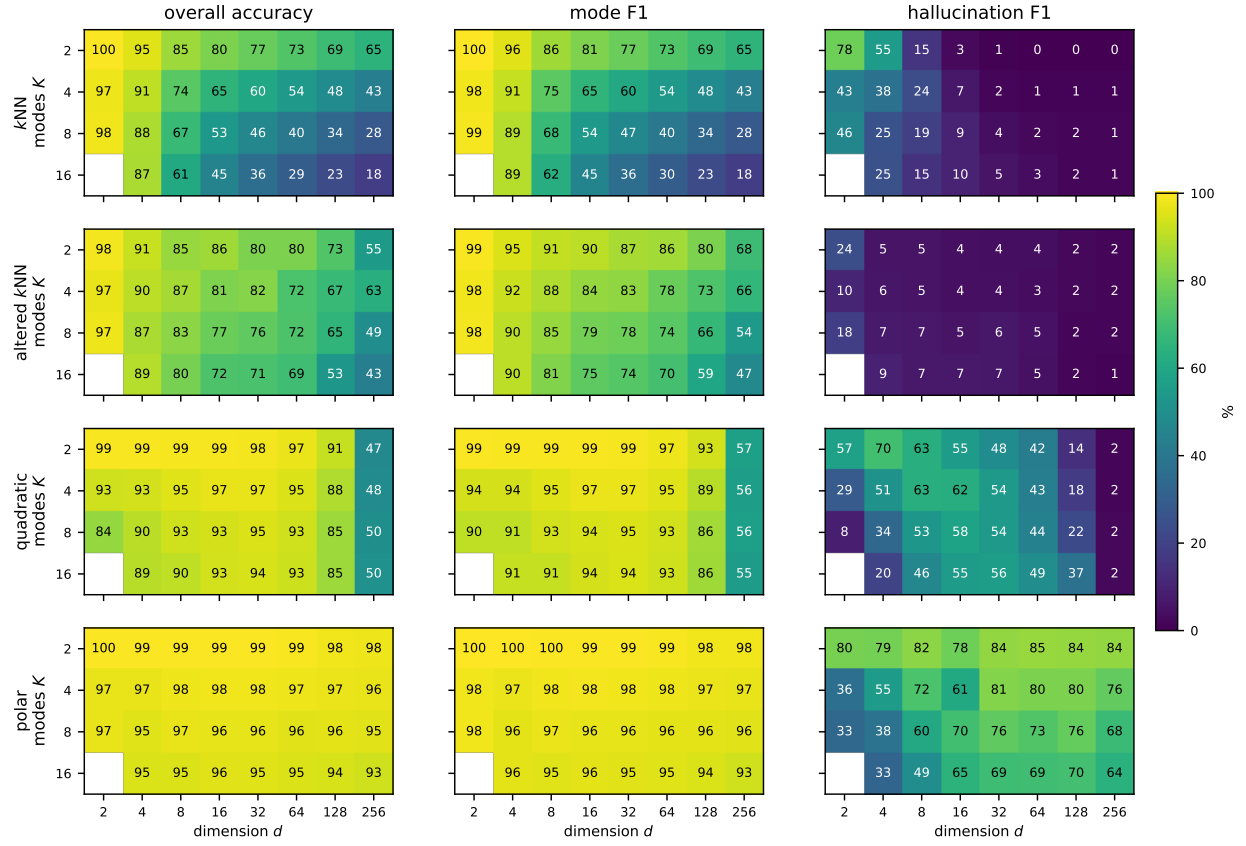


Figure 17: **Unweighted DDIM, $T = 750$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

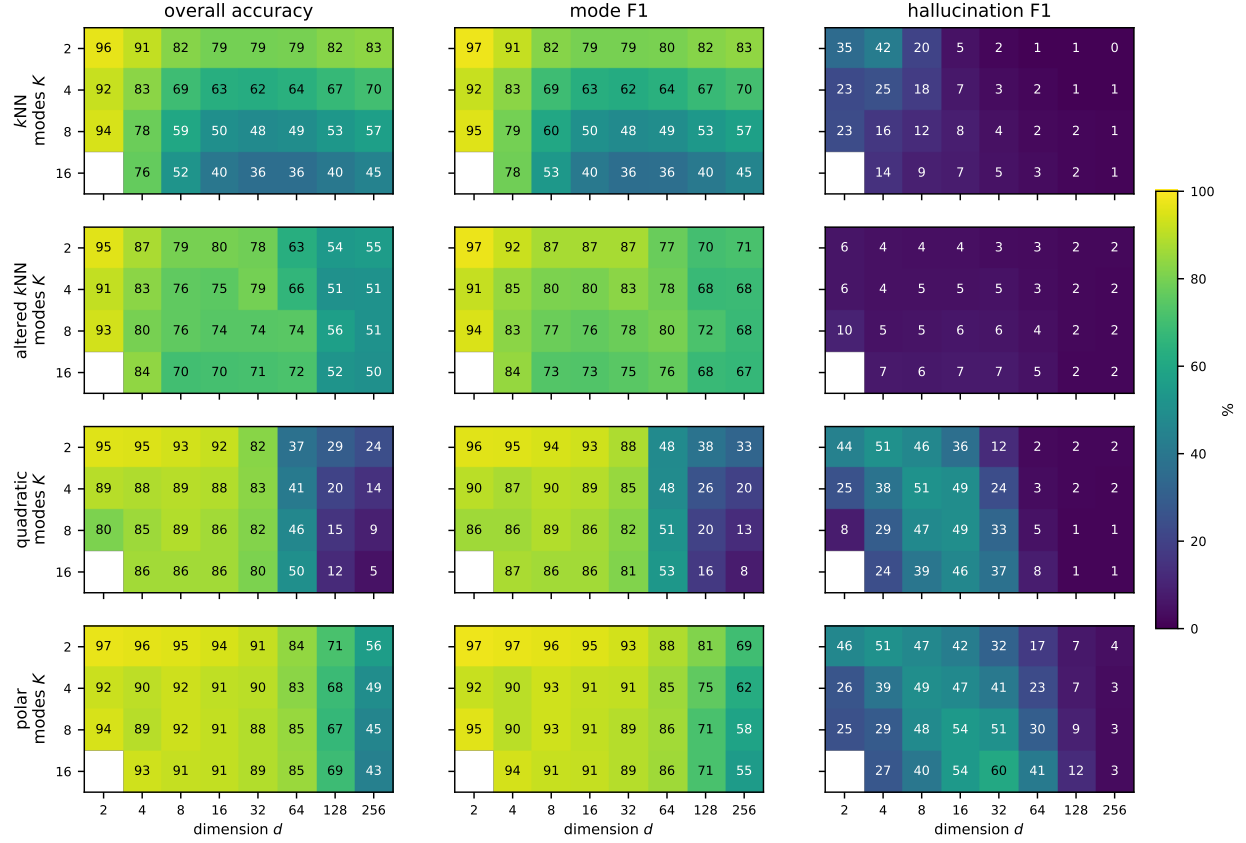


Figure 18: **Weighted DDIM, $T = 250$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.4 Weighted DDIM, $T = 250$

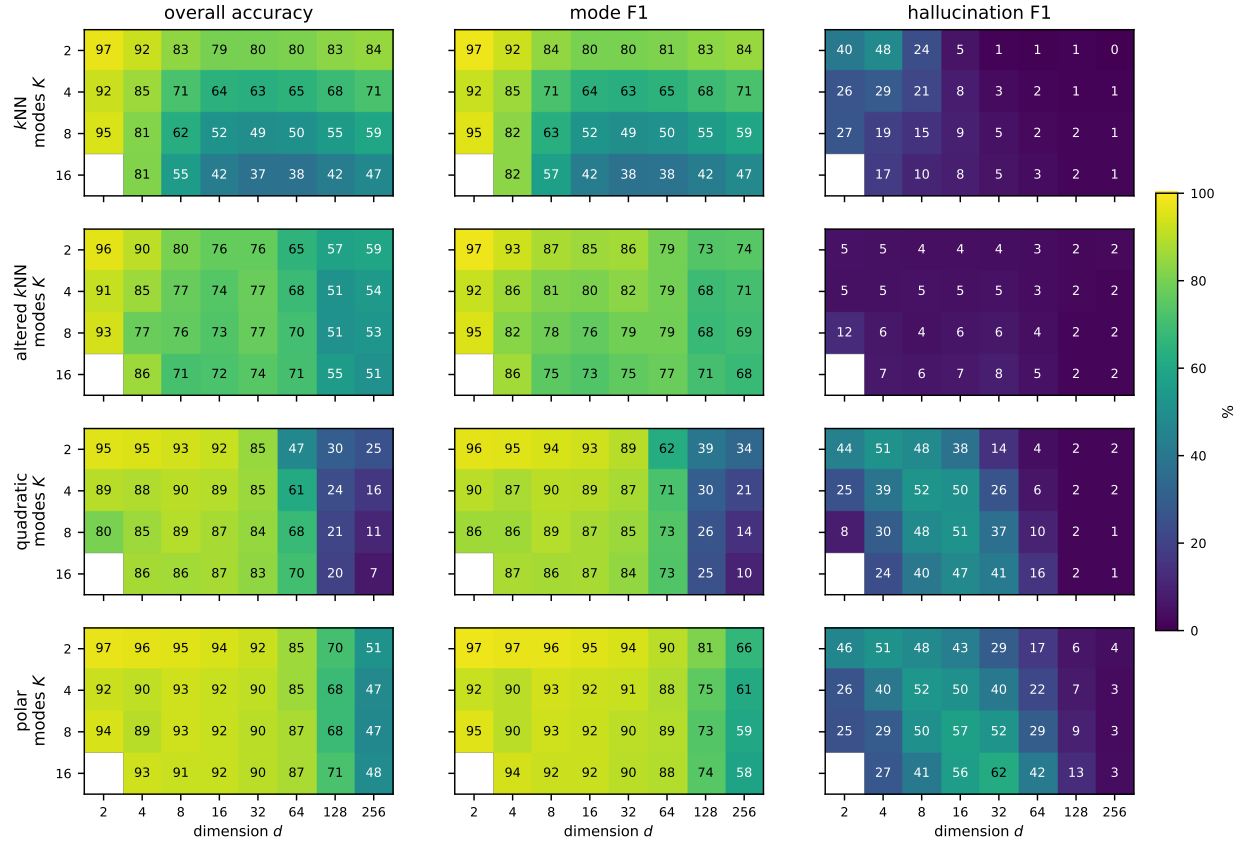


Figure 19: **Weighted DDIM, $T = 250$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

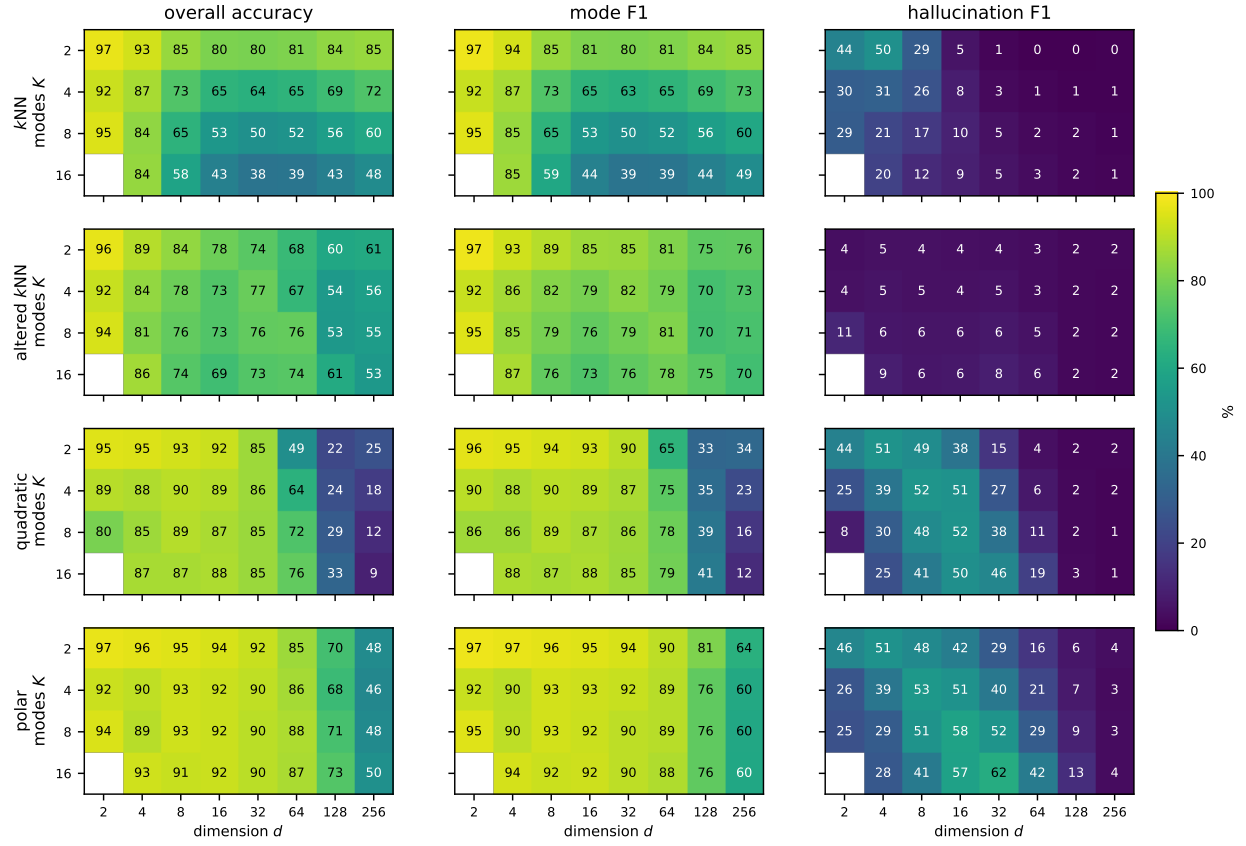


Figure 20: **Weighted DDIM, $T = 250$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

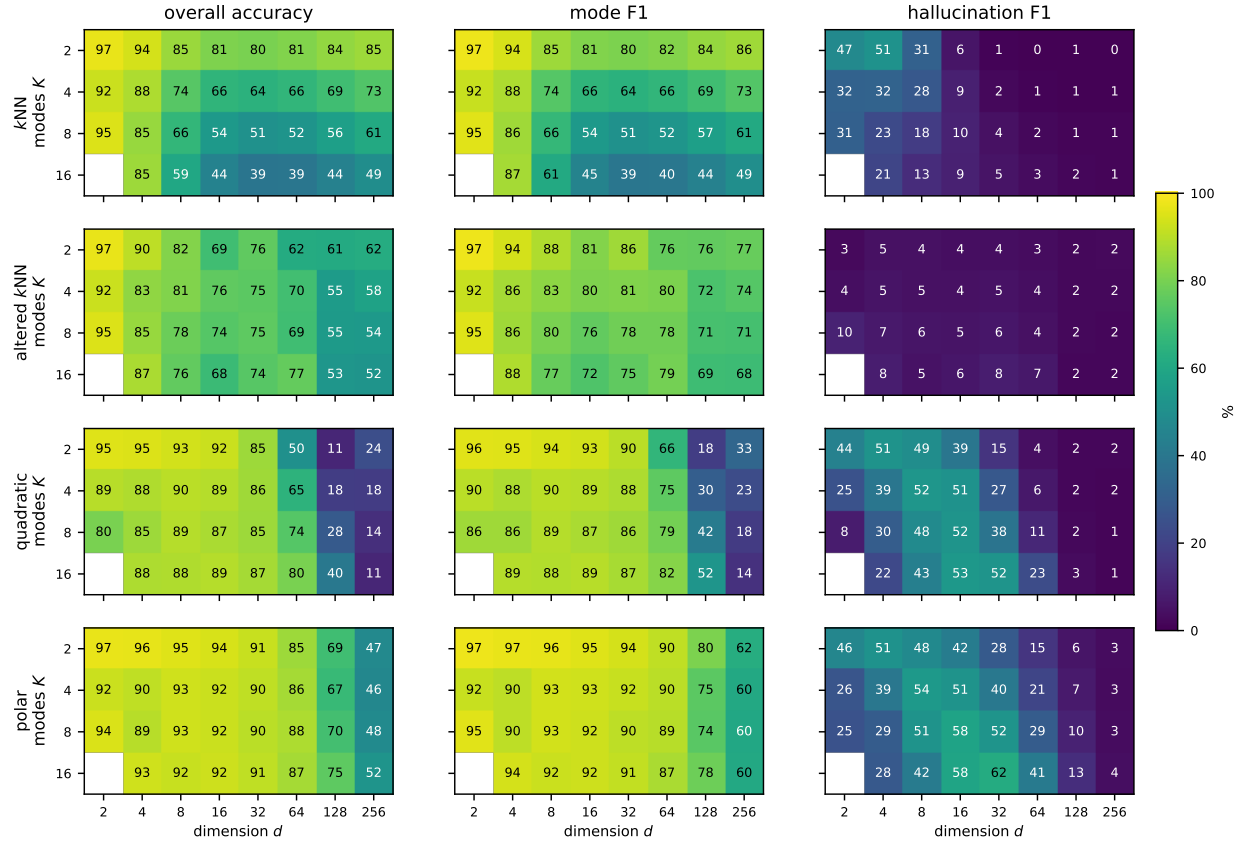


Figure 21: **Weighted DDIM, $T = 250$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

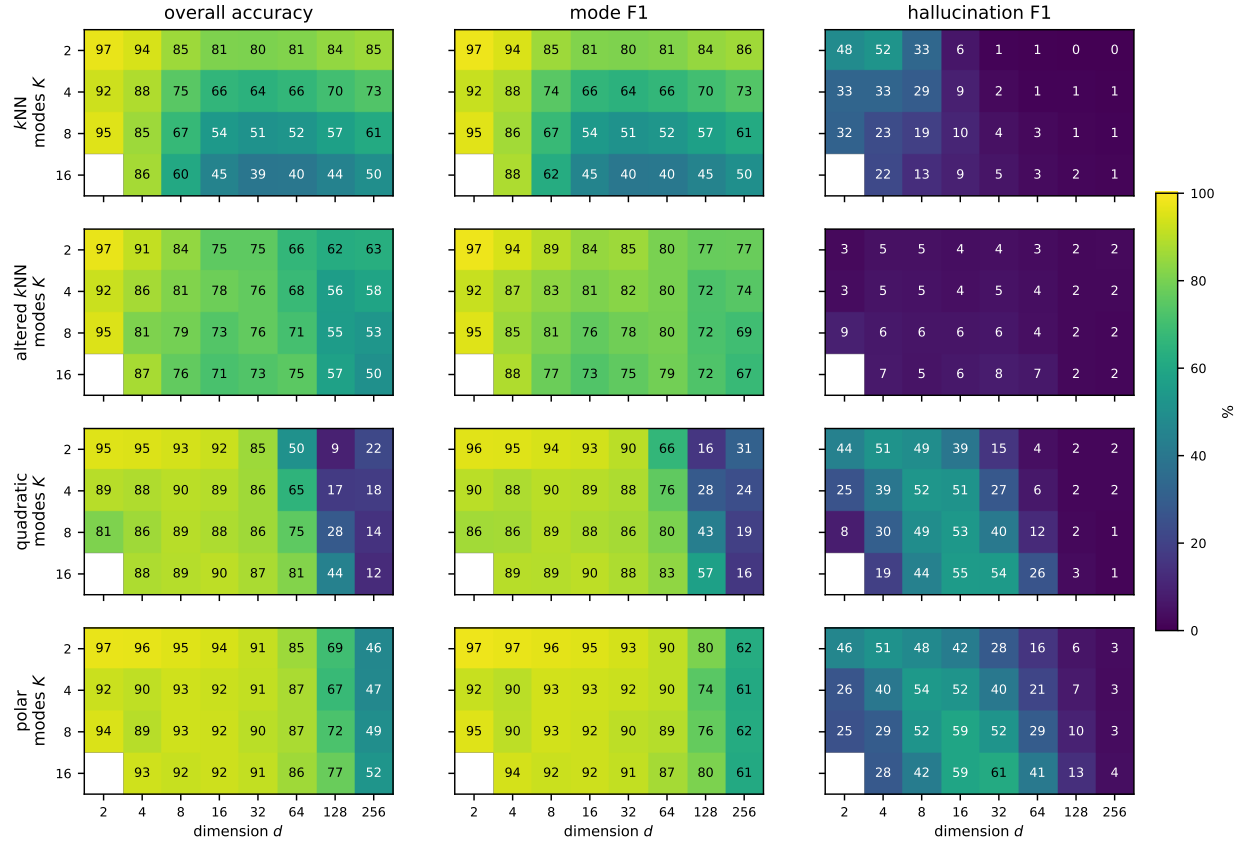


Figure 22: **Weighted DDIM, $T = 250$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

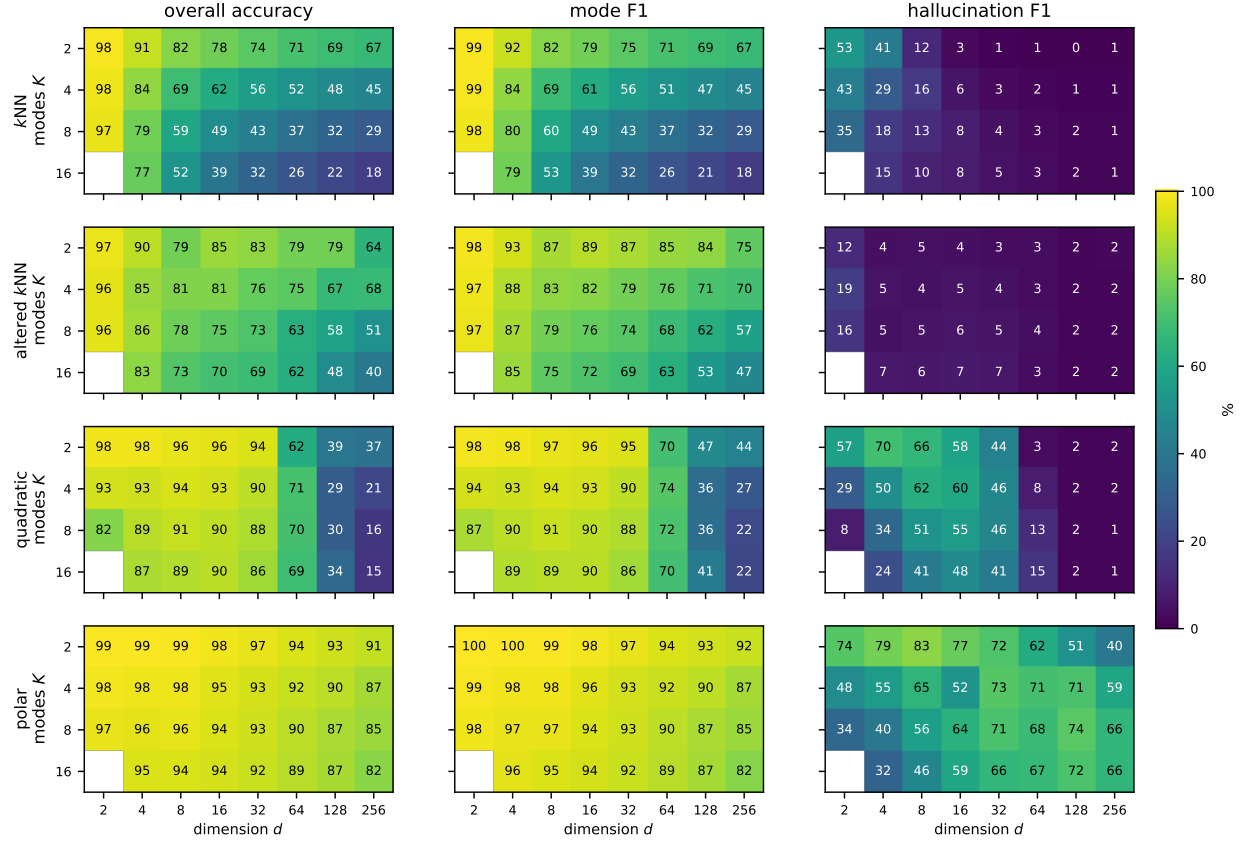


Figure 23: **Weighted DDIM, $T = 500$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.5 Weighted DDIM, $T = 500$

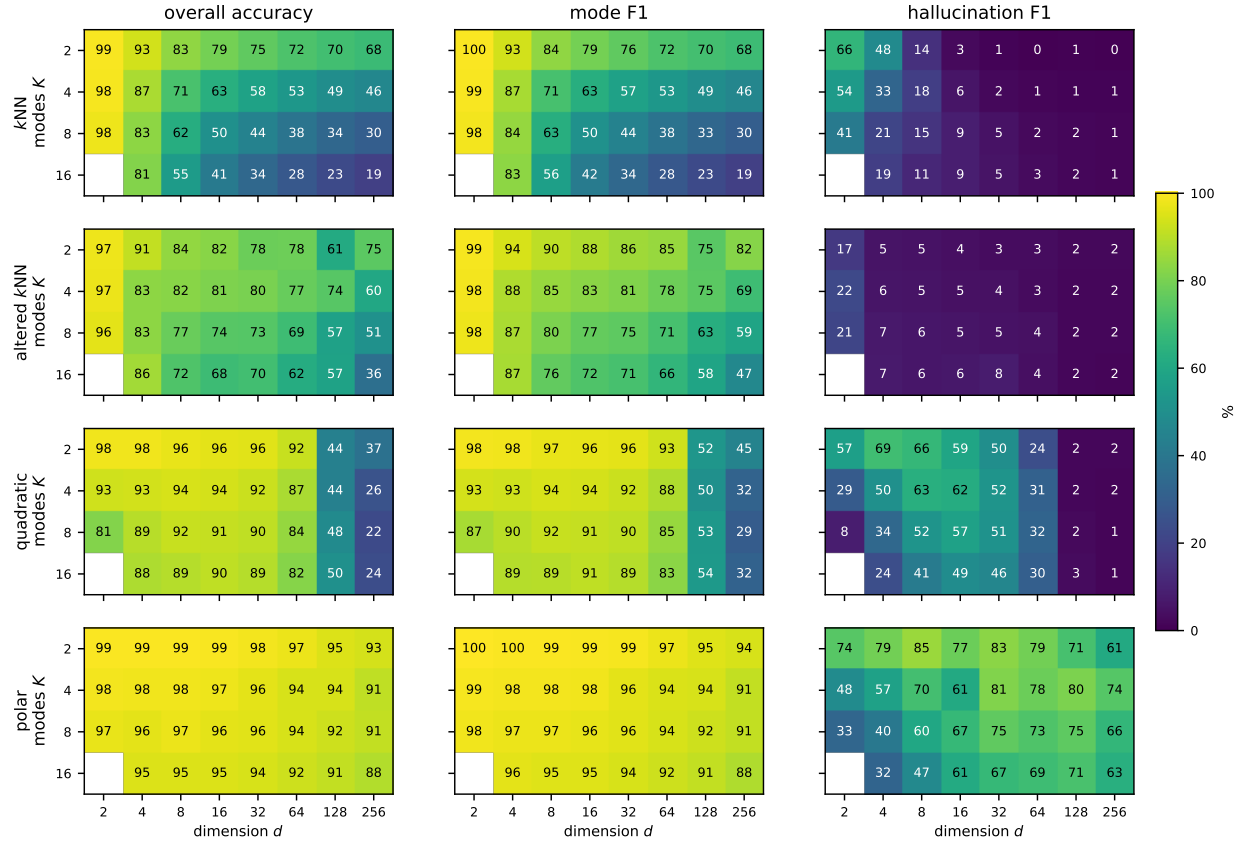


Figure 24: **Weighted DDIM, $T = 500$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

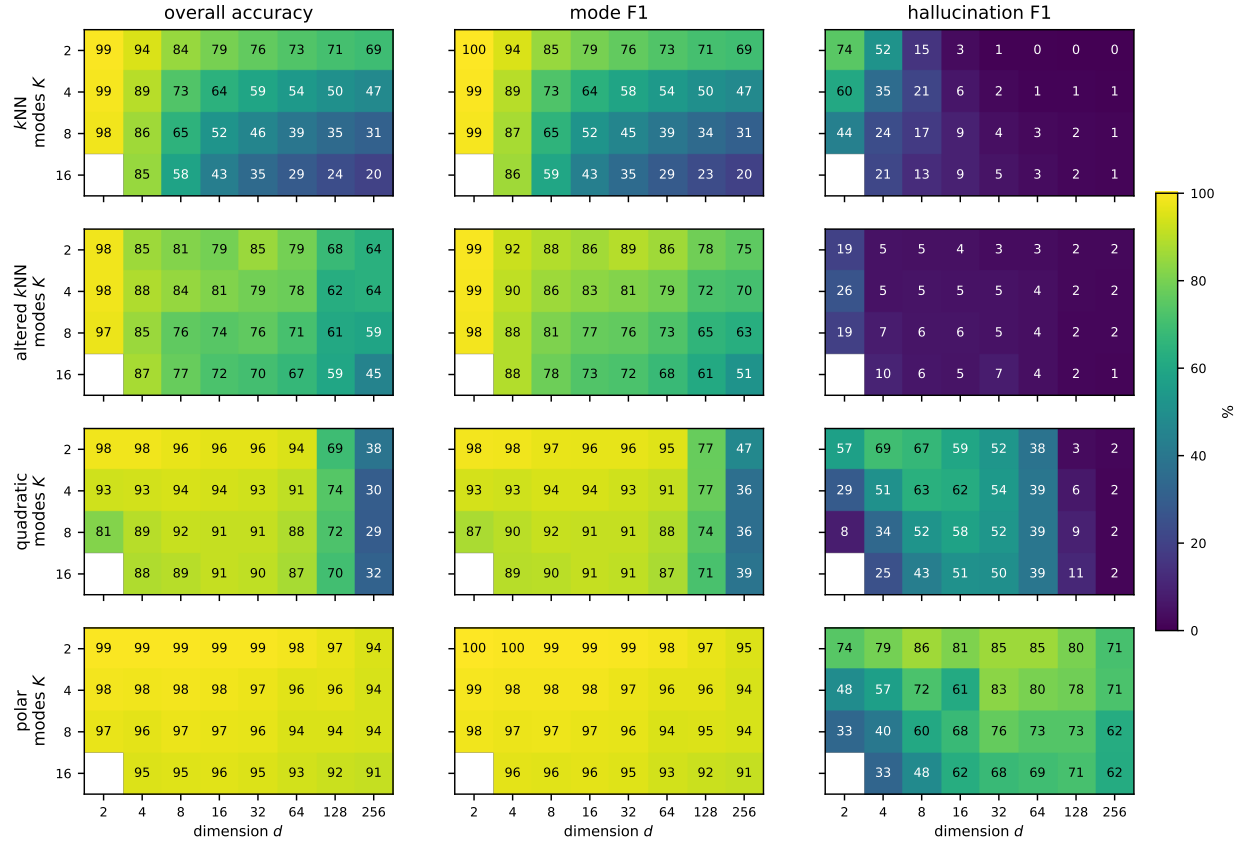


Figure 25: **Weighted DDIM, $T = 500$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

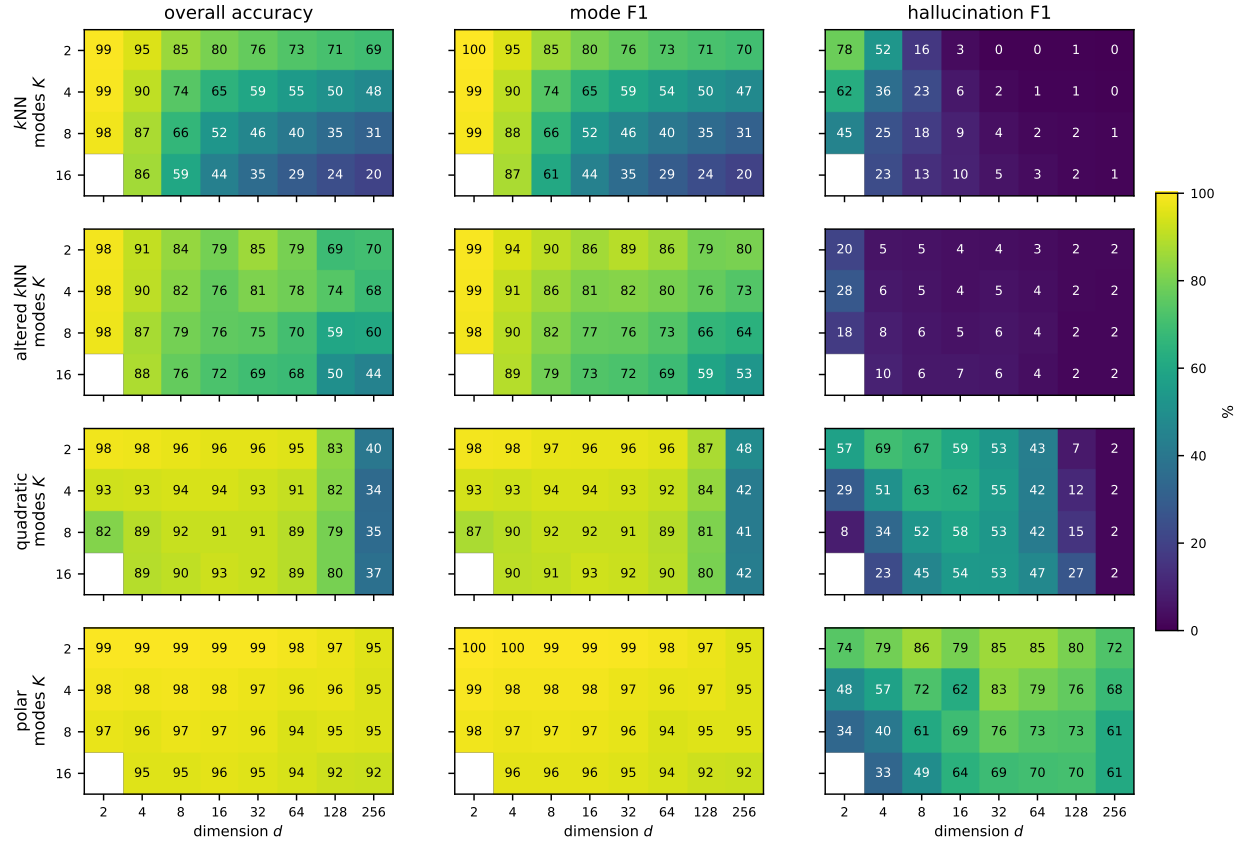


Figure 26: **Weighted DDIM, $T = 500$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

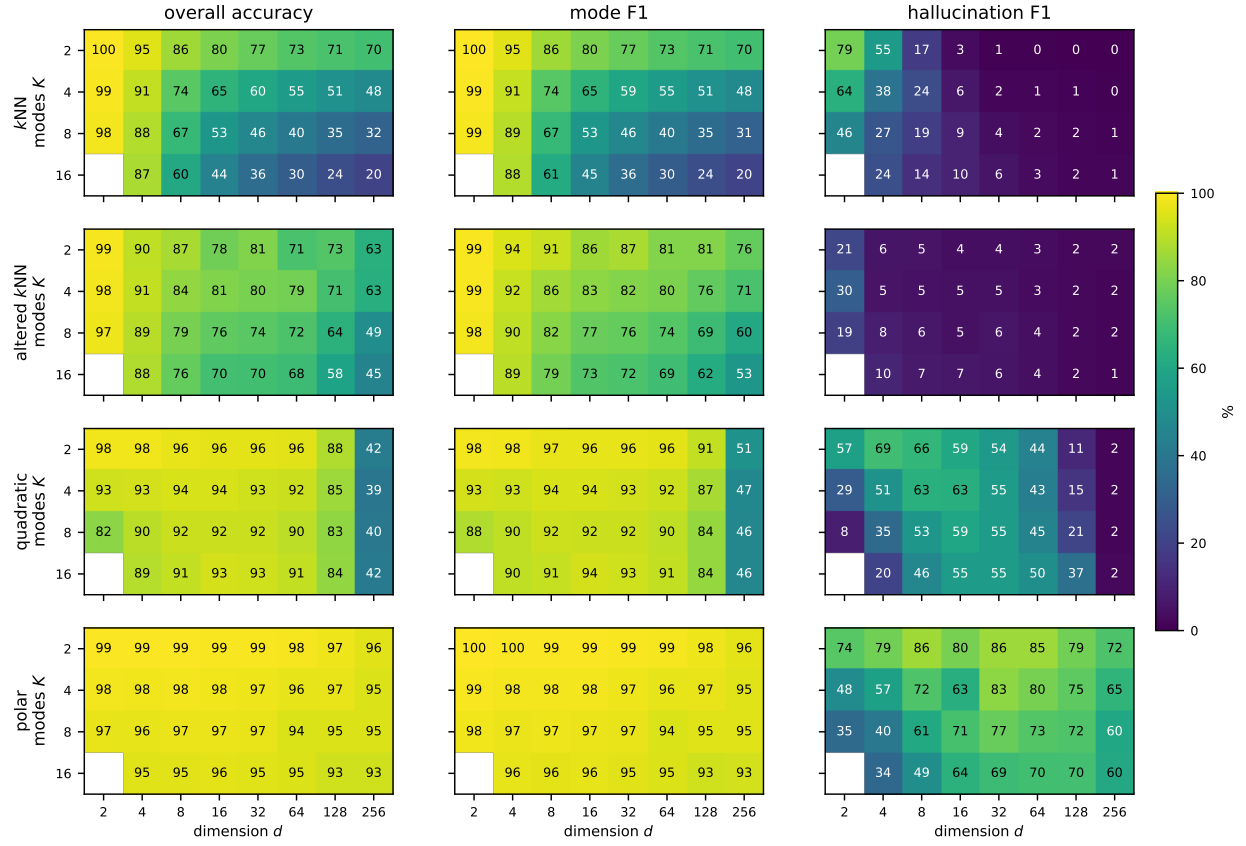


Figure 27: **Weighted DDIM, $T = 500$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

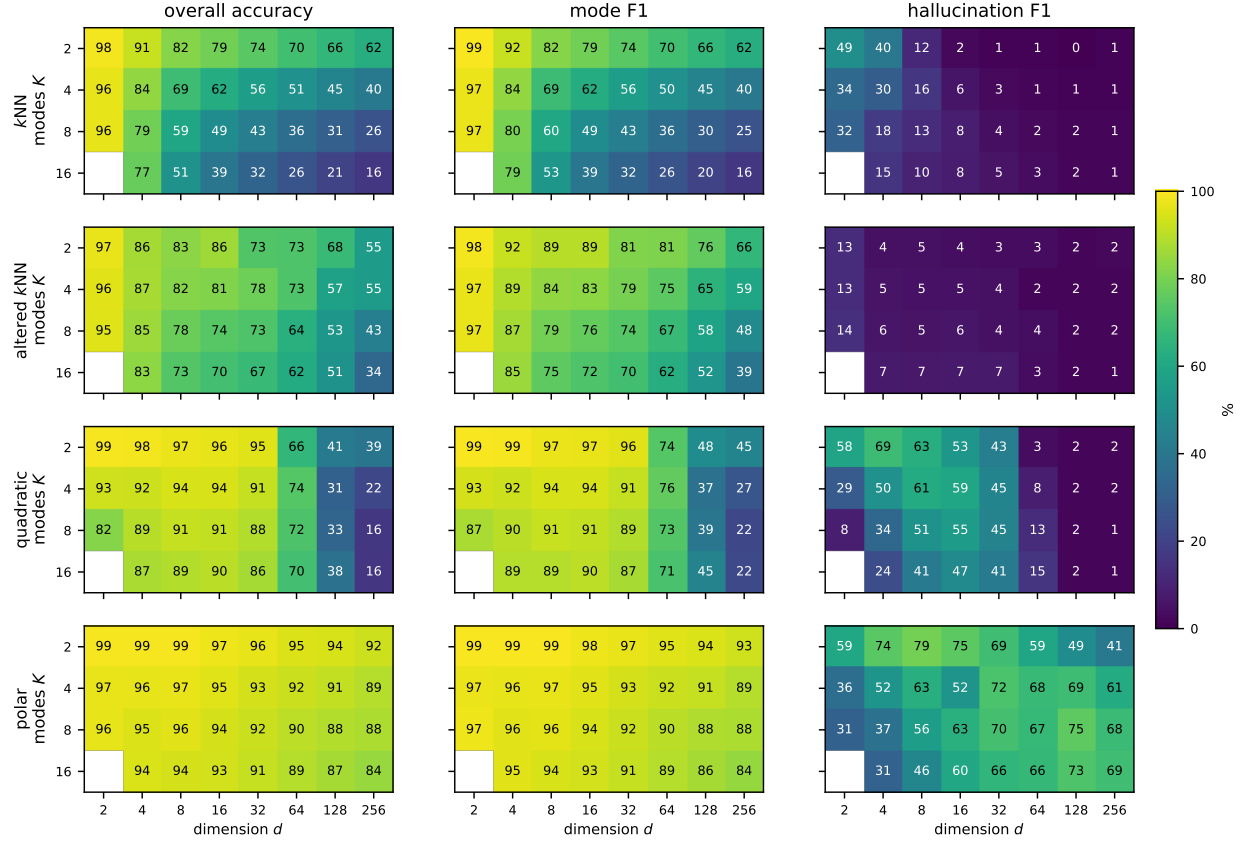


Figure 28: **Weighted DDIM, $T = 750$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.6 Weighted DDIM, $T = 750$

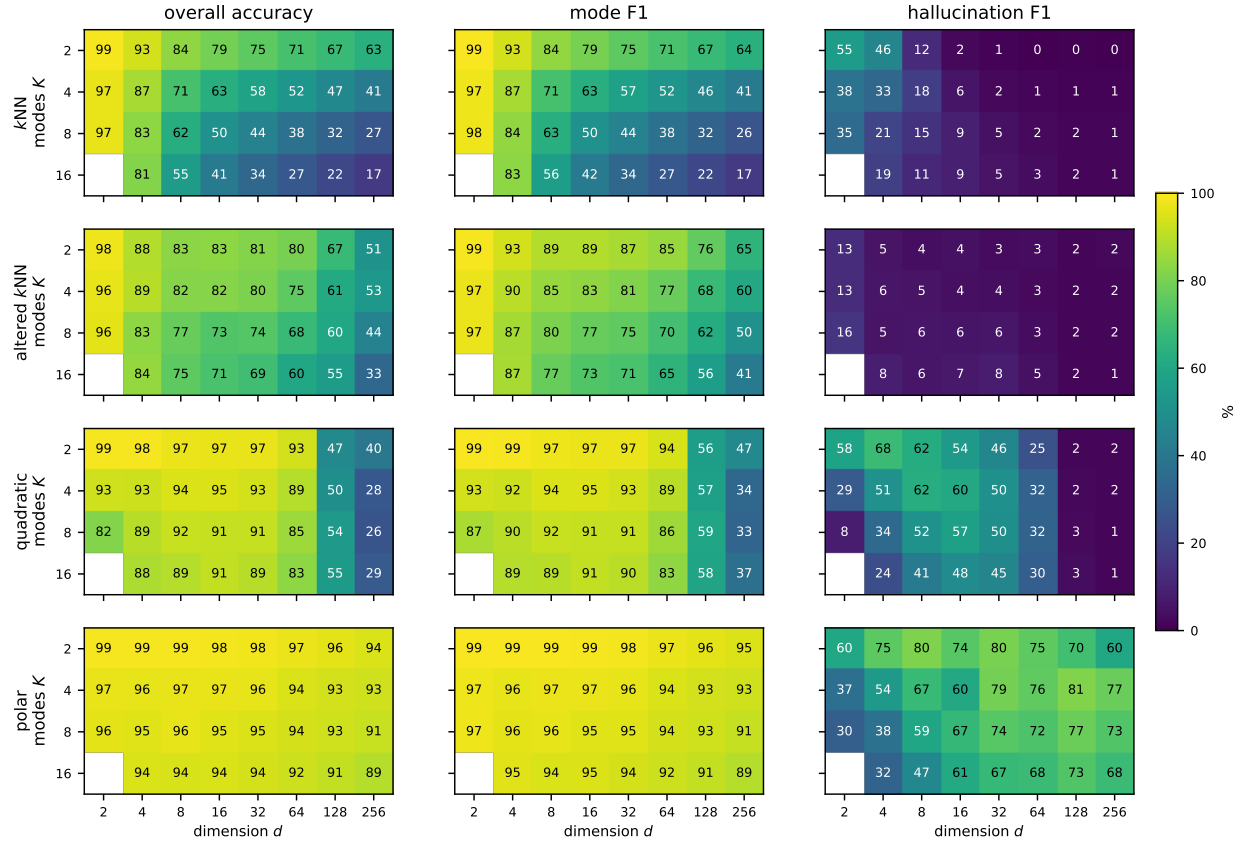


Figure 29: **Weighted DDIM, $T = 750$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

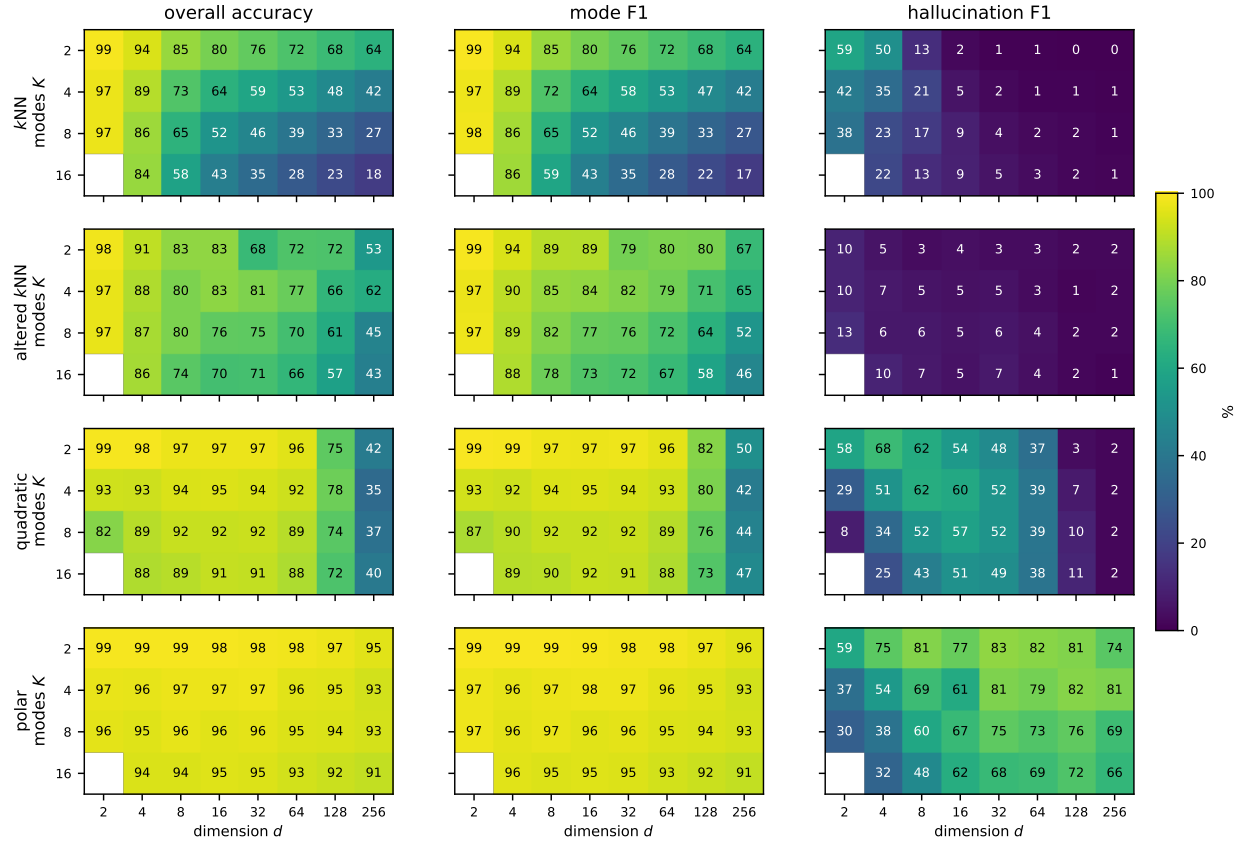


Figure 30: **Weighted DDIM, $T = 750$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

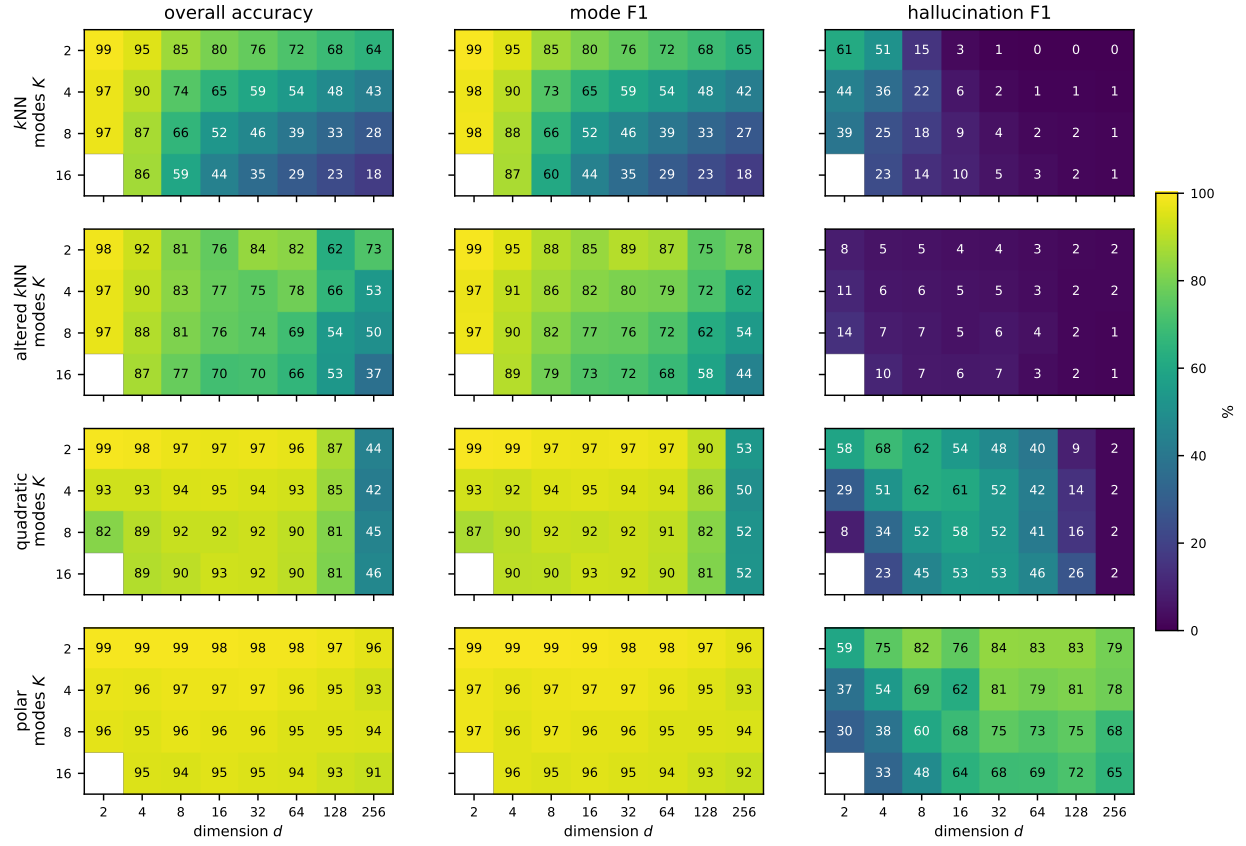


Figure 31: **Weighted DDIM, $T = 750$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

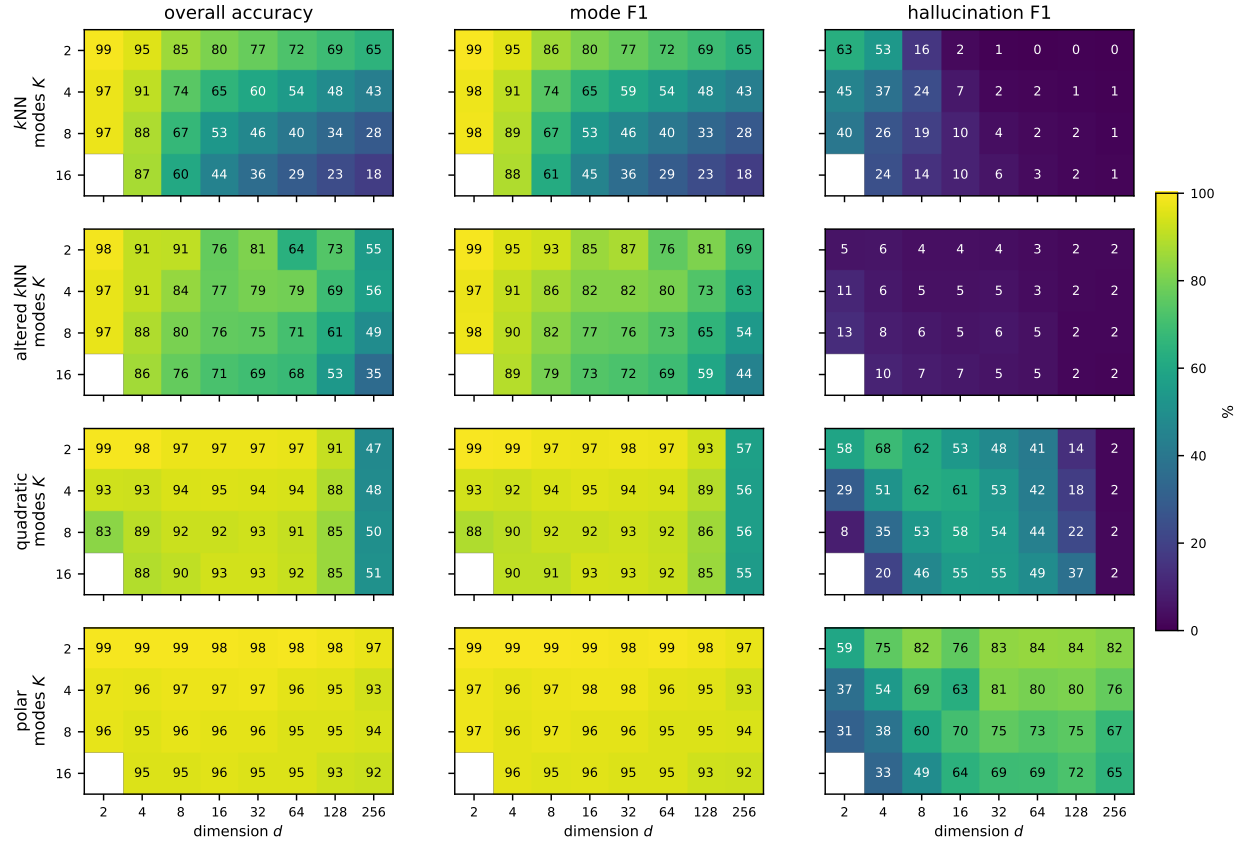


Figure 32: **Weighted DDIM, $T = 750$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

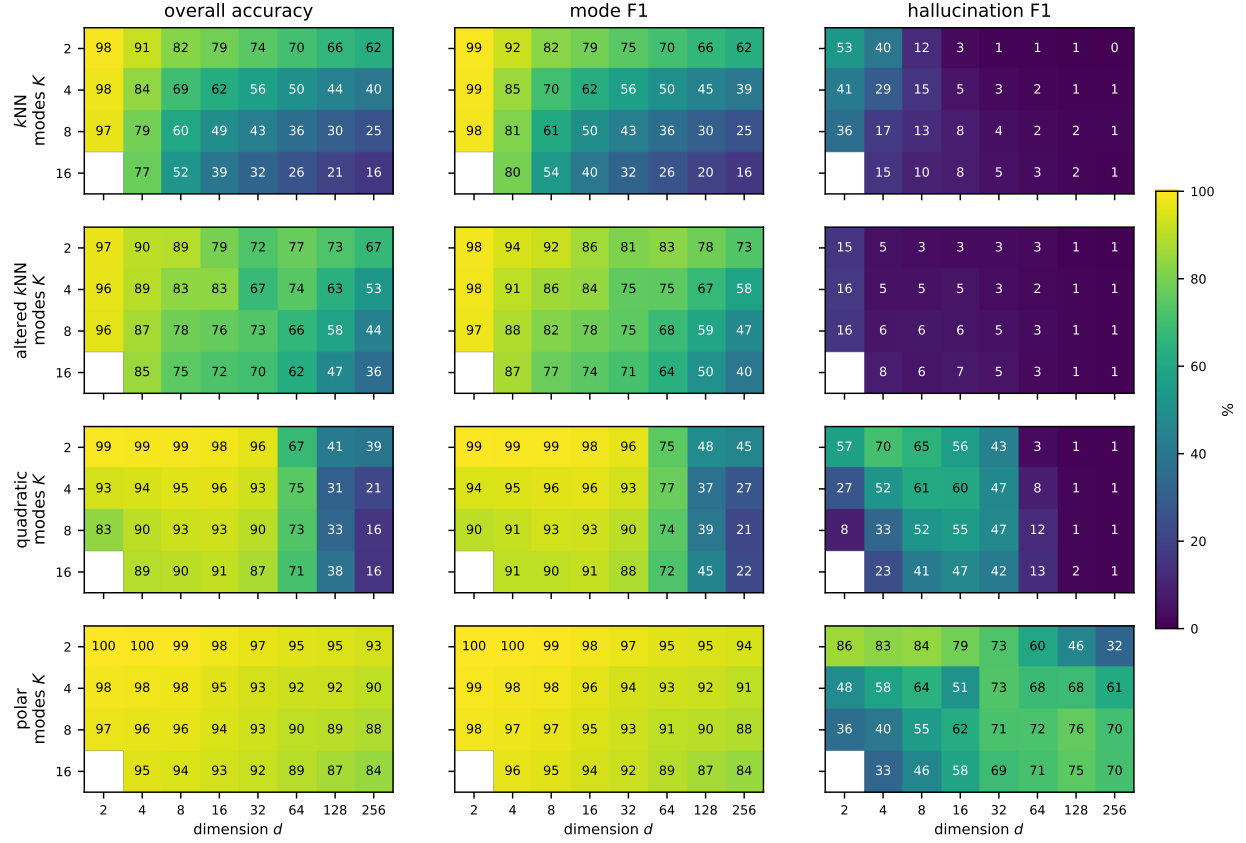


Figure 33: **Unweighted Flow matching, $T = 250$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.7 Unweighted Flow matching, $T = 250$

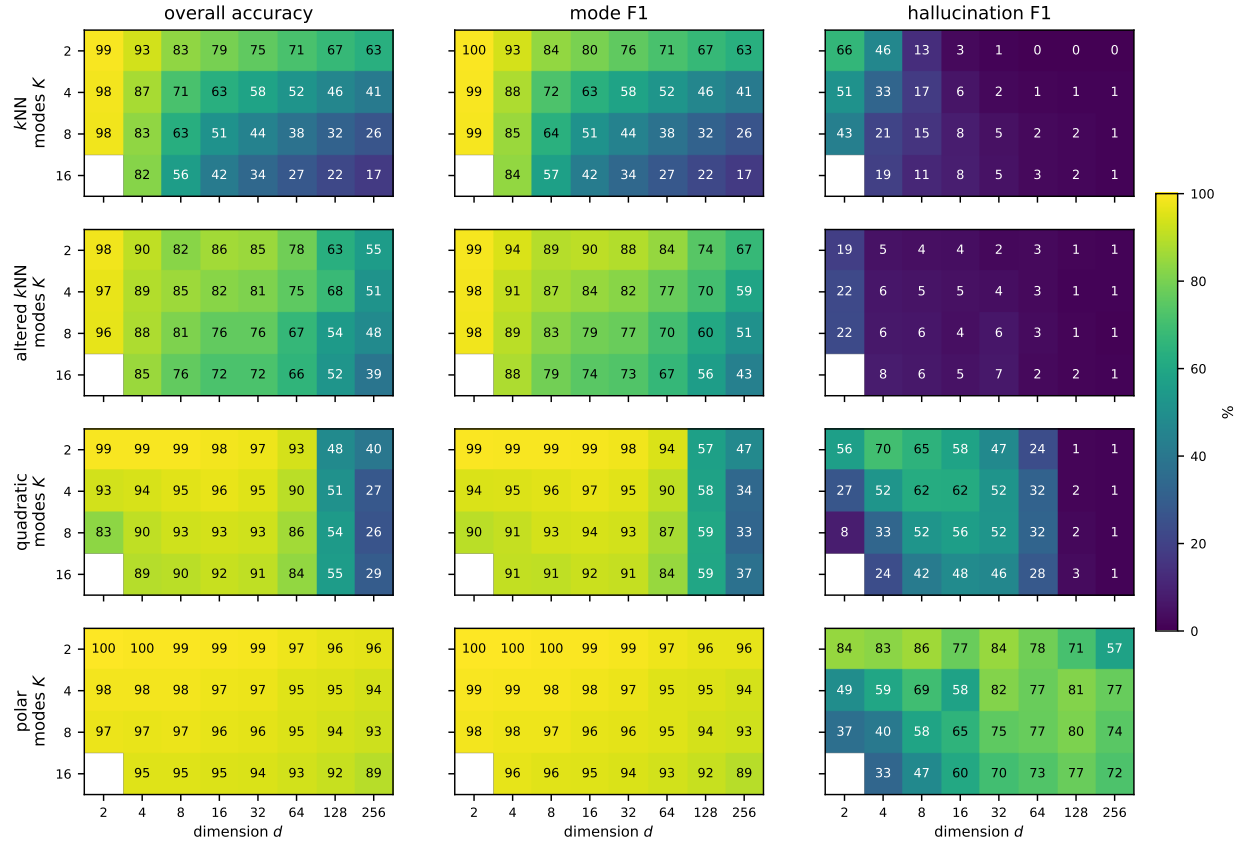


Figure 34: **Unweighted Flow matching, $T = 250$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

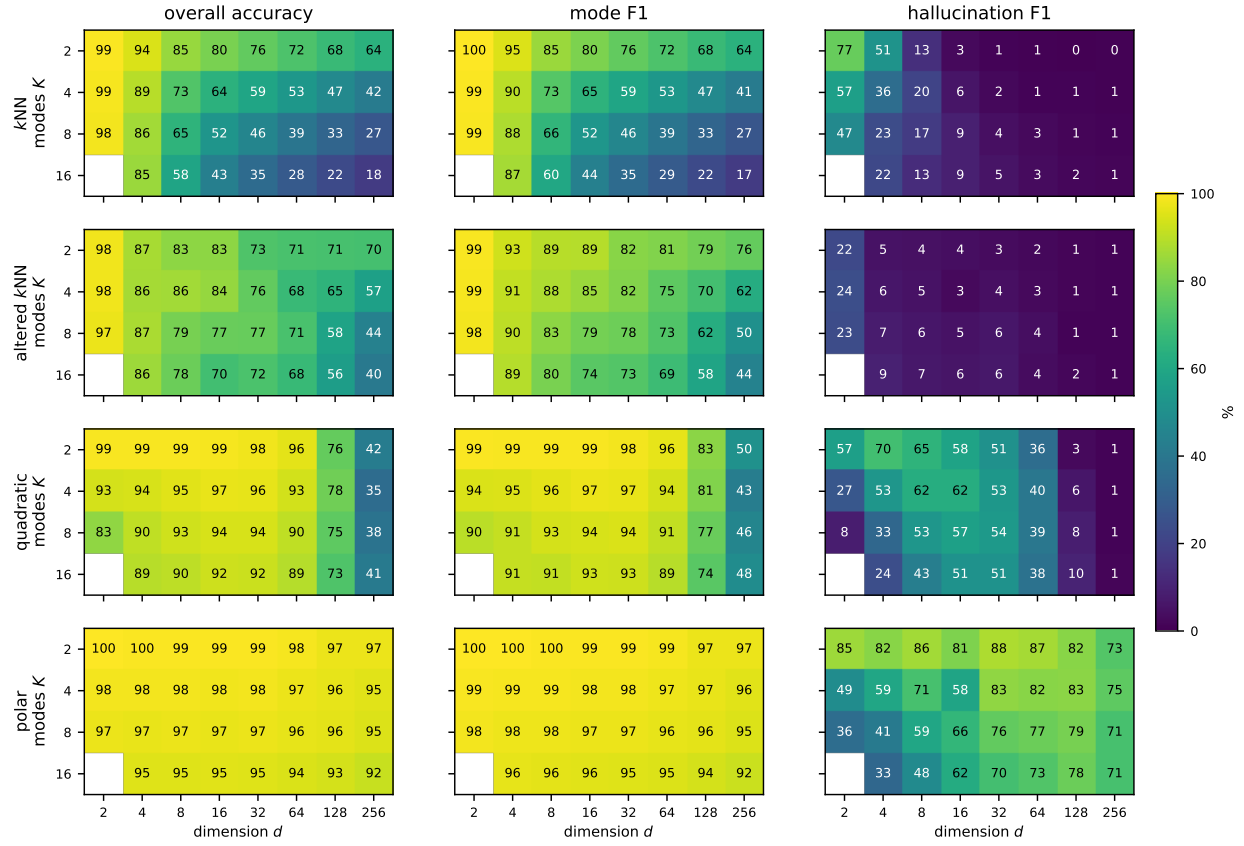


Figure 35: **Unweighted Flow matching**, $T = 250$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

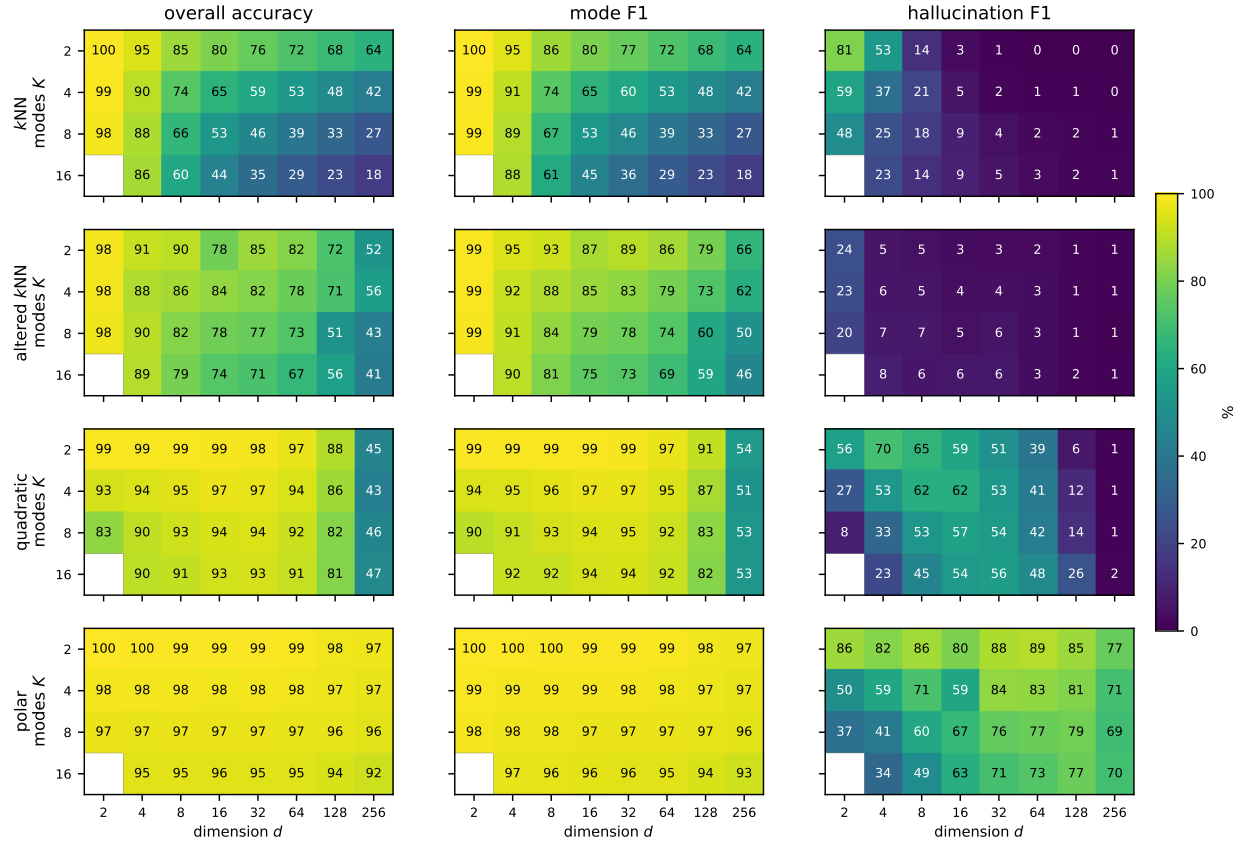


Figure 36: **Unweighted Flow matching, $T = 250$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

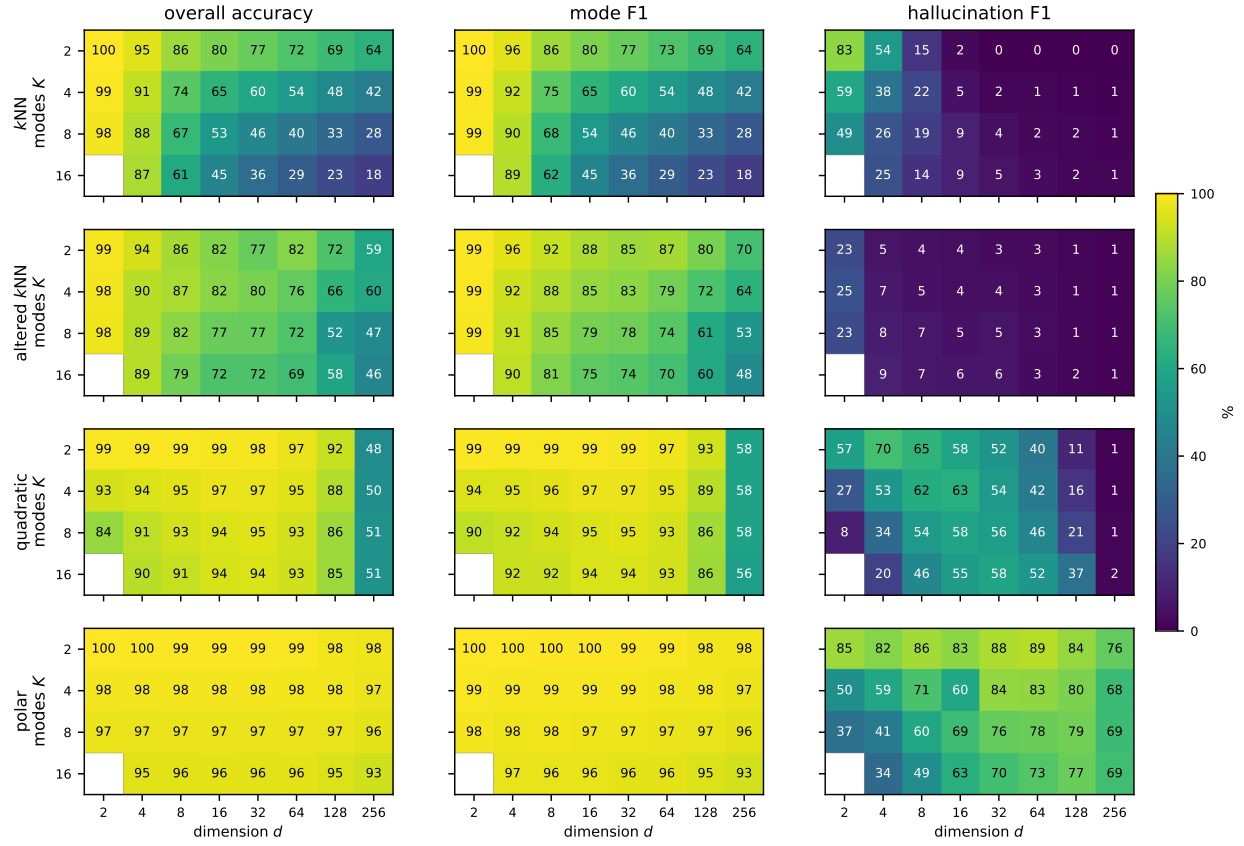


Figure 37: **Unweighted Flow matching, $T = 250$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

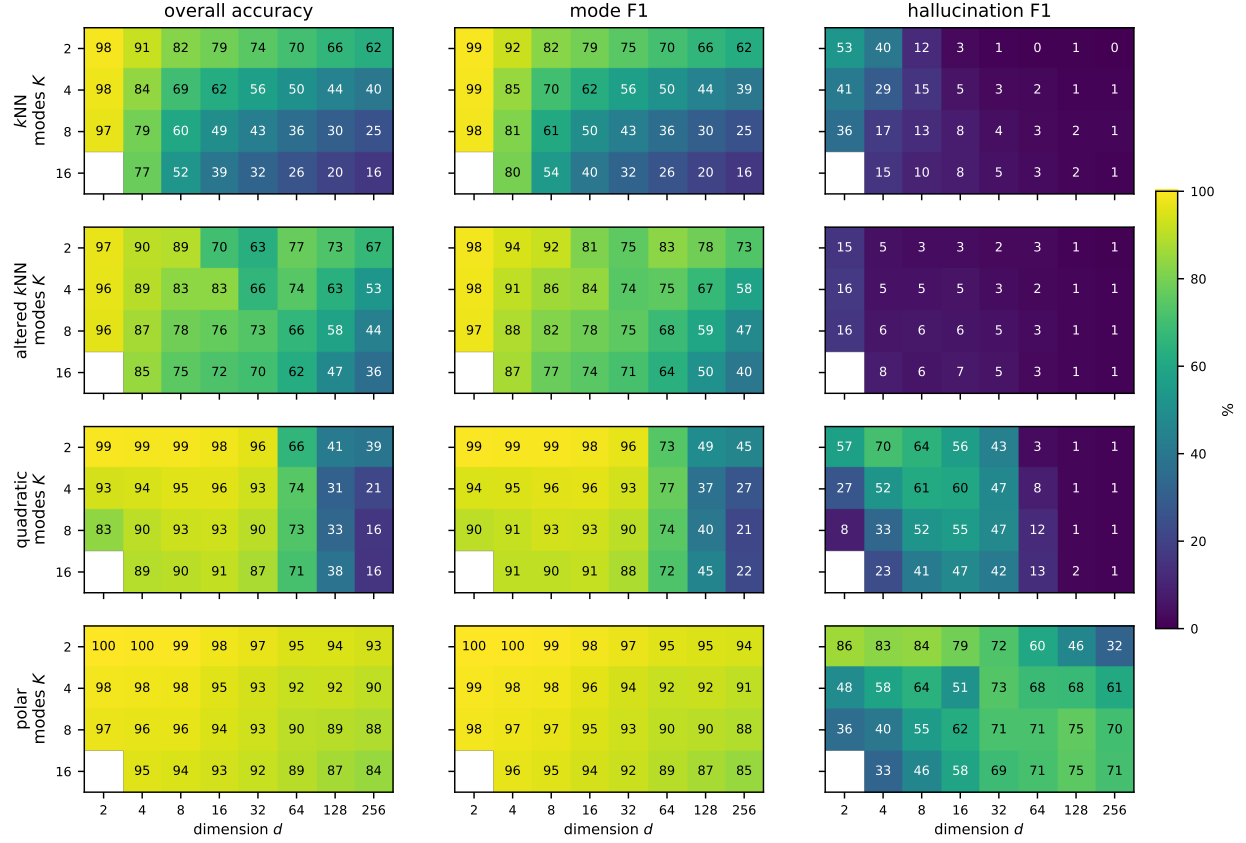


Figure 38: **Unweighted Flow matching, $T = 500$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.8 Unweighted Flow matching, $T = 500$

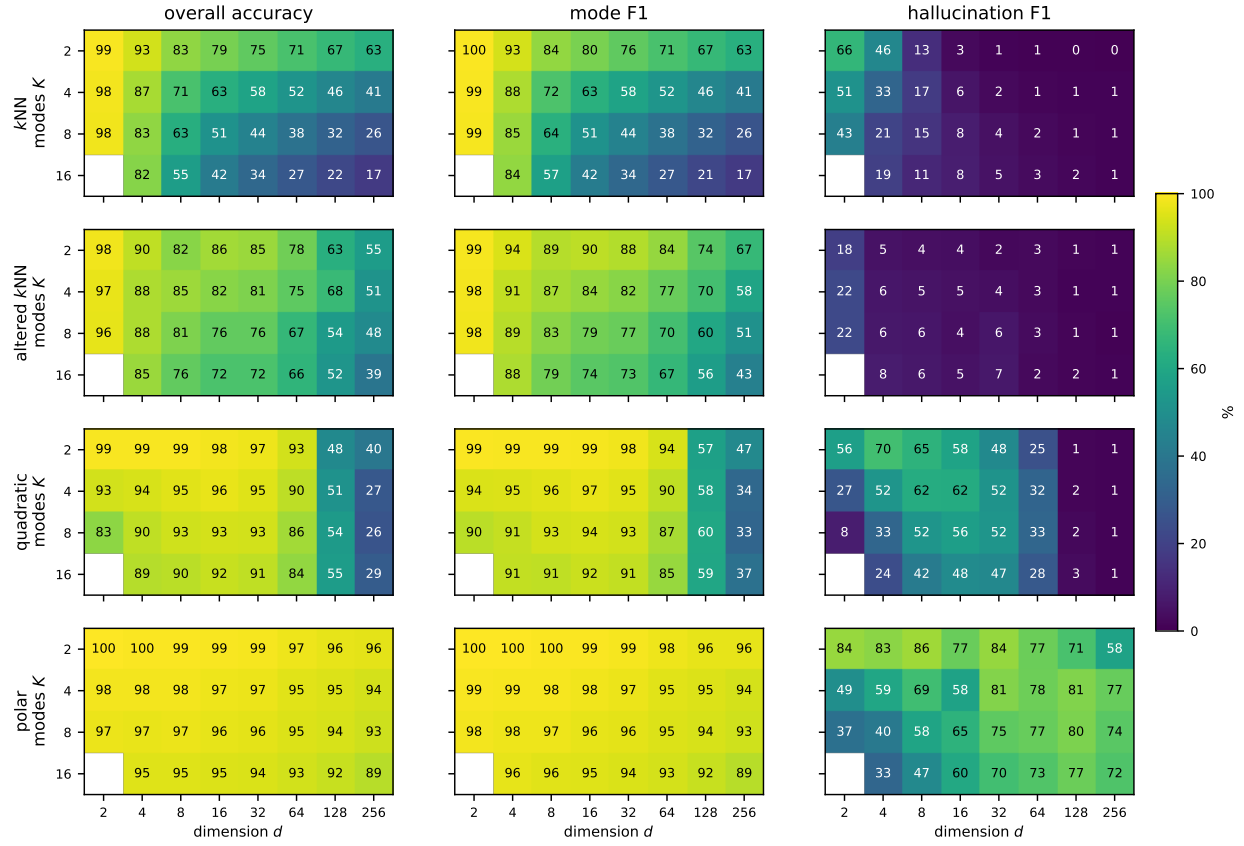


Figure 39: **Unweighted Flow matching, $T = 500$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

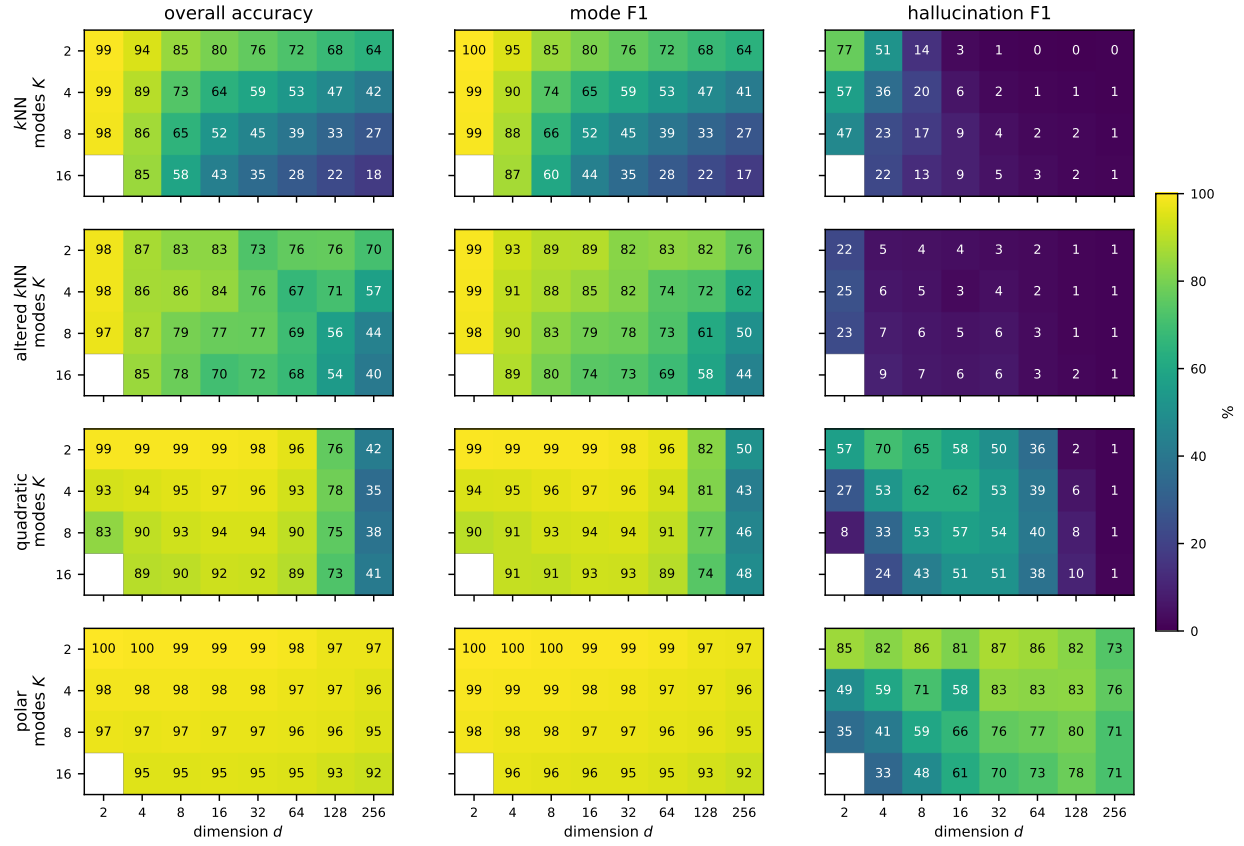


Figure 40: **Unweighted Flow matching**, $T = 500$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

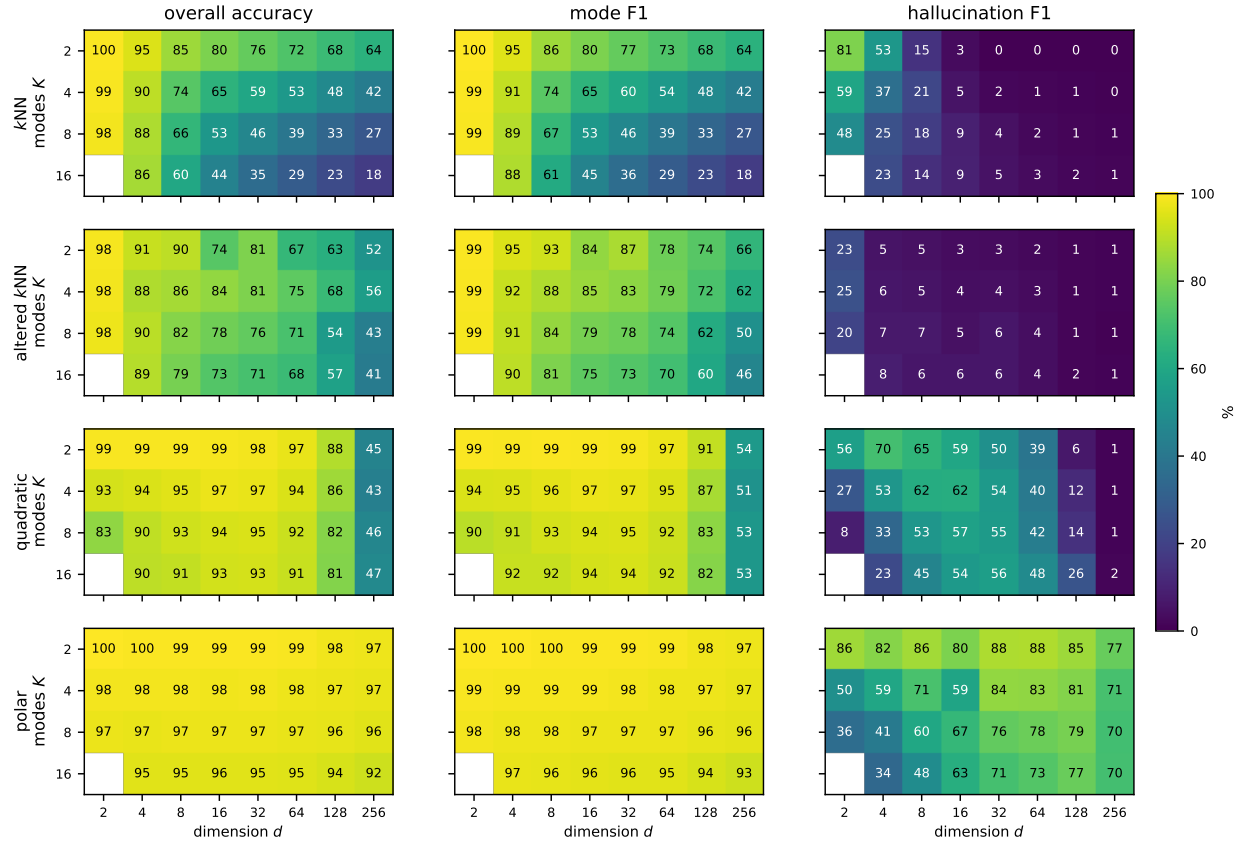


Figure 41: **Unweighted Flow matching, $T = 500$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

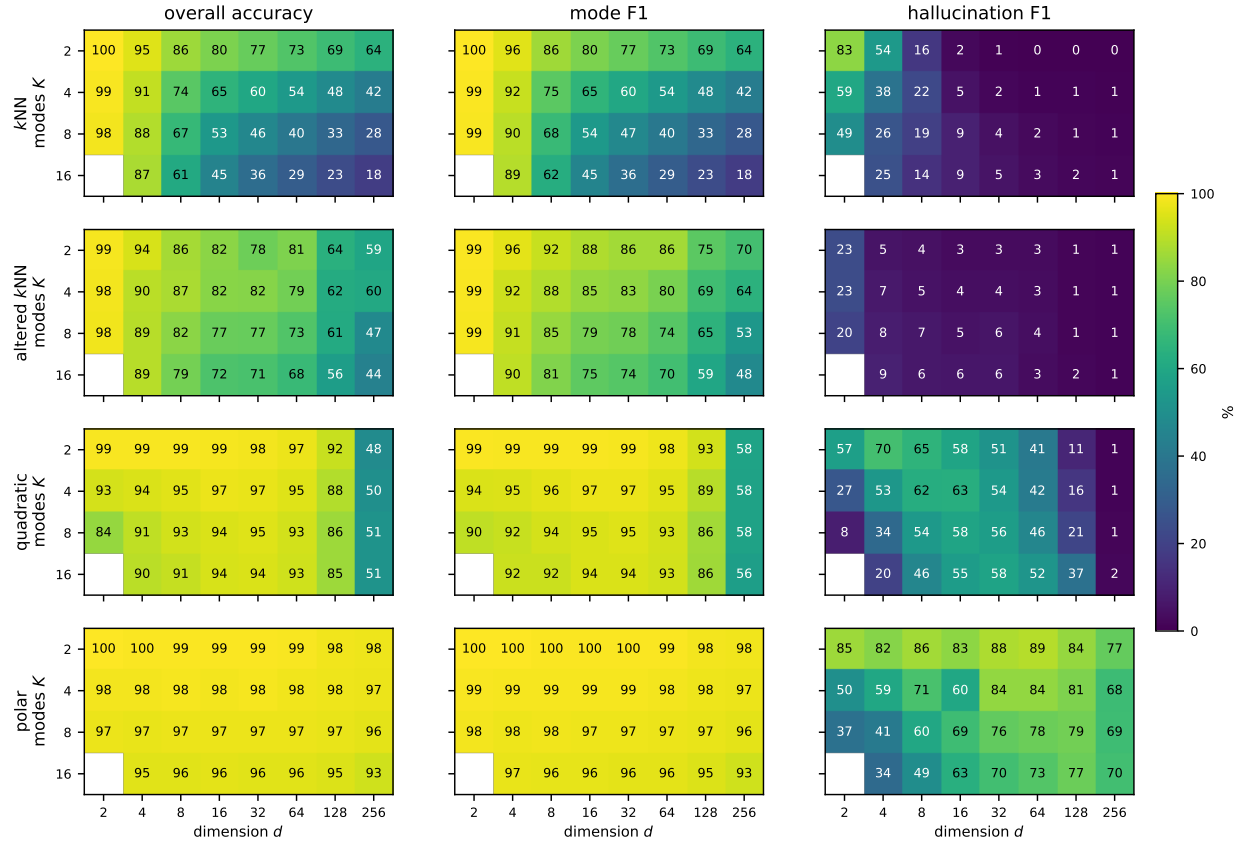


Figure 42: **Unweighted Flow matching, $T = 500$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

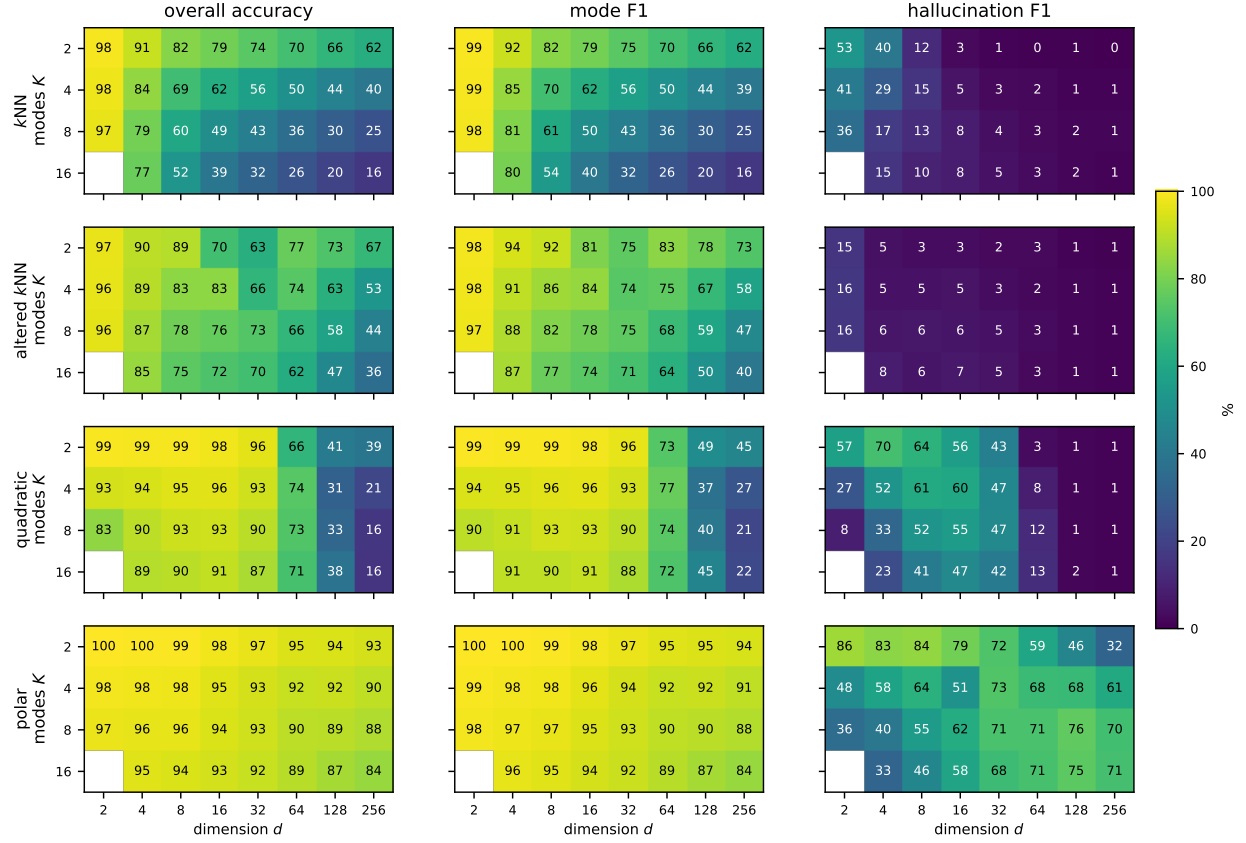


Figure 43: **Unweighted Flow matching, $T = 750$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.9 Unweighted Flow matching, $T = 750$

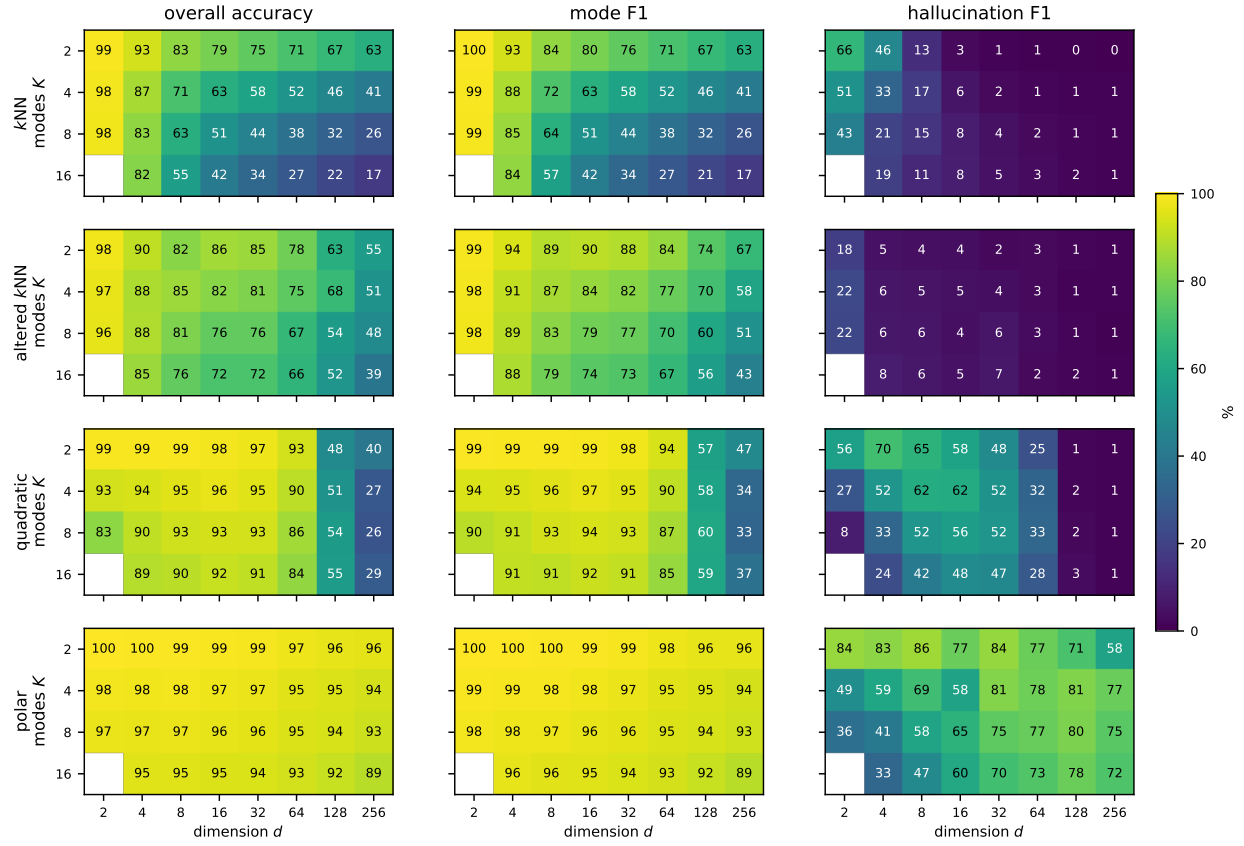


Figure 44: **Unweighted Flow matching, $T = 750$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

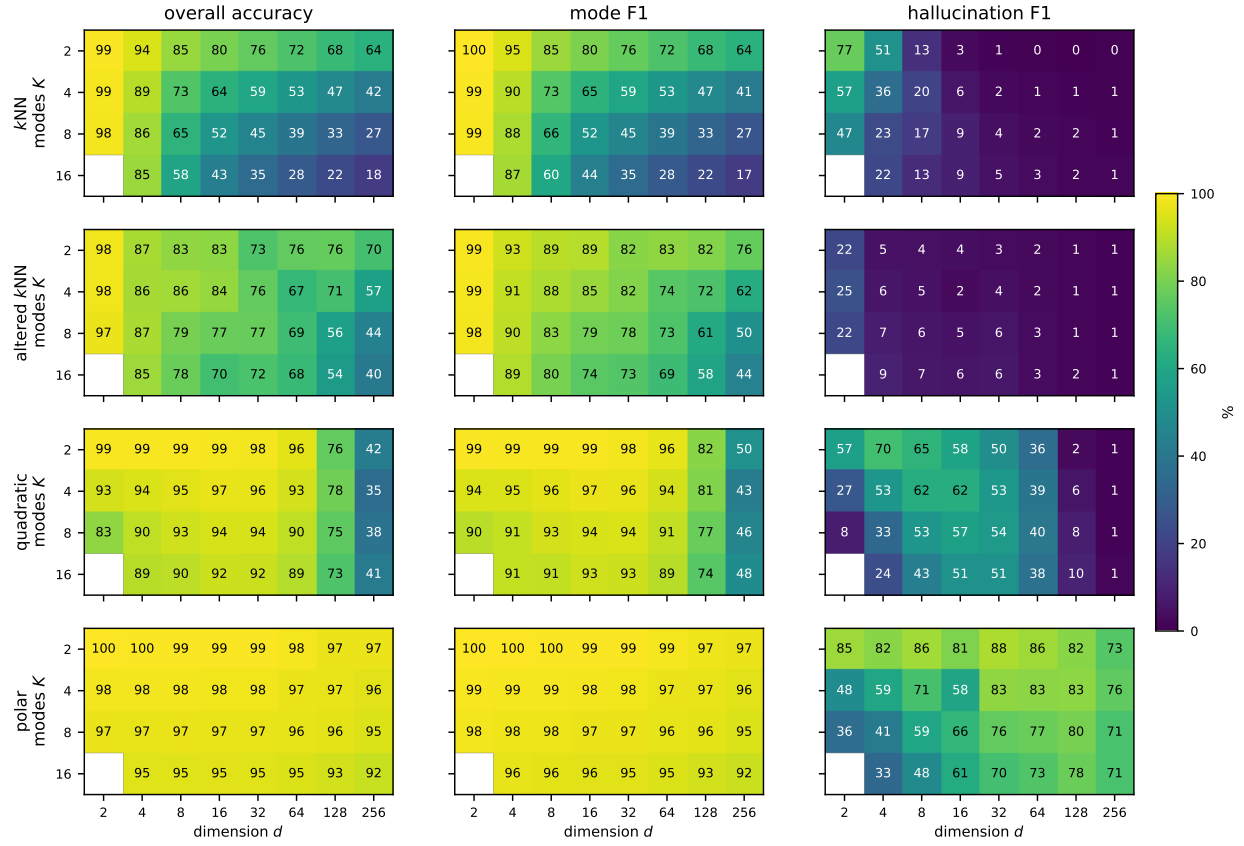


Figure 45: **Unweighted Flow matching, $T = 750$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

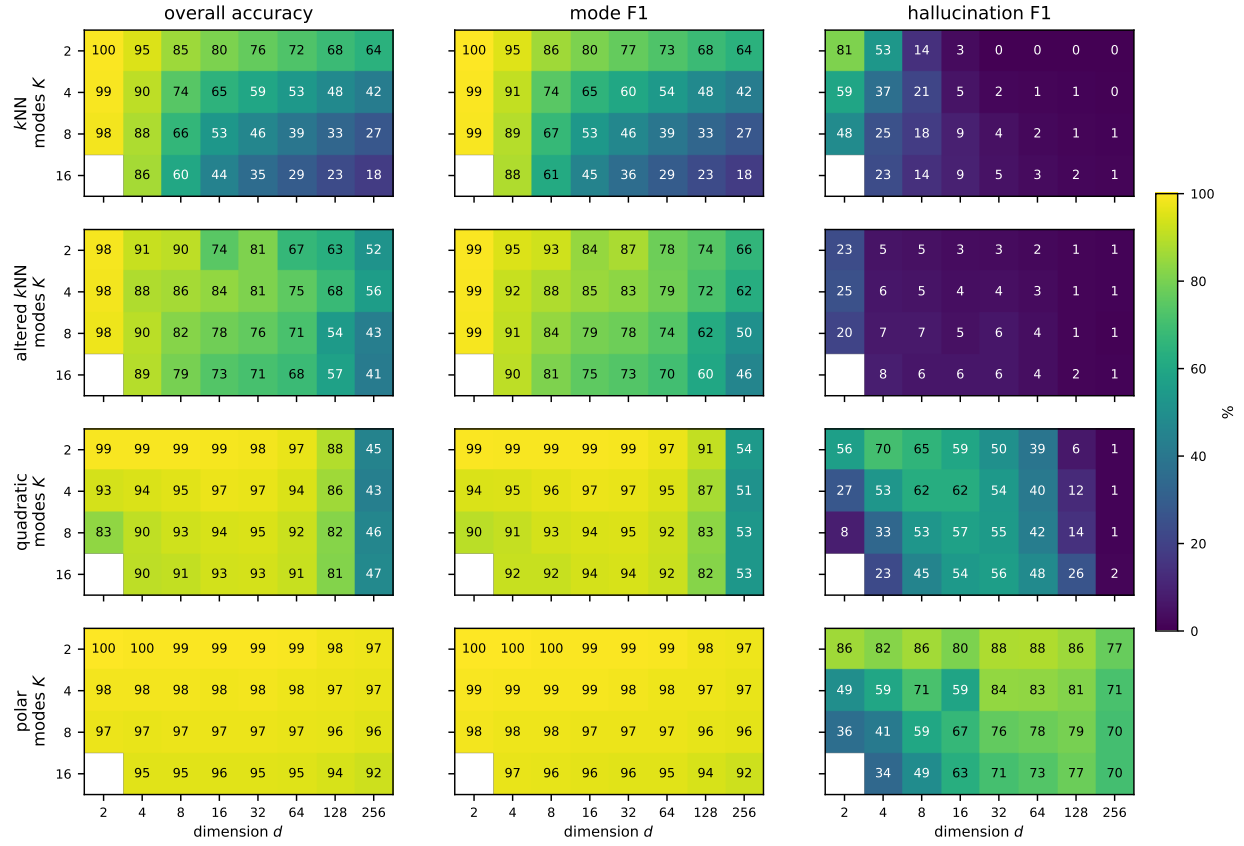


Figure 46: **Unweighted Flow matching, $T = 750$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

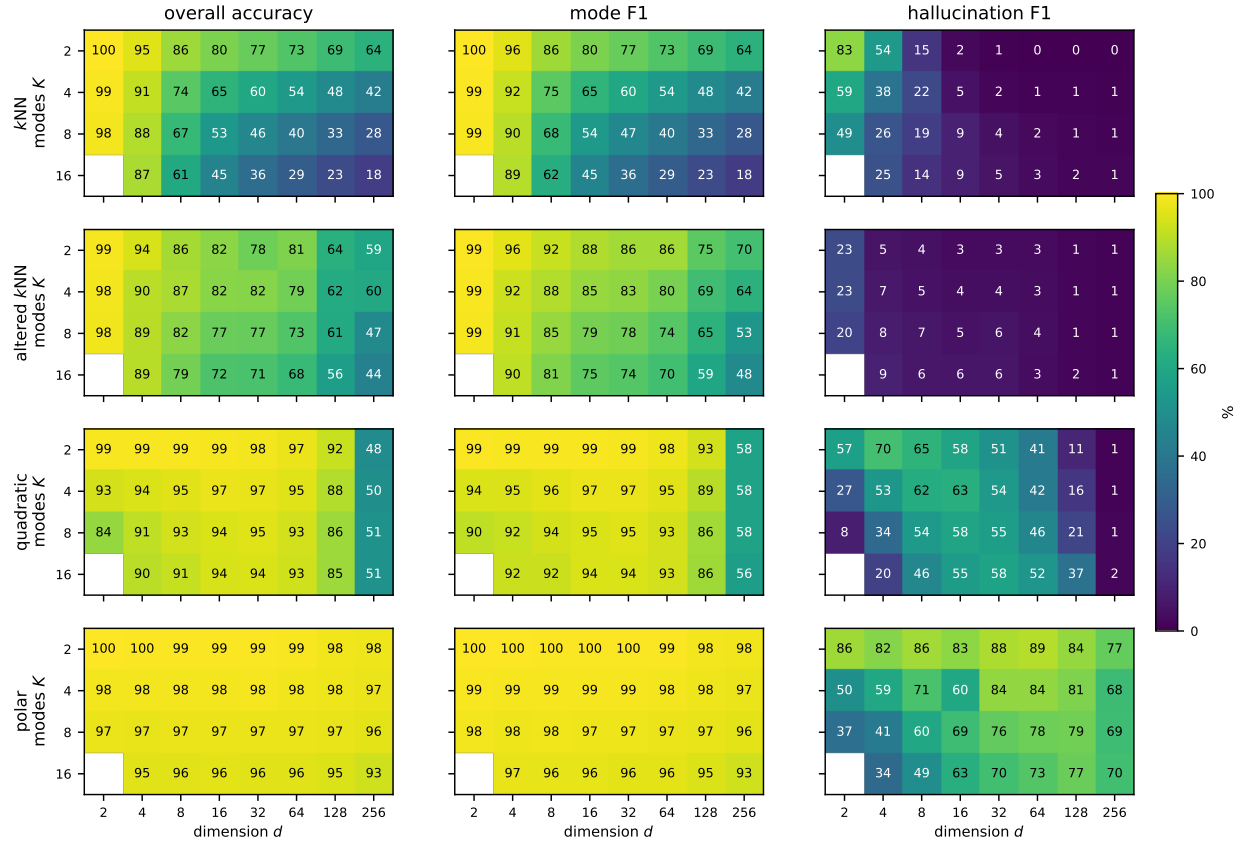


Figure 47: **Unweighted Flow matching, $T = 750$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

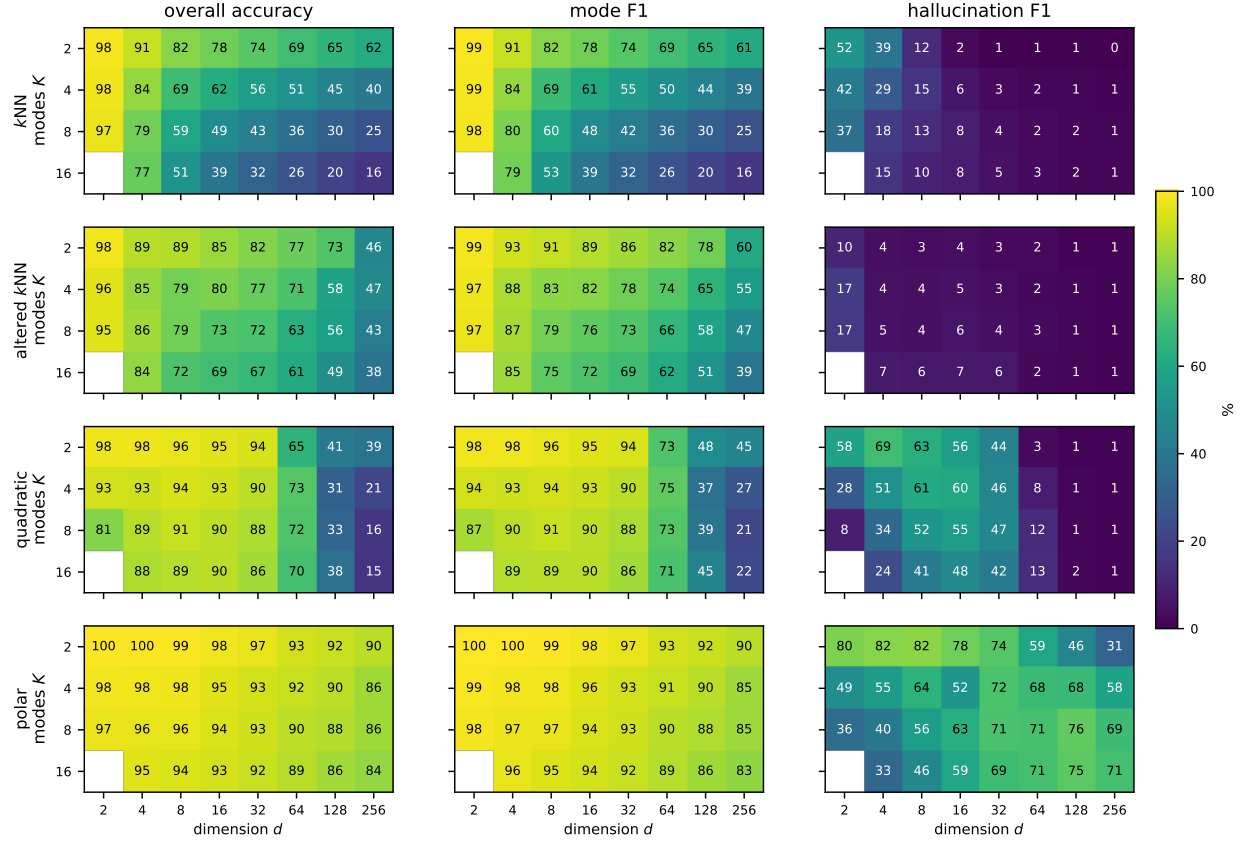


Figure 48: **Weighted Flow matching, $T = 250$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.10 Weighted Flow matching, $T = 250$

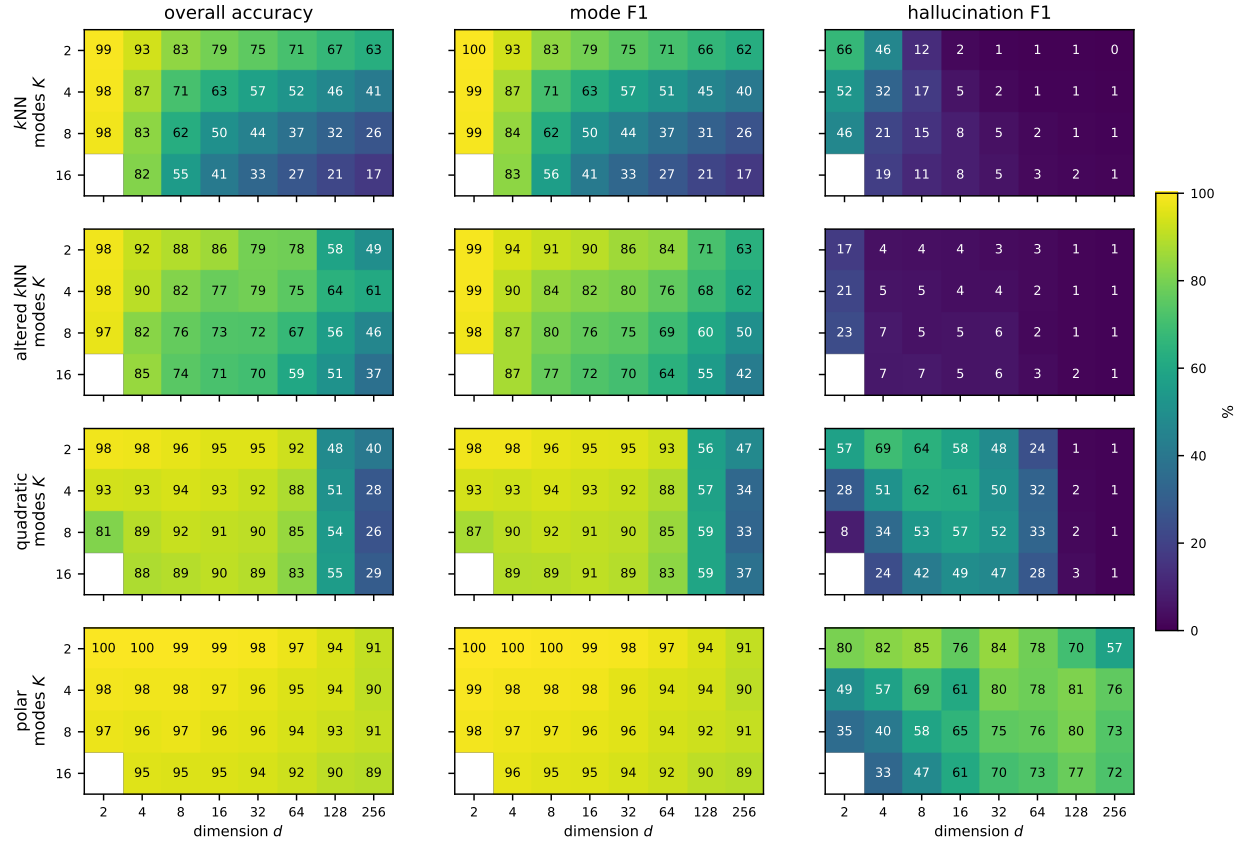


Figure 49: **Weighted Flow matching**, $T = 250$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

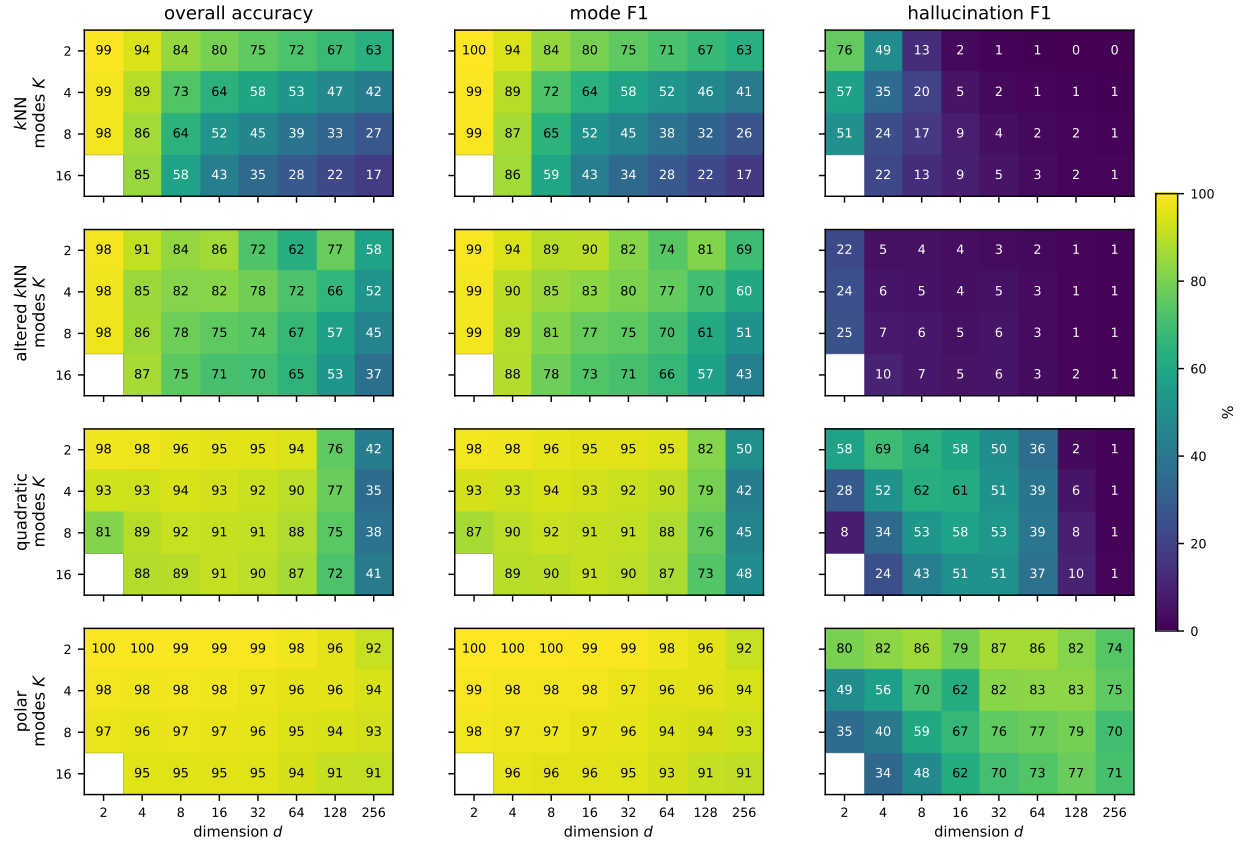


Figure 50: **Weighted Flow matching**, $T = 250$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

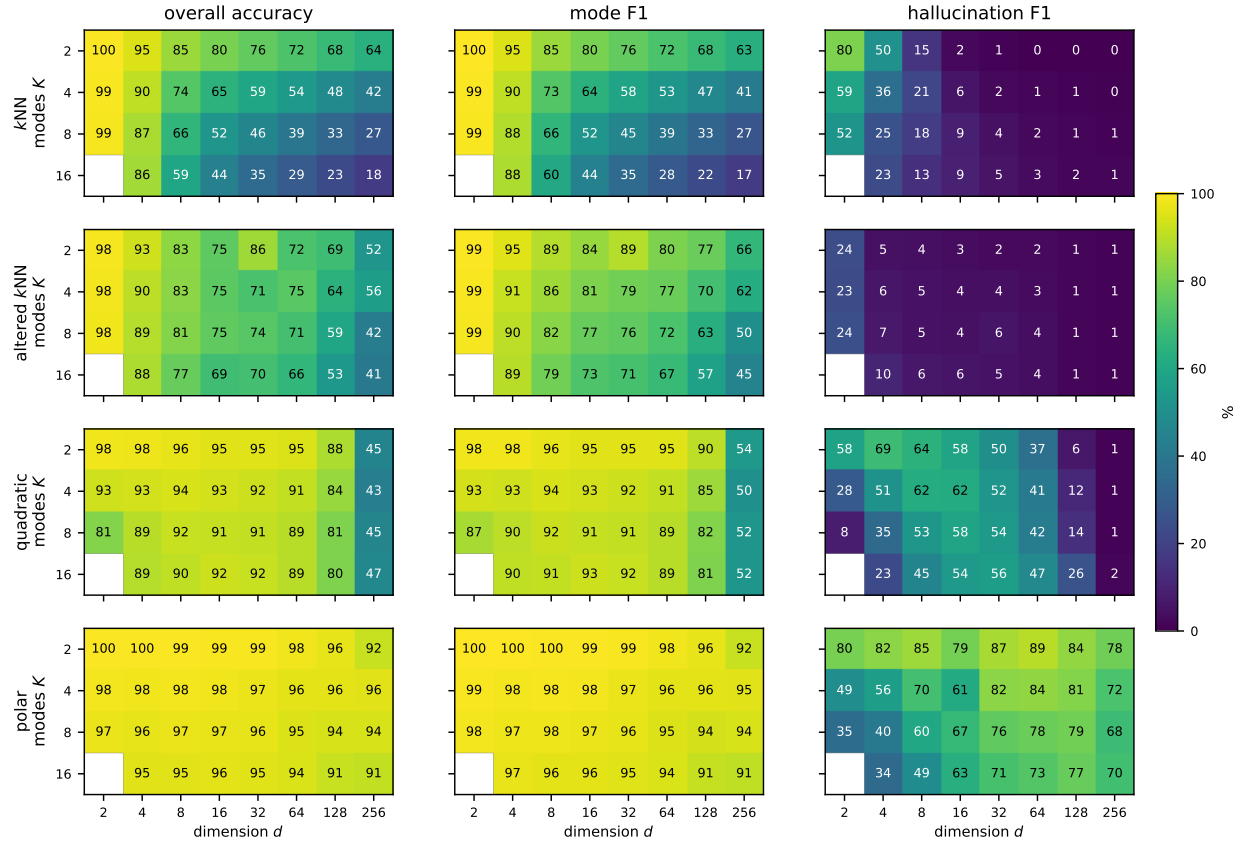


Figure 51: **Weighted Flow matching**, $T = 250$, **15,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

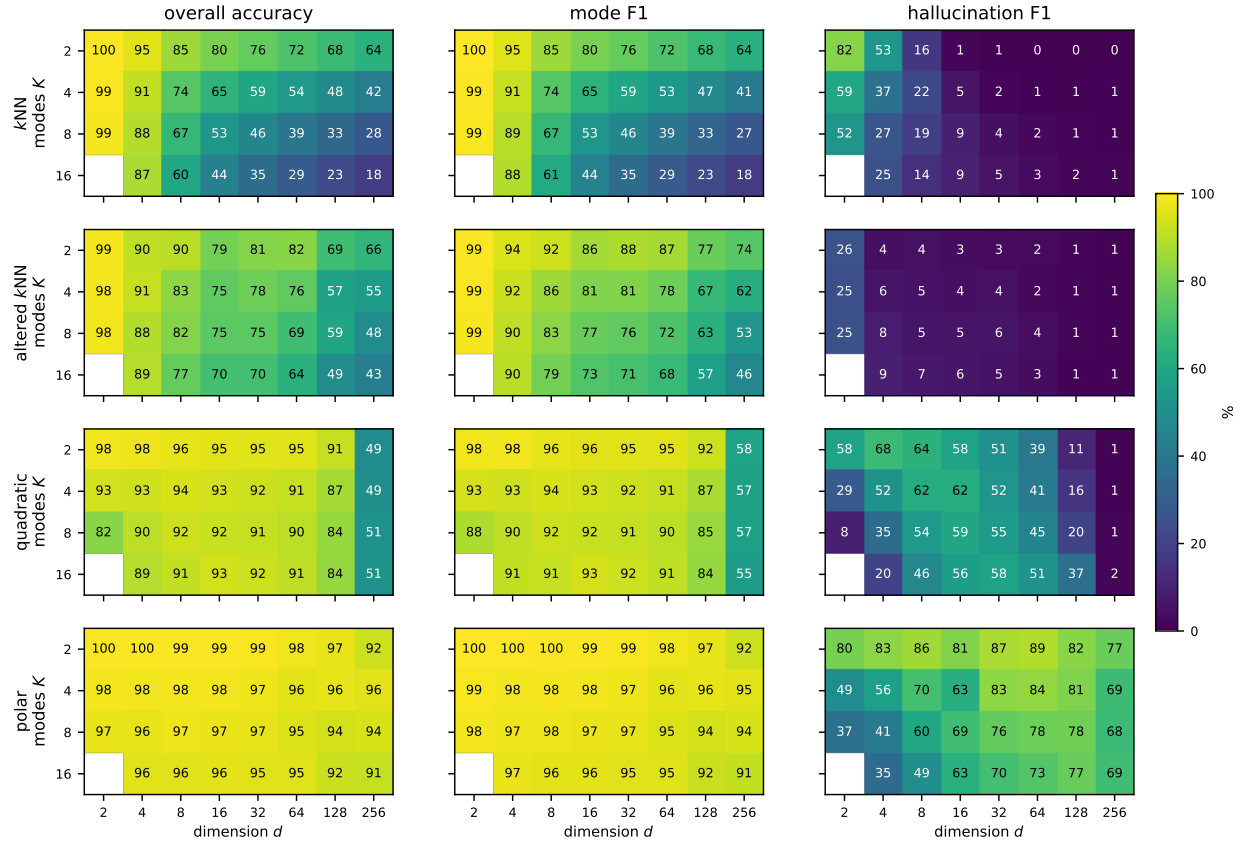


Figure 52: **Weighted Flow matching**, $T = 250$, **20,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

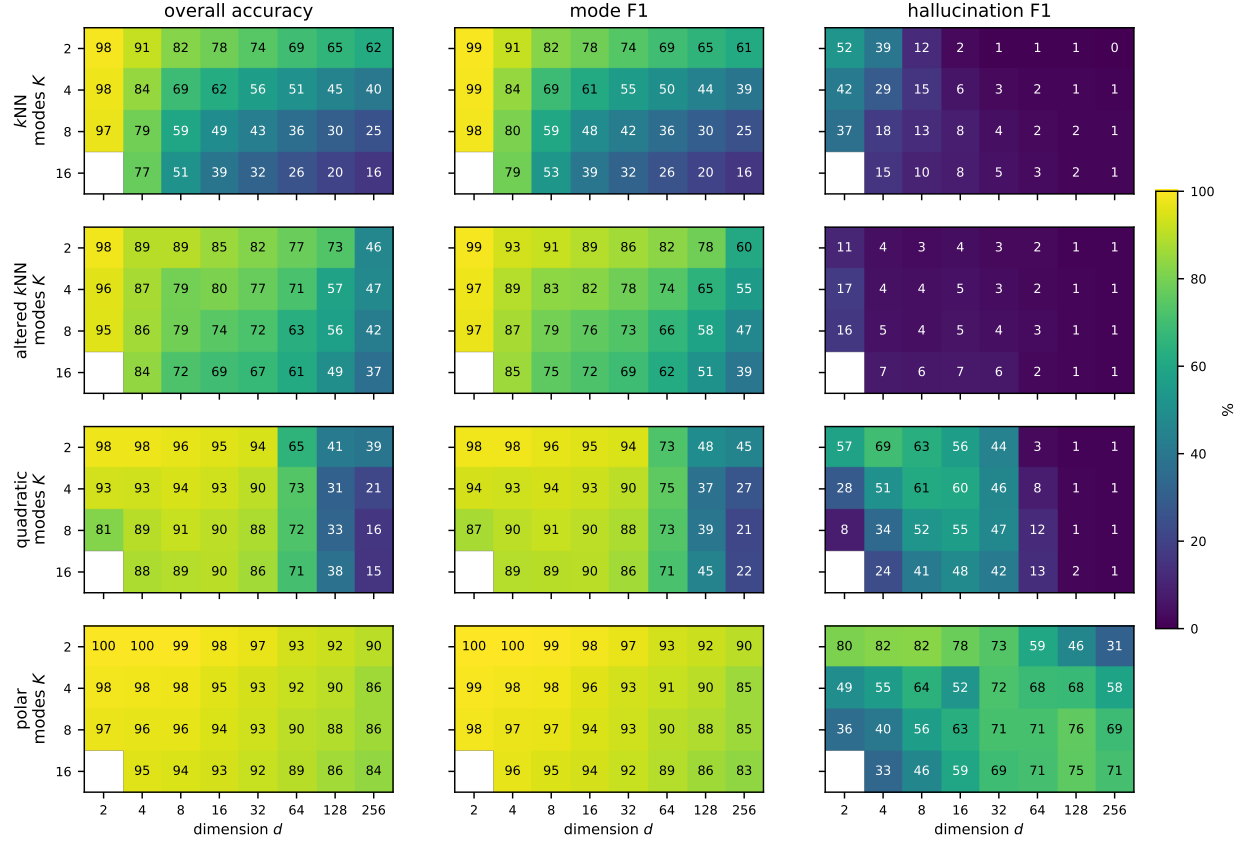


Figure 53: **Weighted Flow matching, $T = 500$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.11 Weighted Flow matching, $T = 500$

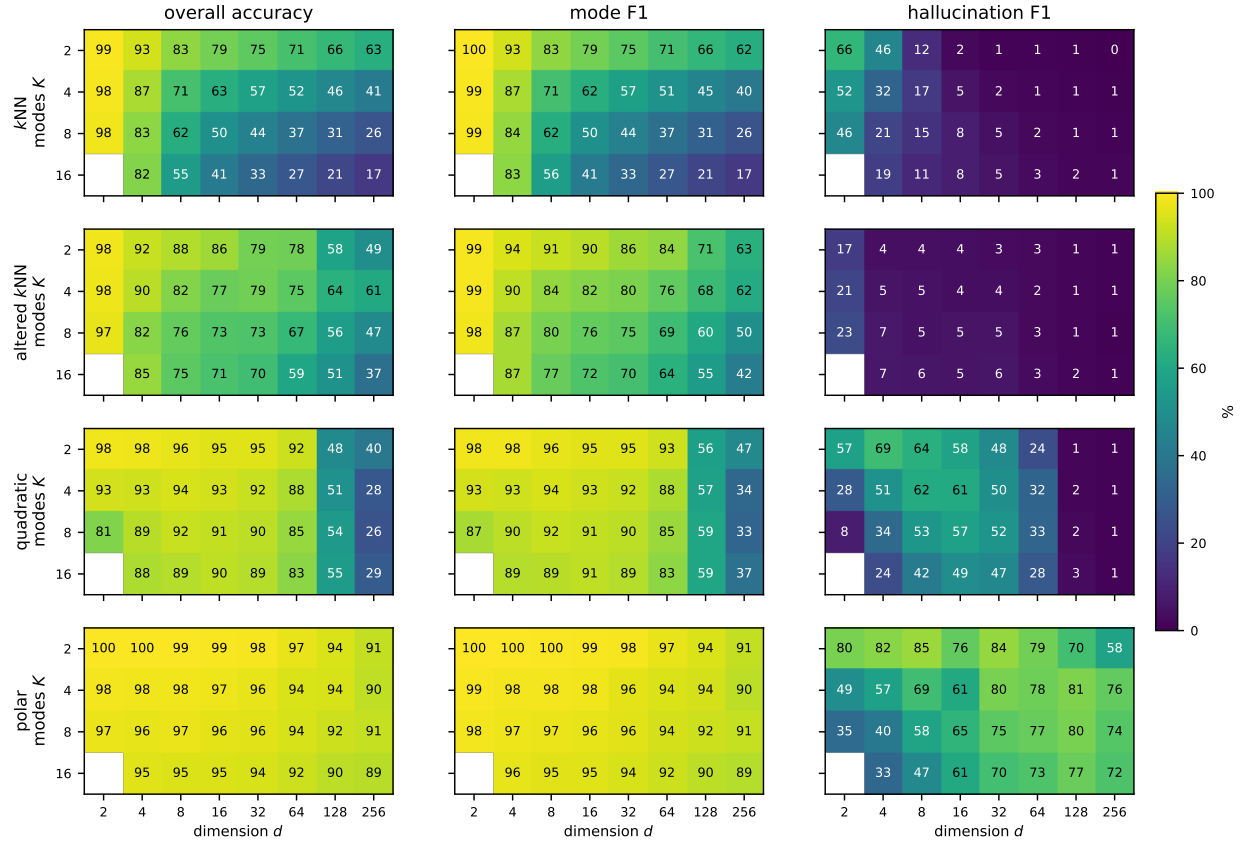


Figure 54: **Weighted Flow matching**, $T = 500$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

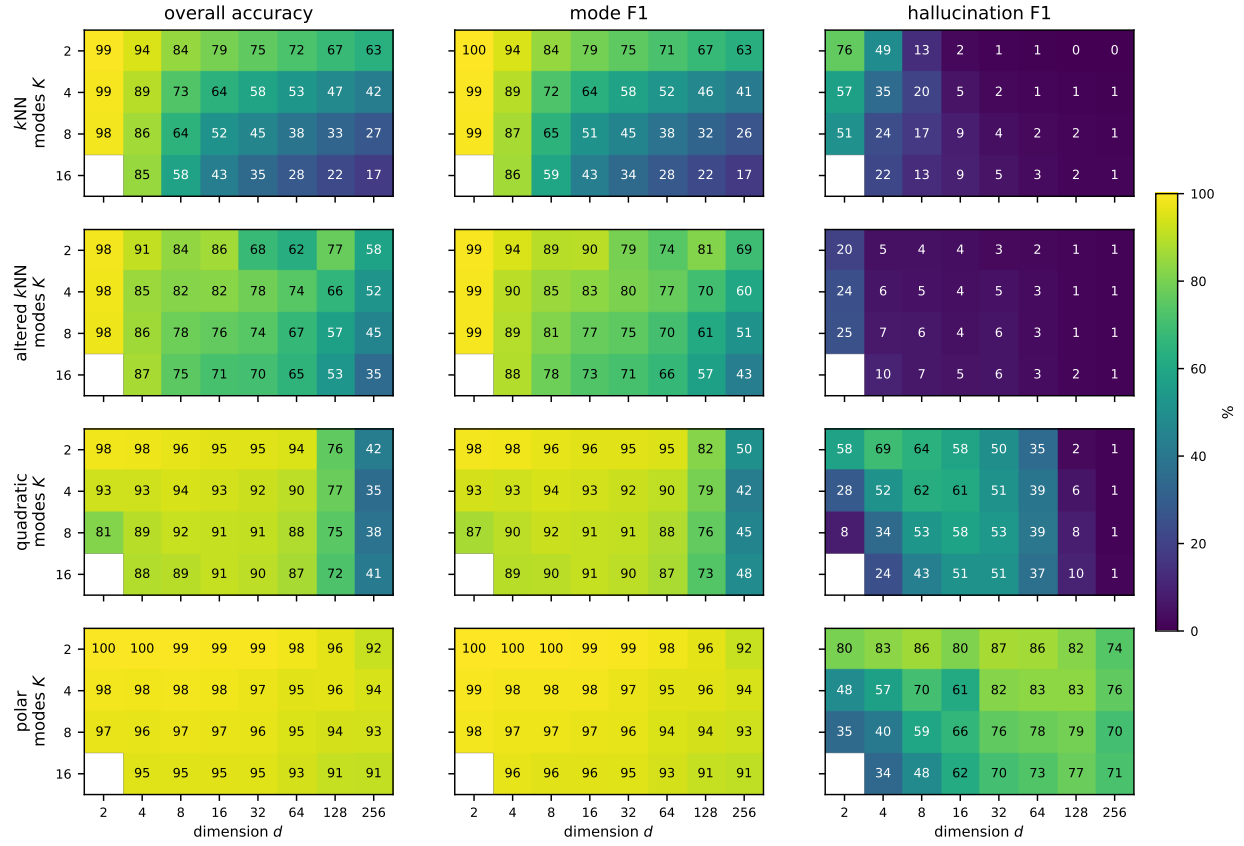


Figure 55: **Weighted Flow matching**, $T = 500$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

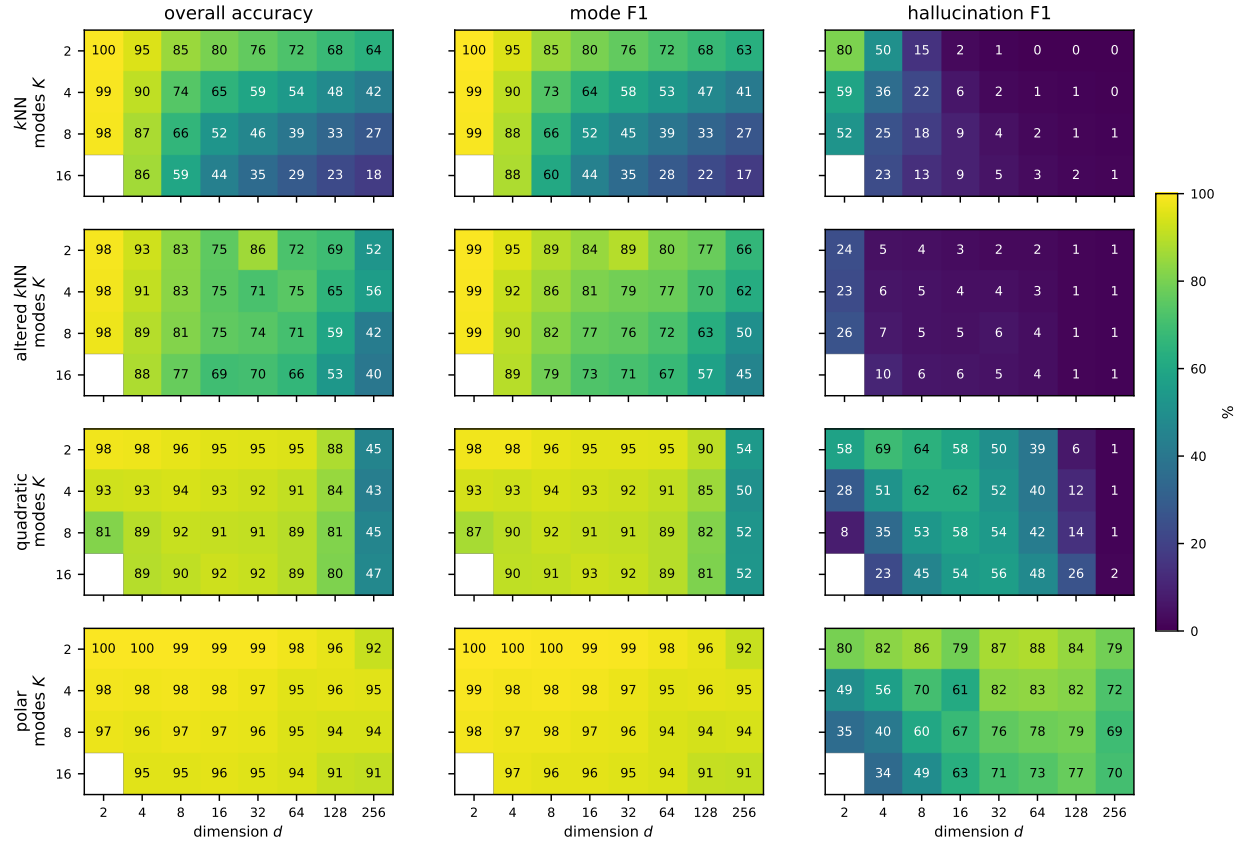


Figure 56: **Weighted Flow matching**, $T = 500$, **15,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

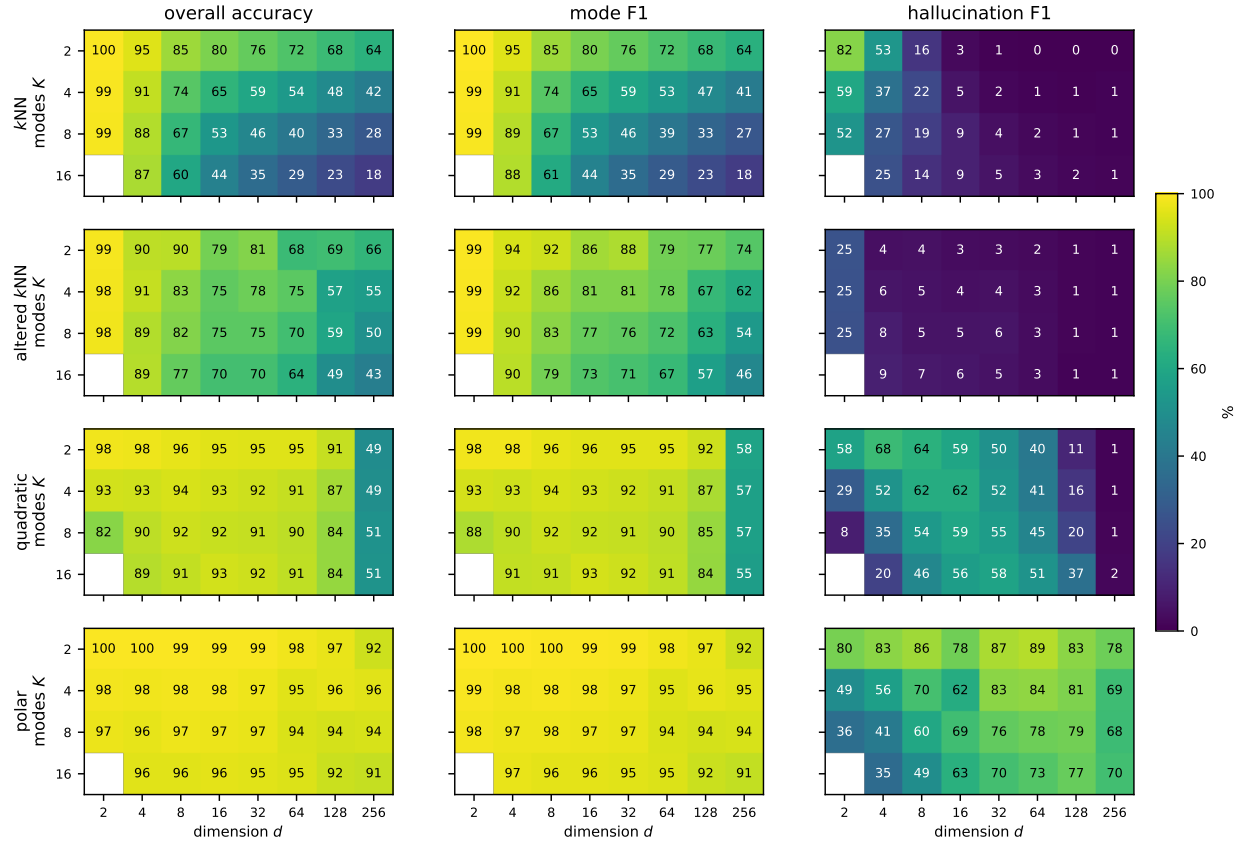


Figure 57: **Weighted Flow matching**, $T = 500$, **20,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

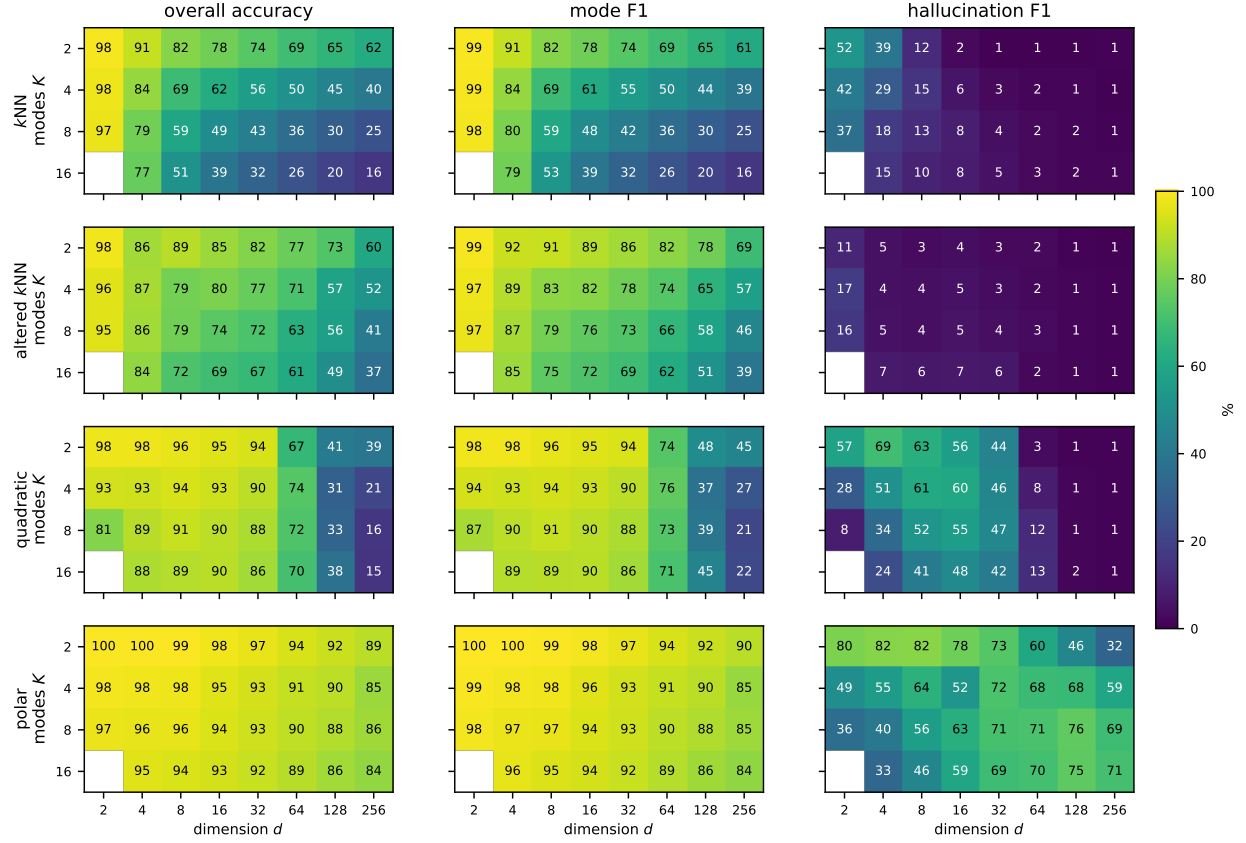


Figure 58: **Weighted Flow matching, $T = 750$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.12 Weighted Flow matching, $T = 750$

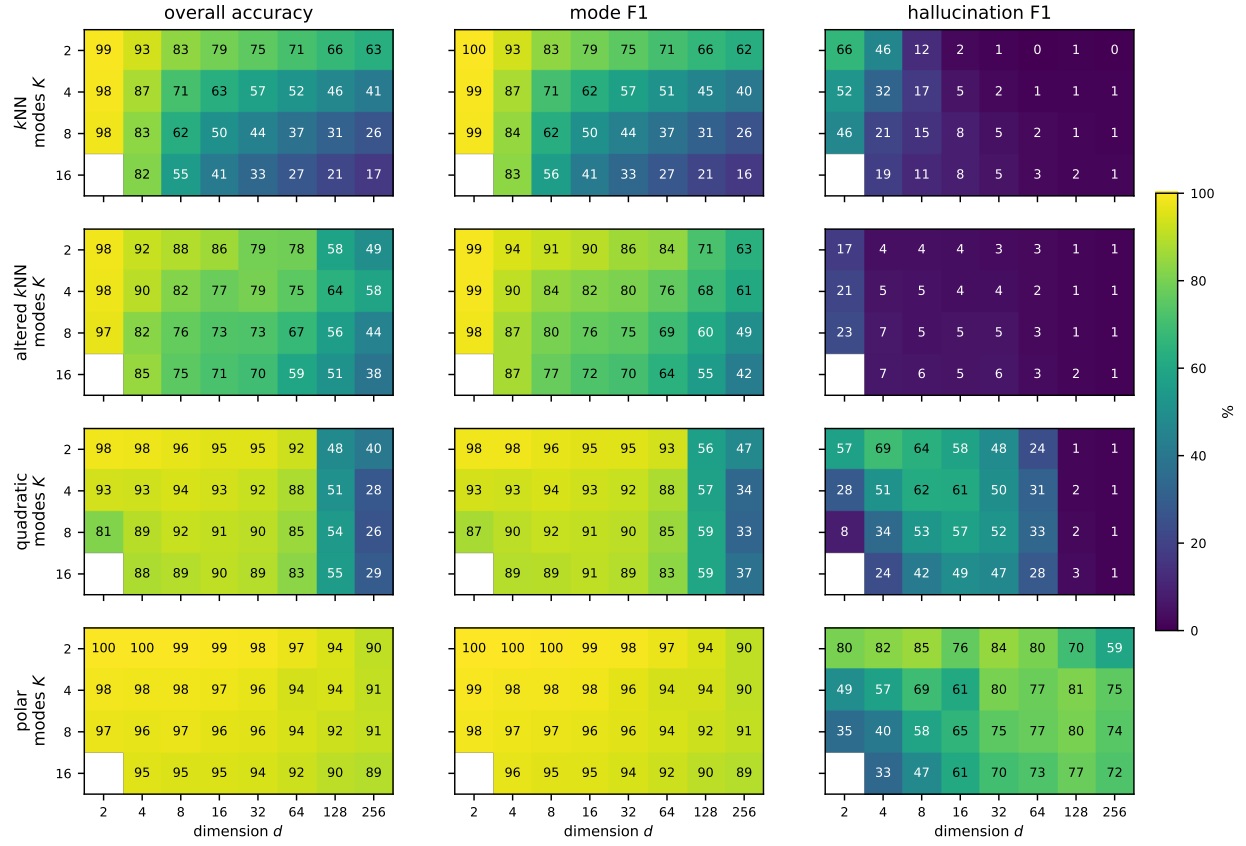


Figure 59: **Weighted Flow matching**, $T = 750$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

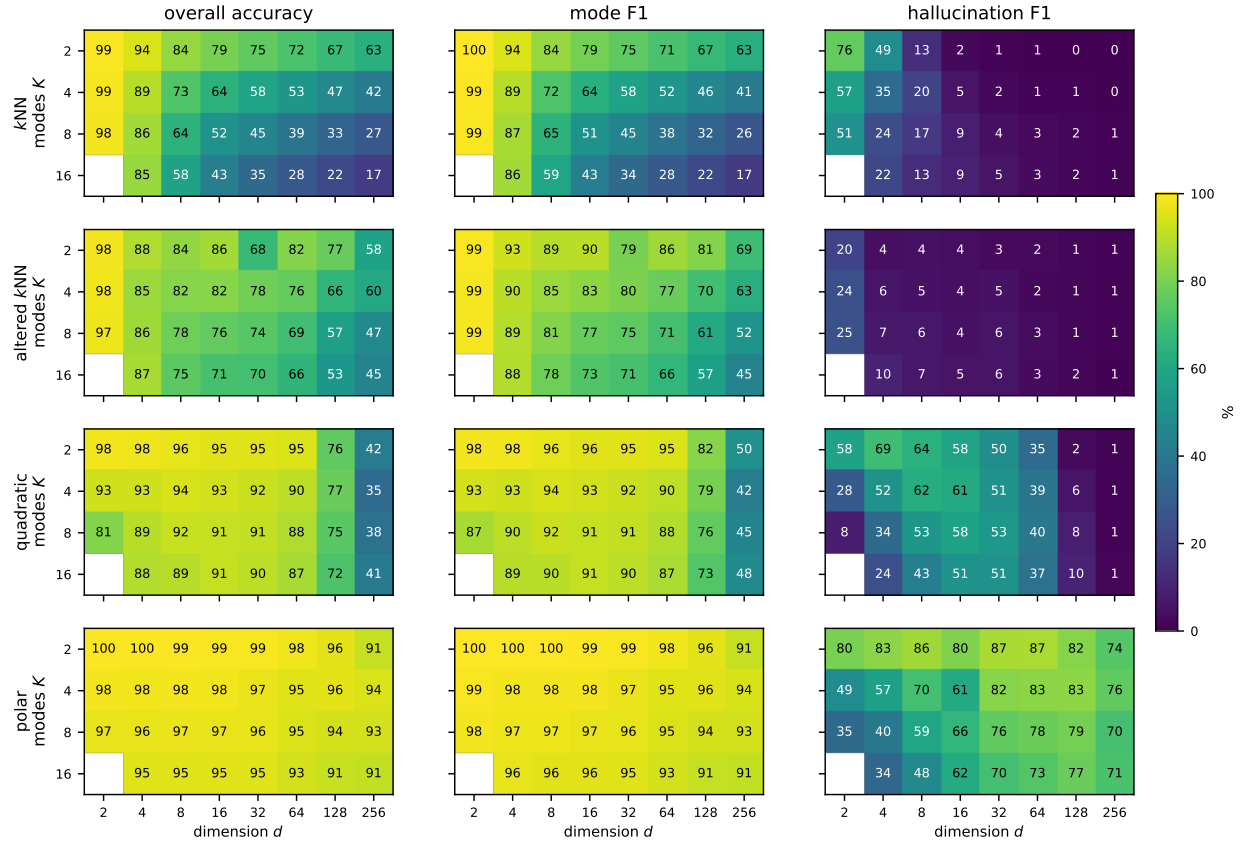


Figure 60: **Weighted Flow matching**, $T = 750$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

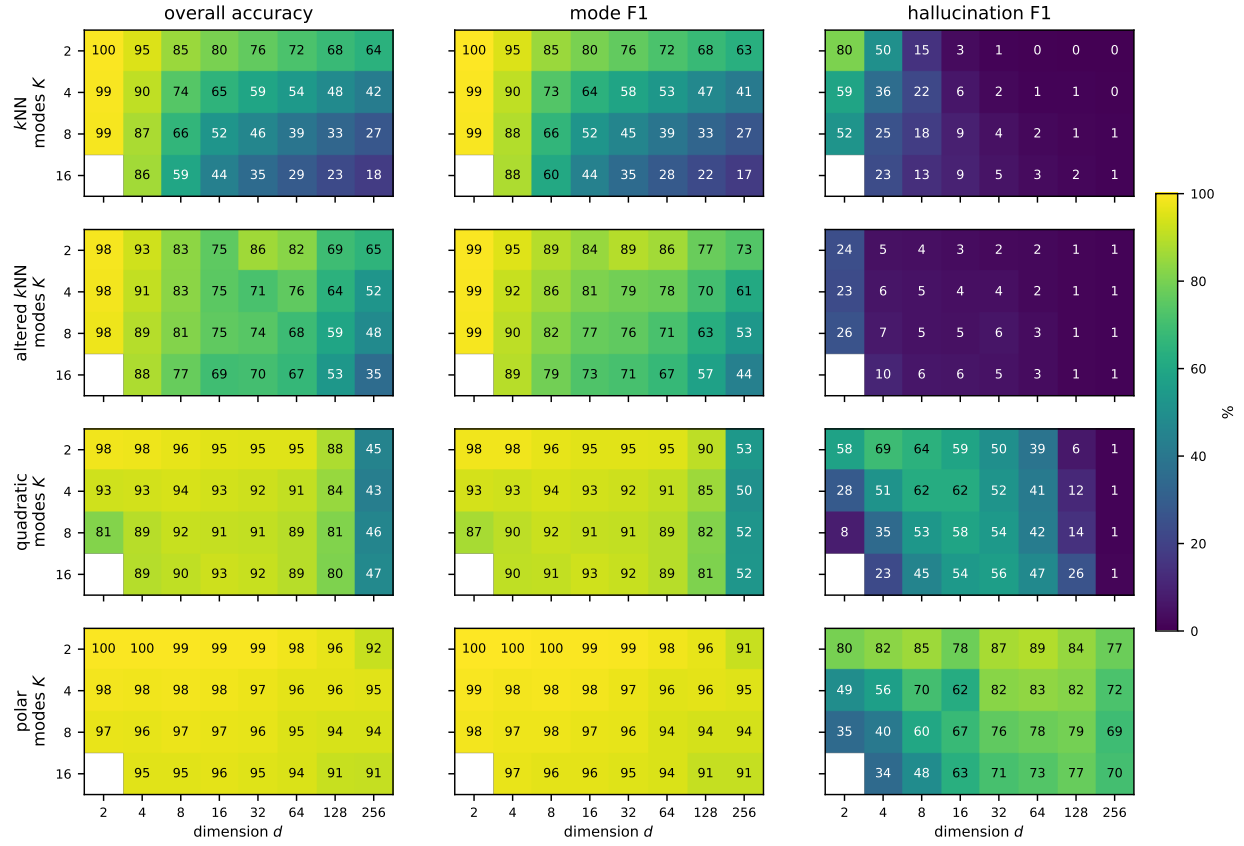


Figure 61: **Weighted Flow matching, $T = 750$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

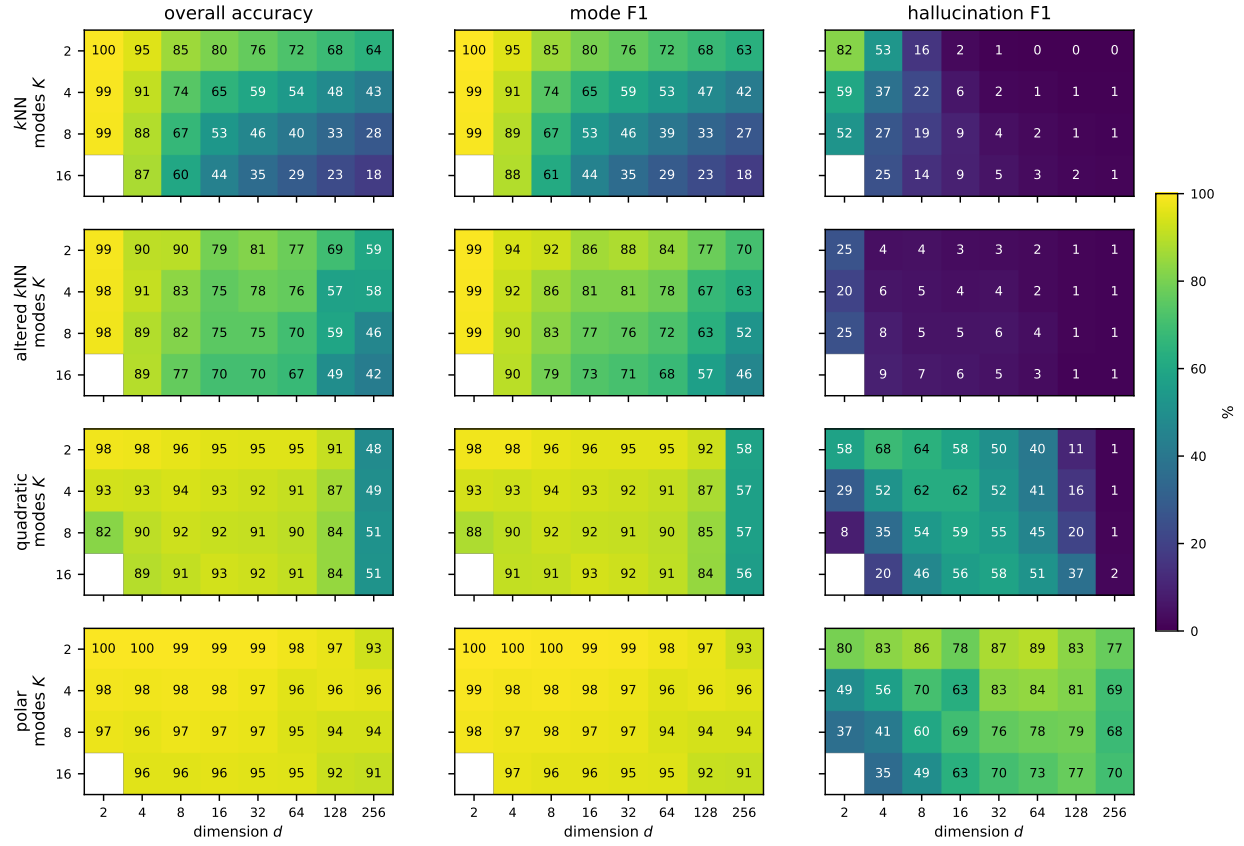


Figure 62: **Weighted Flow matching**, $T = 750$, **20,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

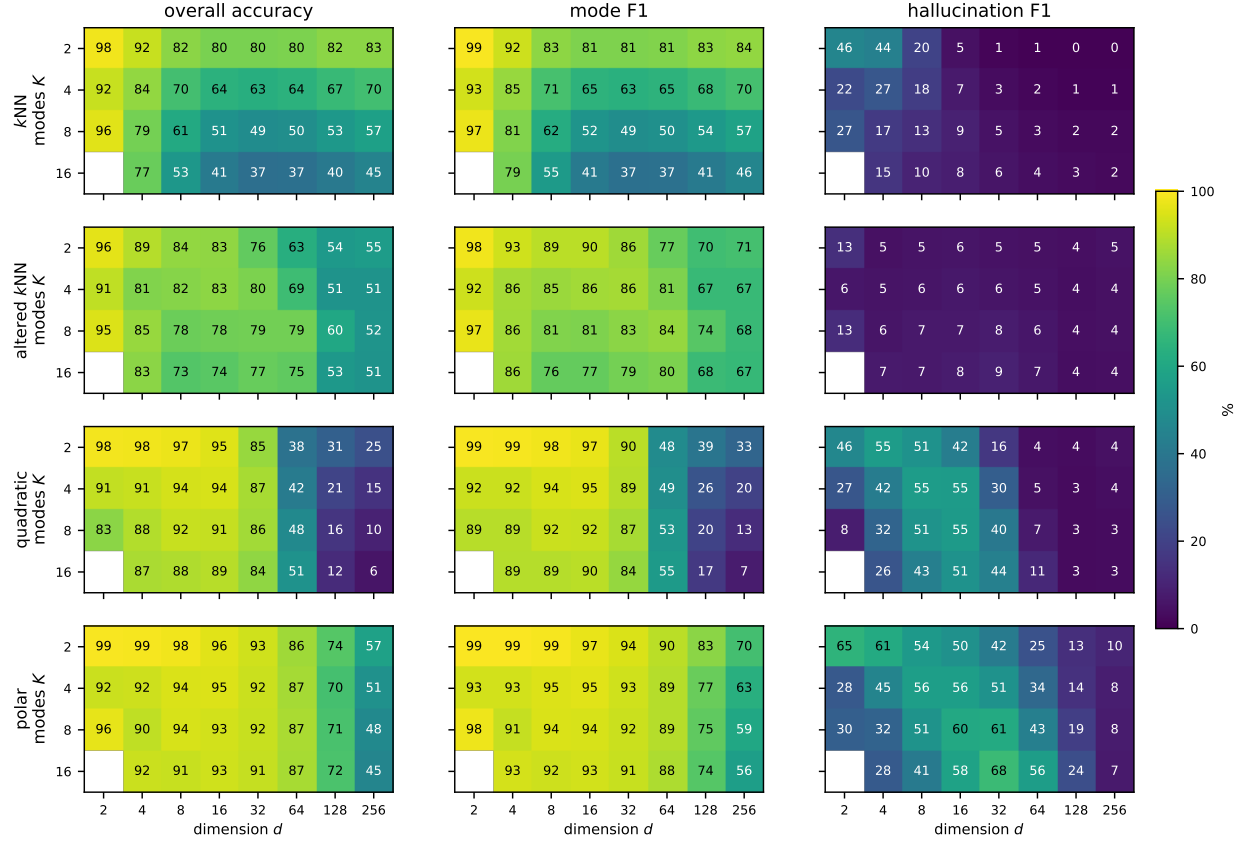


Figure 63: **Unweighted Heun, $T = 250$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.13 Unweighted Heun, $T = 250$

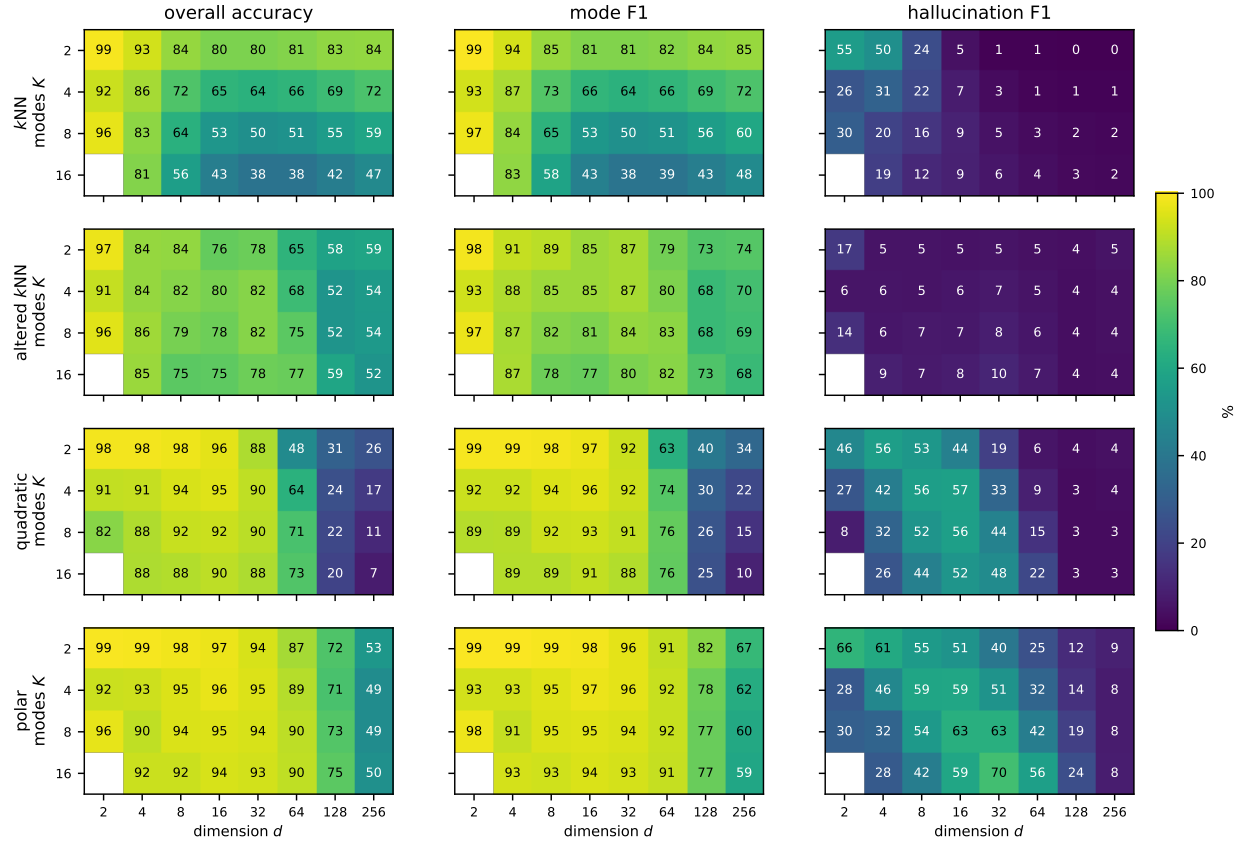


Figure 64: **Unweighted Heun, $T = 250$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

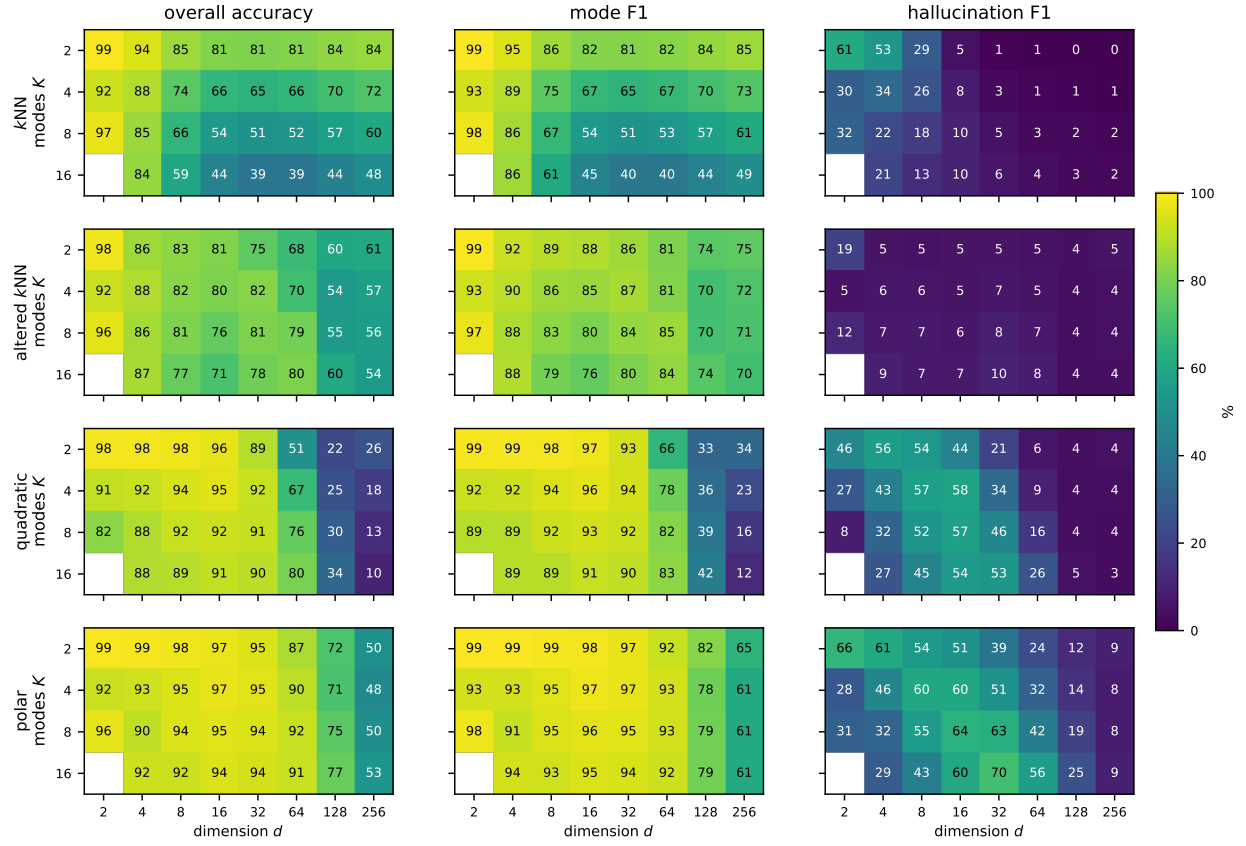


Figure 65: **Unweighted Heun, $T = 250$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

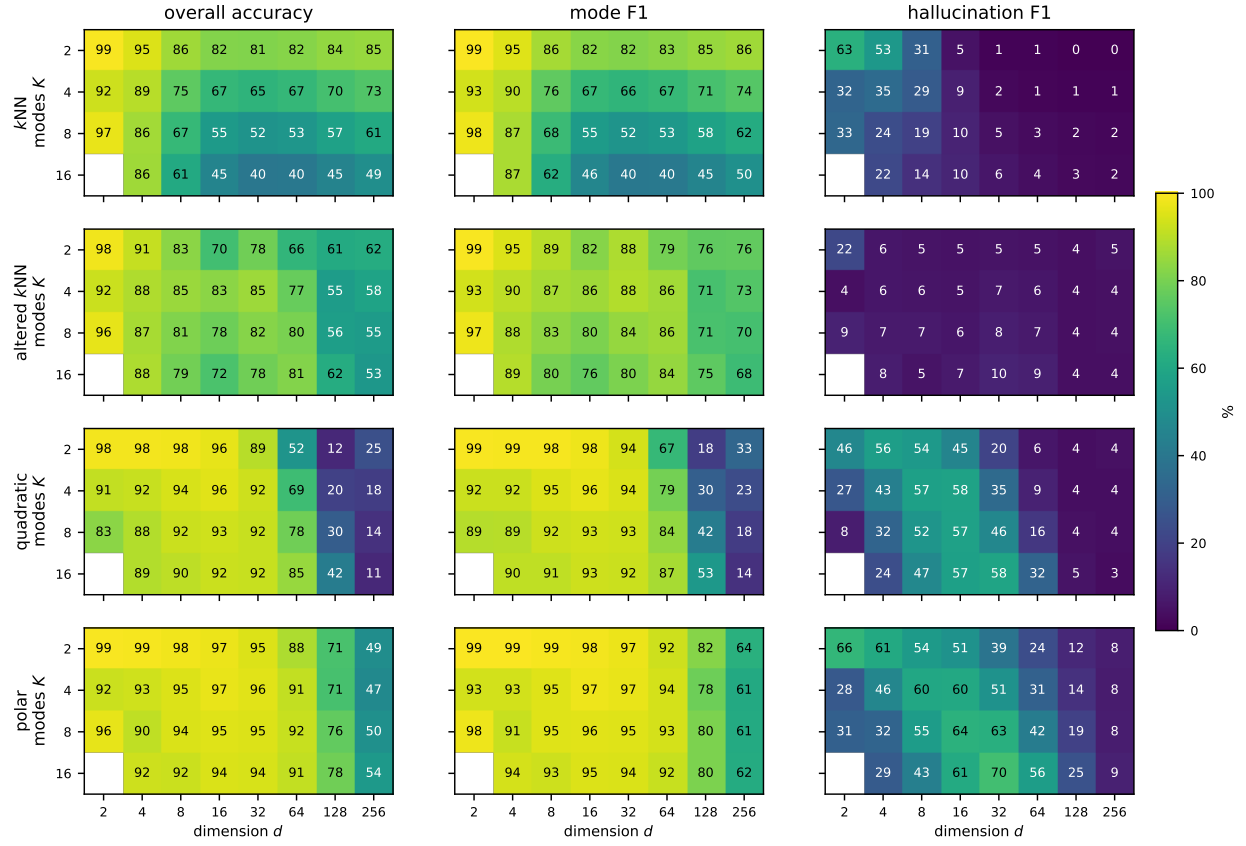


Figure 66: **Unweighted Heun**, $T = 250$, **15,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

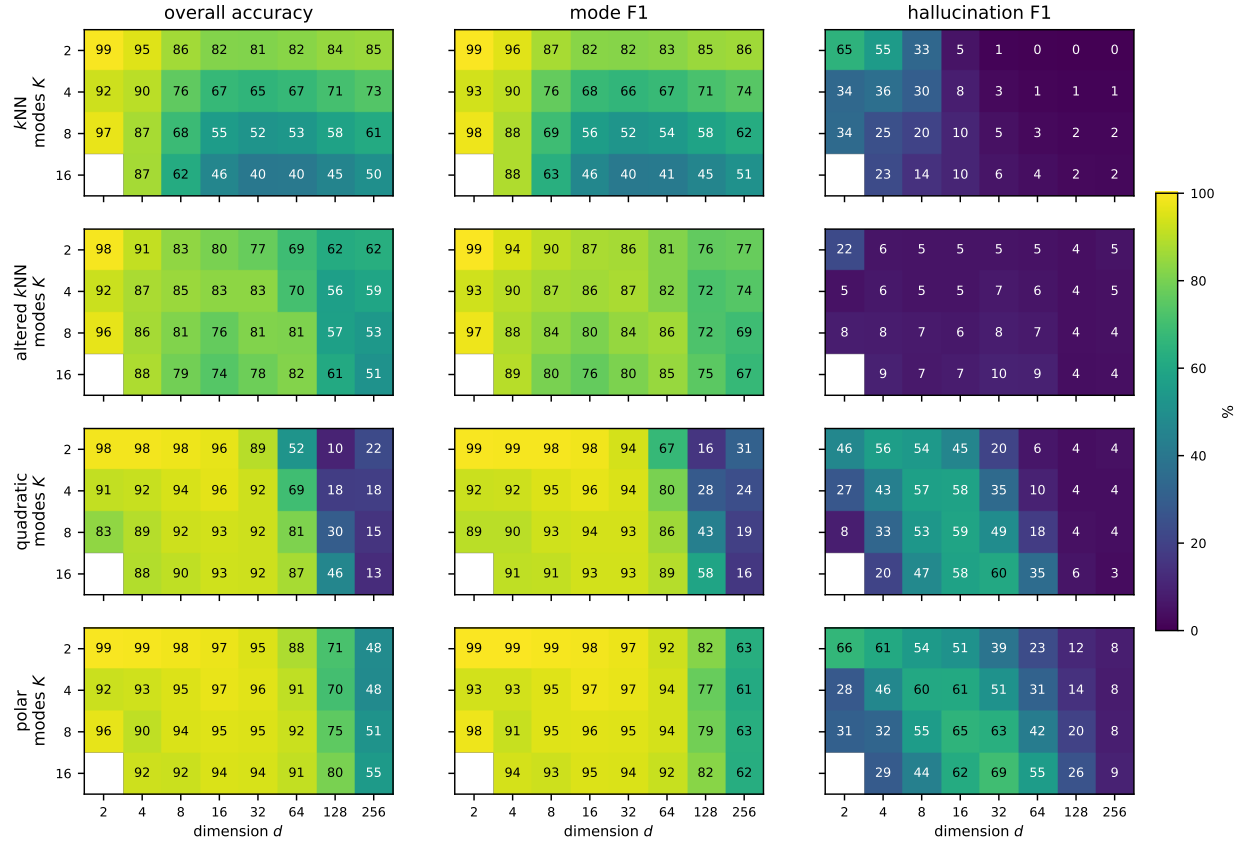


Figure 67: **Unweighted Heun, $T = 250$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

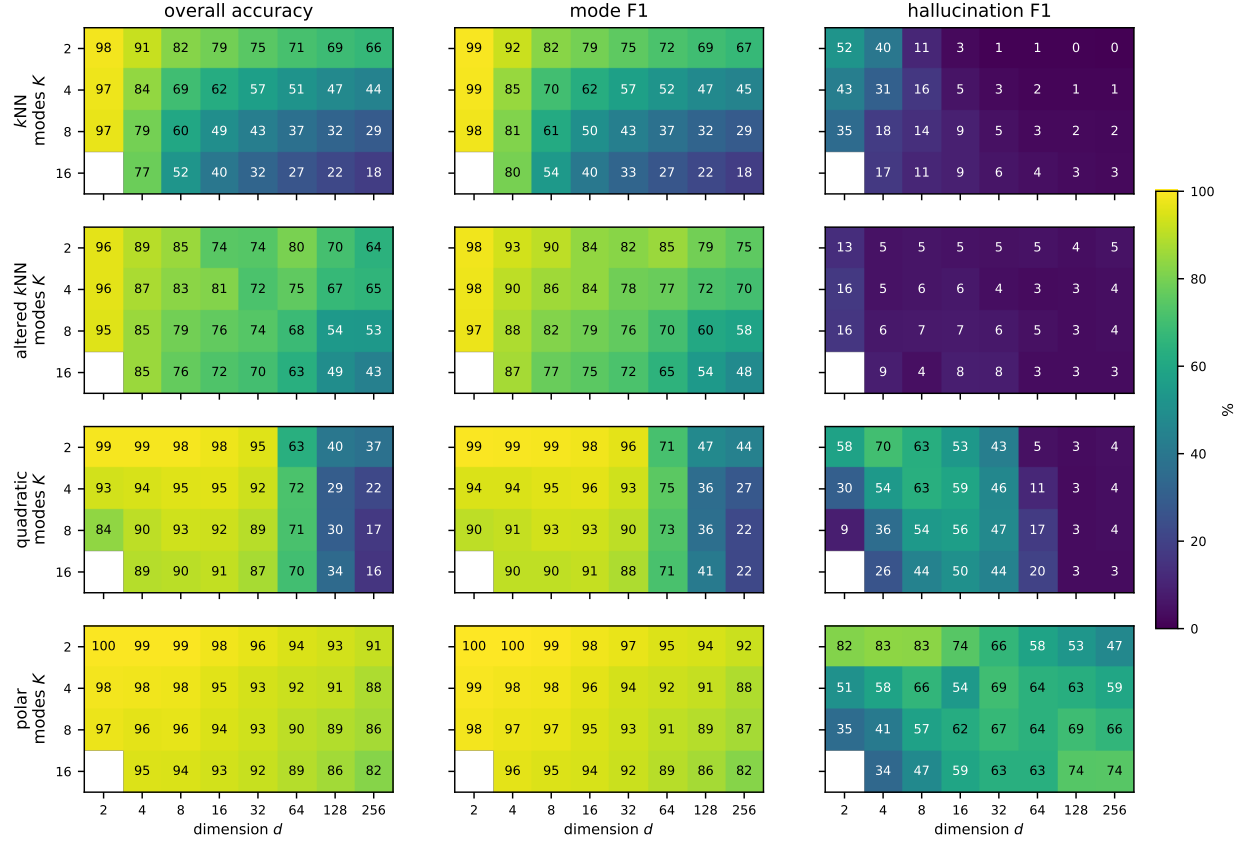


Figure 68: **Unweighted Heun, $T = 500$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.14 Unweighted Heun, $T = 500$

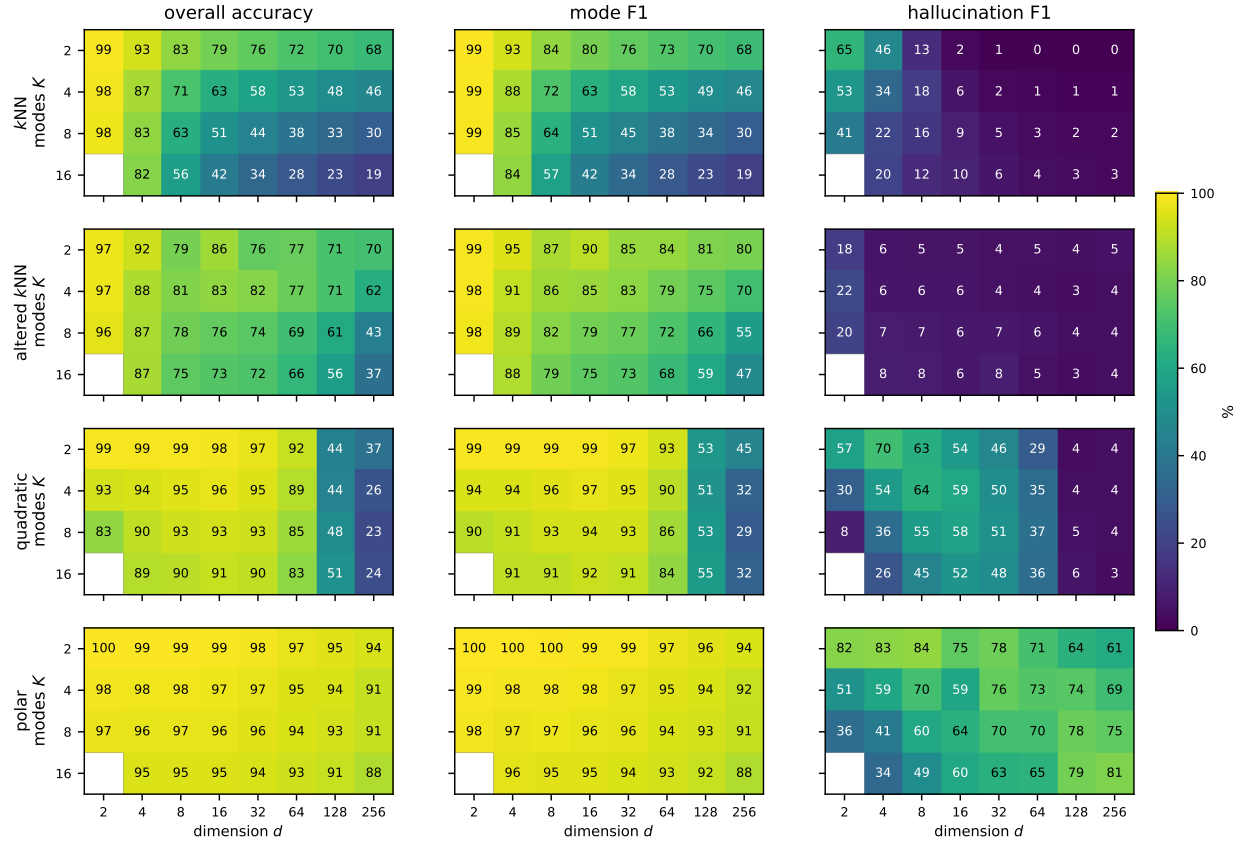


Figure 69: **Unweighted Heun, $T = 500$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

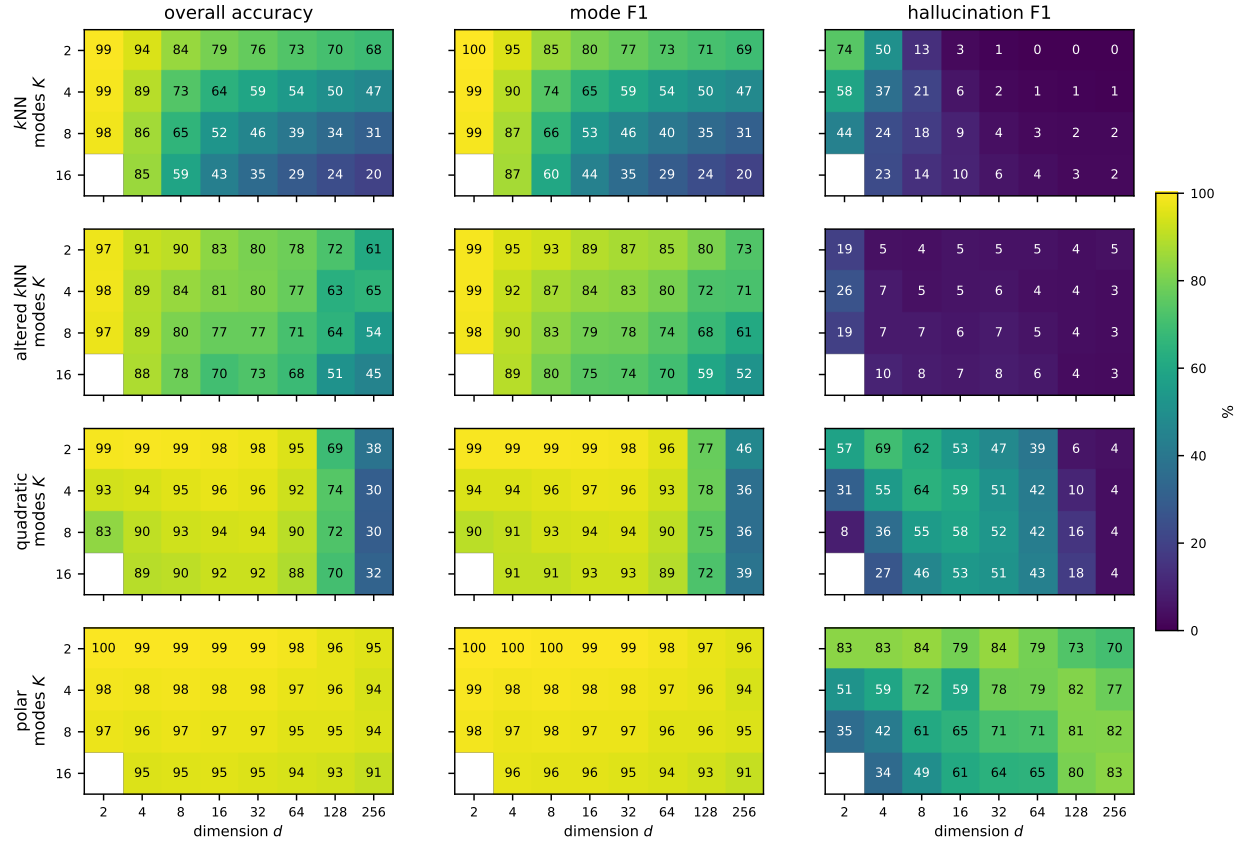


Figure 70: **Unweighted Heun**, $T = 500$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

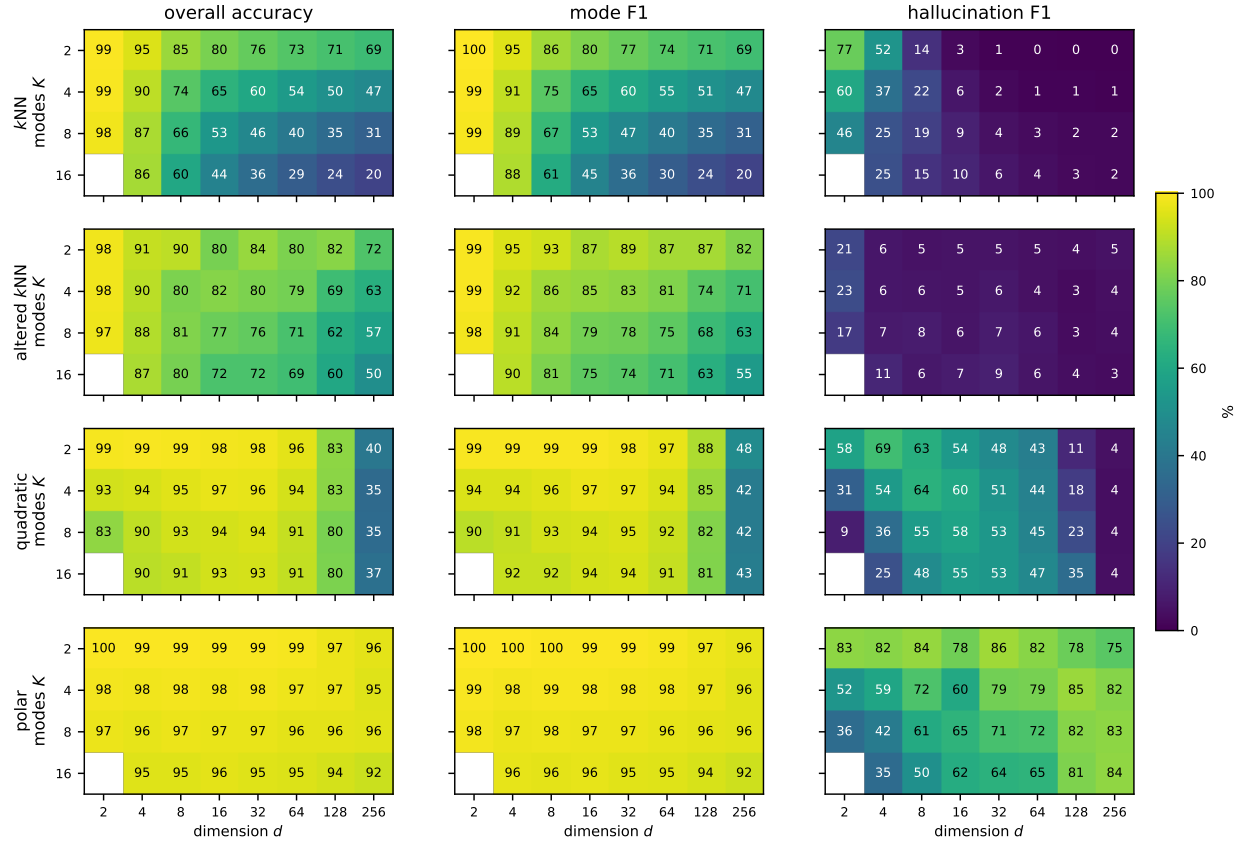


Figure 71: **Unweighted Heun, $T = 500$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

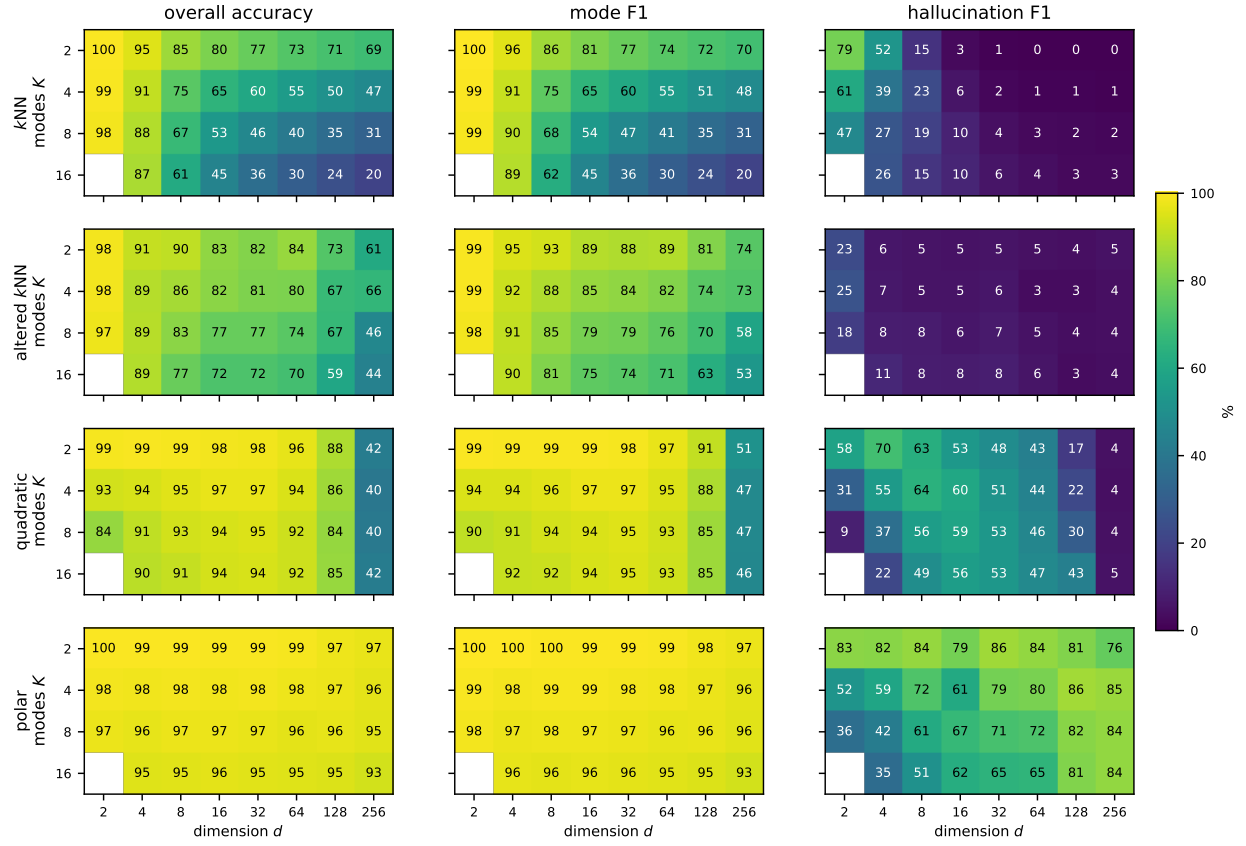


Figure 72: **Unweighted Heun, $T = 500$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

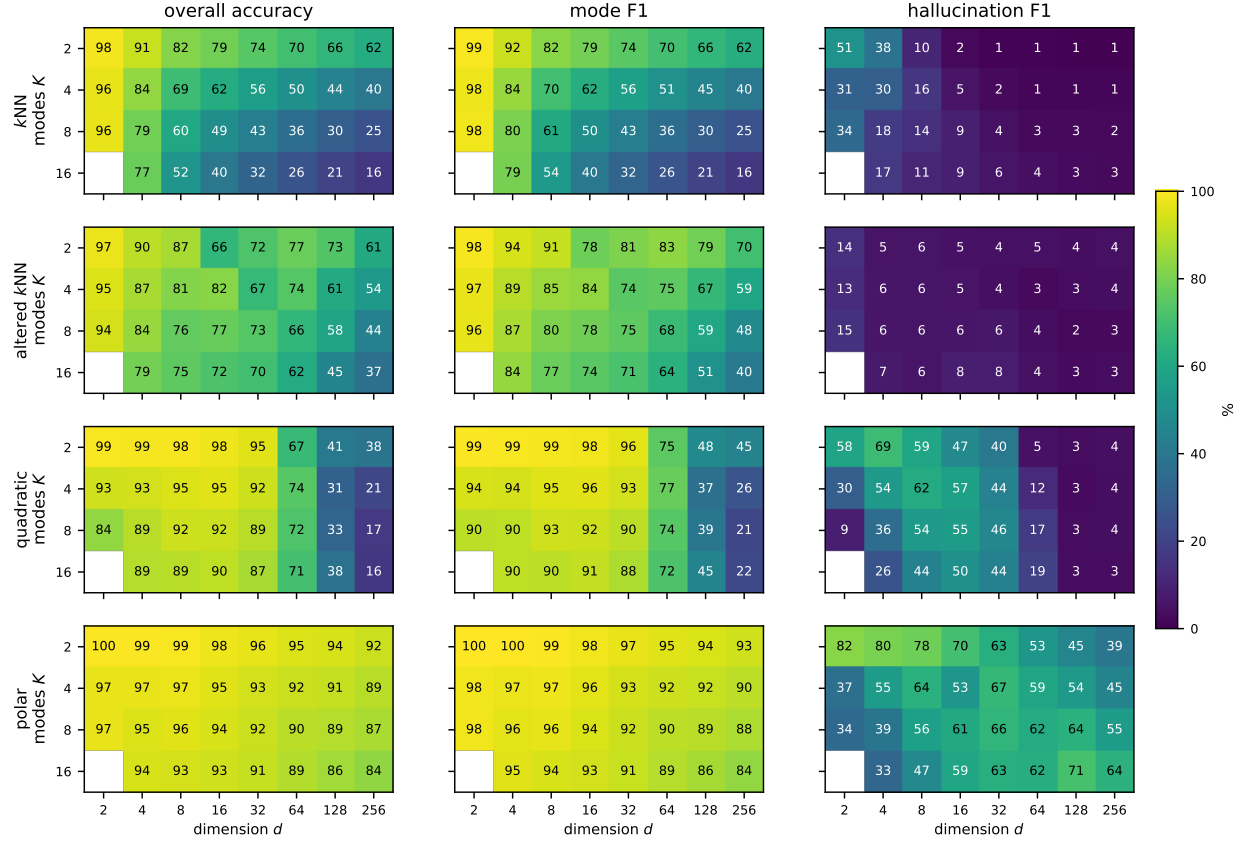


Figure 73: **Unweighted Heun, $T = 750$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.15 Unweighted Heun, $T = 750$

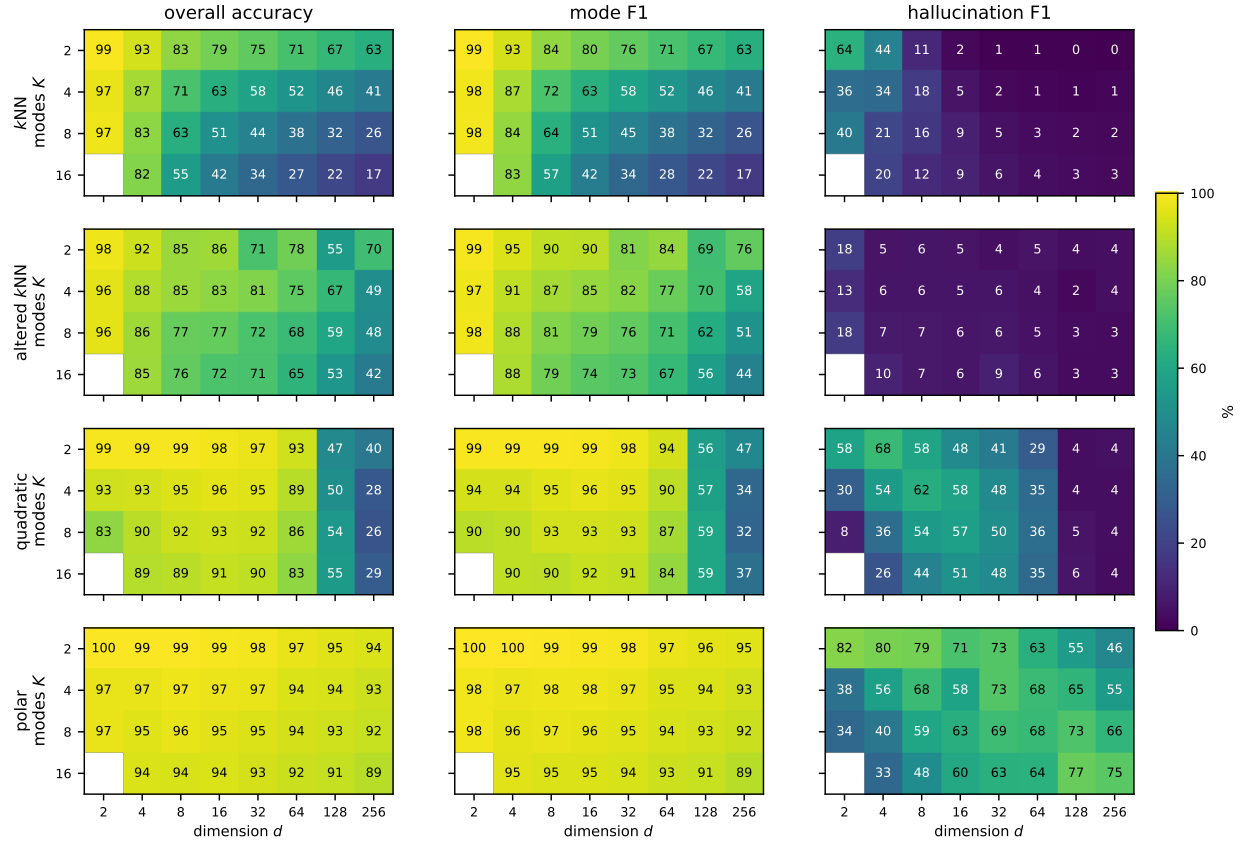


Figure 74: **Unweighted Heun, $T = 750$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

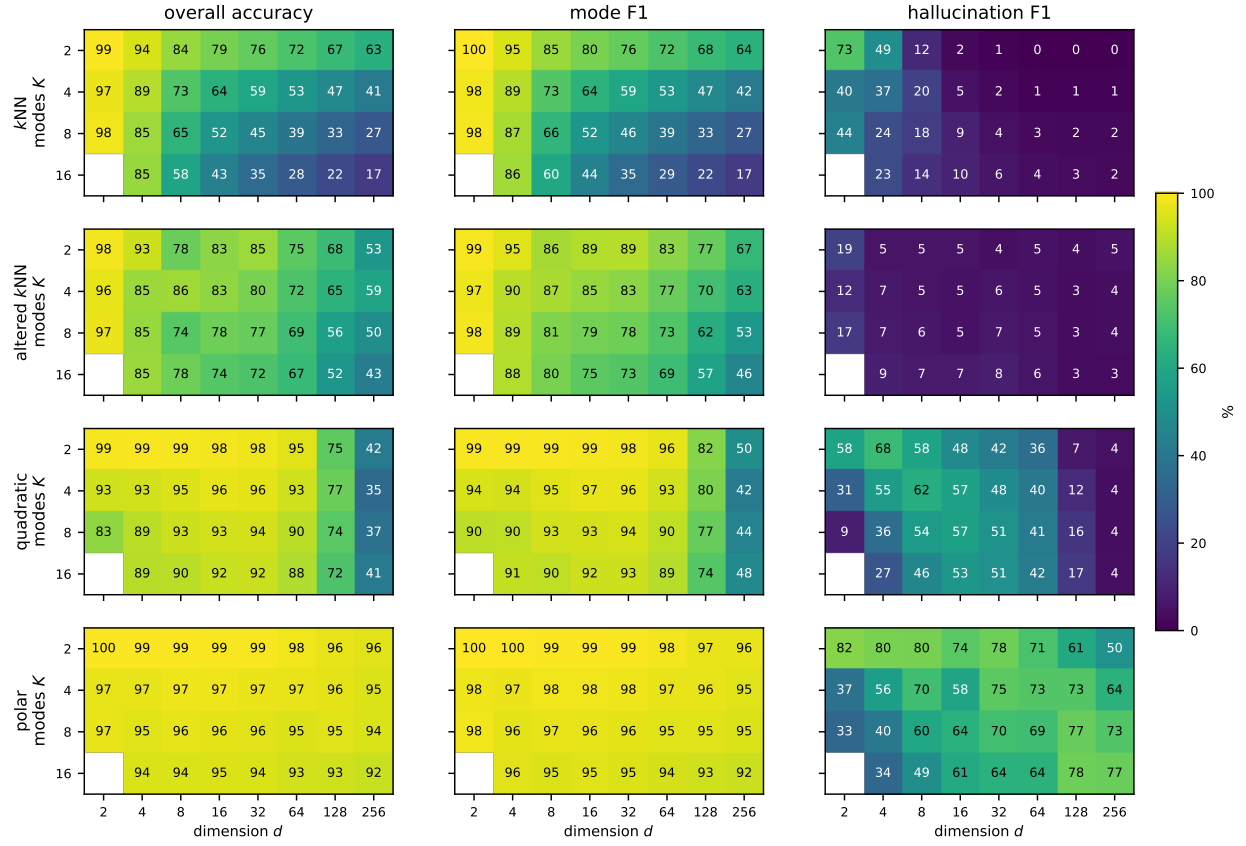


Figure 75: **Unweighted Heun**, $T = 750$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

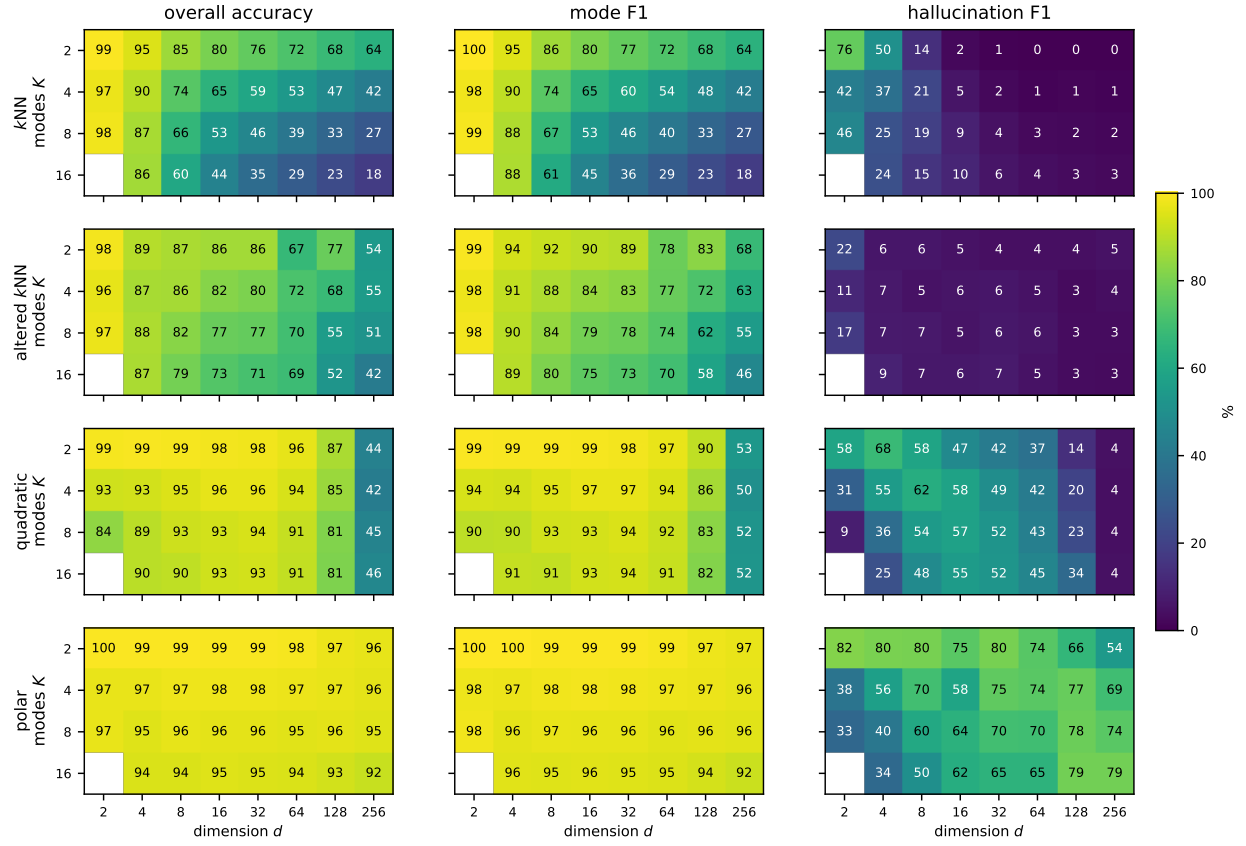


Figure 76: **Unweighted Heun, $T = 750$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

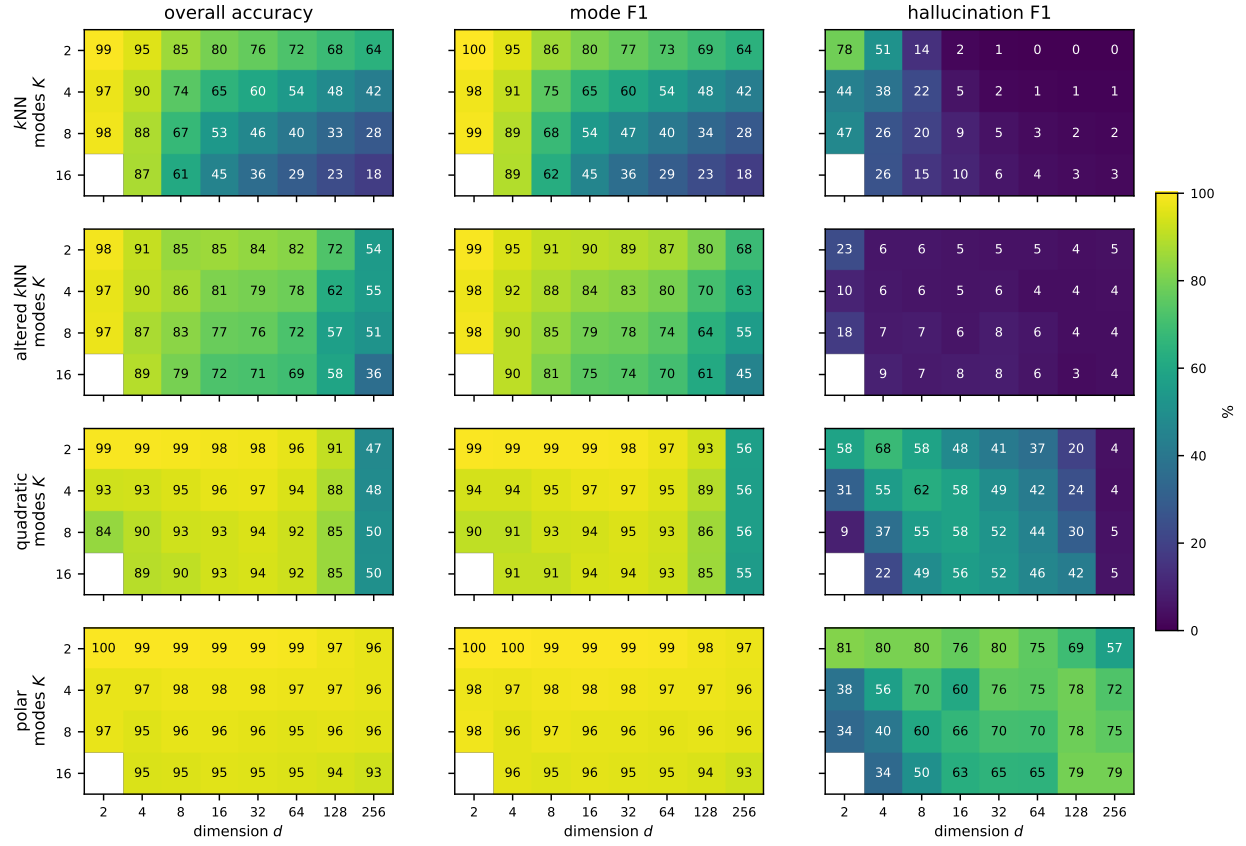


Figure 77: **Unweighted Heun, $T = 750$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

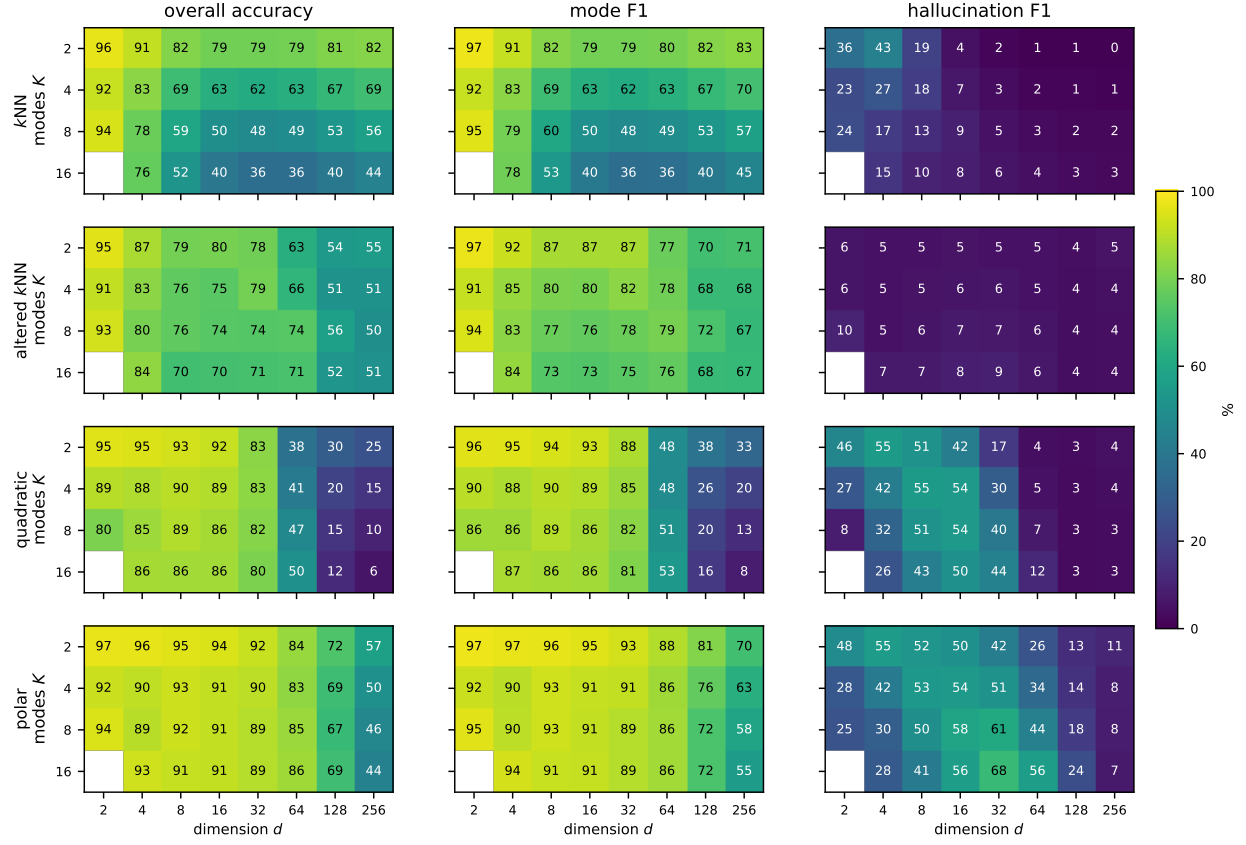


Figure 78: **Weighted Heun, $T = 250$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.16 Weighted Heun, $T = 250$

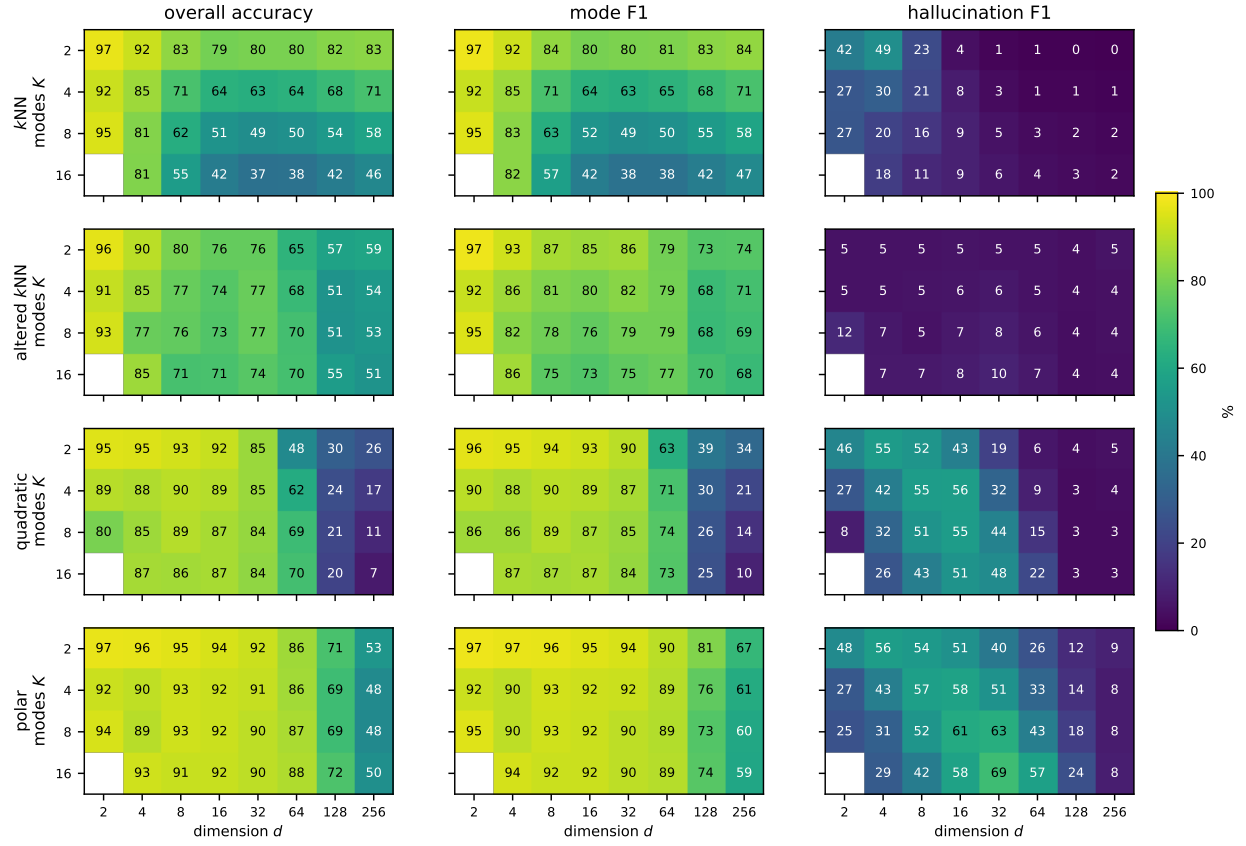


Figure 79: **Weighted Heun**, $T = 250$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

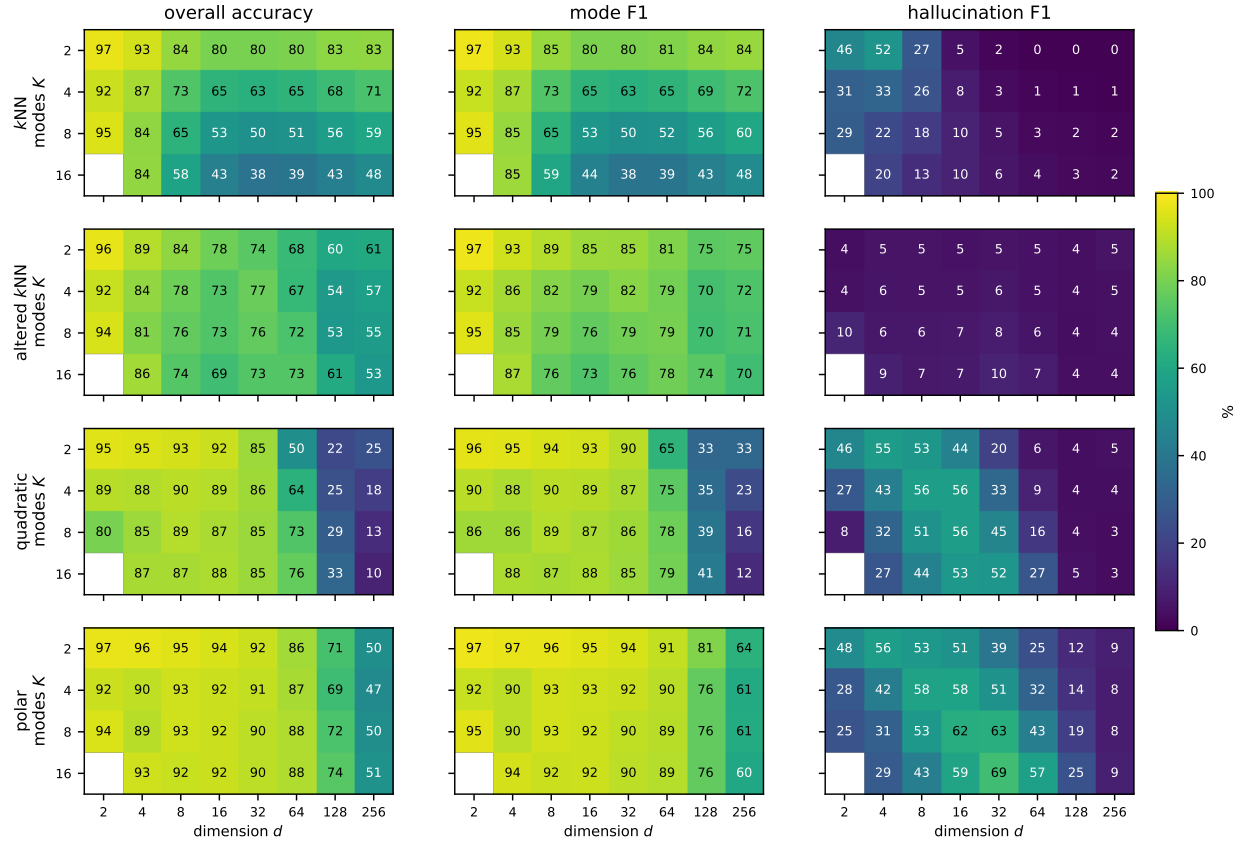


Figure 80: **Weighted Heun**, $T = 250$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

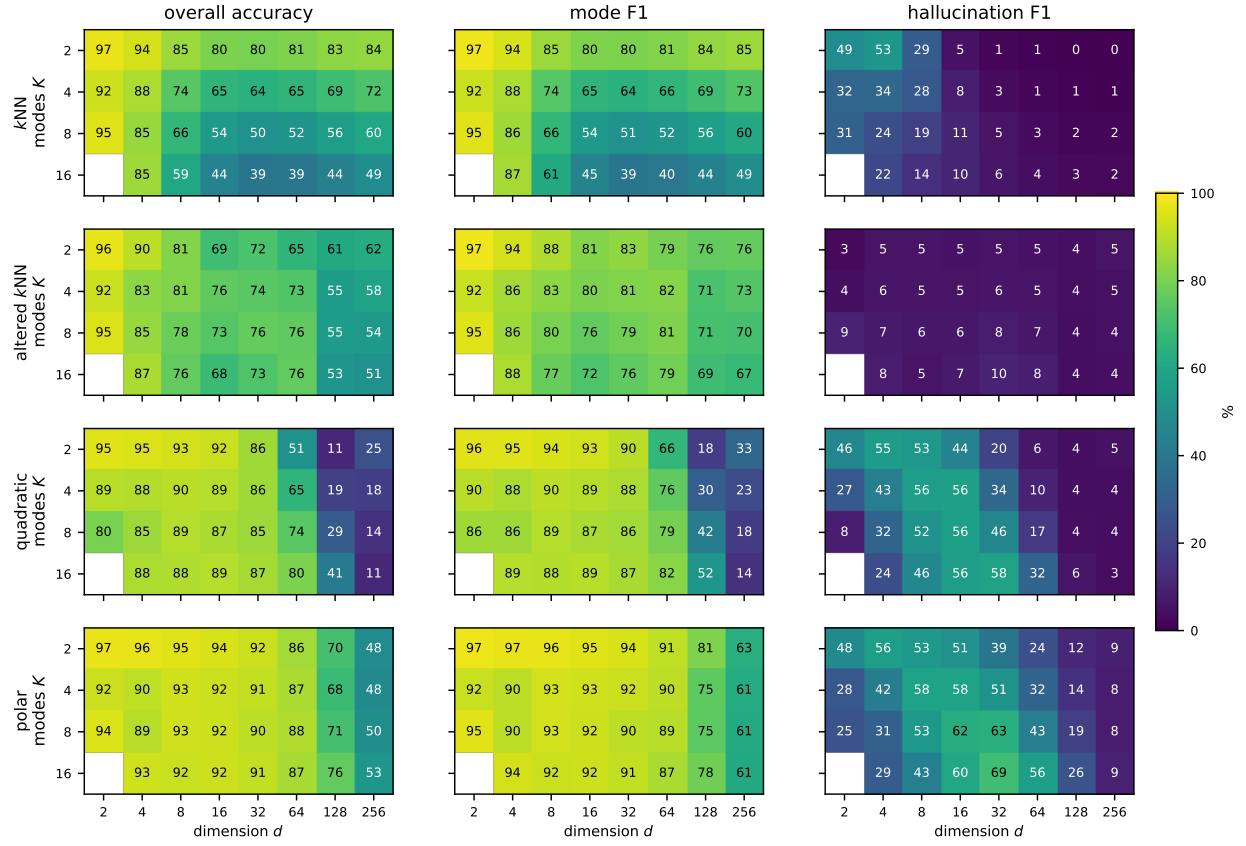


Figure 81: **Weighted Heun**, $T = 250$, **15,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

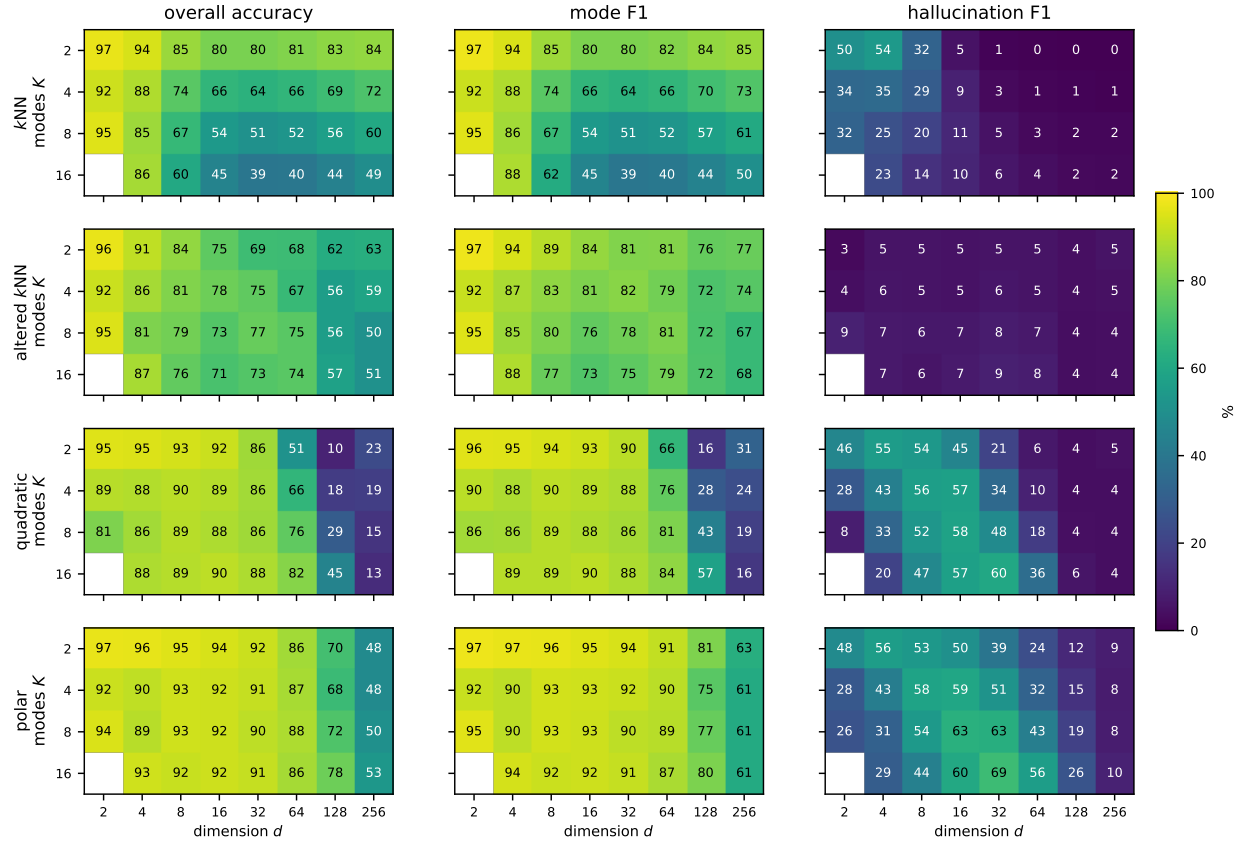


Figure 82: **Weighted Heun**, $T = 250$, **20,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

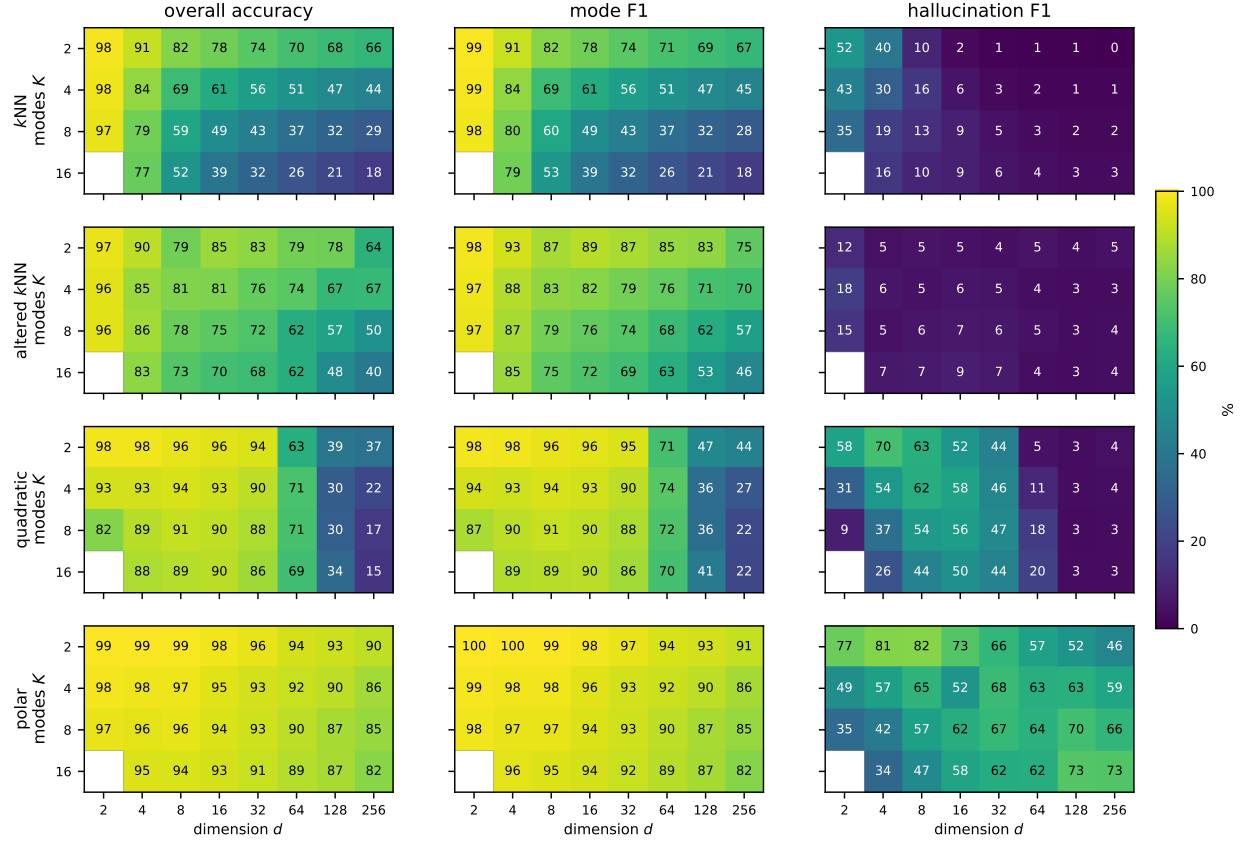


Figure 83: **Weighted Heun, $T = 500$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.17 Weighted Heun, $T = 500$

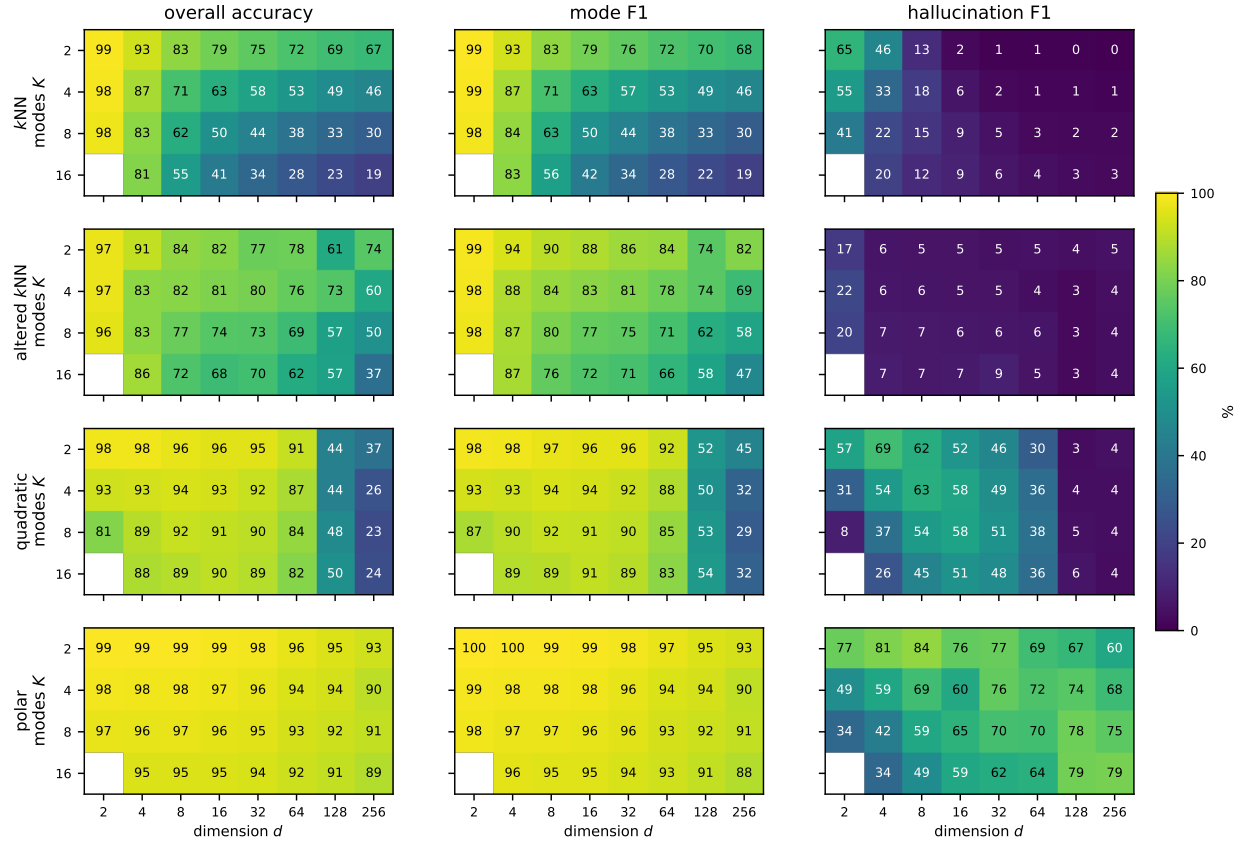


Figure 84: **Weighted Heun**, $T = 500$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

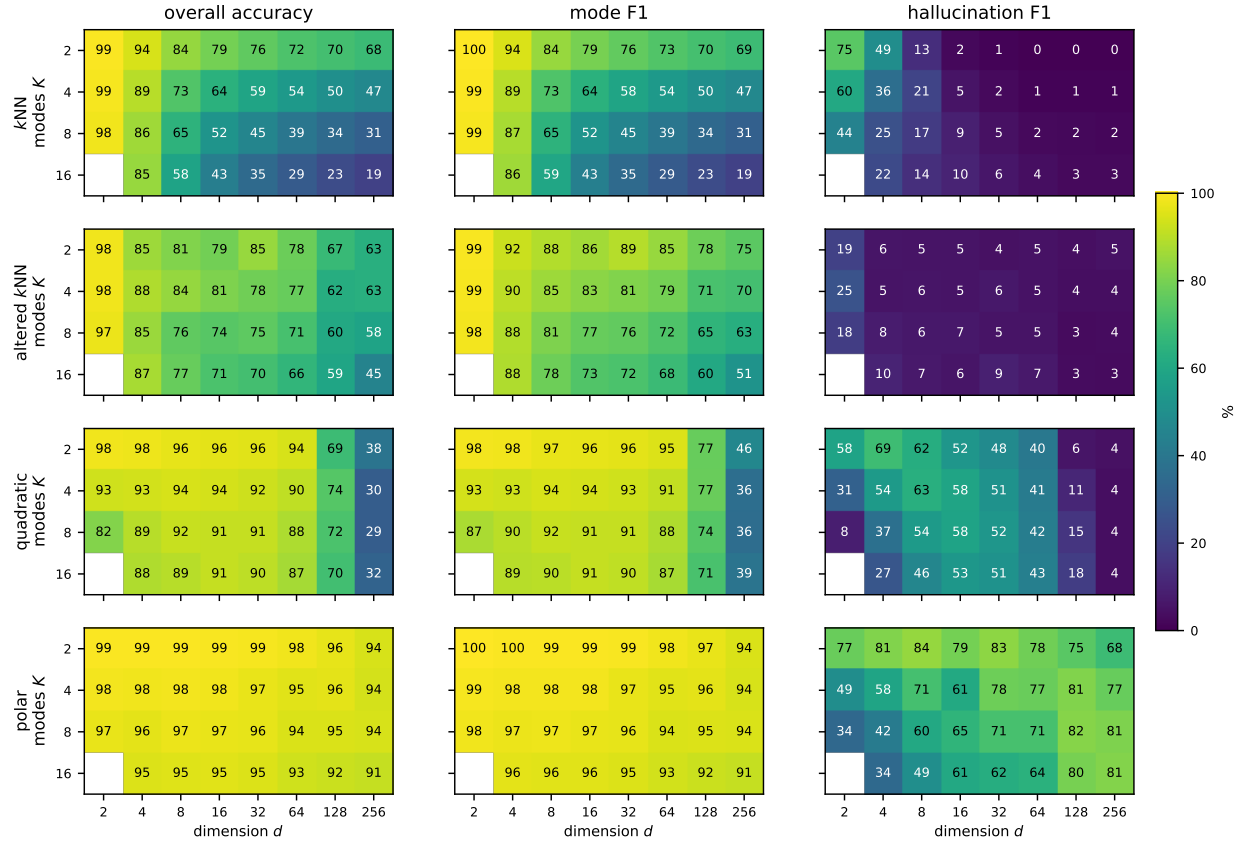


Figure 85: **Weighted Heun**, $T = 500$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

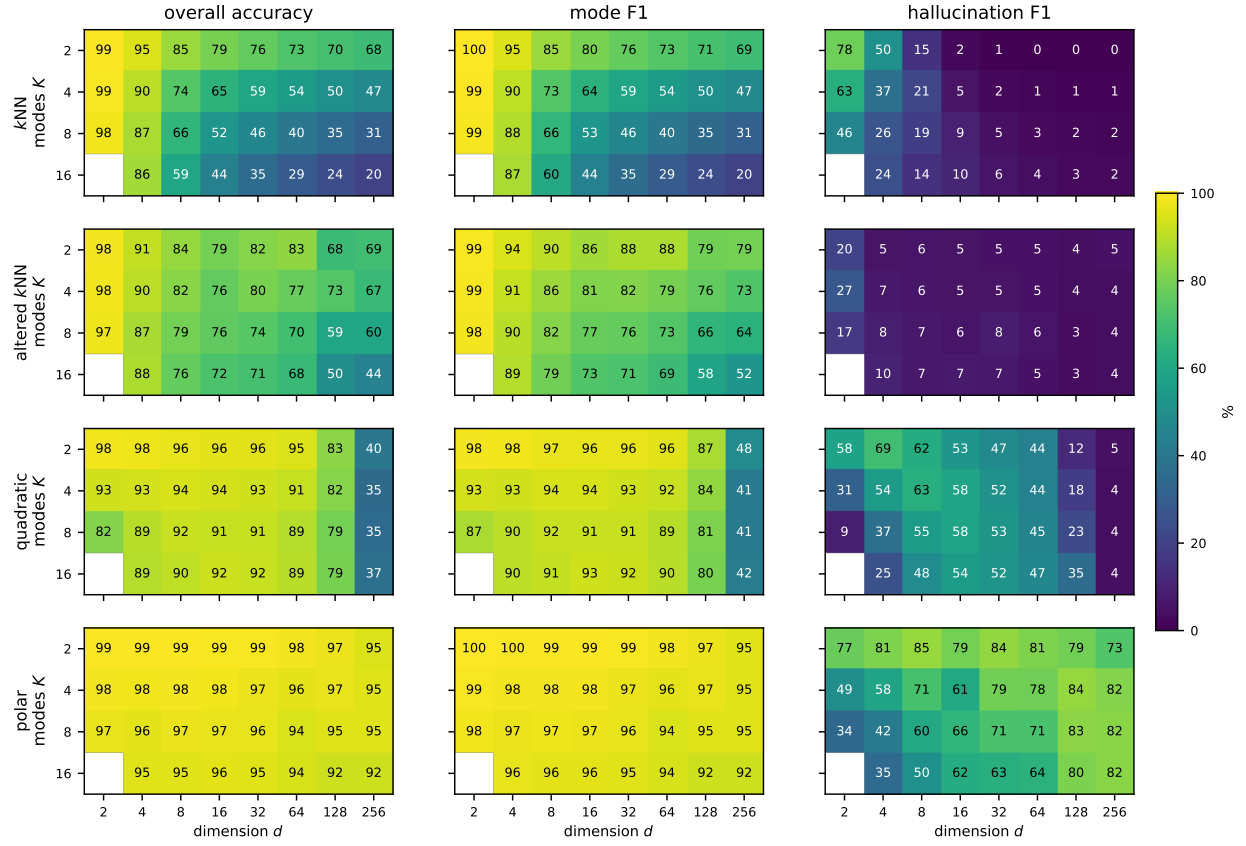


Figure 86: **Weighted Heun, $T = 500$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

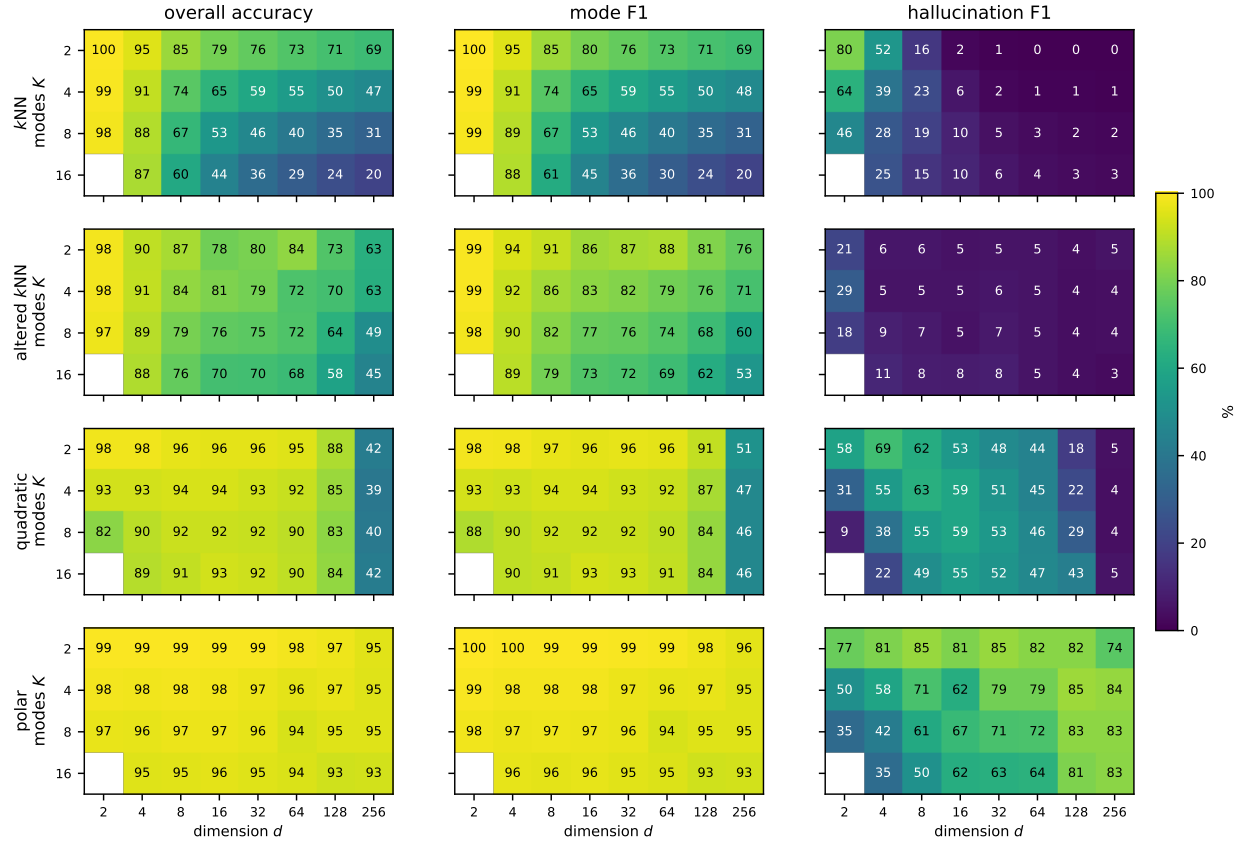


Figure 87: **Weighted Heun, $T = 500$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

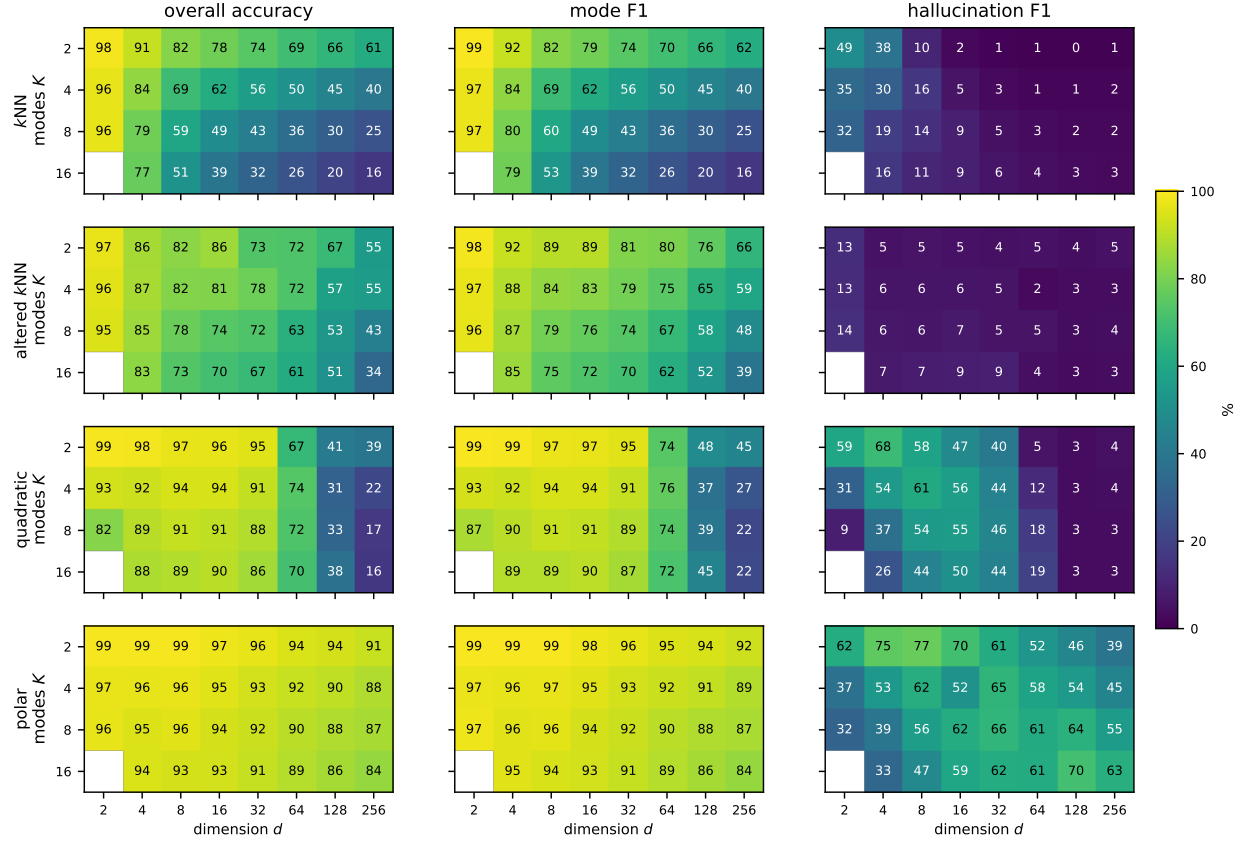


Figure 88: **Weighted Heun, $T = 750$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.18 Weighted Heun, $T = 750$

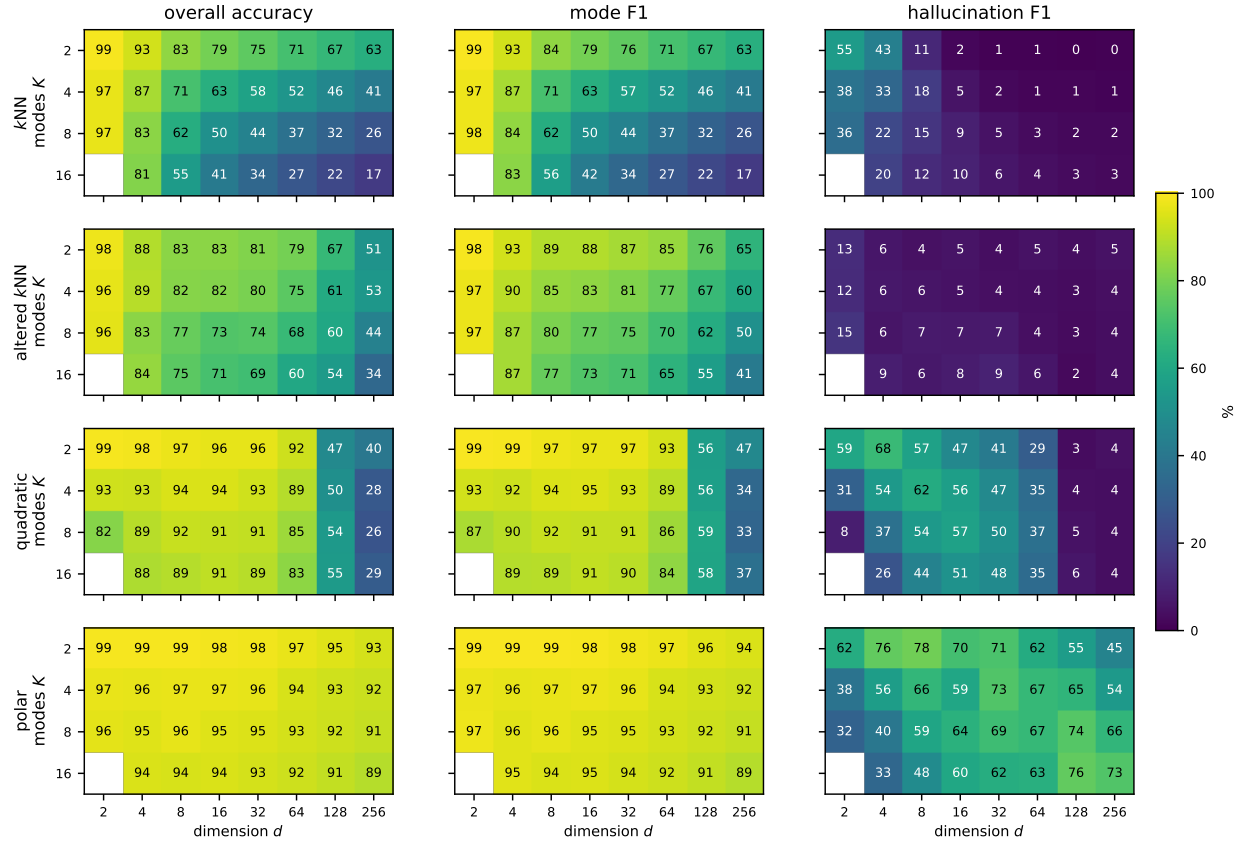


Figure 89: **Weighted Heun, $T = 750$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

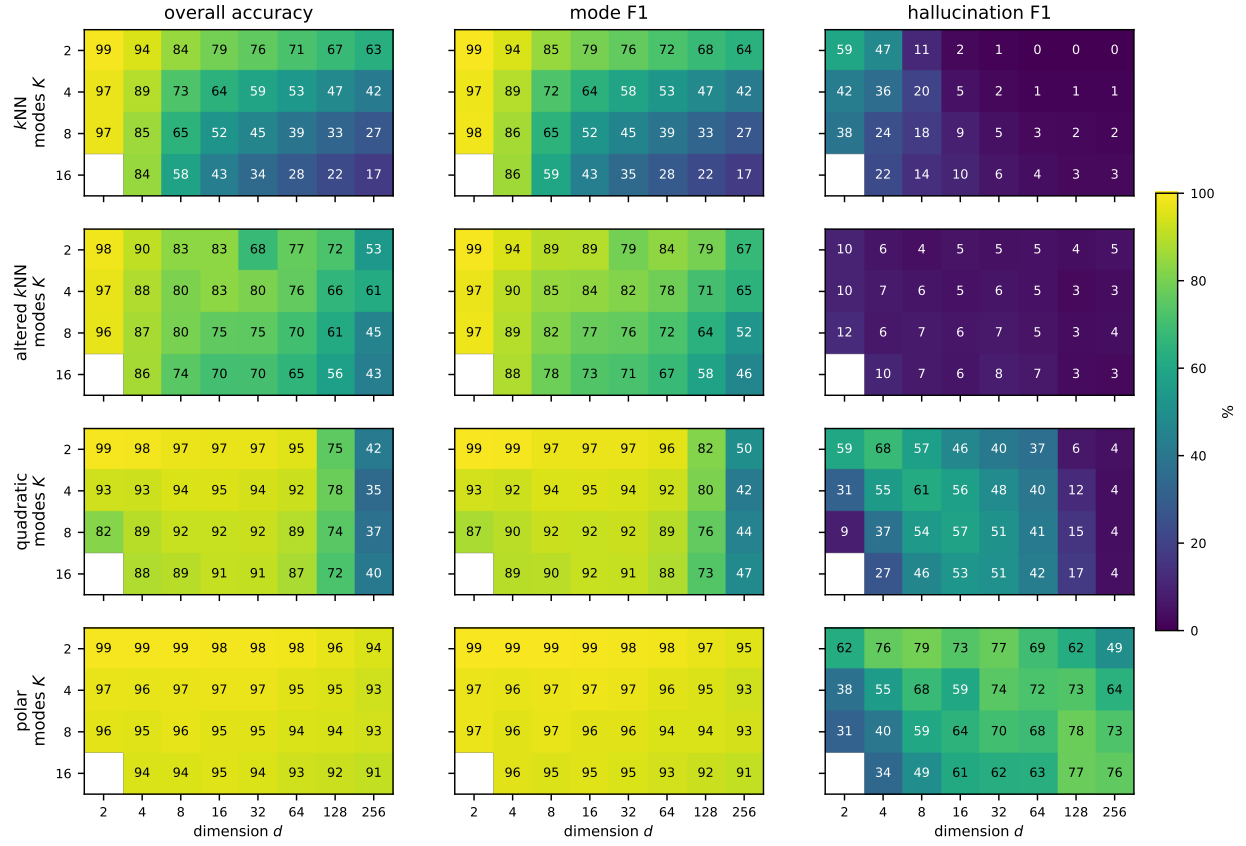


Figure 90: **Weighted Heun, $T = 750$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

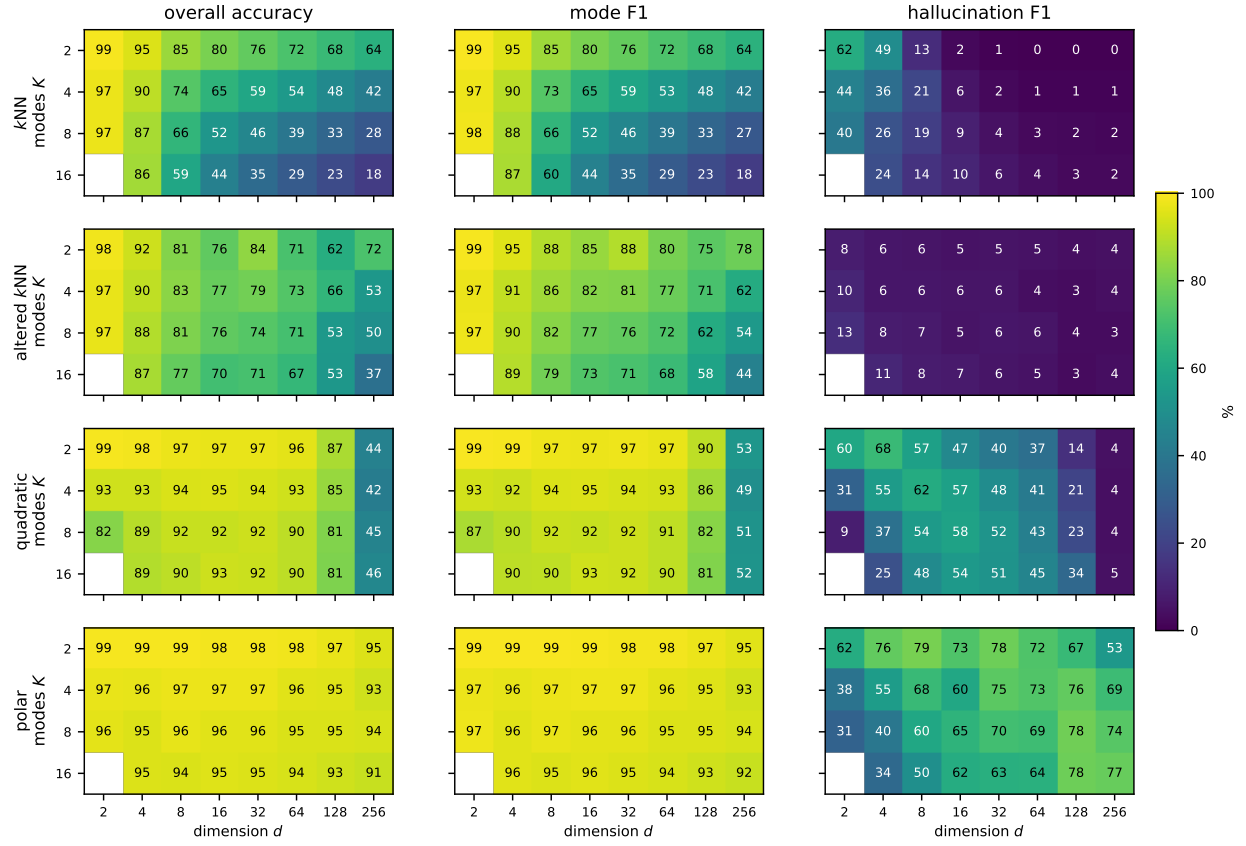


Figure 91: **Weighted Heun, $T = 750$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

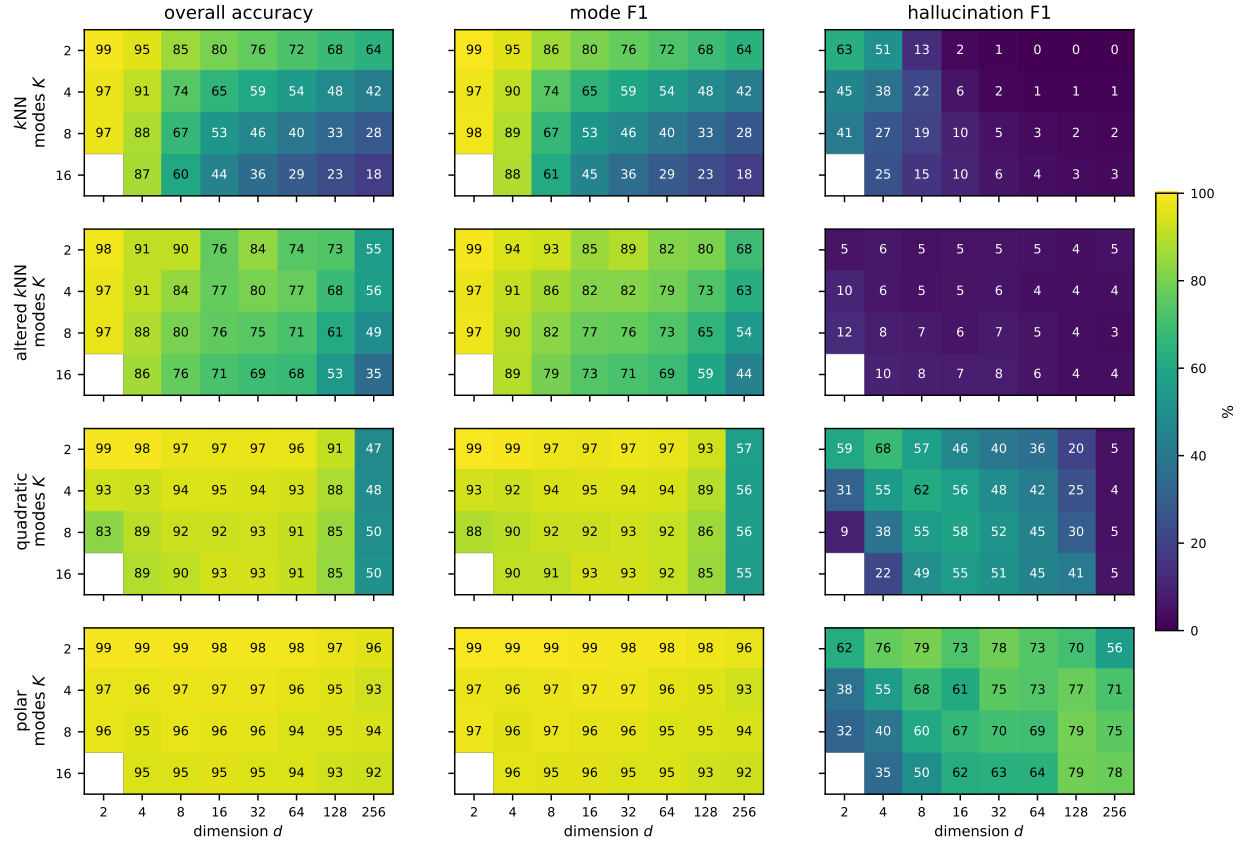


Figure 92: **Weighted Heun, $T = 750$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

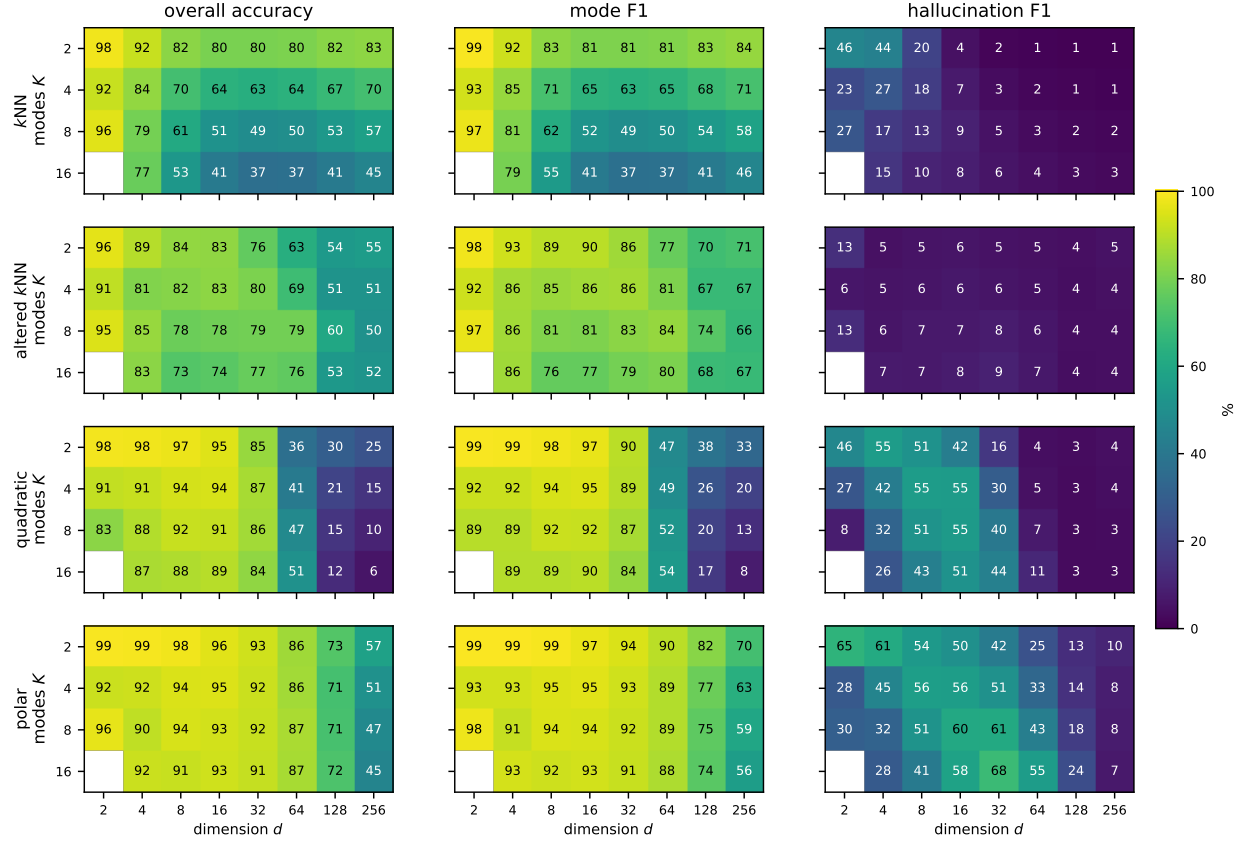


Figure 93: **Unweighted RK45, $T = 250$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.19 Unweighted RK45, $T = 250$

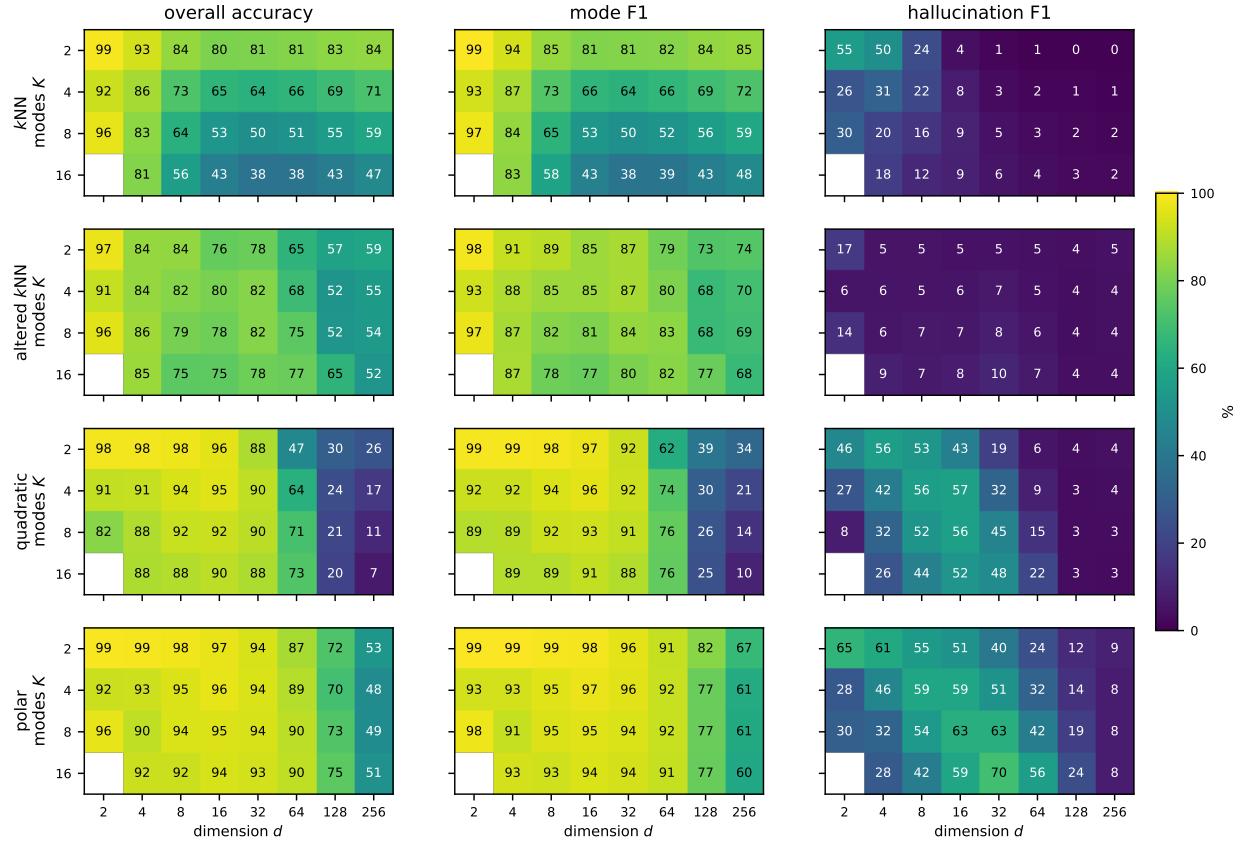


Figure 94: **Unweighted RK45**, $T = 250$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

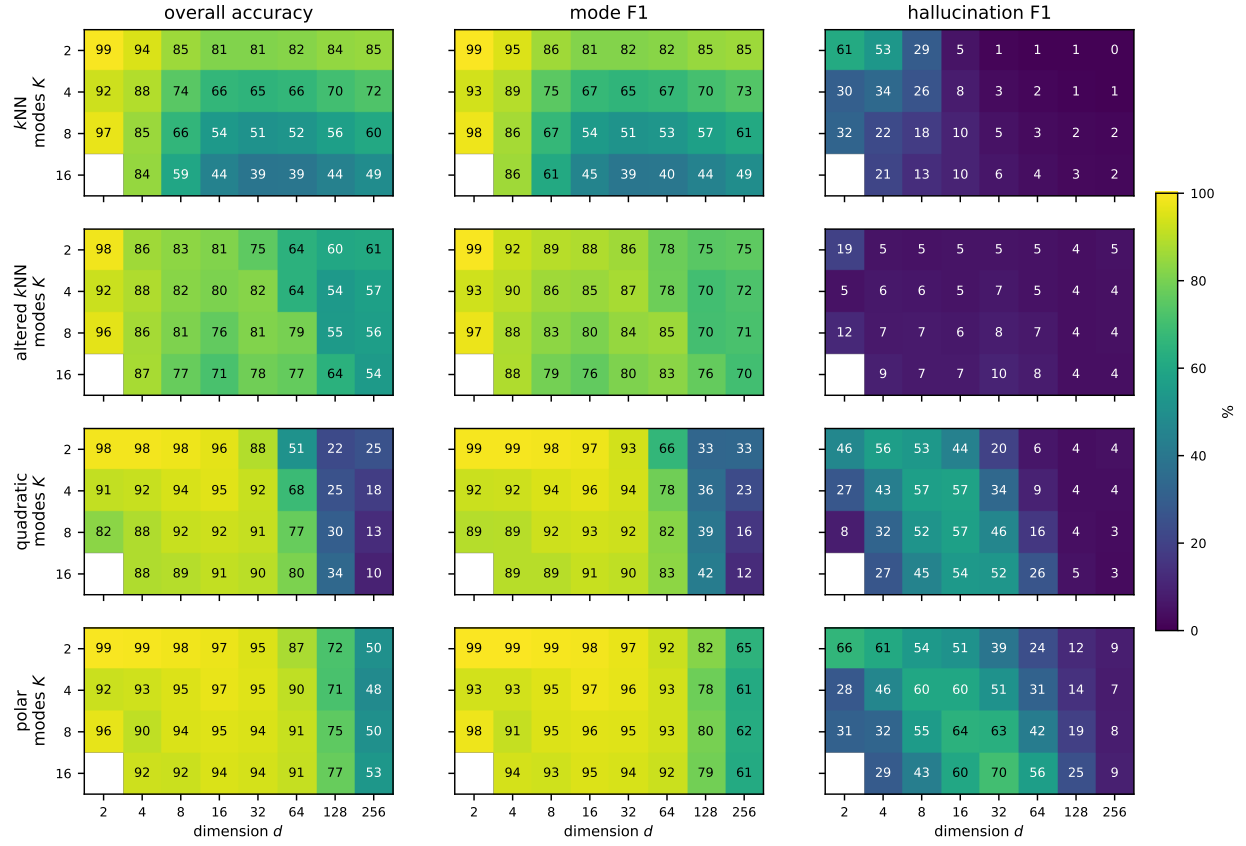


Figure 95: **Unweighted RK45**, $T = 250$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

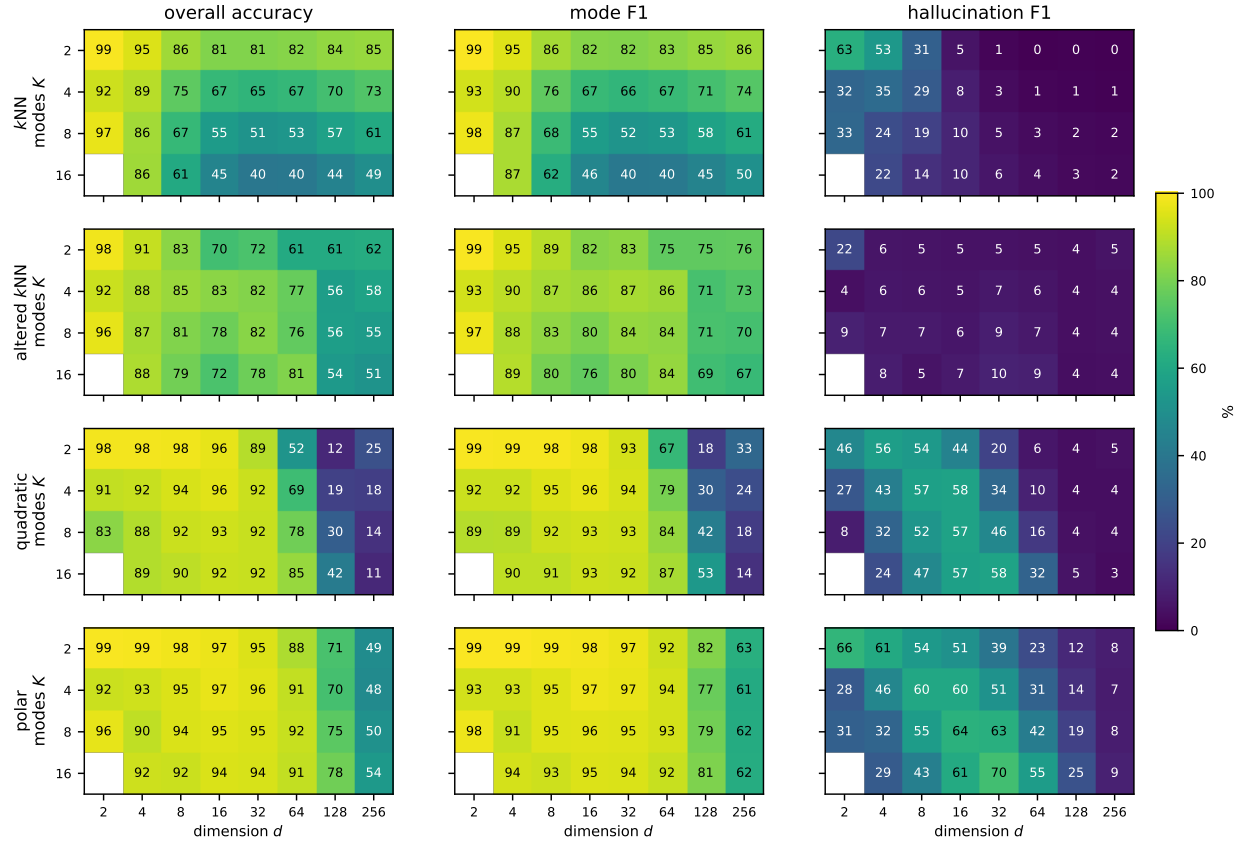


Figure 96: **Unweighted RK45**, $T = 250$, **15,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

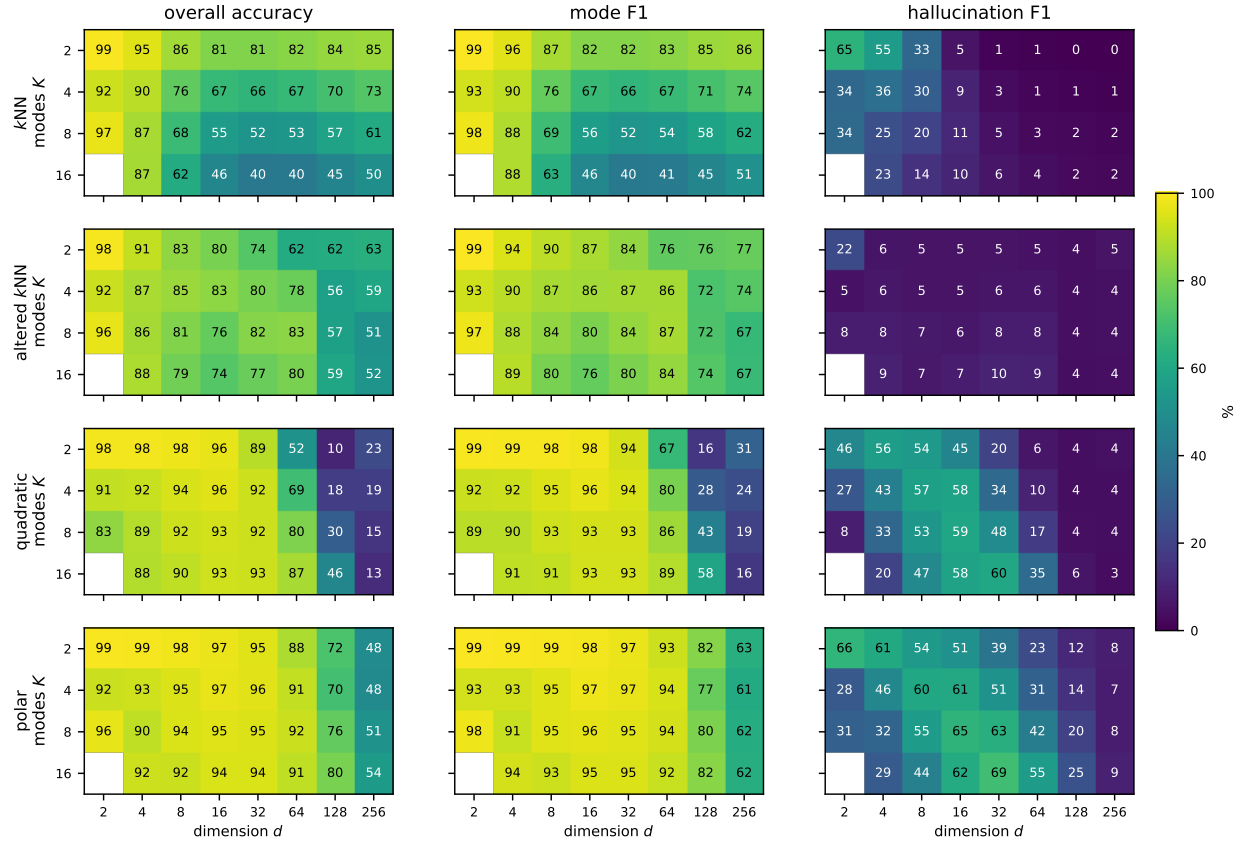


Figure 97: **Unweighted RK45**, $T = 250$, **20,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

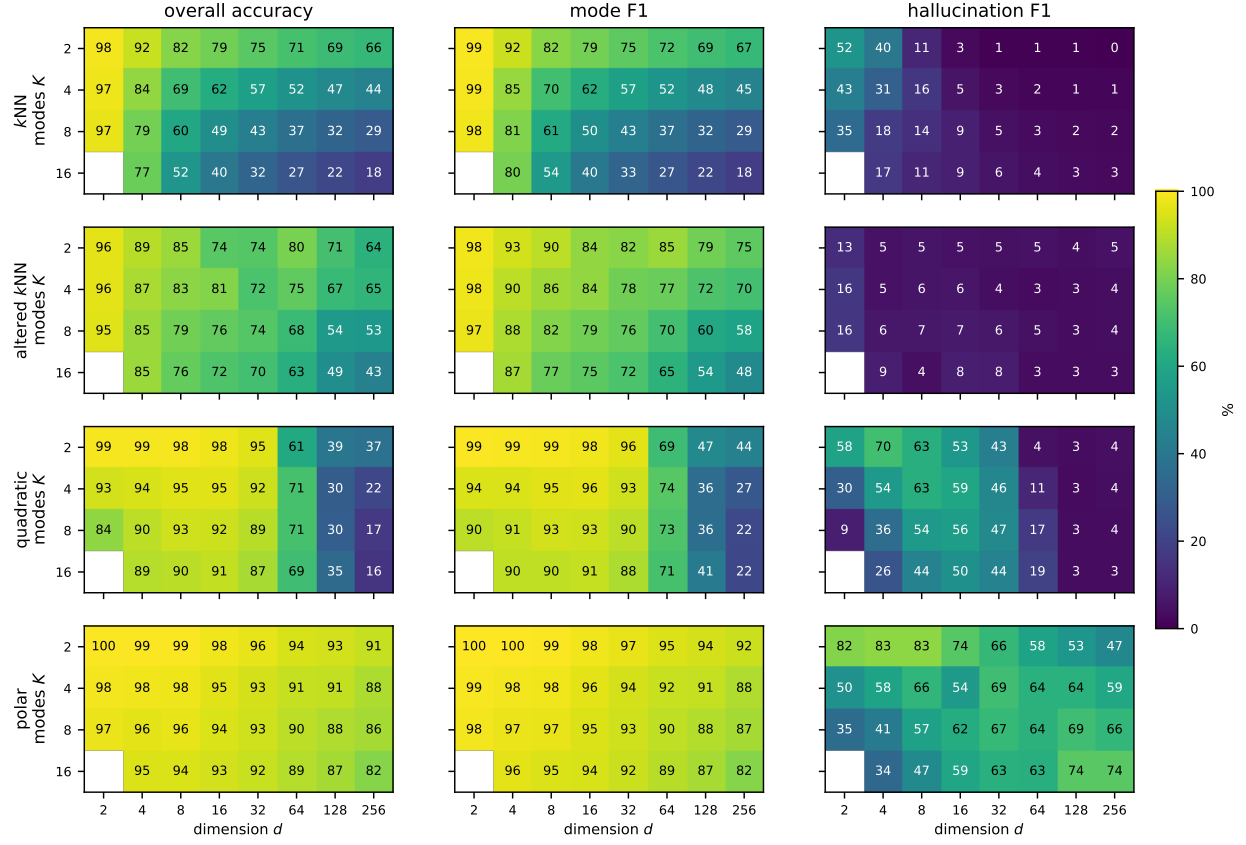


Figure 98: **Unweighted RK45, $T = 500$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.20 Unweighted RK45, $T = 500$

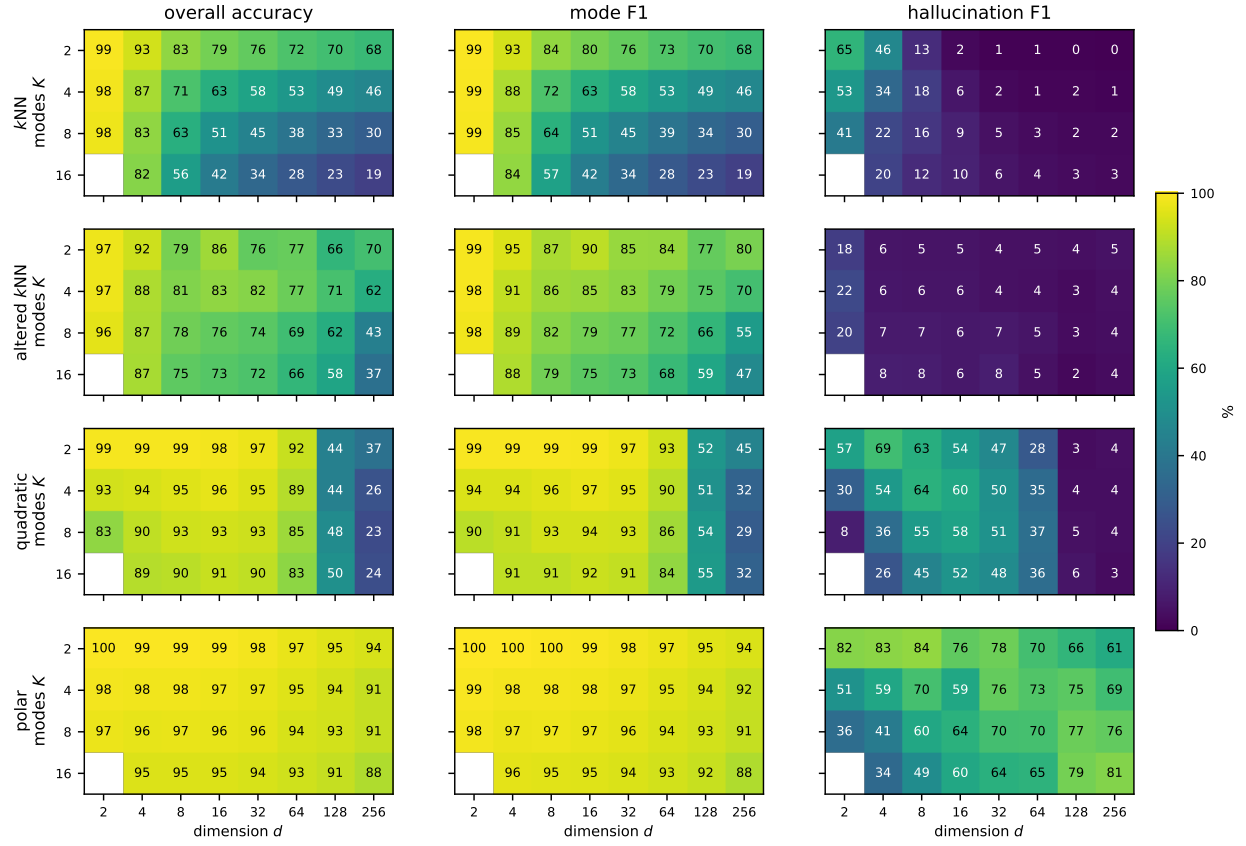


Figure 99: **Unweighted RK45**, $T = 500$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

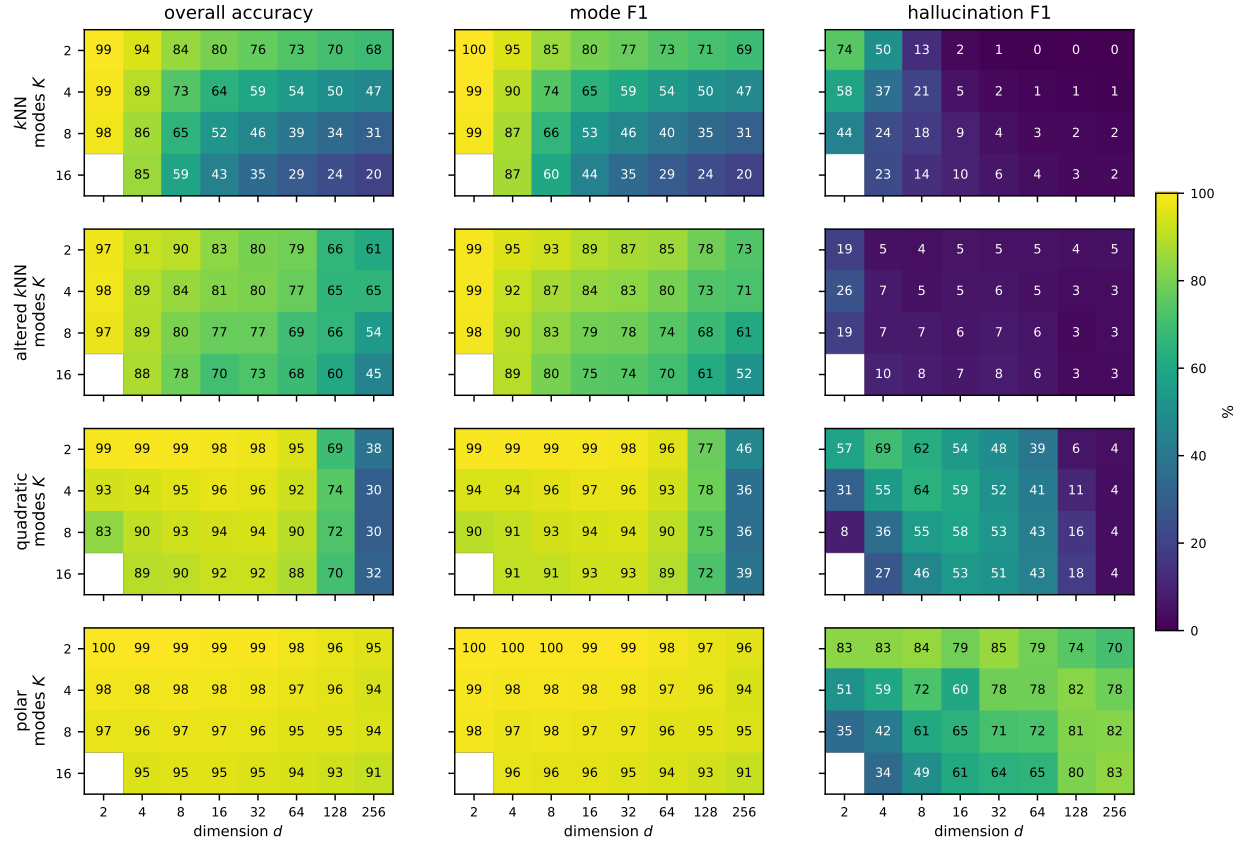


Figure 100: **Unweighted RK45**, $T = 500$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

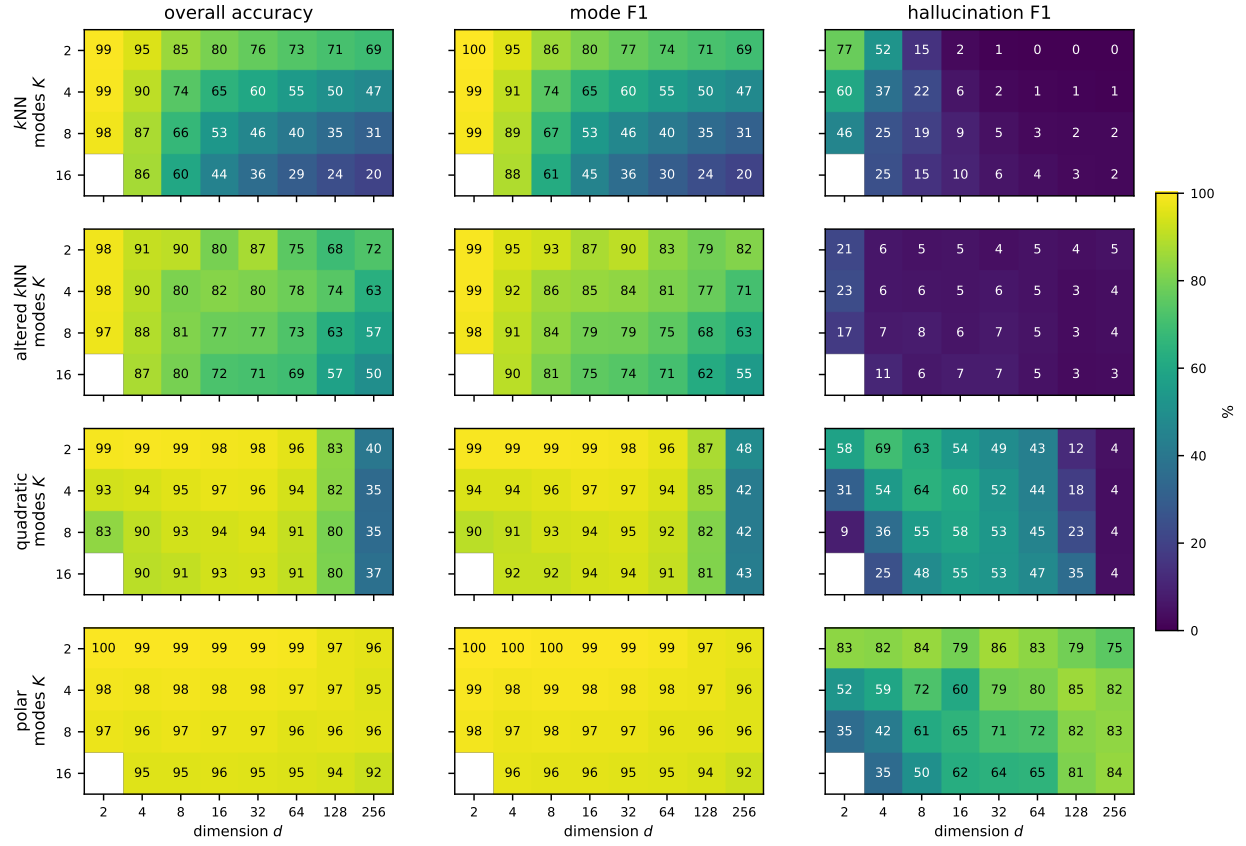


Figure 101: **Unweighted RK45**, $T = 500$, **15,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

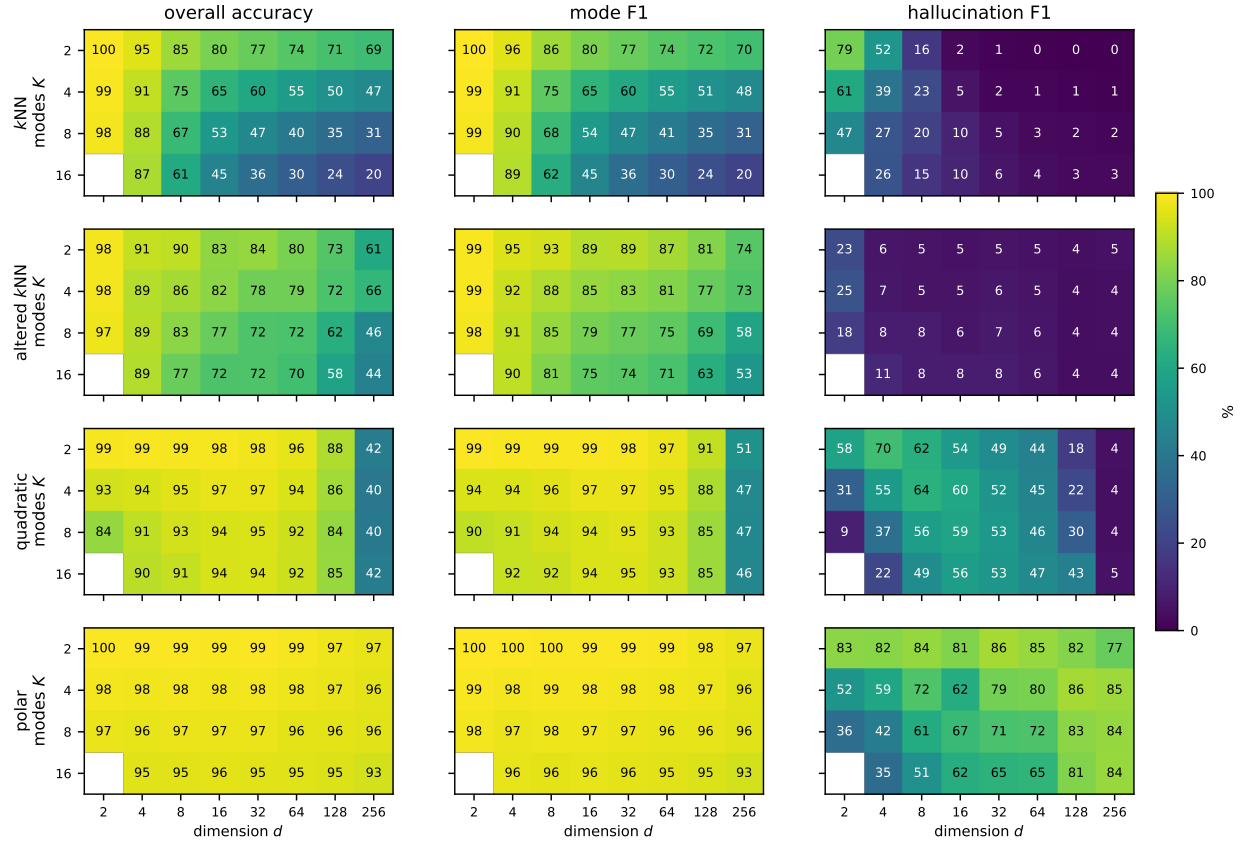


Figure 102: **Unweighted RK45**, $T = 500$, **20,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

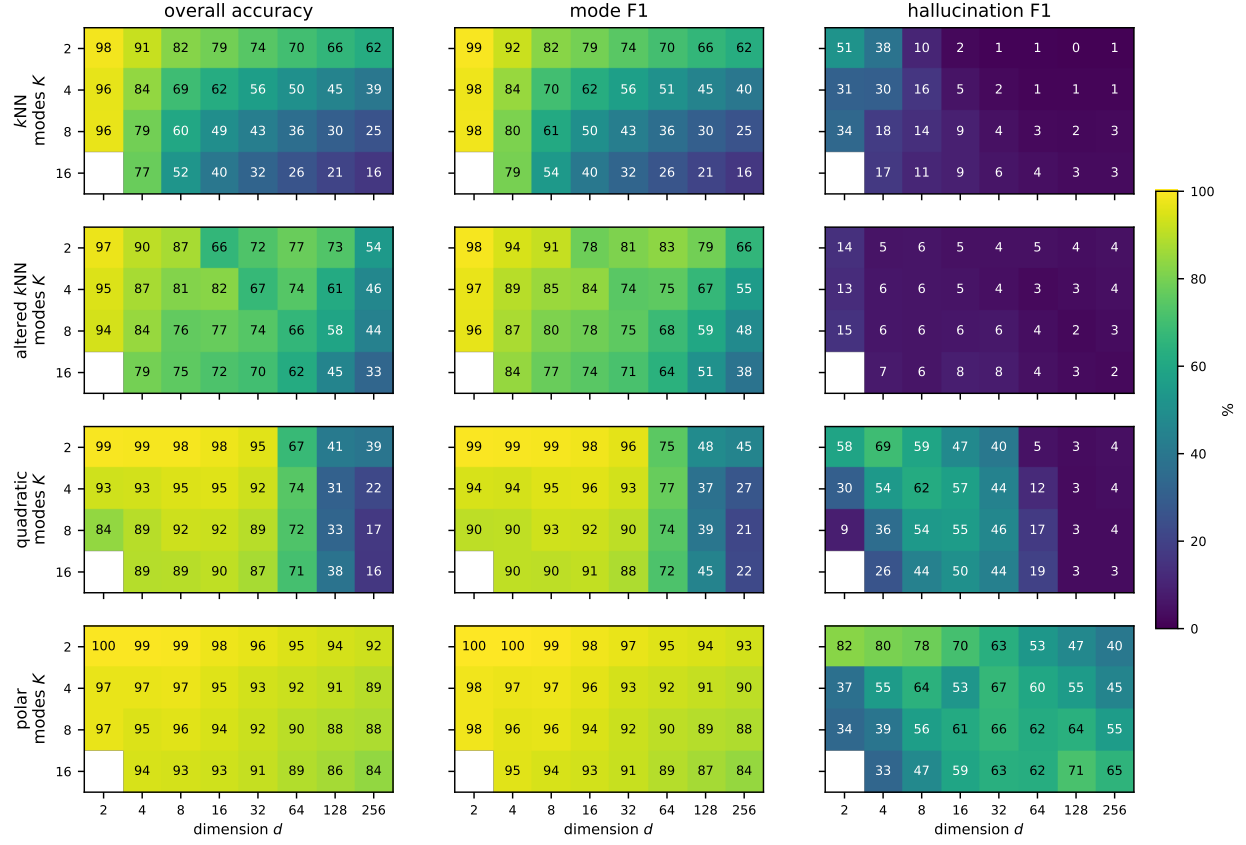


Figure 103: **Unweighted RK45, $T = 750$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.21 Unweighted RK45, $T = 750$

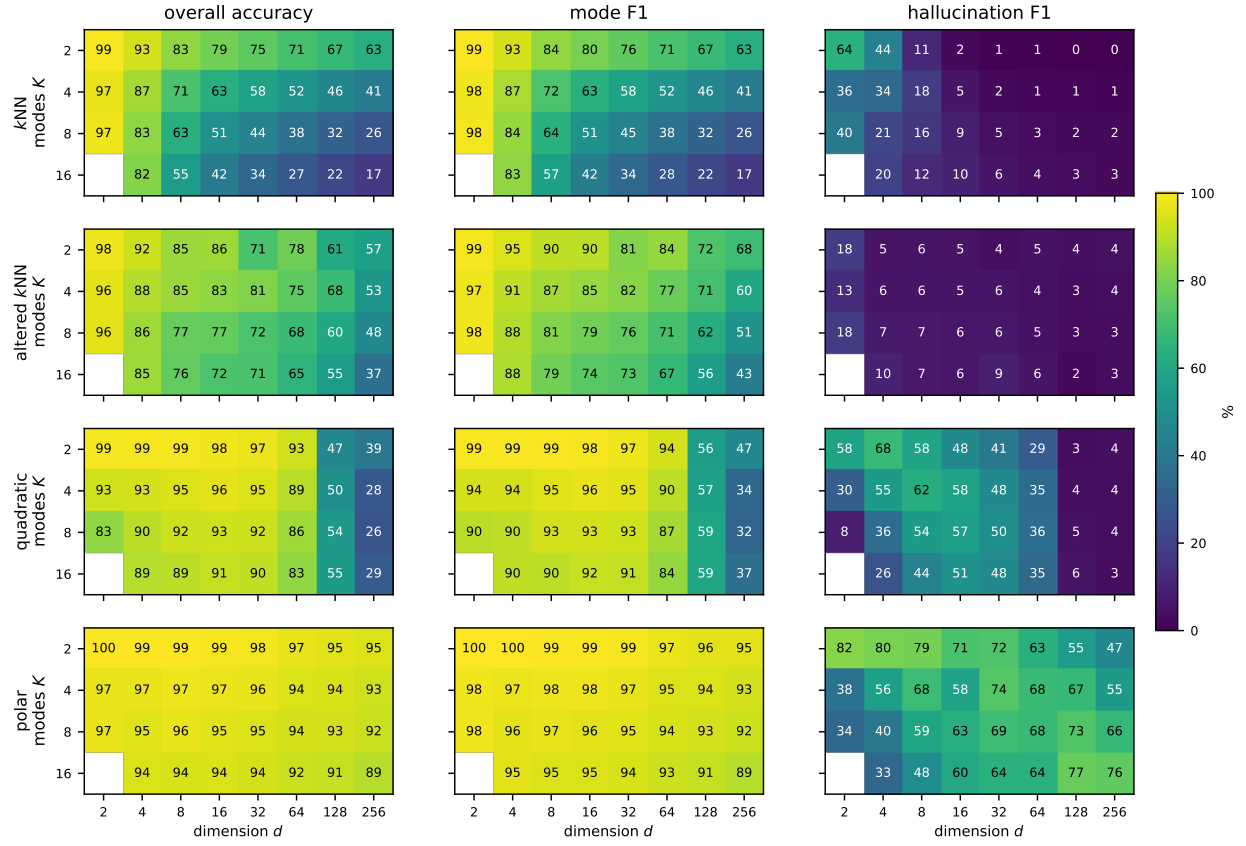


Figure 104: **Unweighted RK45**, $T = 750$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

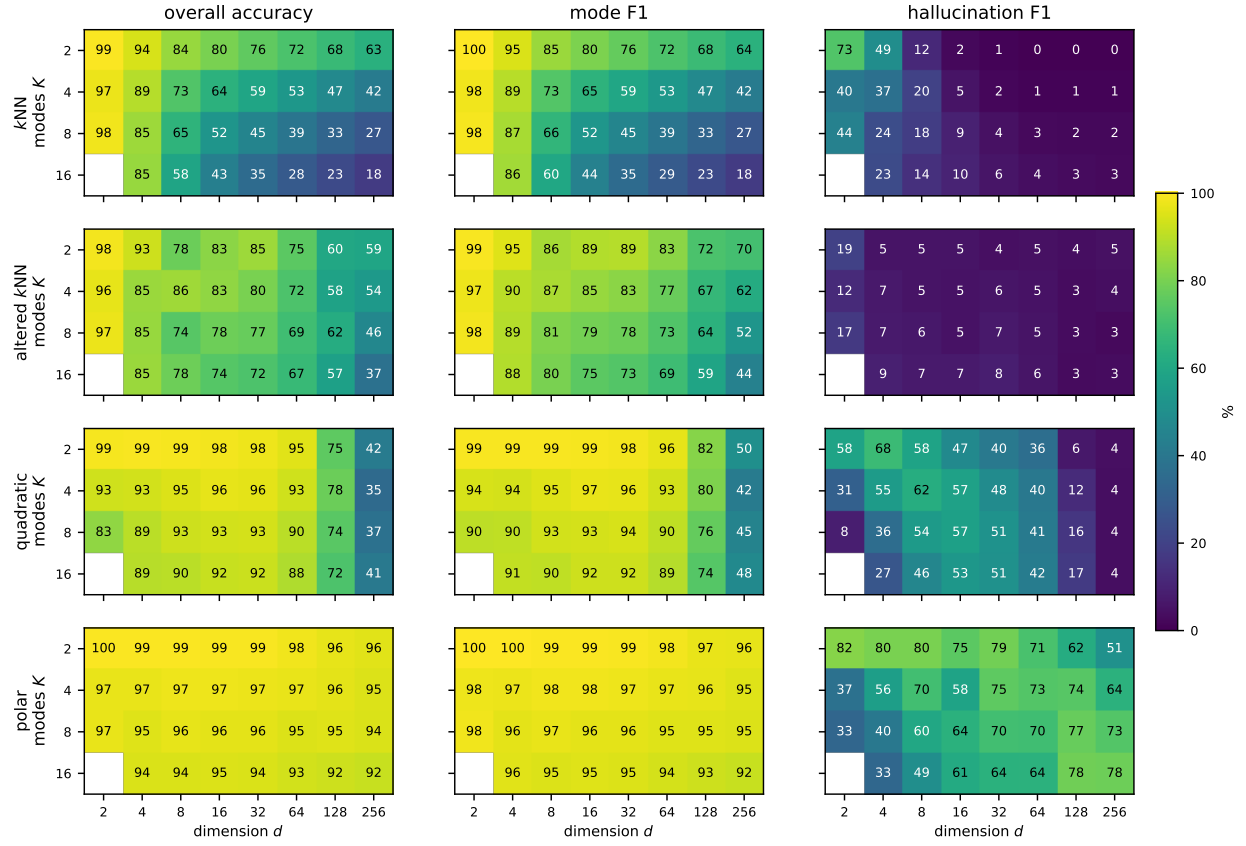


Figure 105: **Unweighted RK45**, $T = 750$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

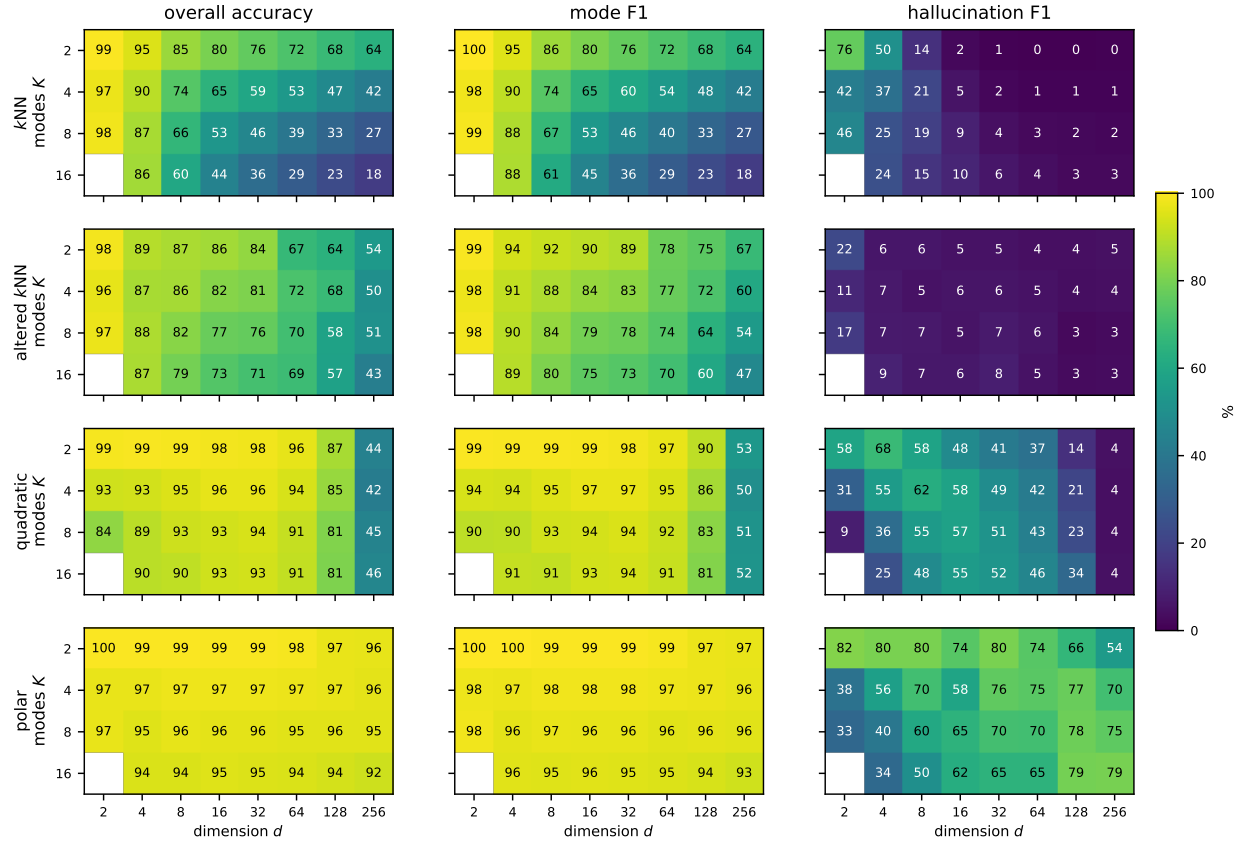


Figure 106: **Unweighted RK45**, $T = 750$, **15,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

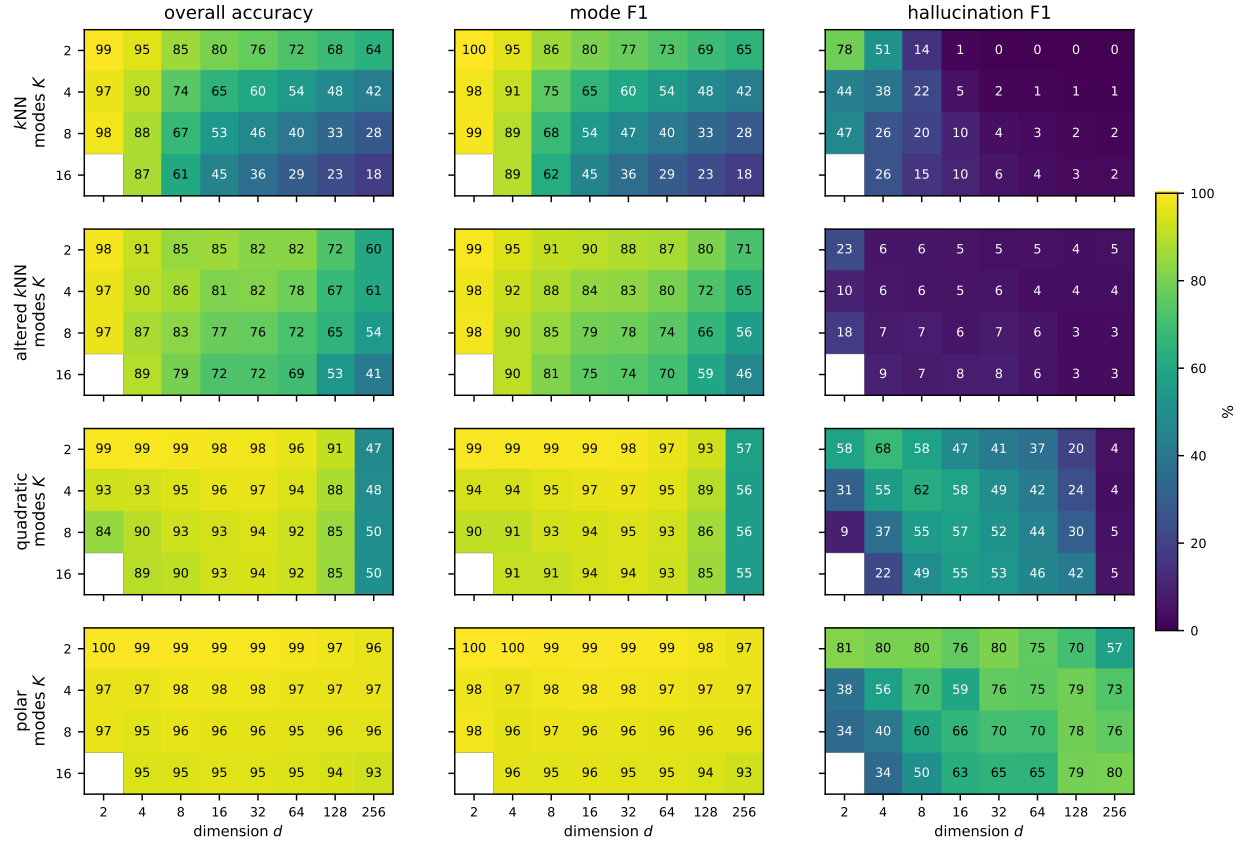


Figure 107: **Unweighted RK45**, $T = 750$, **20,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

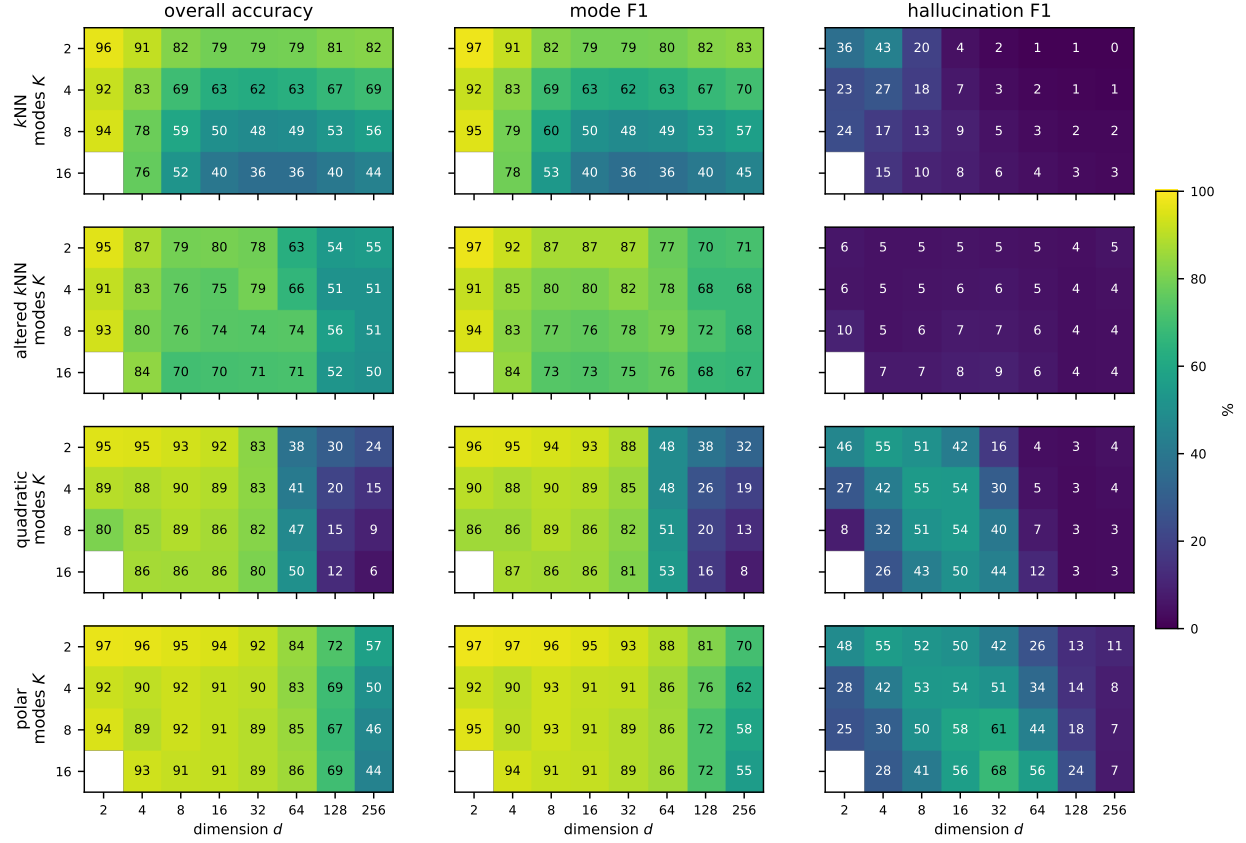


Figure 108: **Weighted RK45, $T = 250$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.22 Weighted RK45, $T = 250$

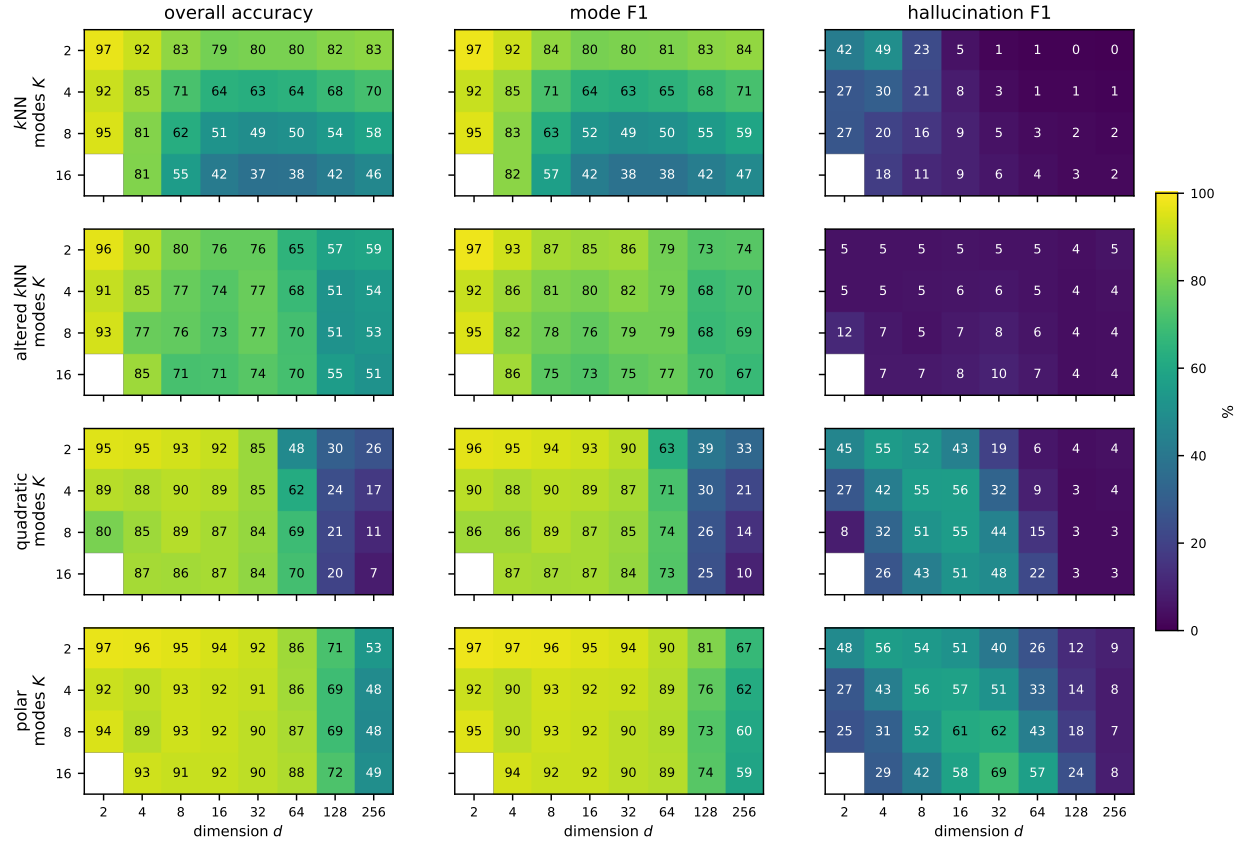


Figure 109: **Weighted RK45**, $T = 250$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

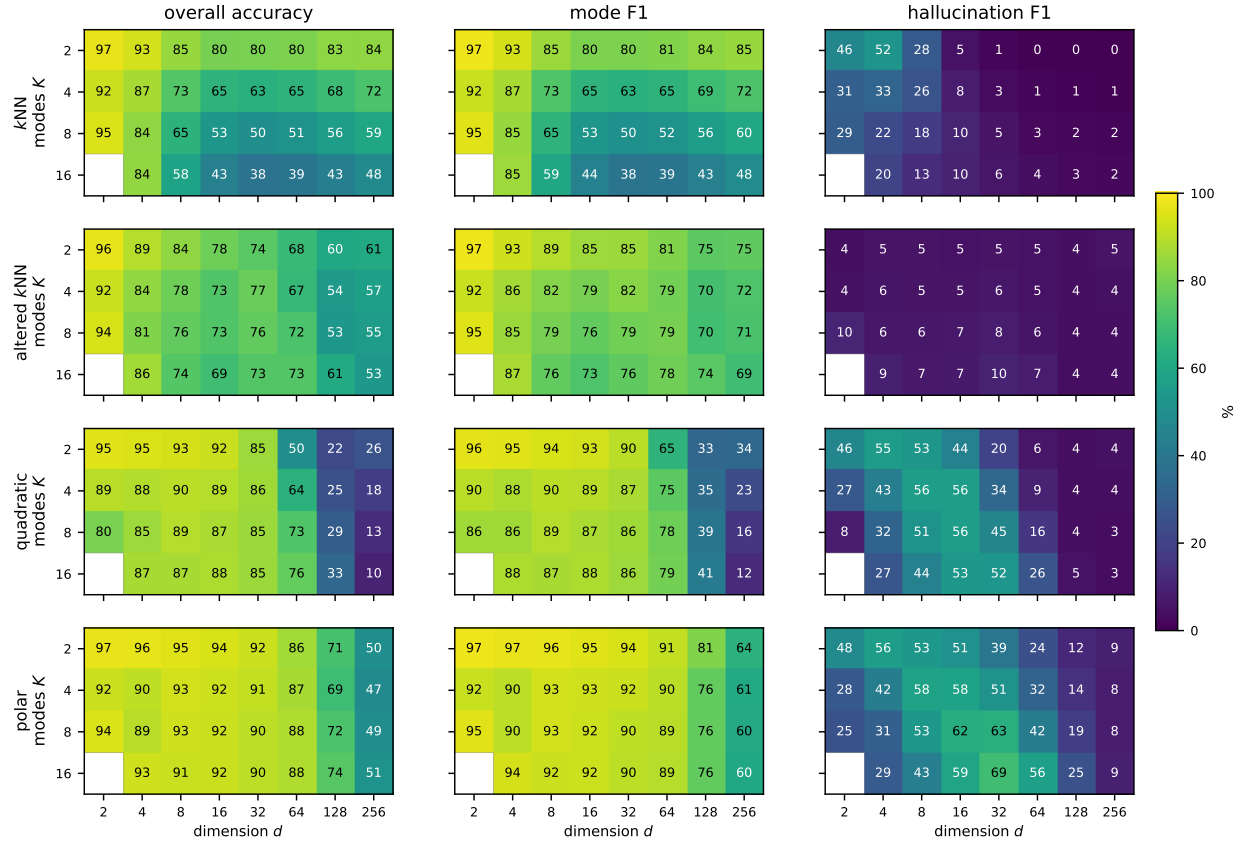


Figure 110: **Weighted RK45**, $T = 250$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

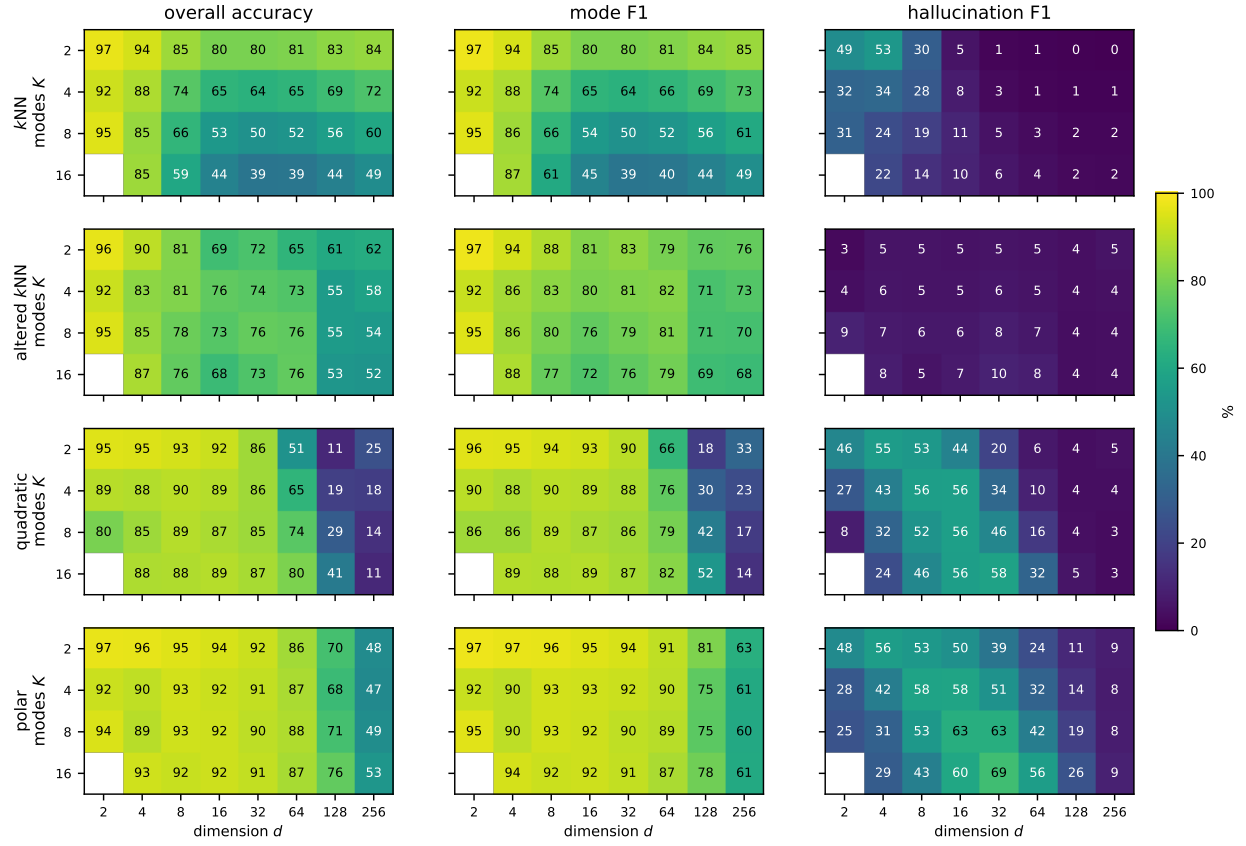


Figure 111: **Weighted RK45, $T = 250$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

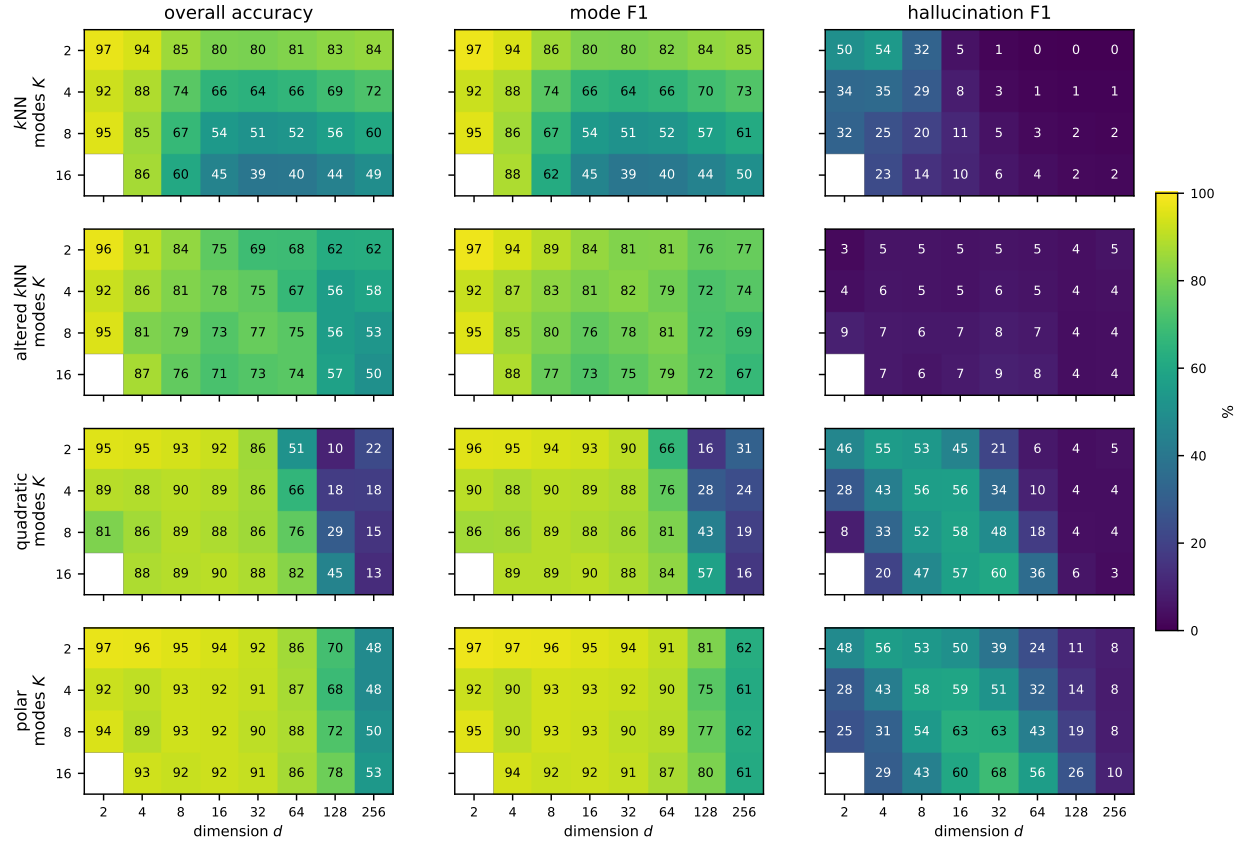


Figure 112: **Weighted RK45, $T = 250$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

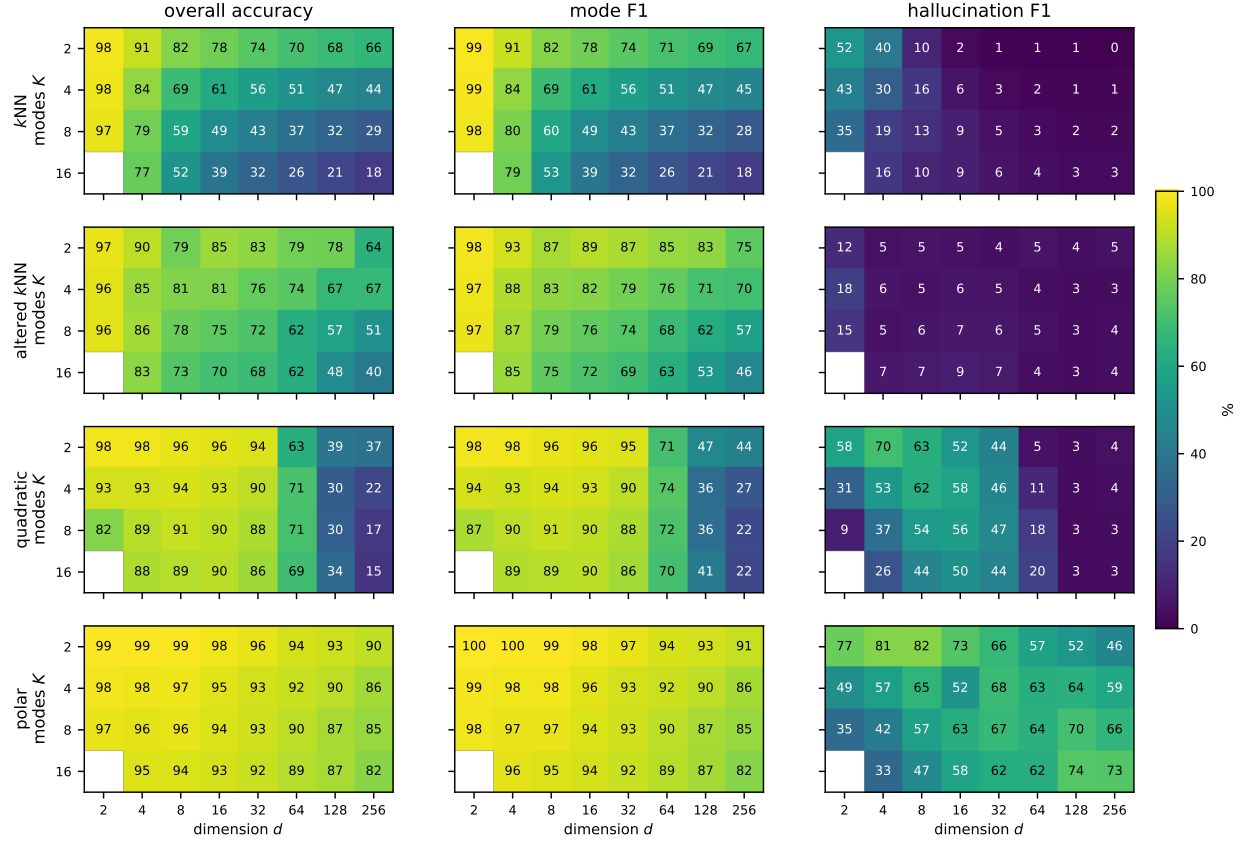


Figure 113: **Weighted RK45, $T = 500$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.23 Weighted RK45, $T = 500$

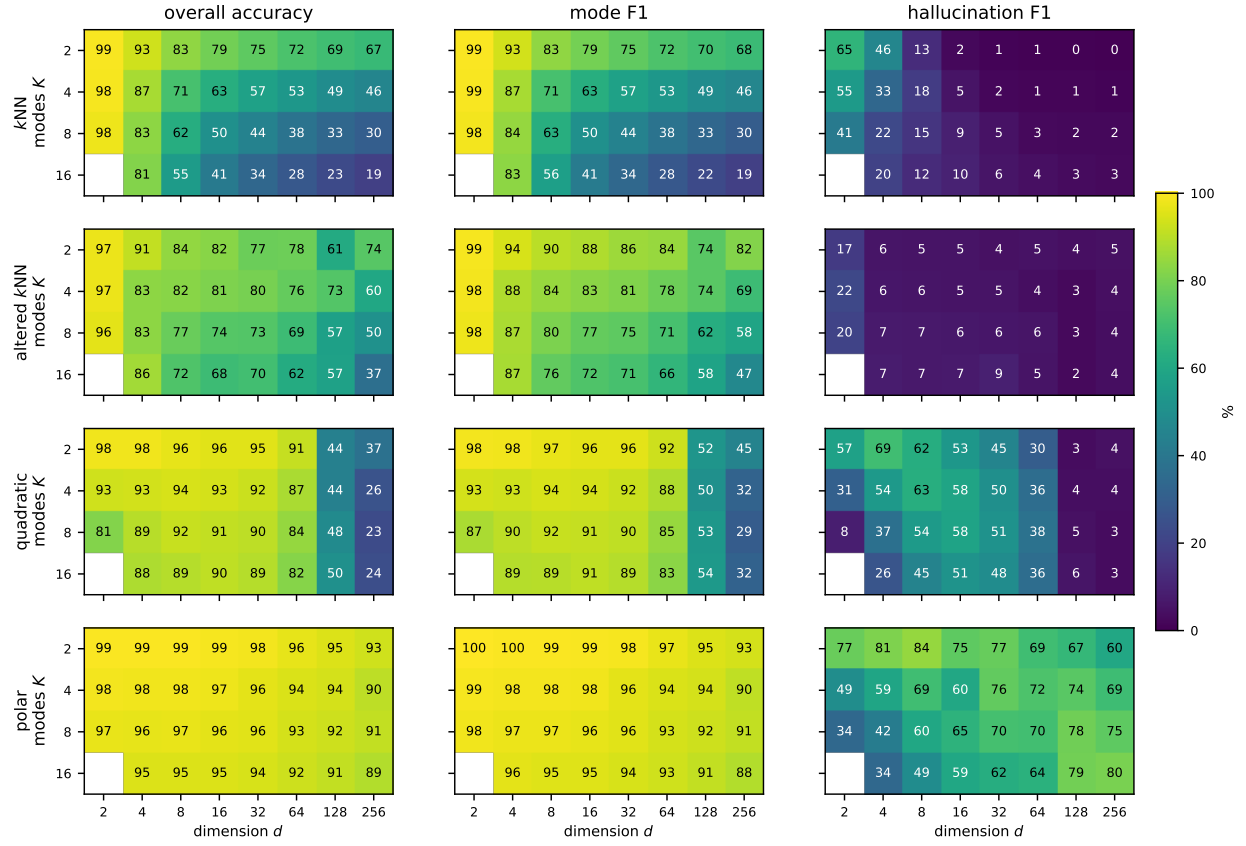


Figure 114: **Weighted RK45**, $T = 500$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

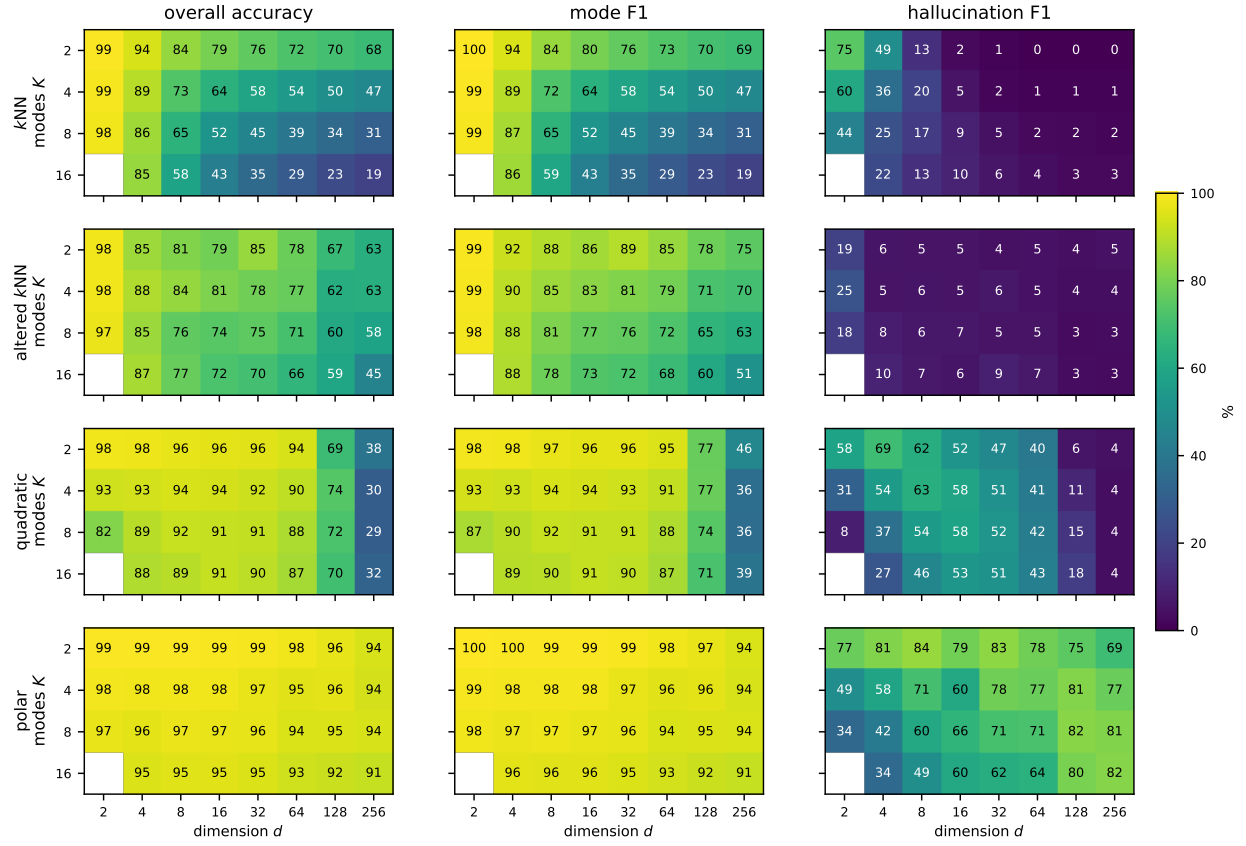


Figure 115: **Weighted RK45, $T = 500$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

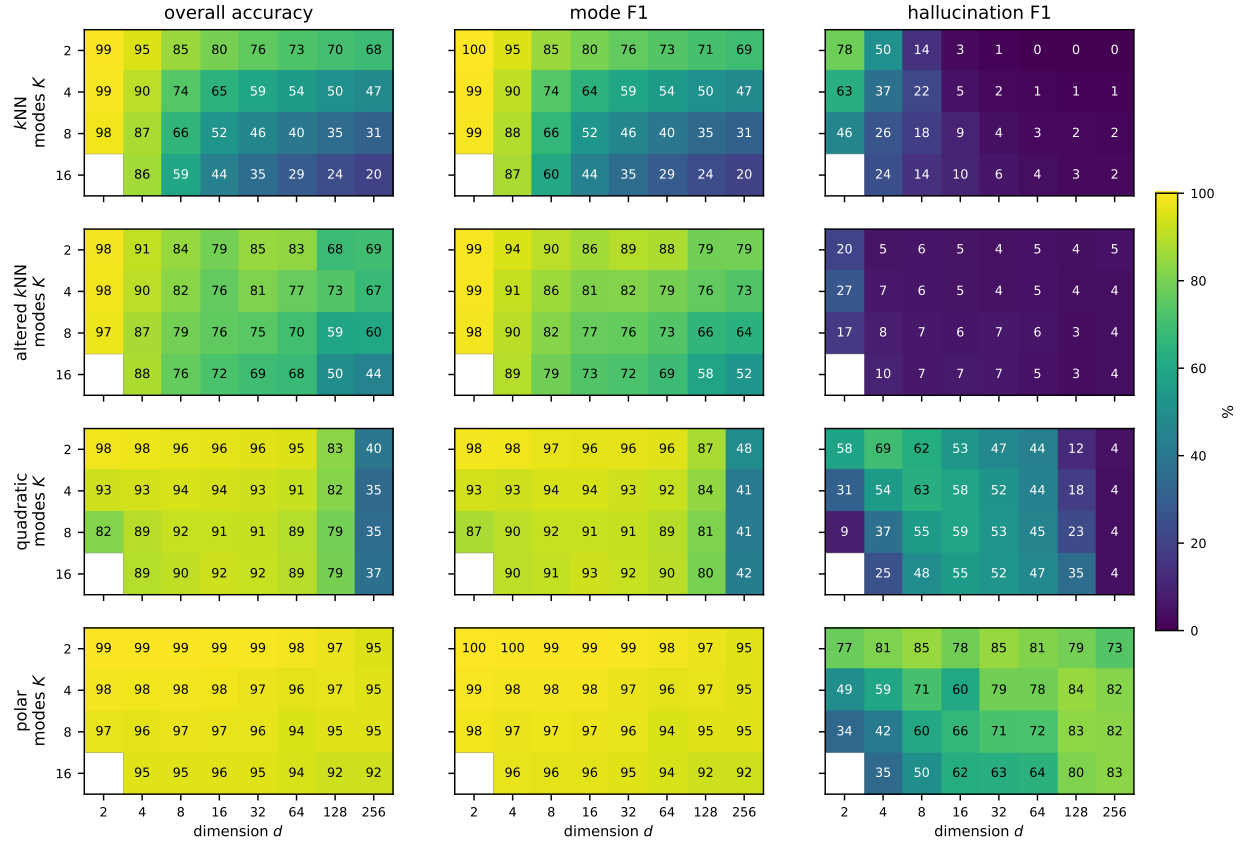


Figure 116: **Weighted RK45, $T = 500$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

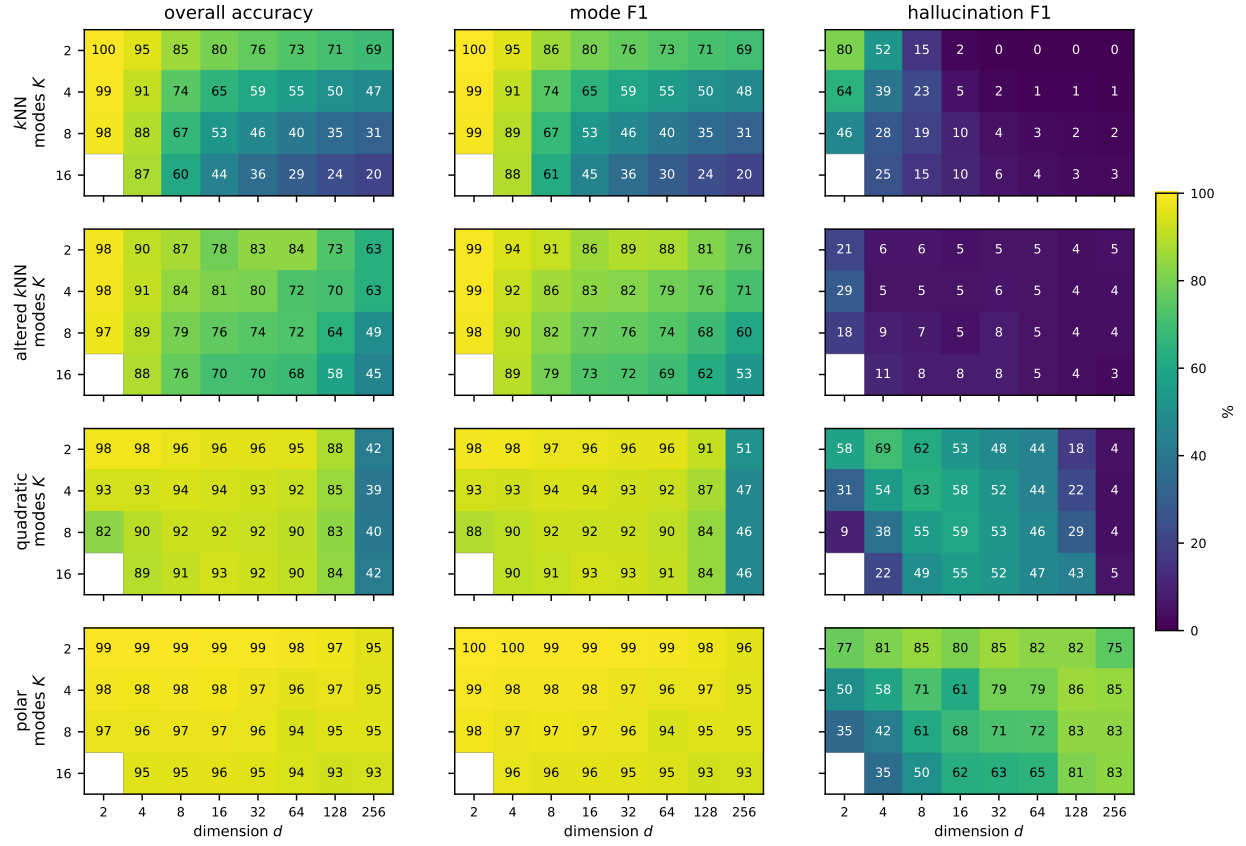


Figure 117: **Weighted RK45, $T = 500$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

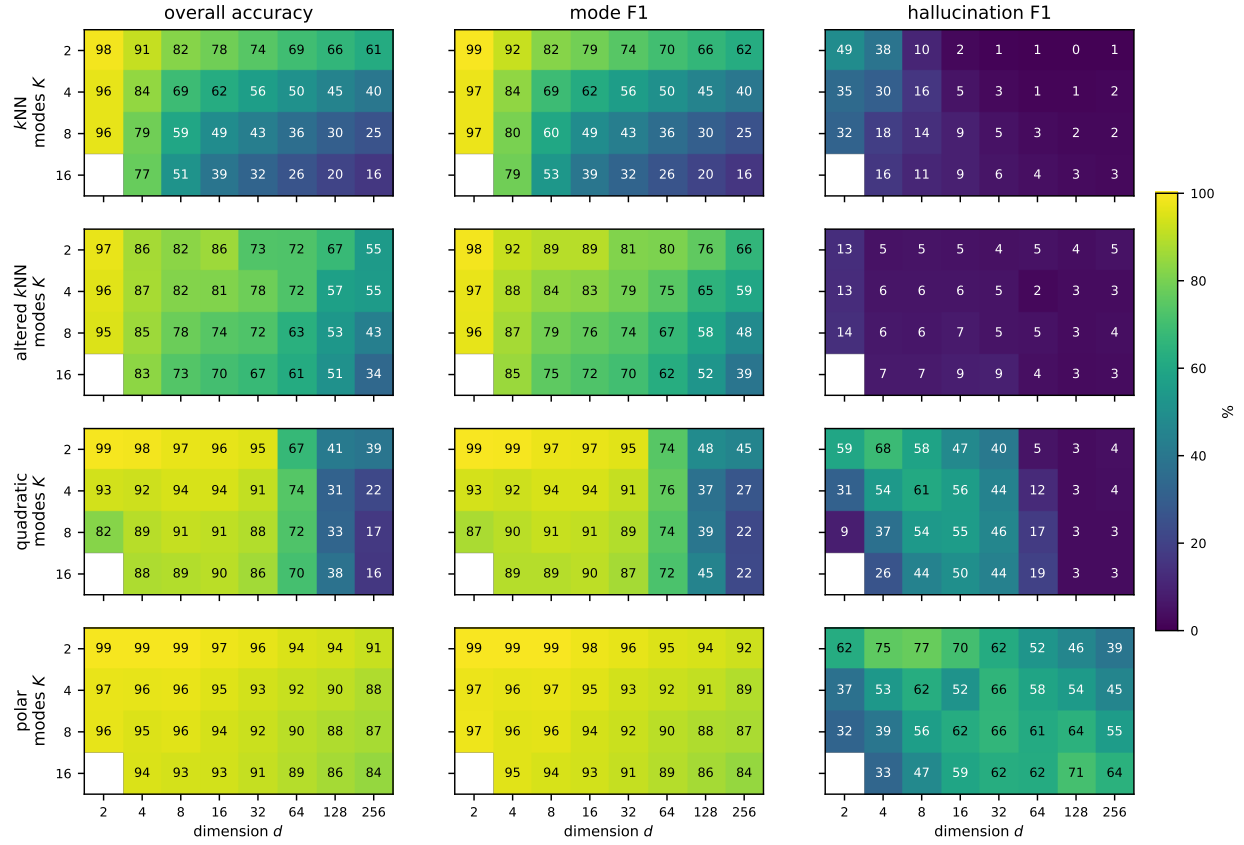


Figure 118: **Weighted RK45, $T = 750$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.24 Weighted RK45, $T = 750$

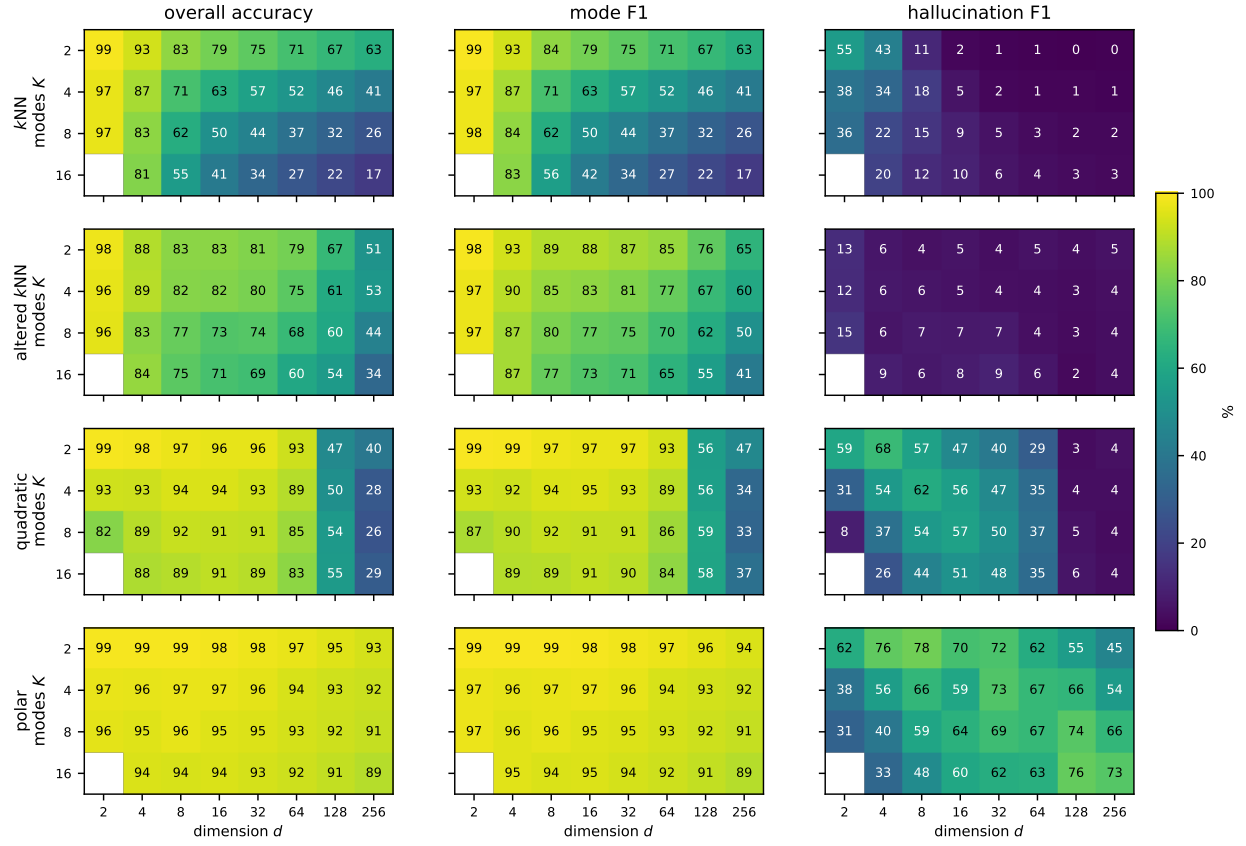


Figure 119: **Weighted RK45, $T = 750$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

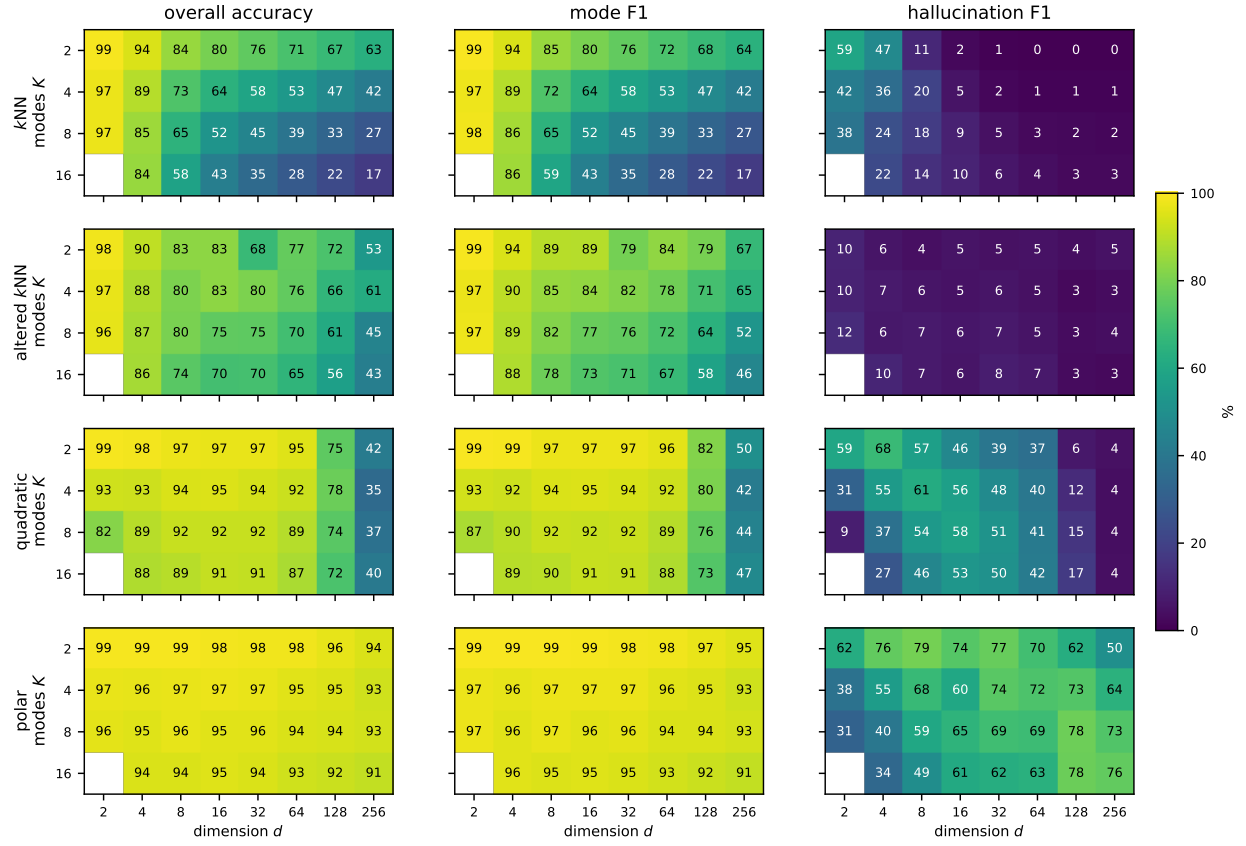


Figure 120: **Weighted RK45, $T = 750$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

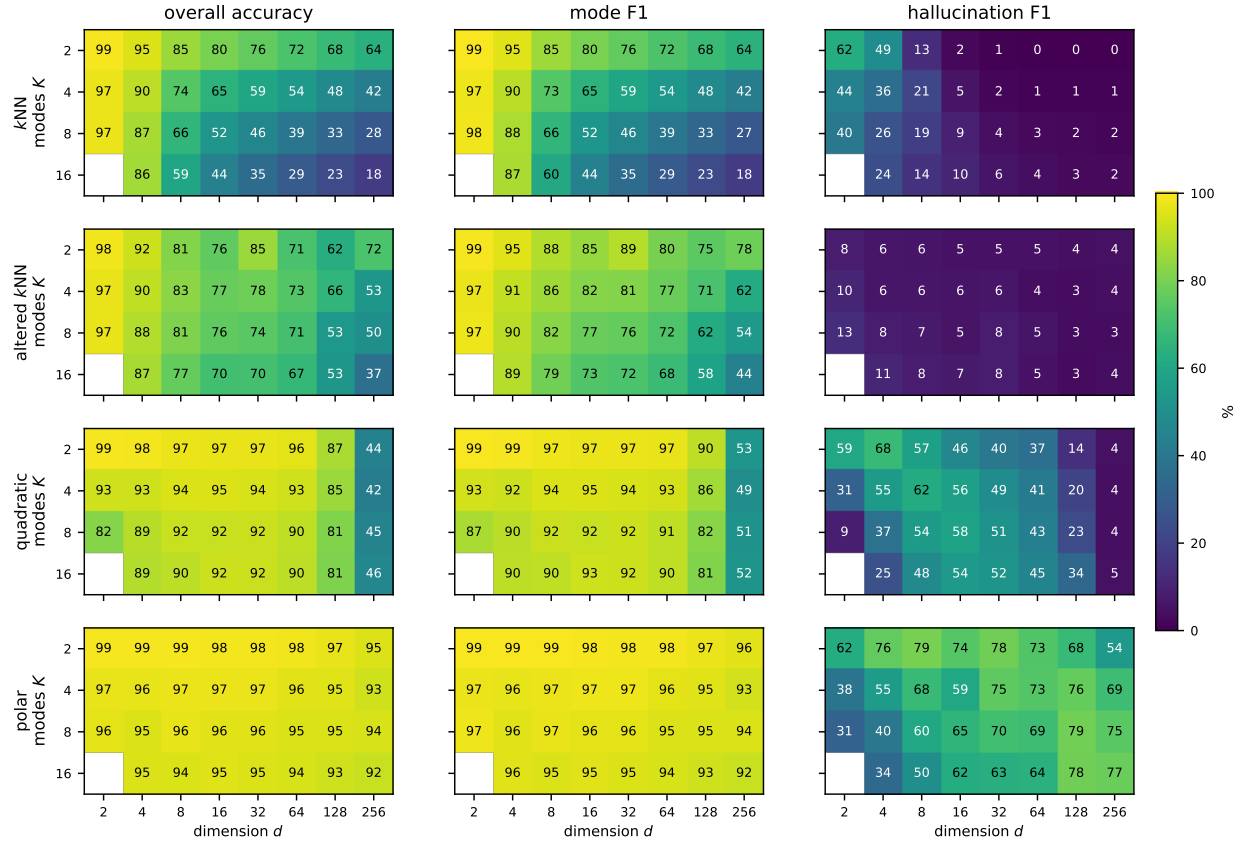


Figure 121: **Weighted RK45, $T = 750$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

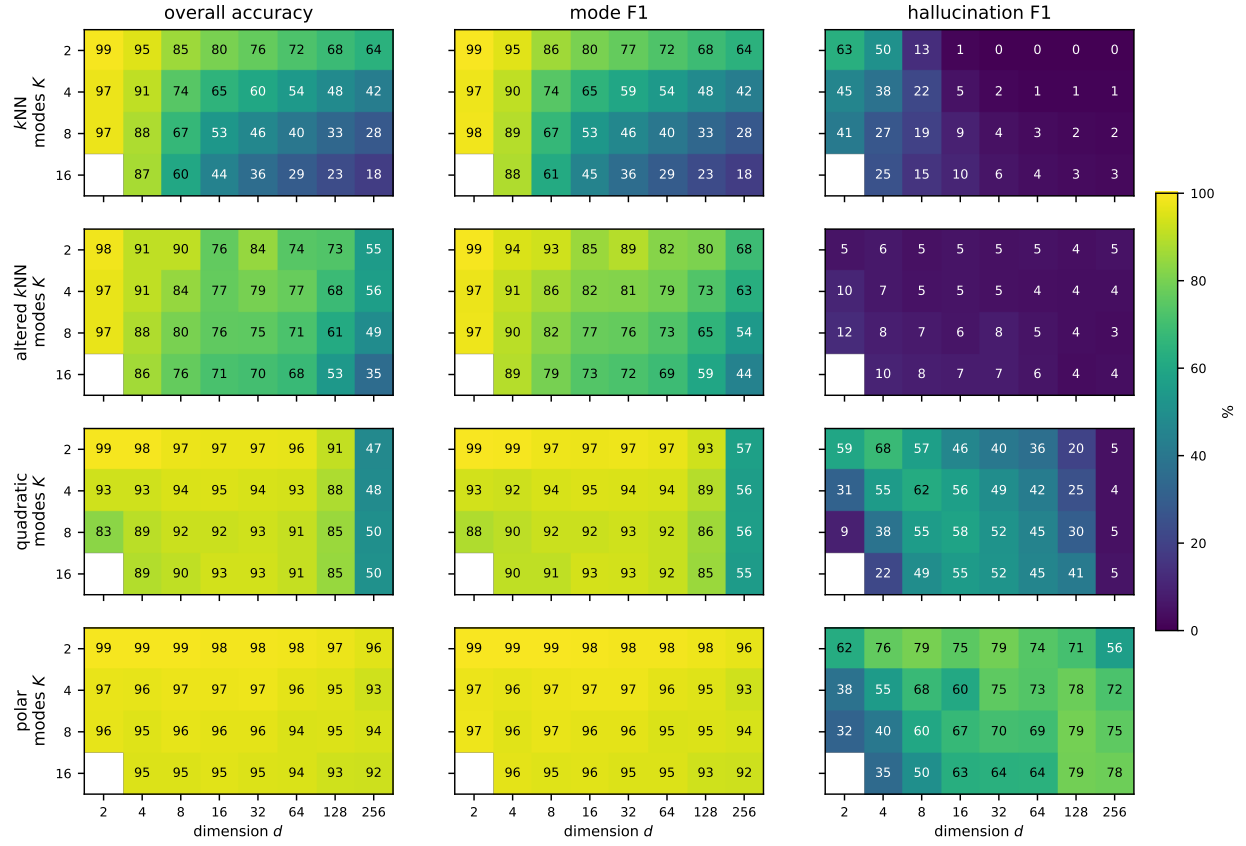


Figure 122: **Weighted RK45, $T = 750$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

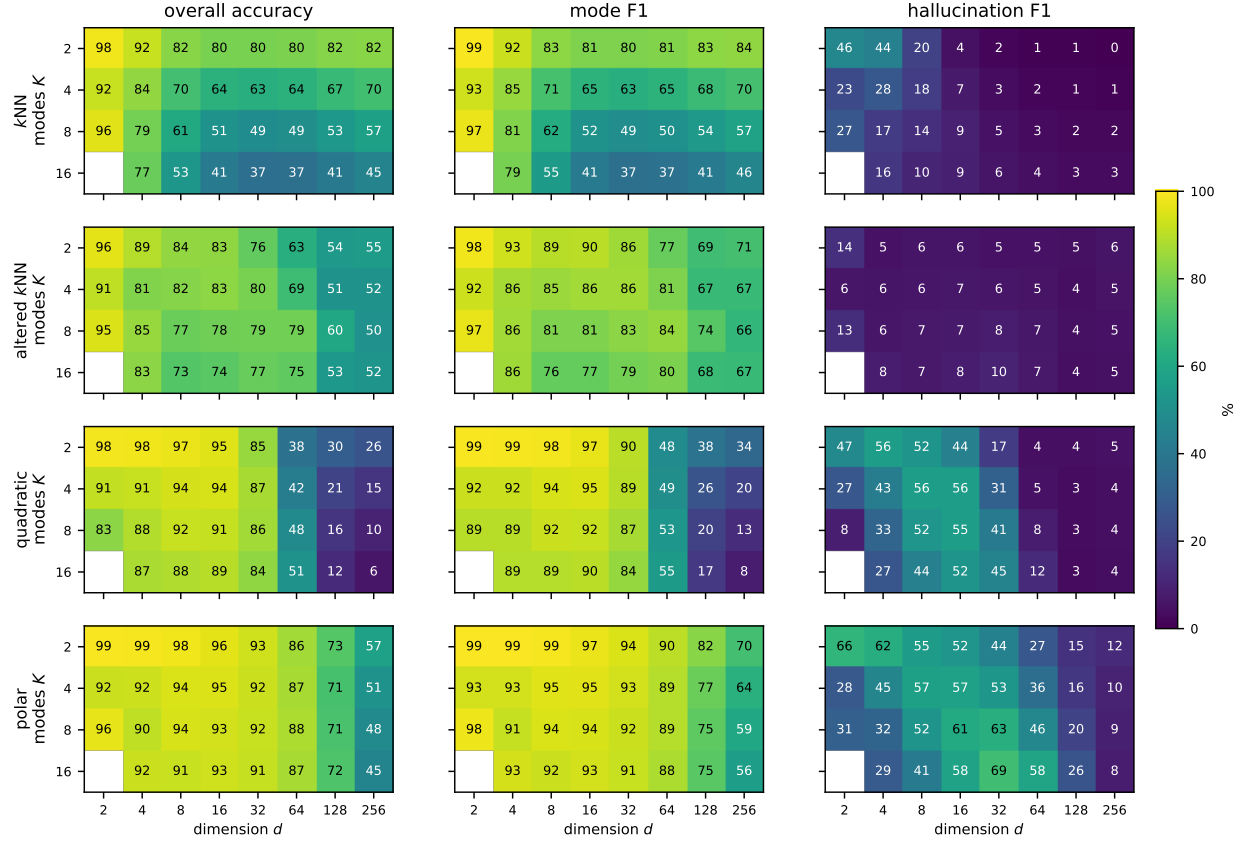


Figure 123: **Unweighted DPM-Solver++(2M)**, $T = 250$, **2,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.25 Unweighted DPM-Solver++(2M), $T = 250$

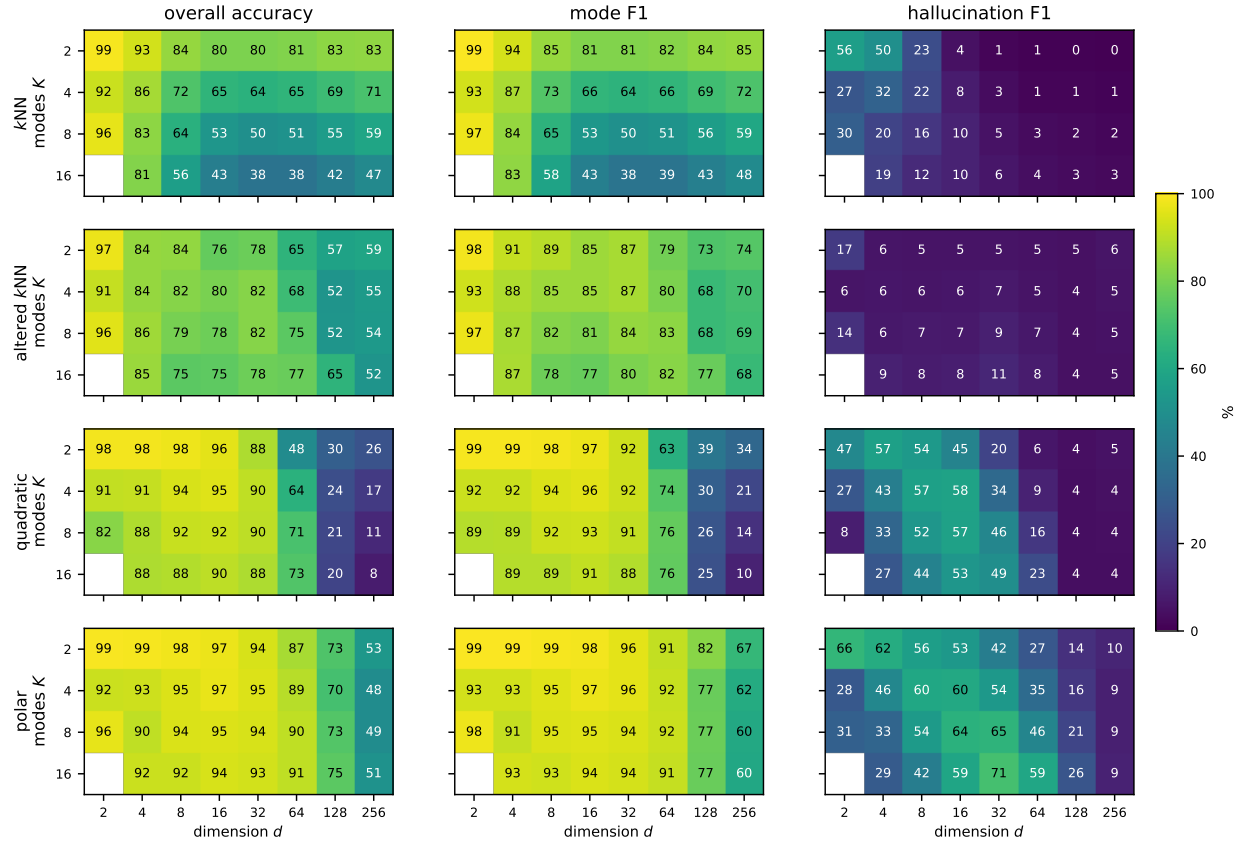


Figure 124: **Unweighted DPM-Solver++(2M), $T = 250$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

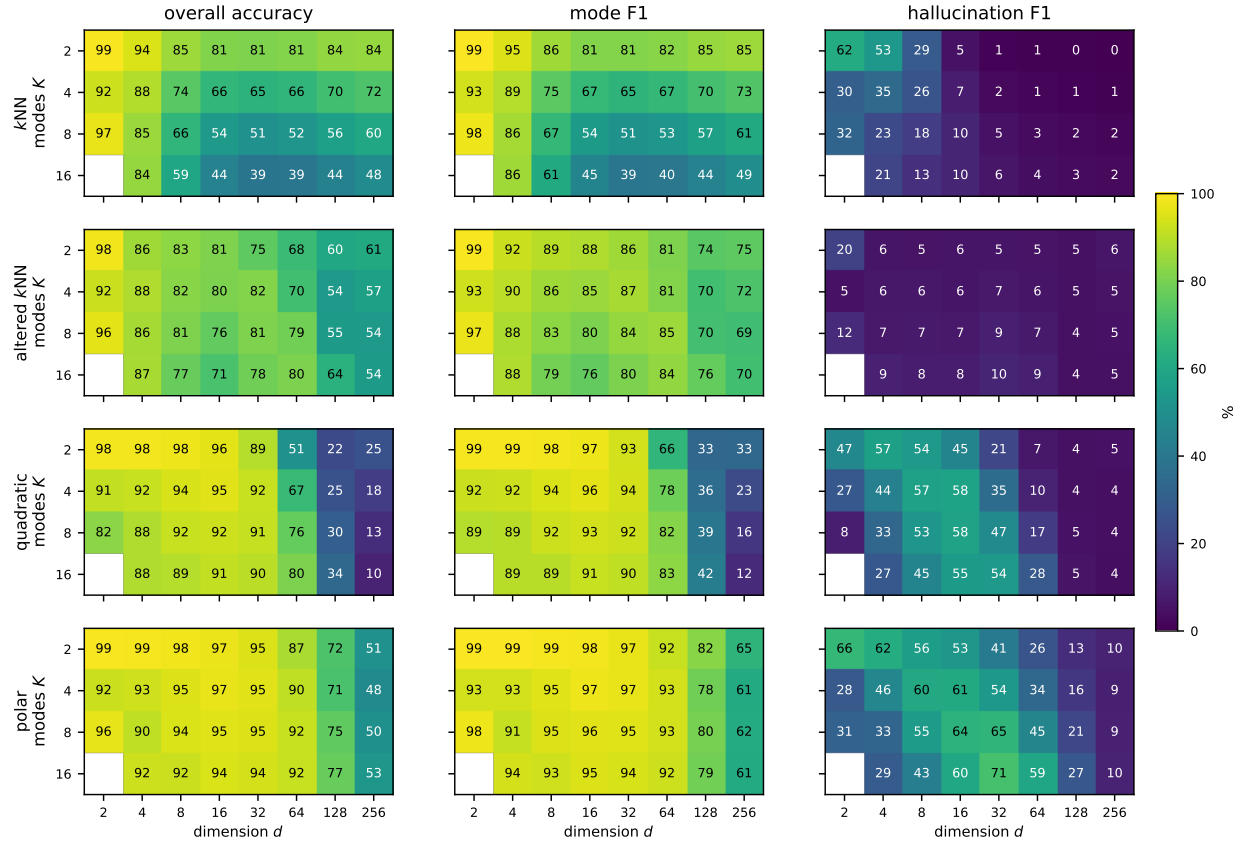


Figure 125: **Unweighted DPM-Solver++(2M), $T = 250$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

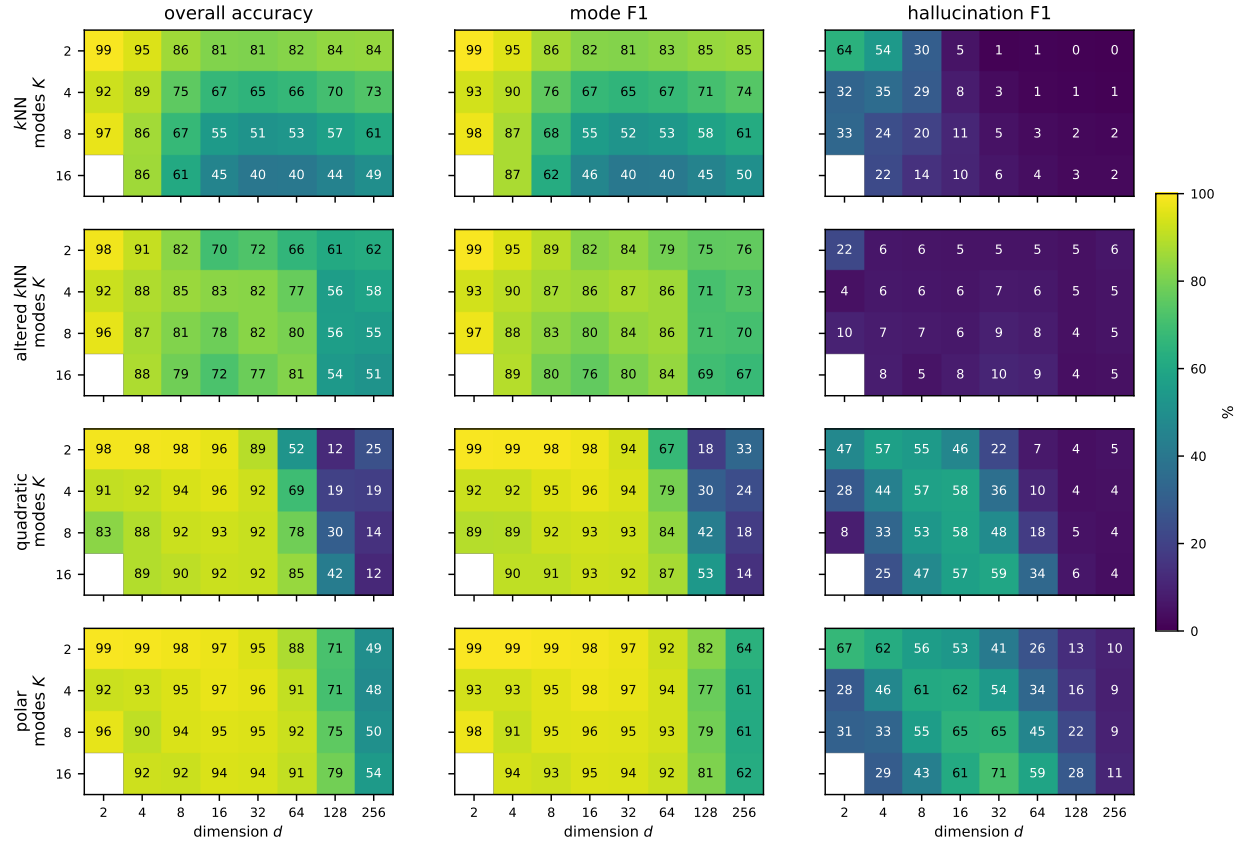


Figure 126: **Unweighted DPM-Solver++(2M), $T = 250$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

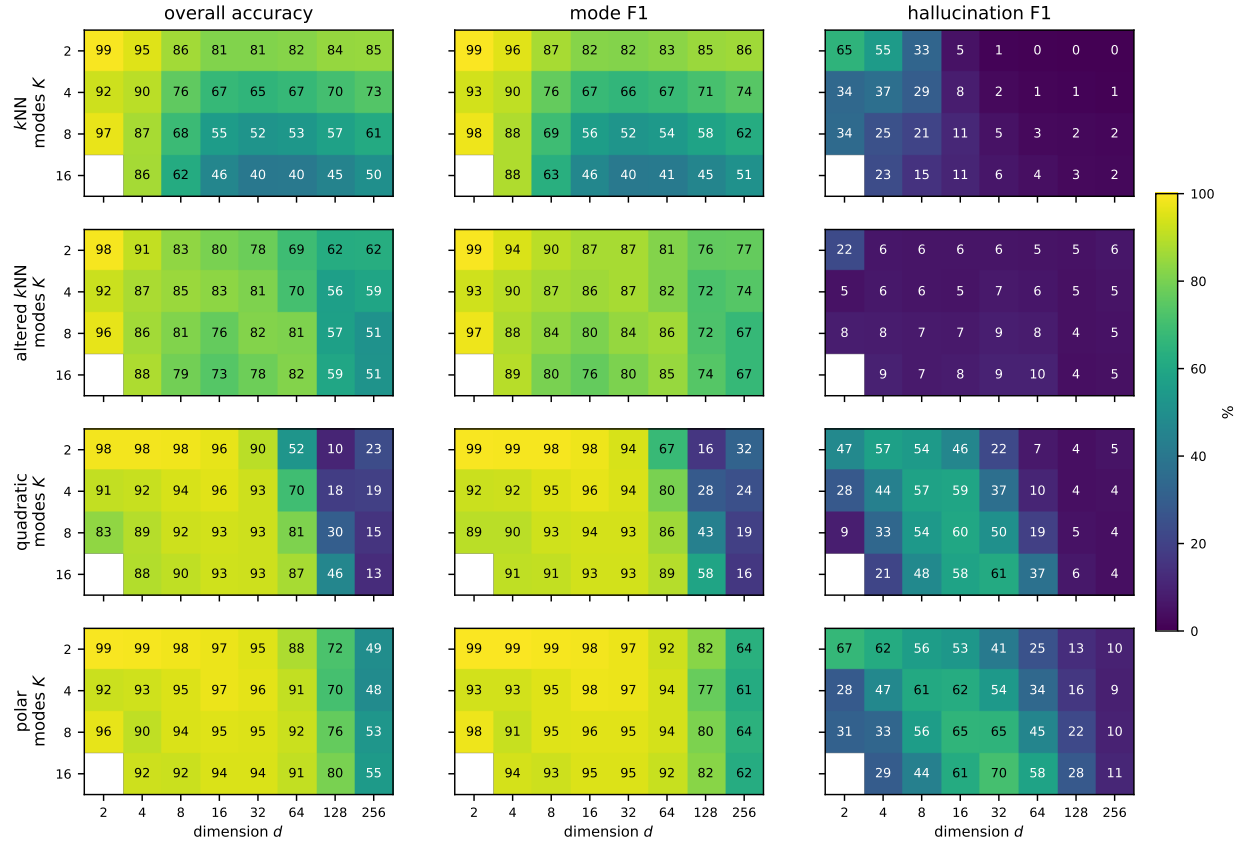


Figure 127: **Unweighted DPM-Solver++(2M), $T = 250$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

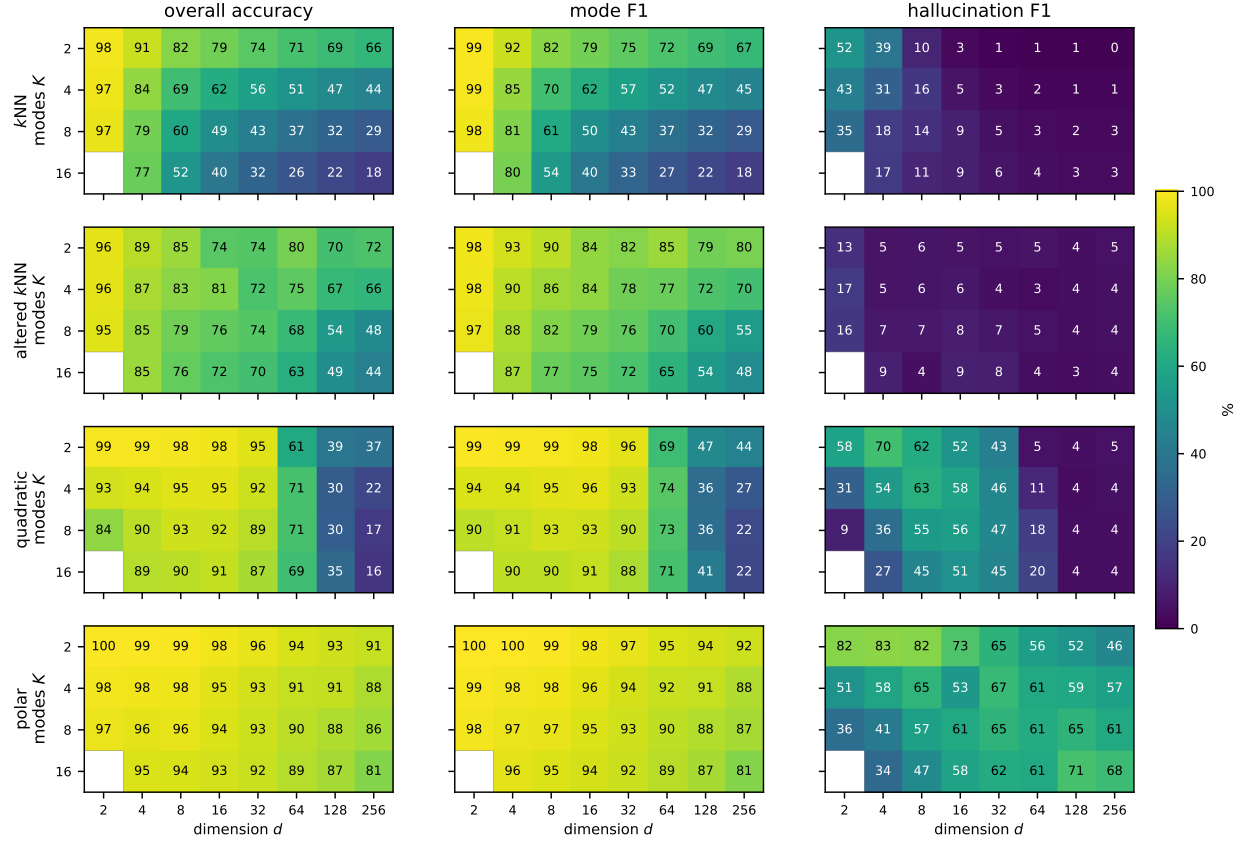


Figure 128: **Unweighted DPM-Solver++(2M), $T = 500$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.26 Unweighted DPM-Solver++(2M), $T = 500$

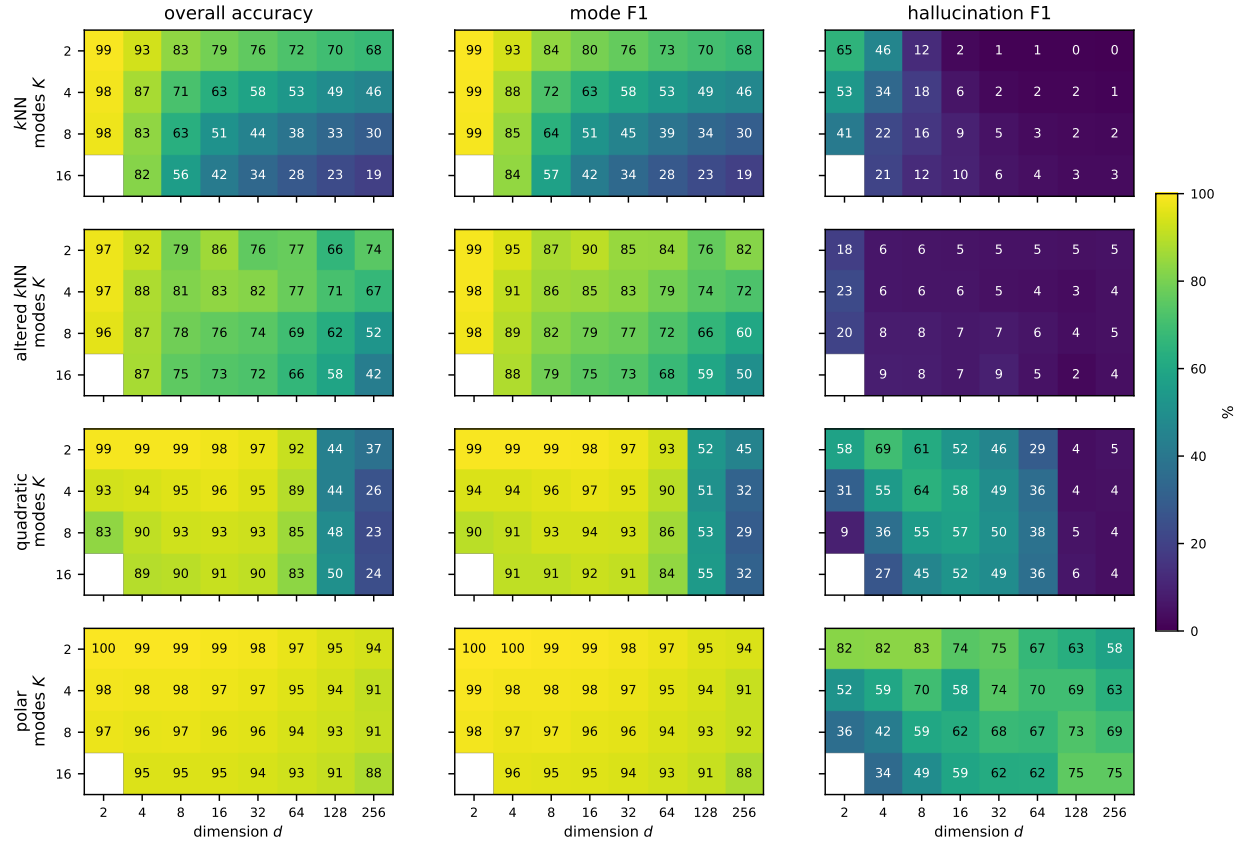


Figure 129: **Unweighted DPM-Solver++(2M), $T = 500$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

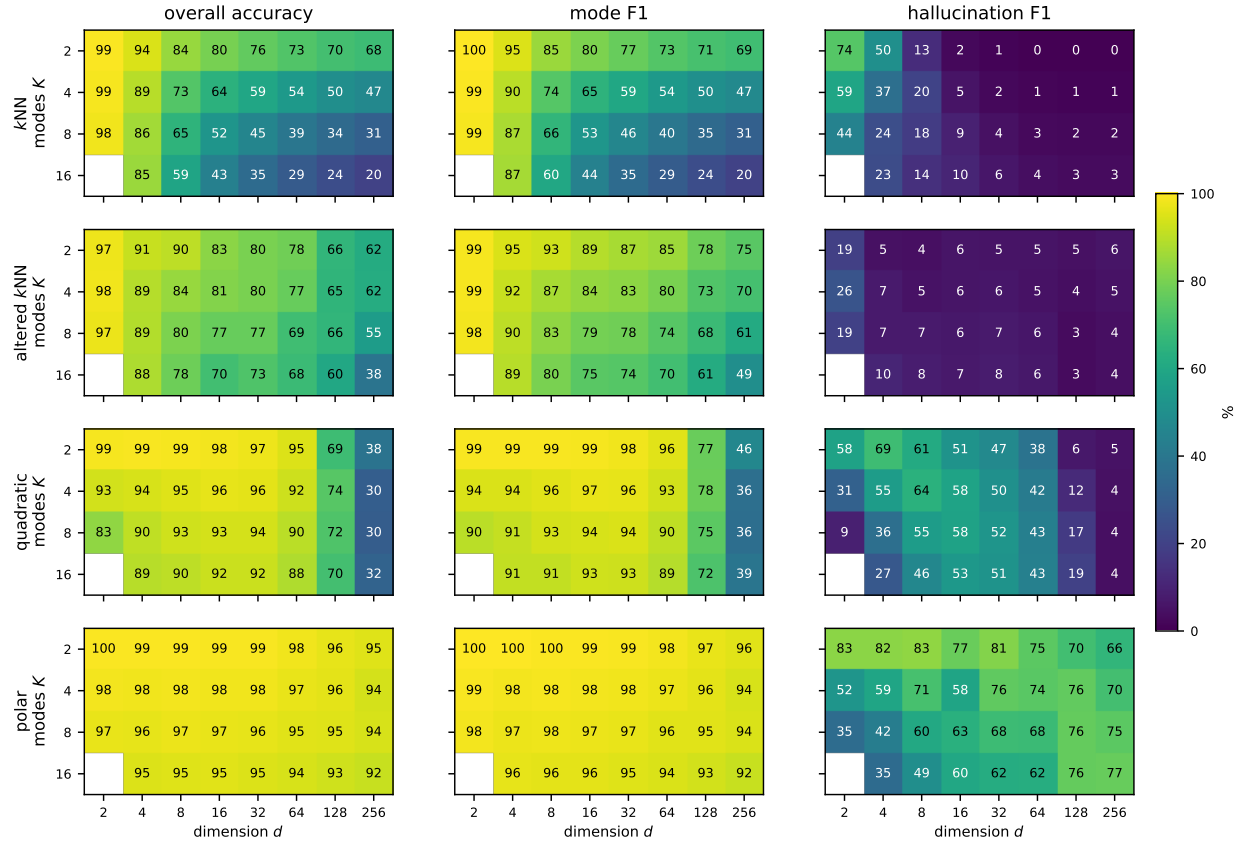


Figure 130: **Unweighted DPM-Solver++(2M), $T = 500$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

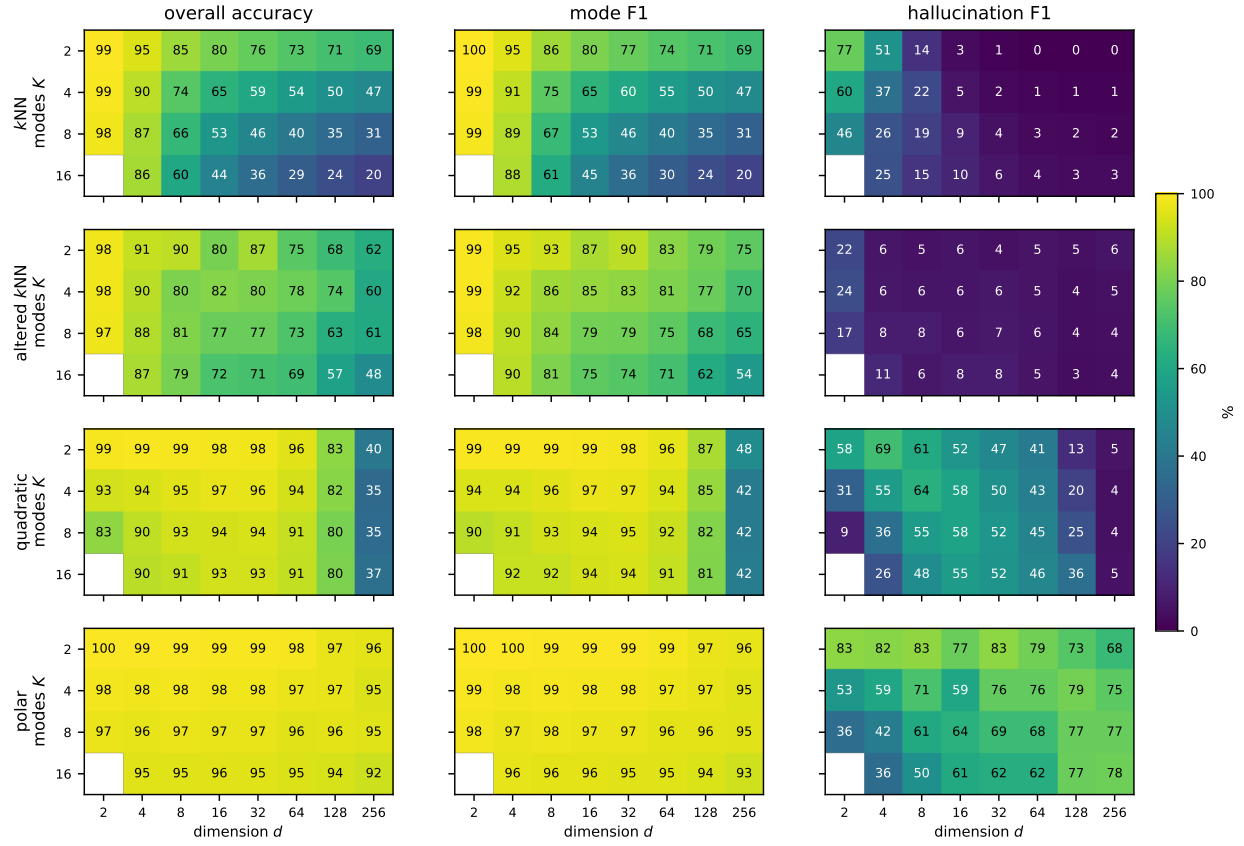


Figure 131: **Unweighted DPM-Solver++(2M), $T = 500$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

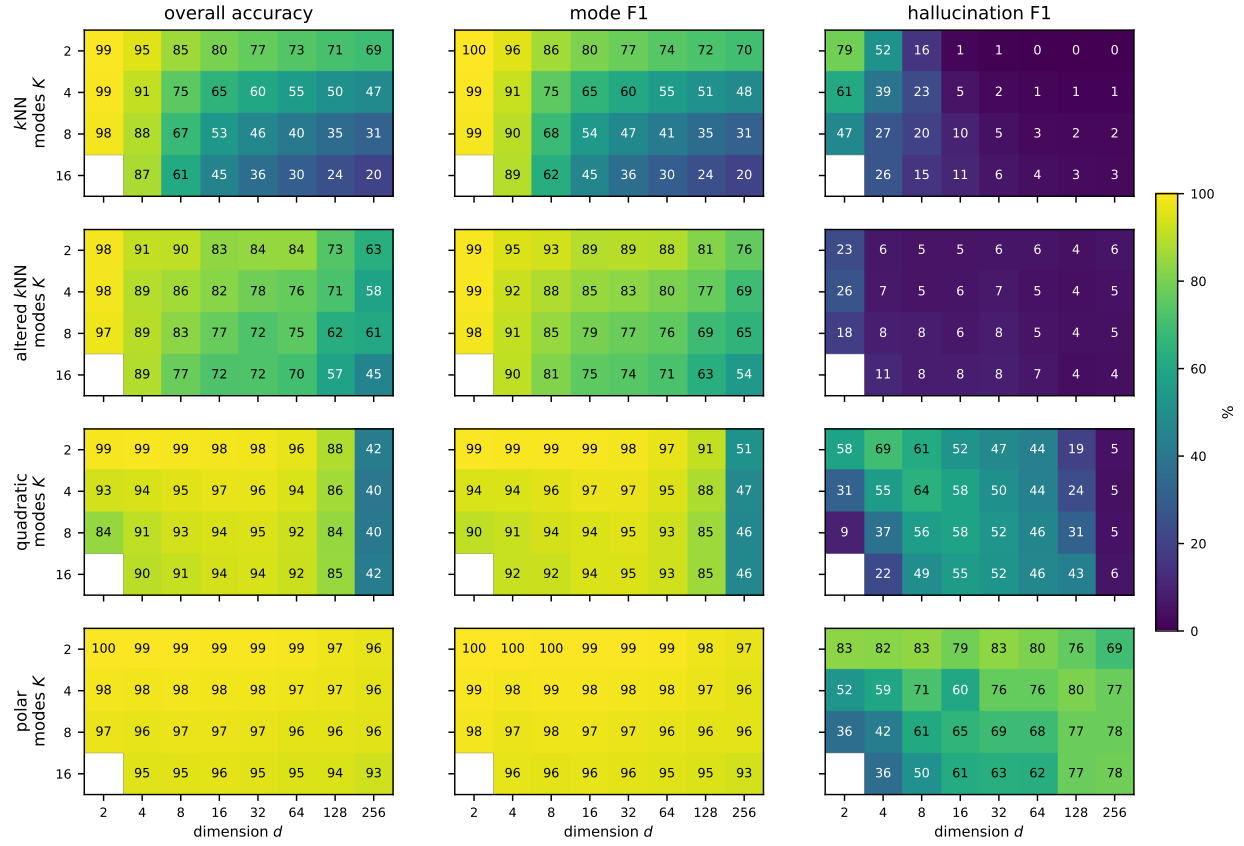


Figure 132: **Unweighted DPM-Solver++(2M), $T = 500$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

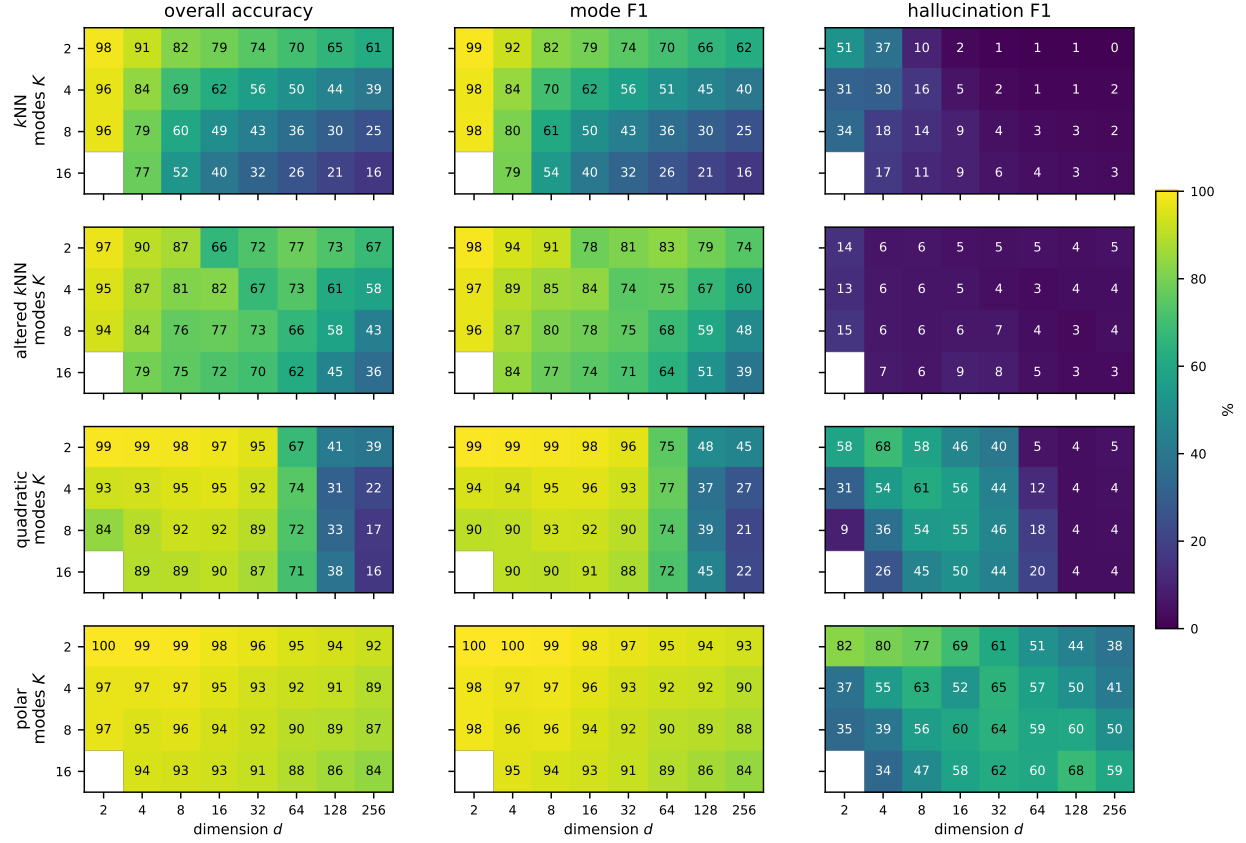


Figure 133: **Unweighted DPM-Solver++(2M), $T = 750$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.27 Unweighted DPM-Solver++(2M), $T = 750$

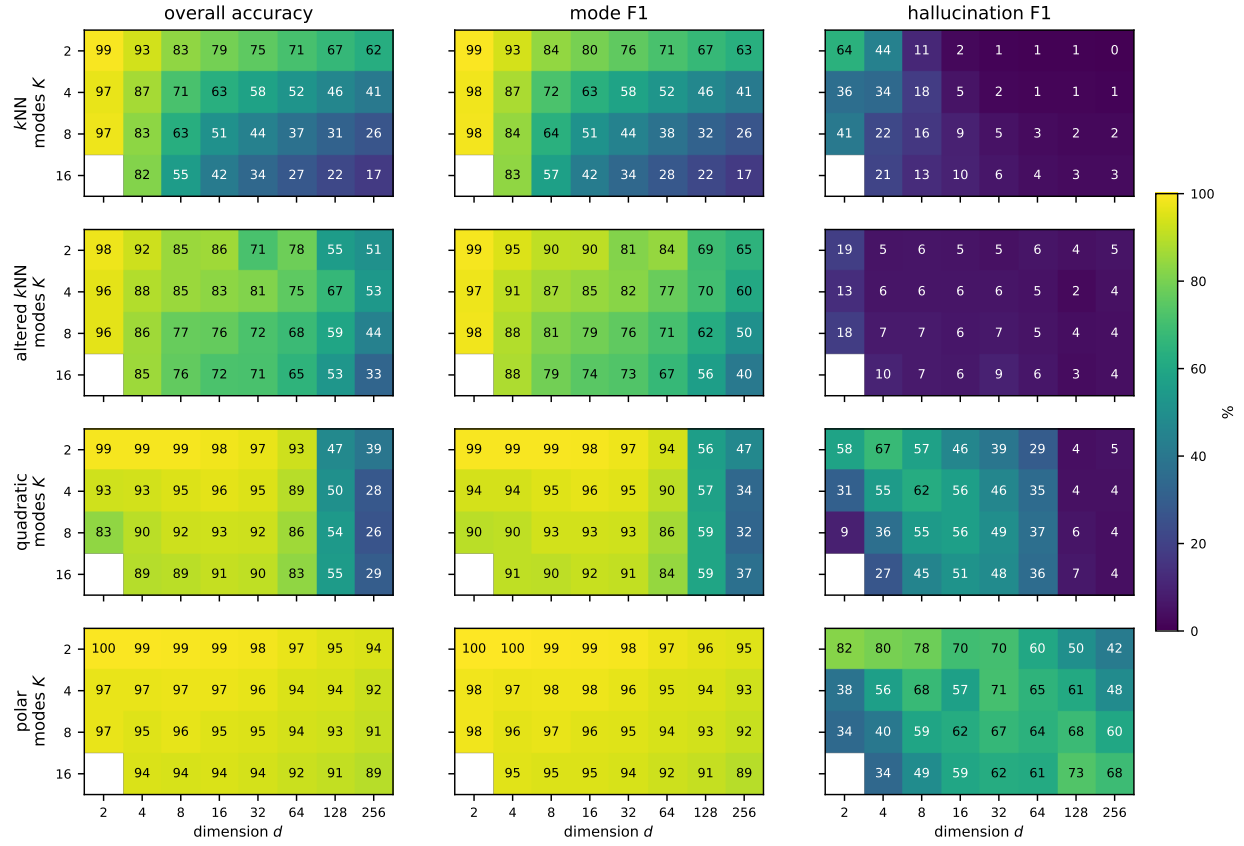


Figure 134: **Unweighted DPM-Solver++(2M), $T = 750$, 5,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

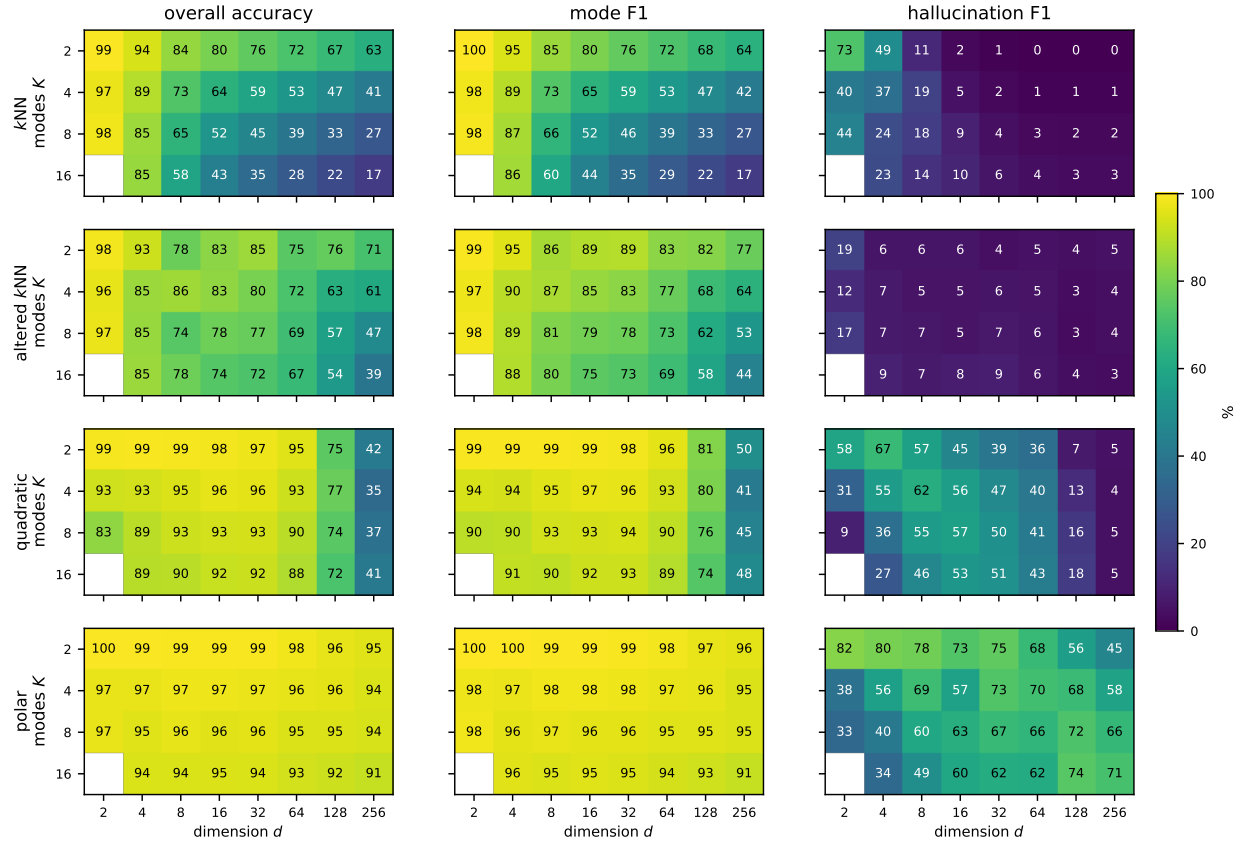


Figure 135: **Unweighted DPM-Solver++(2M), $T = 750$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

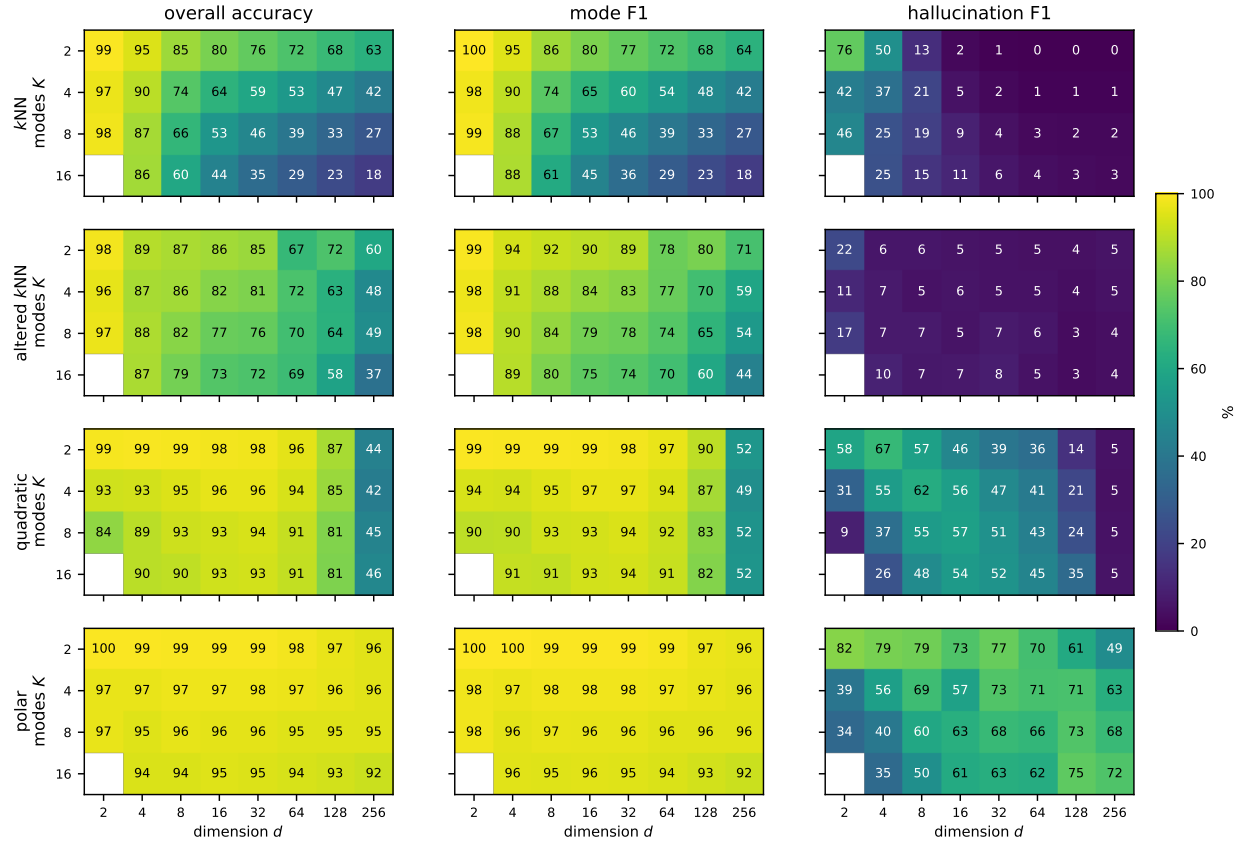


Figure 136: **Unweighted DPM-Solver++(2M), $T = 750$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

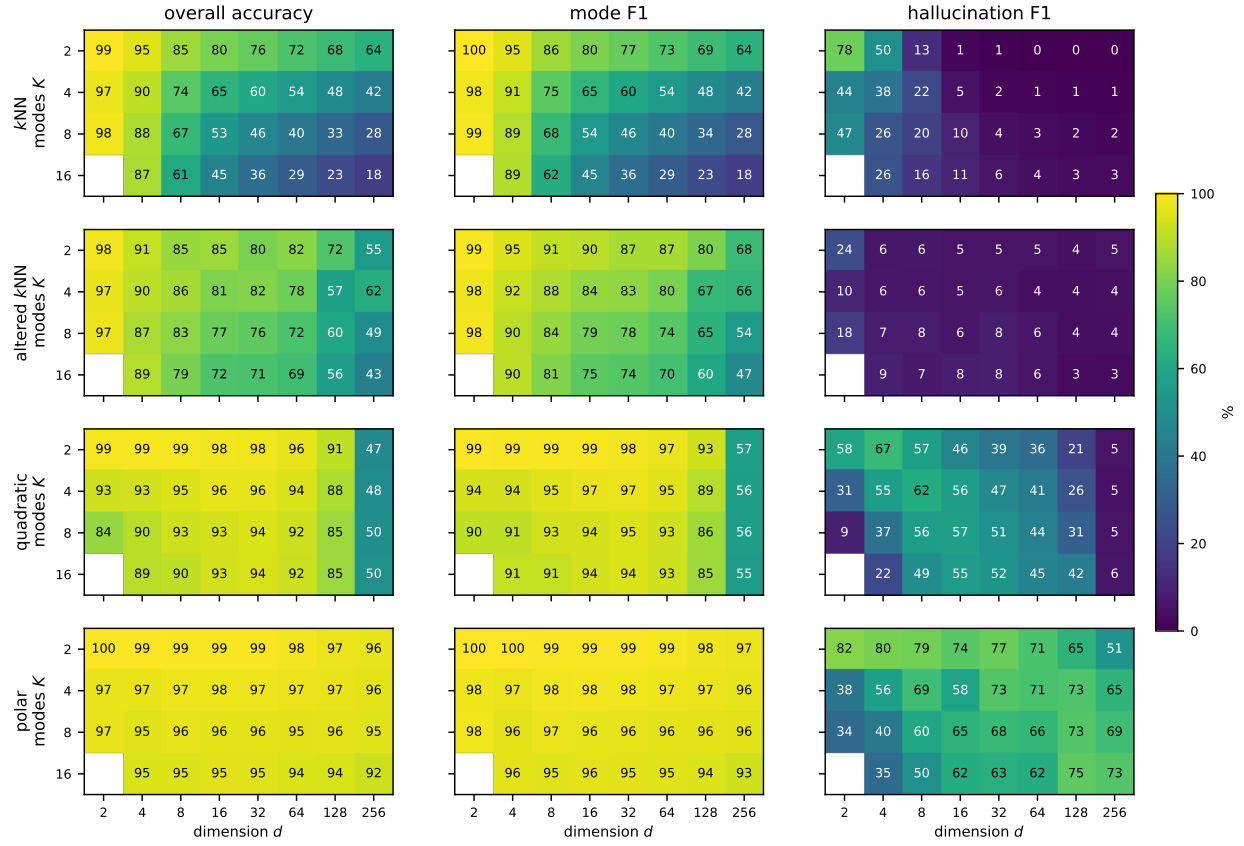


Figure 137: **Unweighted DPM-Solver++(2M), $T = 750$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

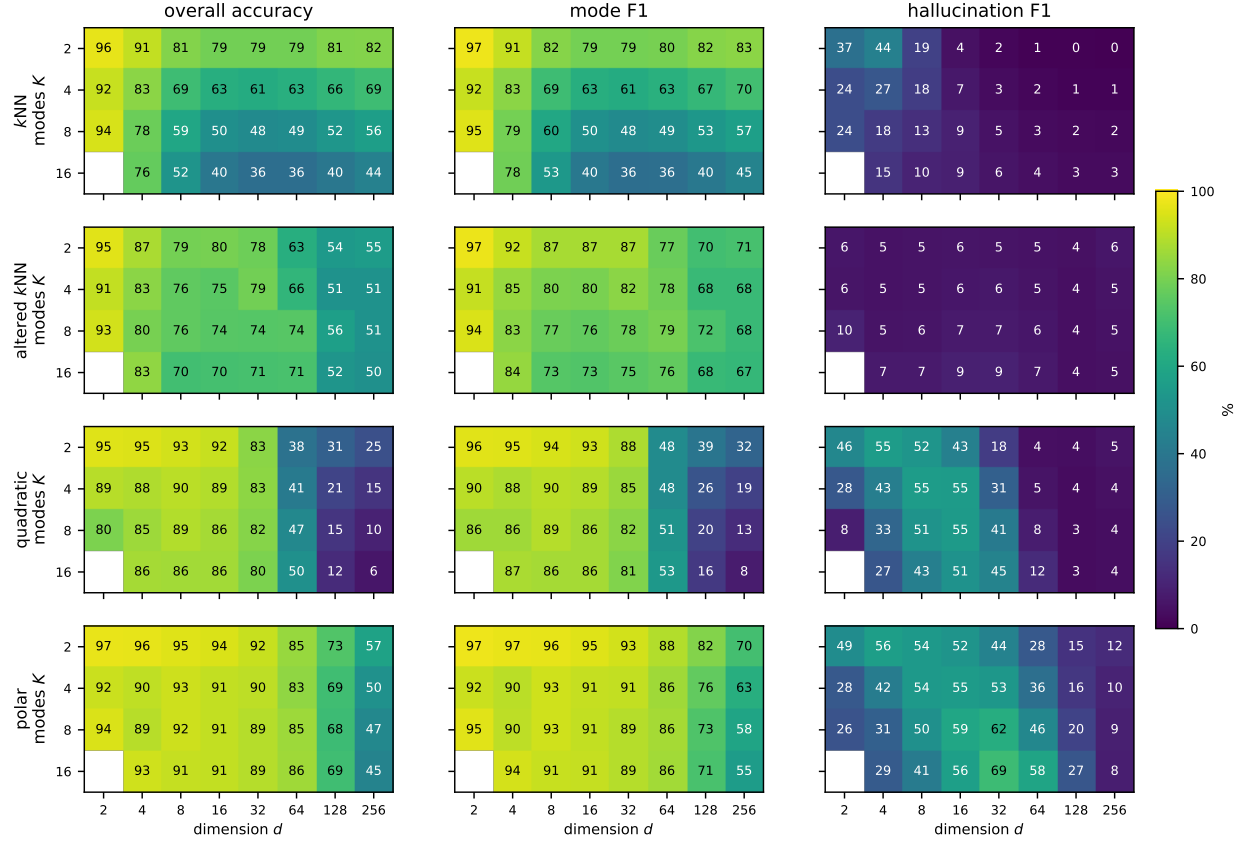


Figure 138: **Weighted DPM-Solver++(2M)**, $T = 250$, **2,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.28 Weighted DPM-Solver++(2M), $T = 250$

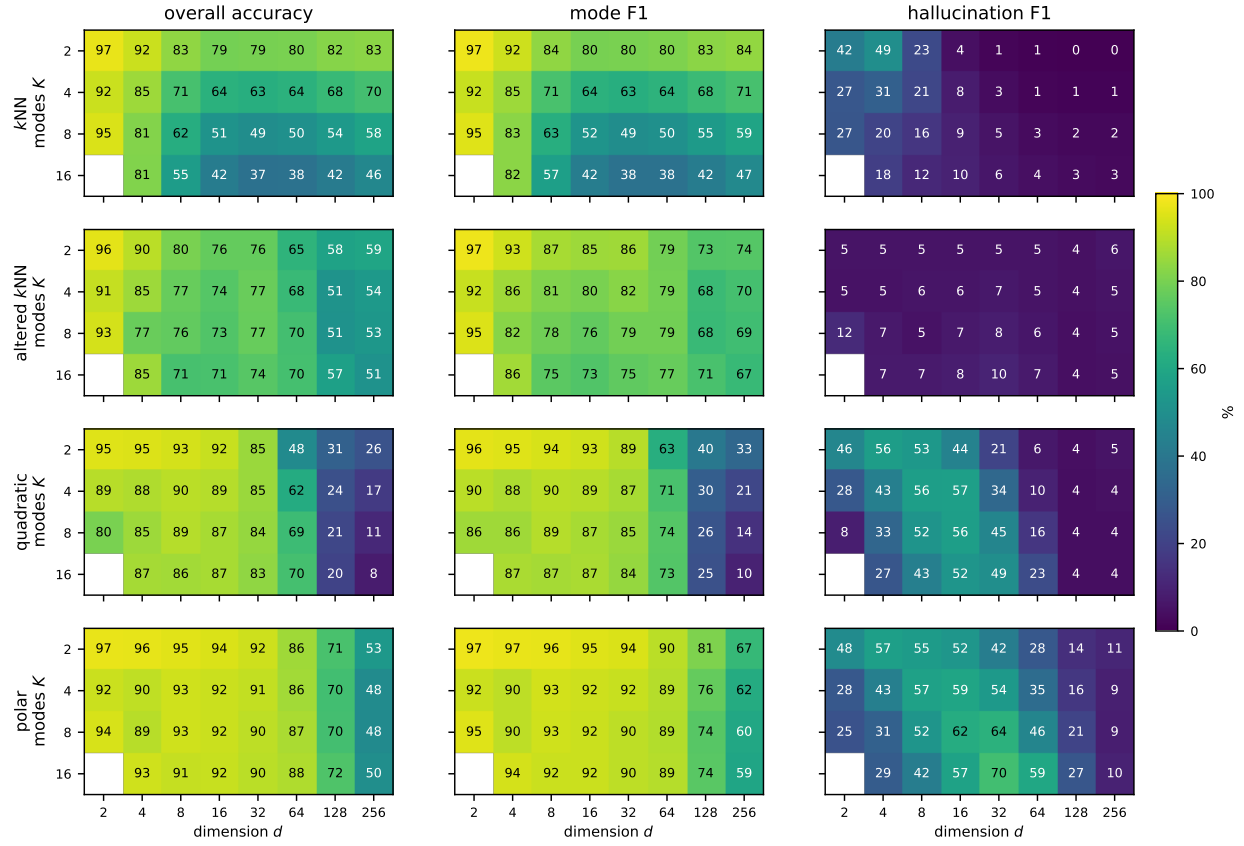


Figure 139: **Weighted DPM-Solver++(2M)**, $T = 250$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

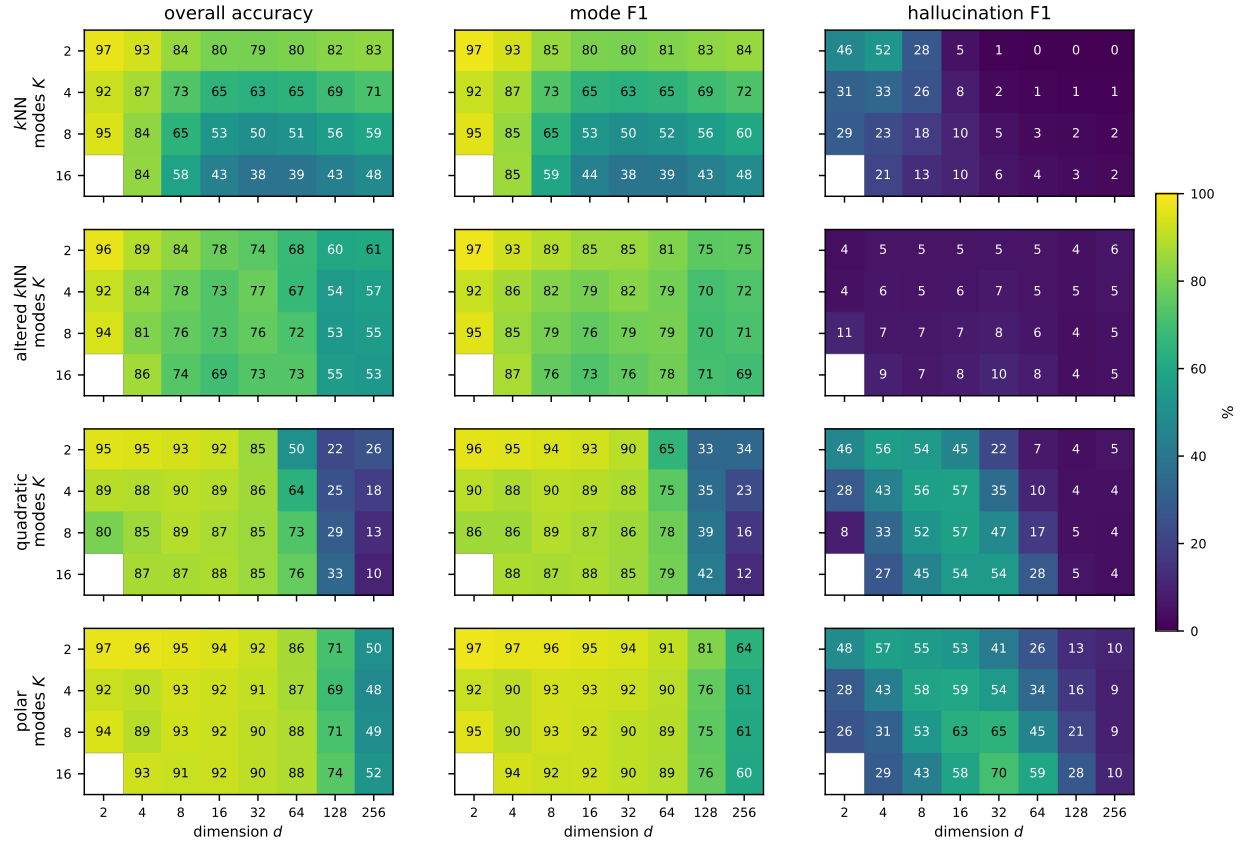


Figure 140: **Weighted DPM-Solver++(2M)**, $T = 250$, **10,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

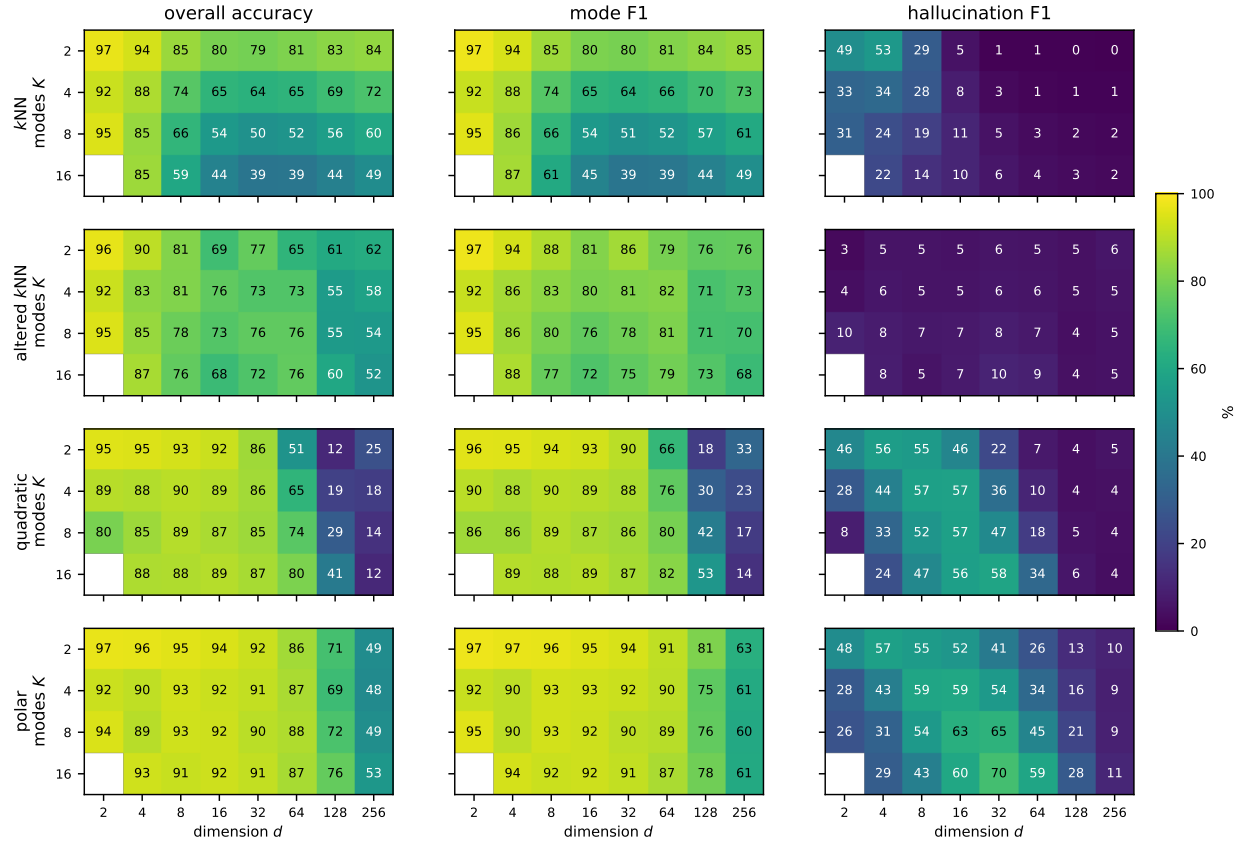


Figure 141: **Weighted DPM-Solver++(2M)**, $T = 250$, **15,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

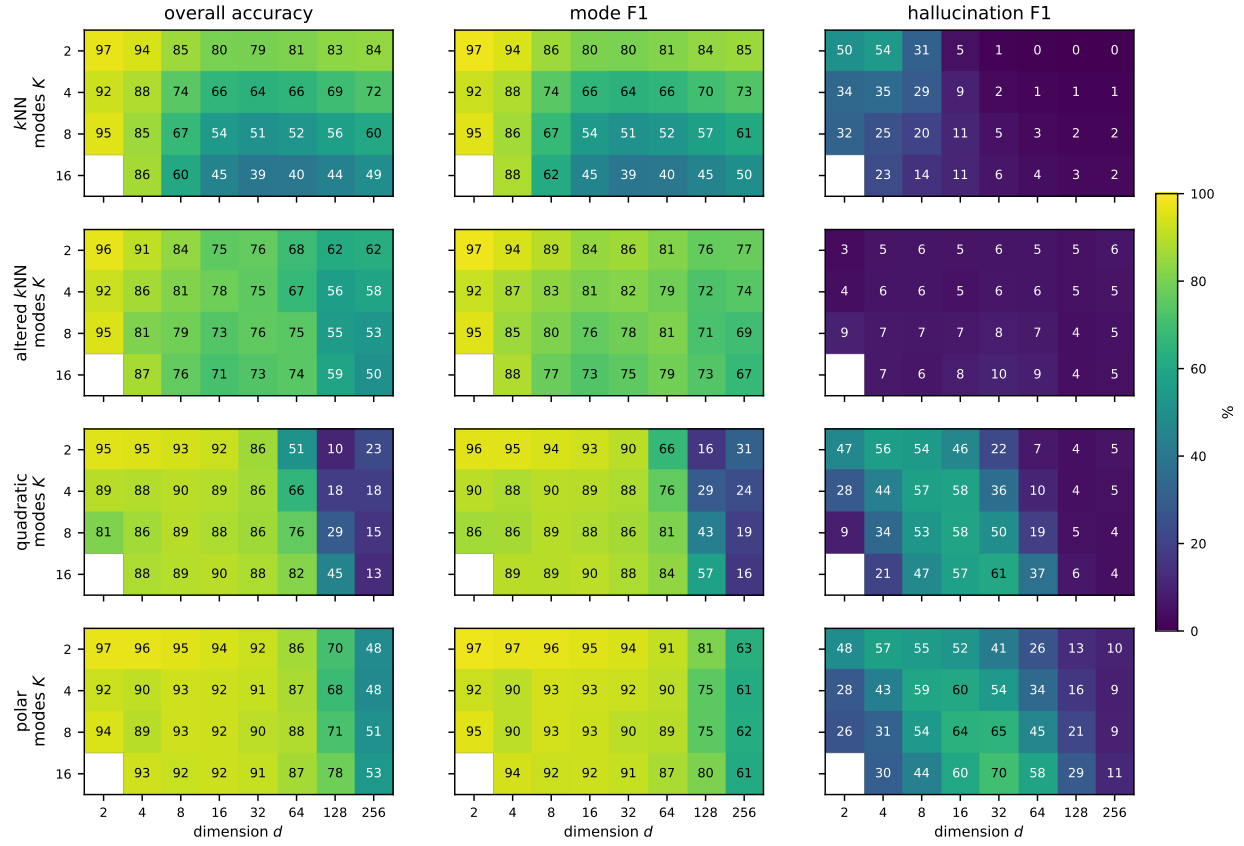


Figure 142: **Weighted DPM-Solver++(2M), $T = 250$, 20,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

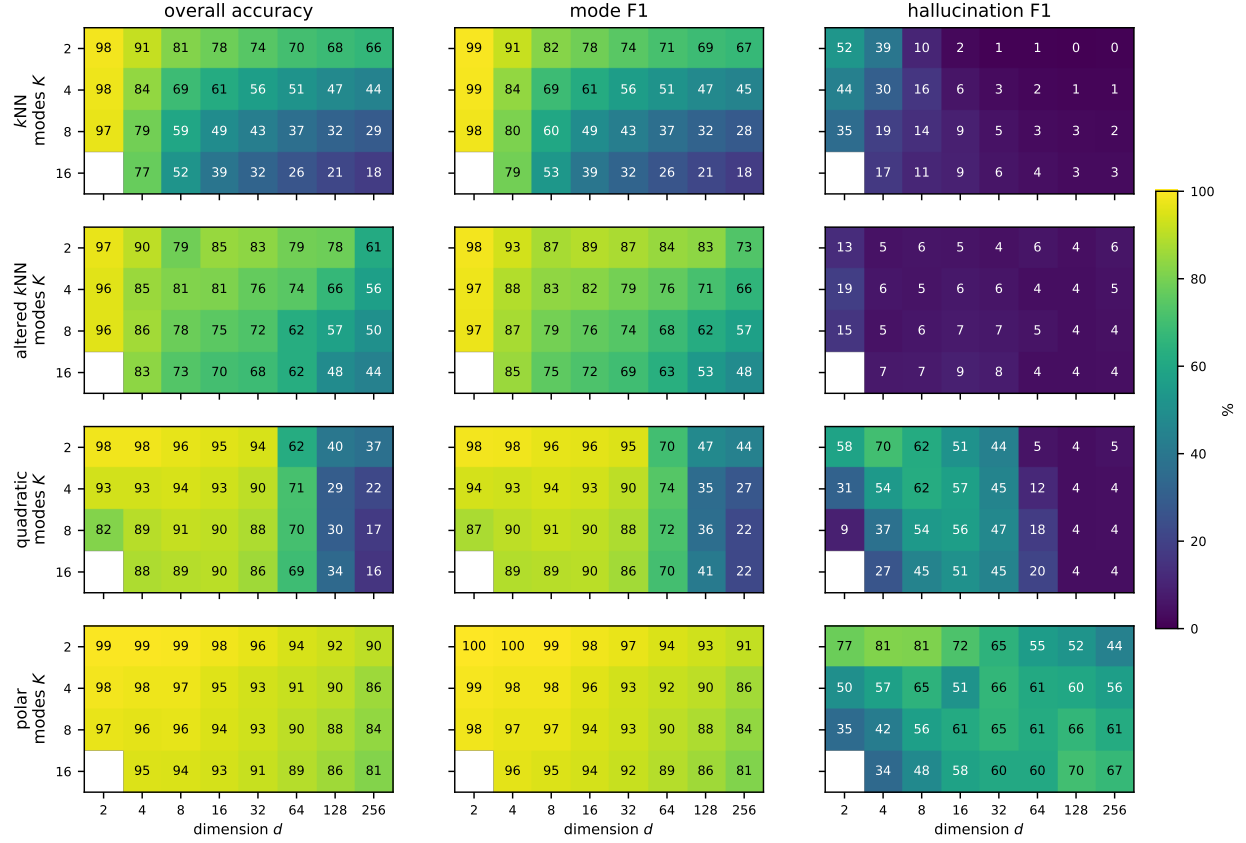


Figure 143: **Weighted DPM-Solver++(2M), $T = 500$, 2,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.29 Weighted DPM-Solver++(2M), $T = 500$

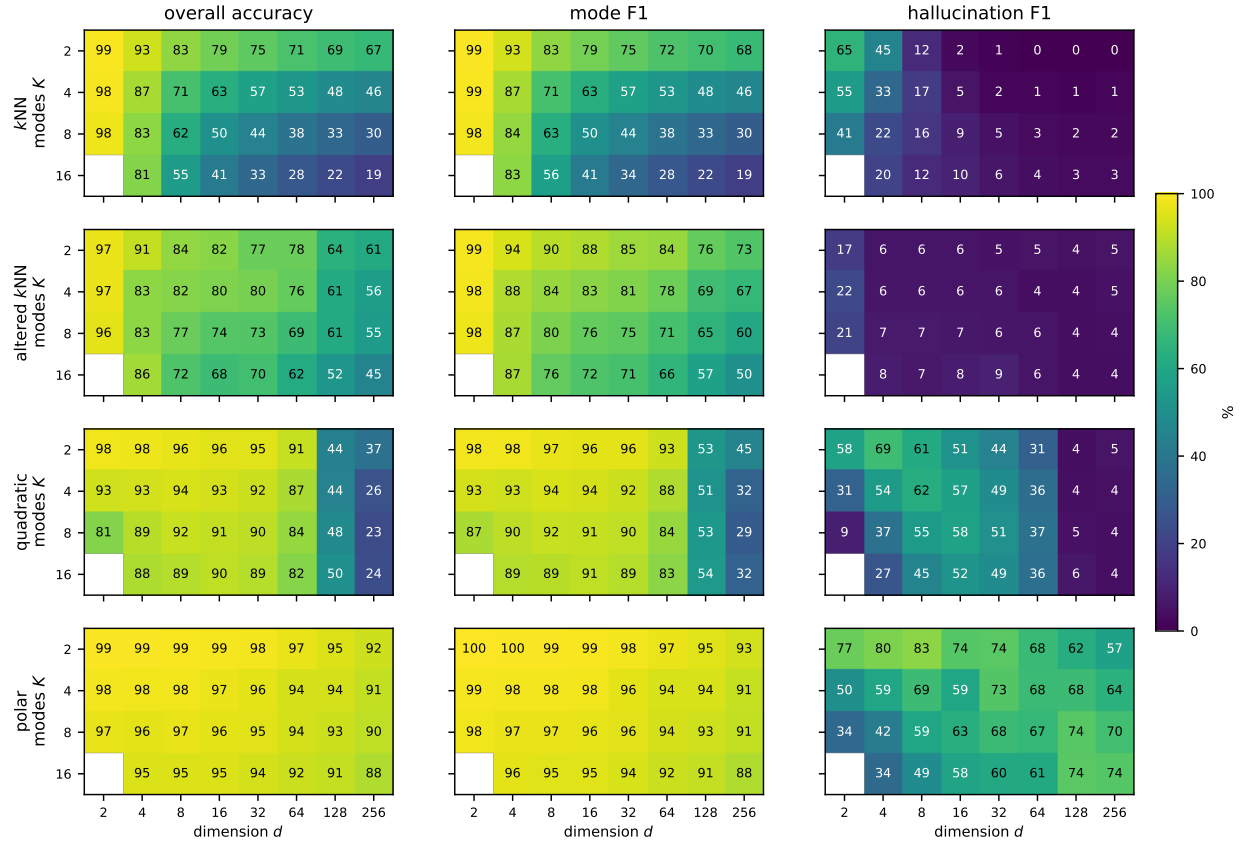


Figure 144: **Weighted DPM-Solver++(2M)**, $T = 500$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

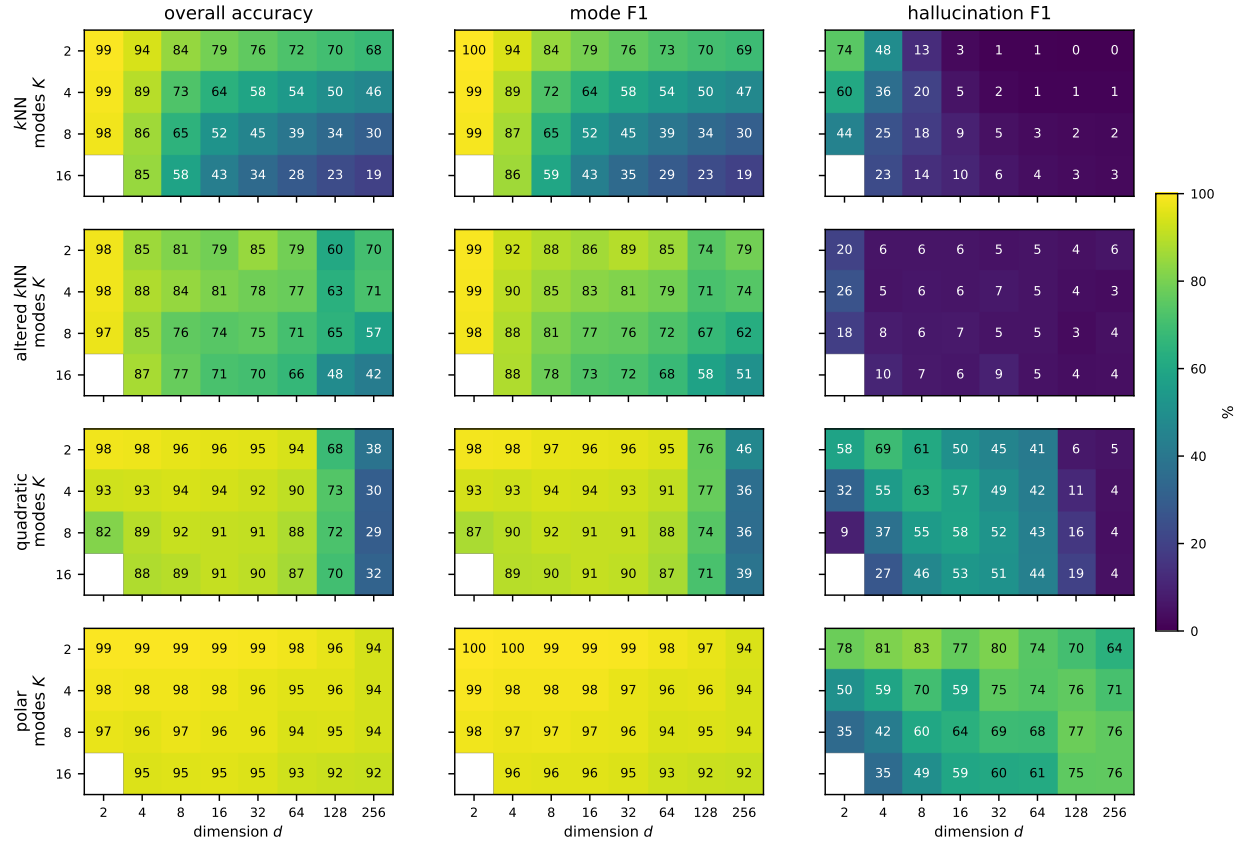


Figure 145: **Weighted DPM-Solver++(2M), $T = 500$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

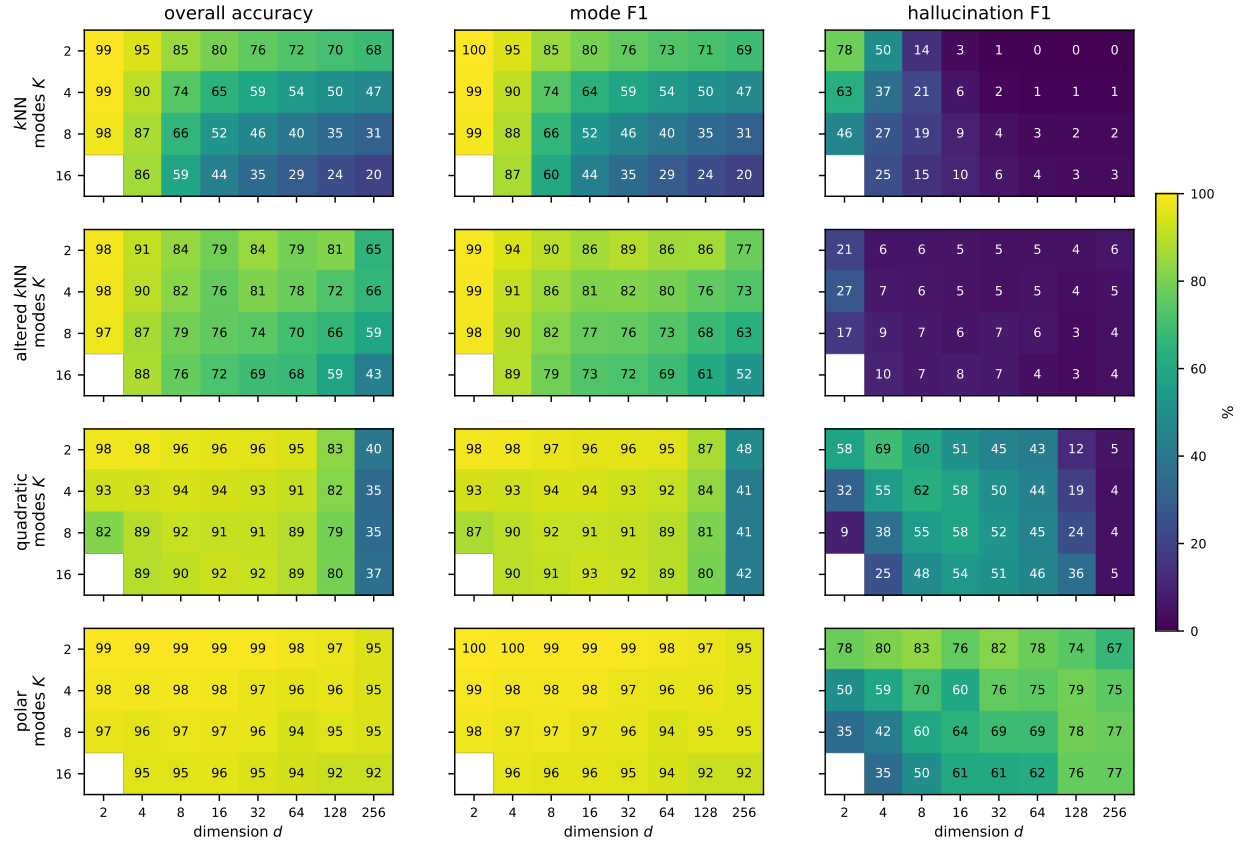


Figure 146: **Weighted DPM-Solver++(2M), $T = 500$, 15,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

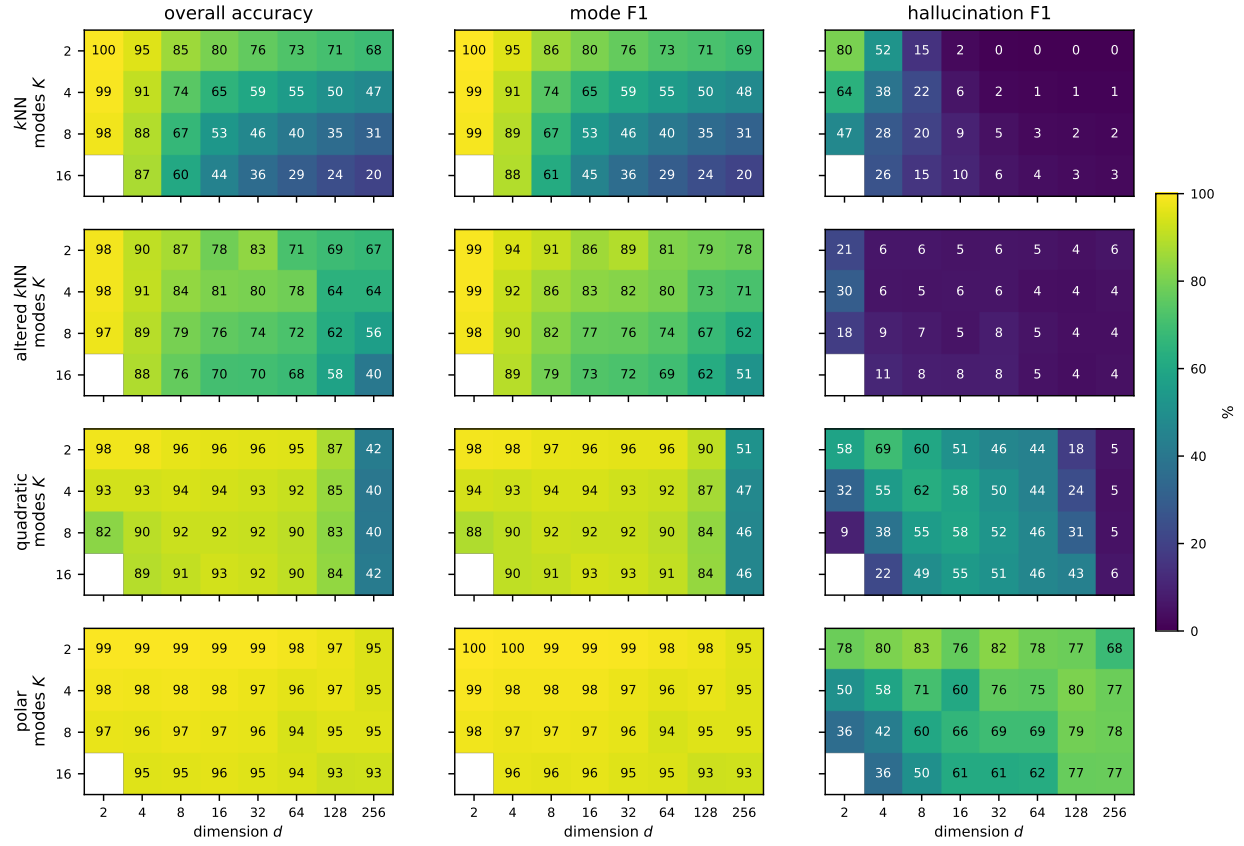


Figure 147: **Weighted DPM-Solver++(2M)**, $T = 500$, **20,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

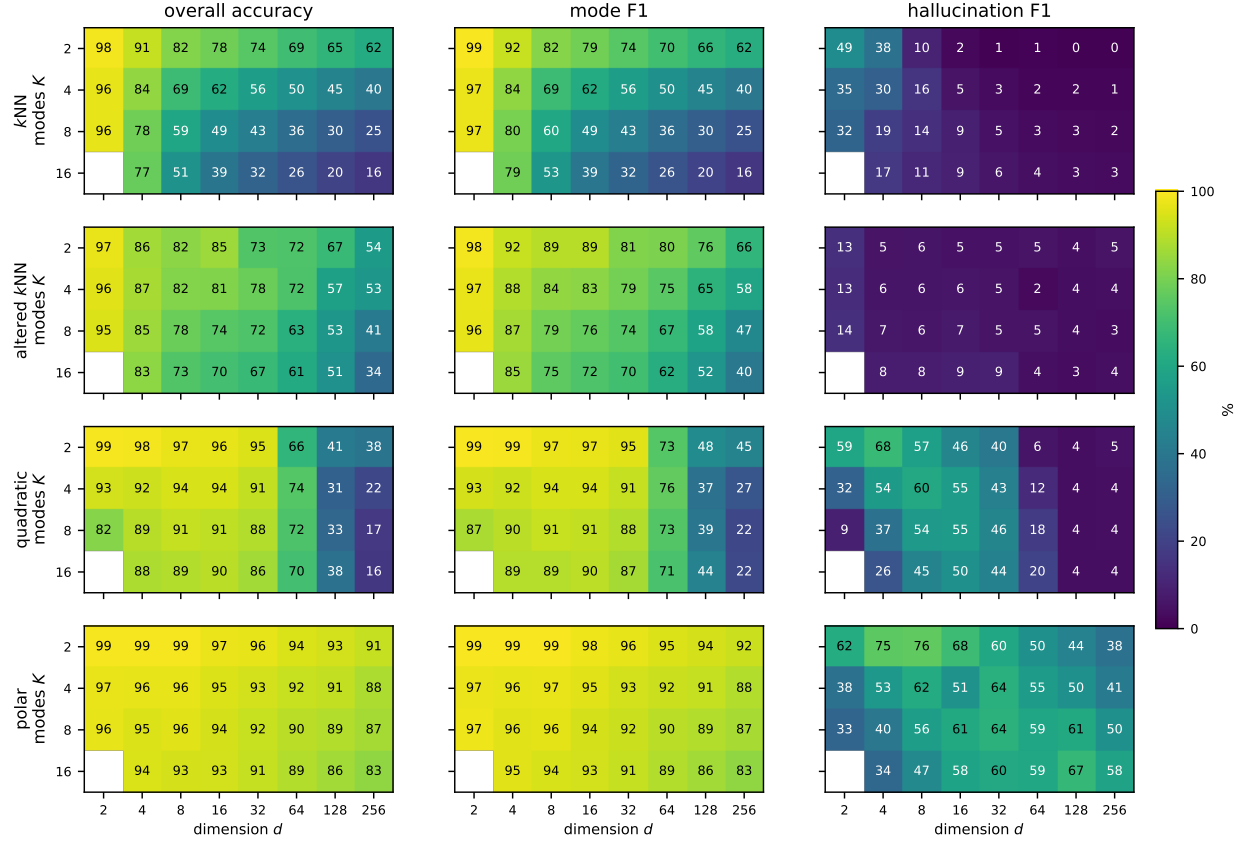


Figure 148: **Weighted DPM-Solver++(2M)**, $T = 750$, **2,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

3.30 Weighted DPM-Solver++(2M), $T = 750$

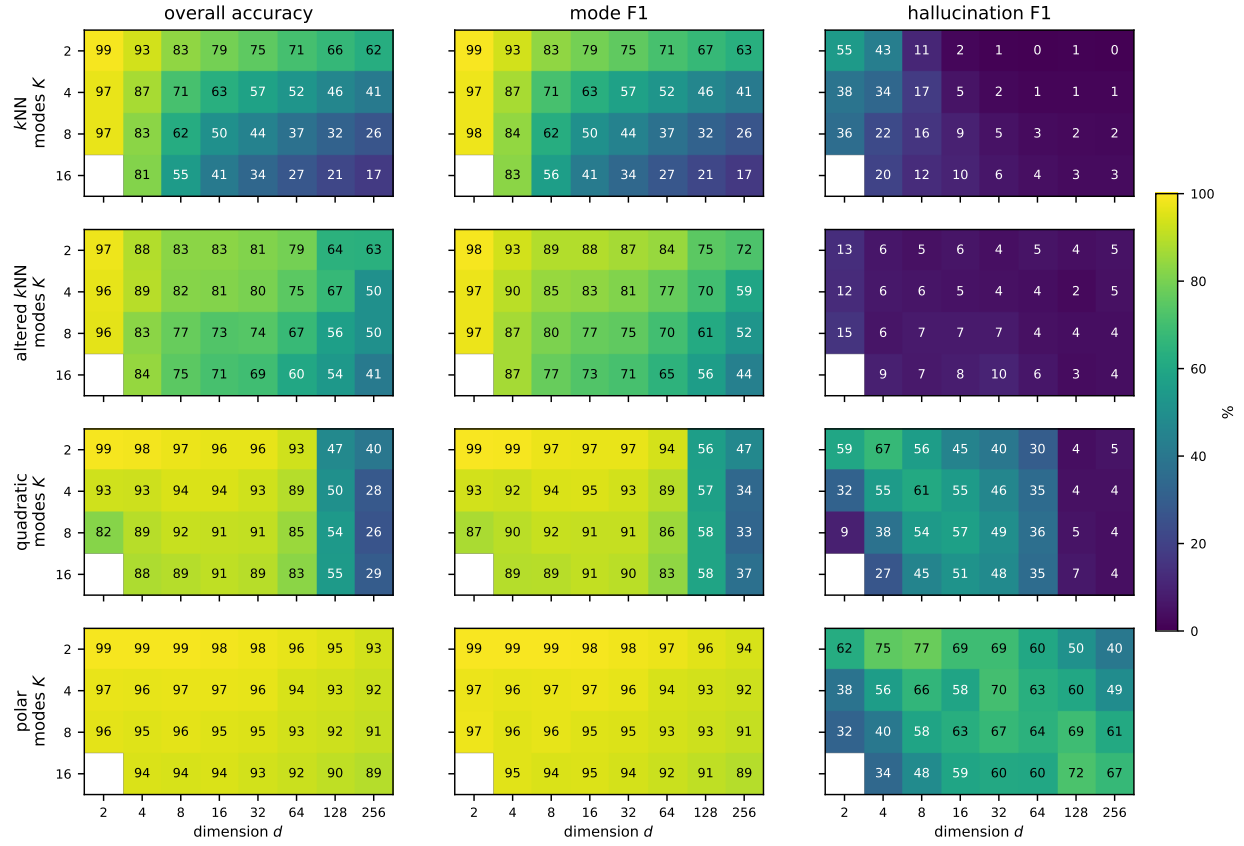


Figure 149: **Weighted DPM-Solver++(2M)**, $T = 750$, **5,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

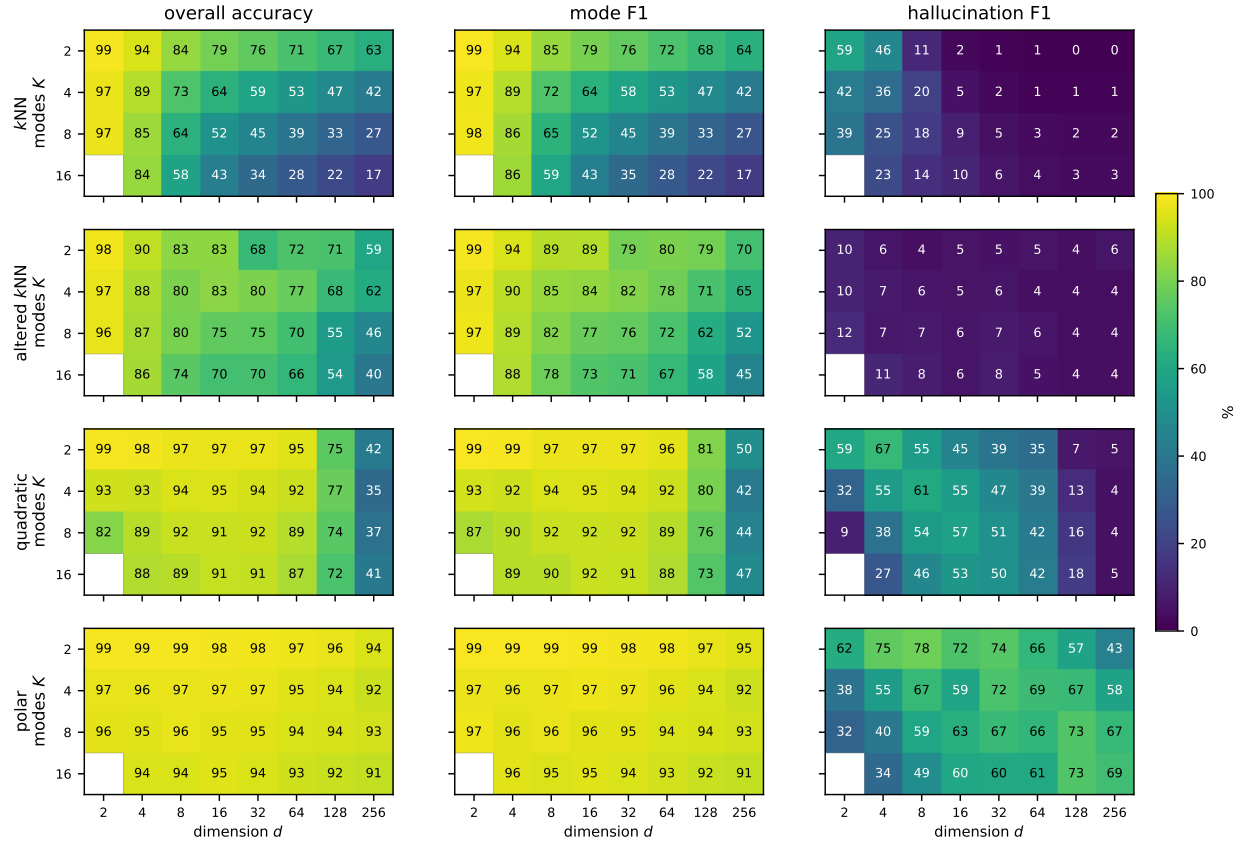


Figure 150: **Weighted DPM-Solver++(2M), $T = 750$, 10,000 anchors per mode.** Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

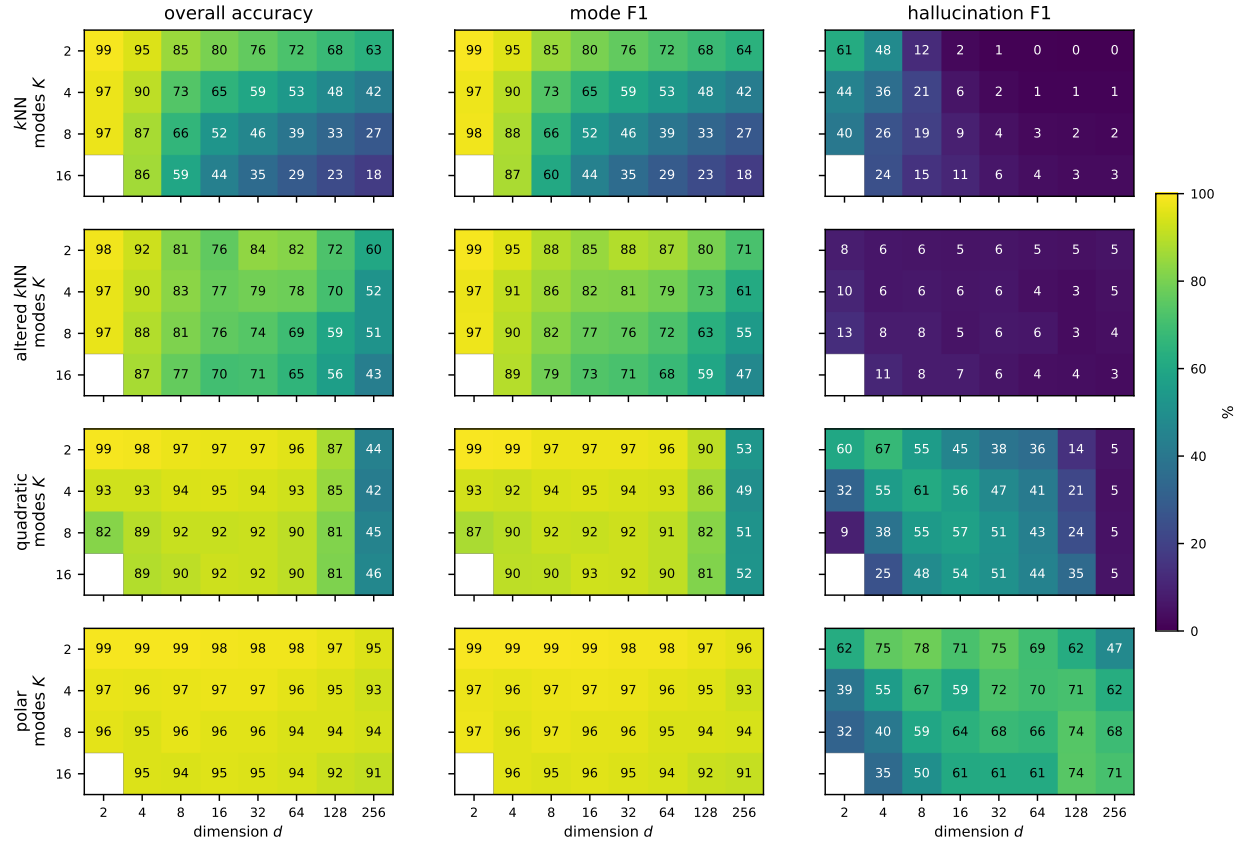


Figure 151: **Weighted DPM-Solver++(2M)**, $T = 750$, **15,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.

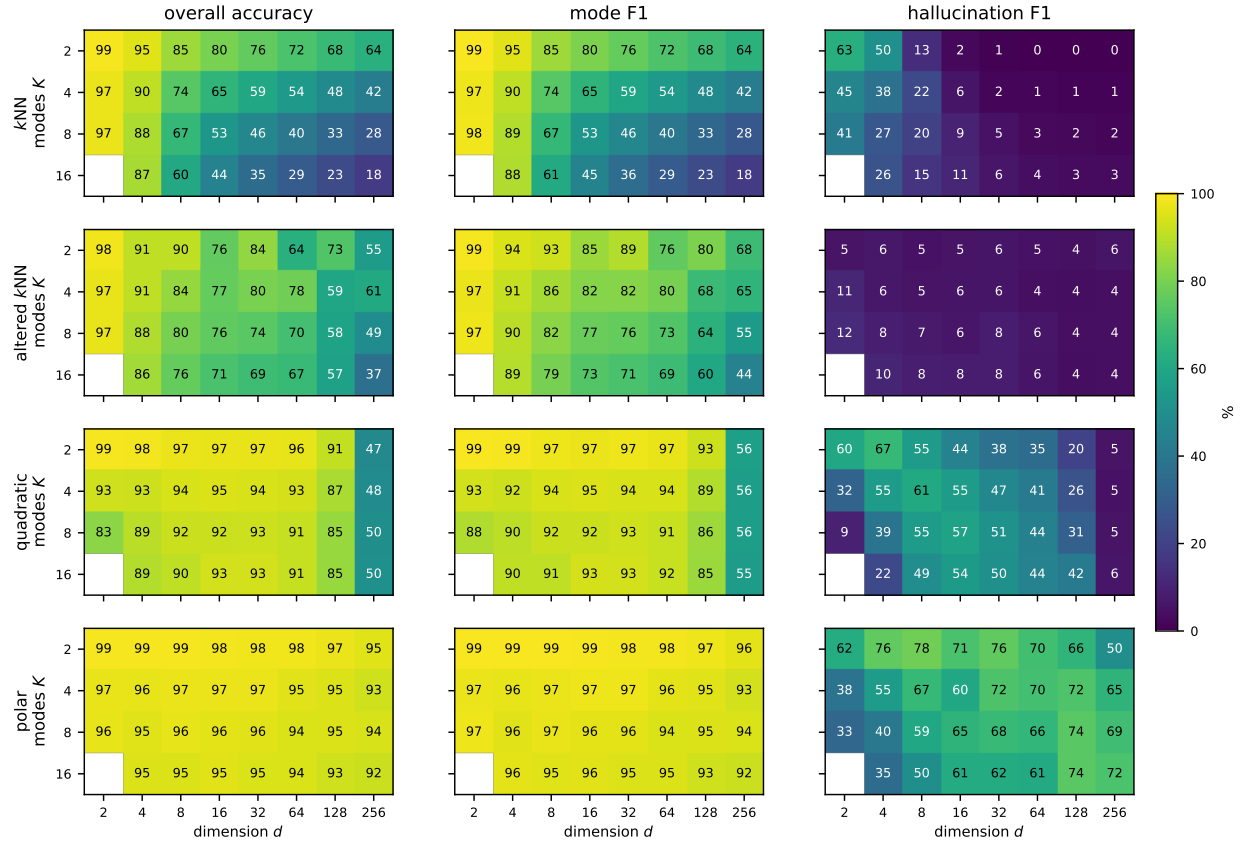


Figure 152: **Weighted DPM-Solver++(2M)**, $T = 750$, **20,000 anchors per mode**. Overall accuracy, mode F1 and hallucination F1 (% , mean over 3 seeds) per predictor (rows) on the (K, d) grid; empty cells were not run.